

Medical Symptoms

Medical Team

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Chapter 1

When to Seek Emergency Medical Care

Seek immediate emergency attention for:

- Chest pain with shortness of breath or radiation to arm/jaw
- Sudden weakness or numbness in face, arm, or leg (especially one-sided)
- Sudden confusion or trouble speaking/understanding
- Sudden vision problems in one or both eyes
- Sudden difficulty walking, dizziness, or loss of balance
- Sudden severe headache ("worst headache of life")
- High fever ($>40^{\circ}\text{C}$) or fever in infants under 3 months
- Difficulty breathing at rest
- Unconsciousness or altered consciousness
- Seizures
- Heavy/uncontrollable bleeding
- Serious burns or major trauma
- Poisoning or overdose

These symptoms may indicate life-threatening conditions including heart attack, stroke, sepsis, pulmonary embolism, meningitis. Do not delay seeking professional medical care.

Chapter 2

Abdominal Pain

Definition

Pain anywhere between chest and groin; one of most common reasons for medical consultation.

Pathophysiology

Location, character, timing provide diagnostic clues distinguishing visceral vs somatic pain.

Etiology

- Gastroenteritis
- Appendicitis
- Peptic ulcer disease
- IBS
- Gallbladder disease

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 3 days
- Accompanied by other concerning signs

Related Symptoms

- Nausea
- Vomiting

- Bloating
- Anorexia
- Constipation/Diarrhea

Note: Always consider serious underlying conditions when evaluating persistent abdominal pain.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.
- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 3

Abscess (Skin Abscess)

3.1 Definition

A skin abscess is a localized collection of pus within the dermis and subcutaneous tissue, typically caused by bacterial infection. It is a walled-off pocket of infection that requires drainage for resolution.

3.2 Pathophysiology

Bacteria (most commonly *Staphylococcus aureus*) invade the skin through a hair follicle or break in the skin. The immune system responds with neutrophils that accumulate at the site. As bacteria and dead neutrophils accumulate, pus forms. The body walls off the infection with a fibrous capsule, preventing the spread of bacteria. The increasing pressure causes pain and fluctuation.

3.3 Etiology

- Bacterial infection: *Staphylococcus aureus* (including MRSA)
- *Streptococcus pyogenes*
- Anaerobic bacteria (in areas with poor circulation)
- Blocked sweat gland or hair follicle
- Foreign body
- Intravenous drug use
- Diabetes mellitus
- Immunosuppression
- Poor hygiene

3.4 Indications for Evaluation

- Tender, red, swollen nodule under the skin
- Warmth and erythema of the overlying skin
- Fluctuance on palpation (feeling of fluid)
- Pain and tenderness
- Possible overlying pustule or draining sinus
- Fever and malaise in systemic infection
- Enlarging lymph nodes near the abscess

3.5 Related Symptoms

- Cellulitis (surrounding infection)
- Boil (furuncle, infected hair follicle)
- Carbuncle (coalescence of multiple furuncles)
- Hidradenitis suppurativa (chronic abscesses in apocrine glands)
- Sepsis

3.6 Self-Care/Home Management

- Apply warm compresses to promote drainage
- Do NOT squeeze or puncture the abscess
- Keep the area clean
- Take OTC pain relievers
- Seek medical care if fluctuant or systemic symptoms

3.7 Diagnostic Approach

- Clinical examination
- Ultrasound to confirm fluid collection
- Wound culture and sensitivity
- Blood cultures if systemic infection suspected

3.8 Treatment/Management Overview

- Incision and drainage (I and D) is the primary treatment
- Packing of the wound cavity
- Antibiotics: trimethoprim-sulfamethoxazole or clindamycin (for MRSA coverage)
- Warm compresses and wound care
- Pain management

Chapter 4

Acne

Definition

A common skin condition characterised by the formation of comedones (blackheads and whiteheads), papules, pustules, nodules, and cysts, typically on the face, chest, and back, resulting from pilosebaceous unit inflammation.

Pathophysiology

Acne vulgaris develops from four key processes: increased sebum production driven by androgens (testosterone converted to dihydrotestosterone), follicular hyperkeratinisation leading to comedone formation, *Cutibacterium acnes* colonisation of the follicle, and subsequent inflammation. In perimenopause, declining oestrogen unmasks androgen effects, increasing sebaceous gland activity. Menopausal hormonal shifts can trigger late-onset acne in previously clear skin.

Etiology

- Hormonal fluctuations (perimenopause, menopause, puberty, polycystic ovary syndrome)
- Androgen excess (increased sebum production)
- Genetic predisposition to follicular hyperkeratinisation
- *Cutibacterium acnes* overgrowth
- Comedogenic cosmetics or skincare products
- Medications (corticosteroids, lithium, androgens, certain contraceptives)
- Stress (cortisol-mediated androgen stimulation)
- High glycaemic load diet (controversial, potential insulin/IGF-1 mediated)
- Mechanical occlusion (headbands, helmets, face masks)
- Excessive facial cleansing or scrubbing (irritation)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Acne that is severe, painful, cystic, causing scarring, or unresponsive to over-the-counter treatments for 2-3 months
- Accompanied by other concerning signs

Related Symptoms

- Oily skin
- Hair loss
- Hirsutism
- Irregular periods
- Skin changes

Note: Always consider serious underlying conditions when evaluating persistent acne.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 5

Addiction Withdrawal

Definition

A syndrome of physical and psychological symptoms that occurs when a person stops or reduces use of a substance to which they have developed dependence.

Pathophysiology

Chronic substance use causes neuroadaptation — the brain compensates for the drug's effects by altering neurotransmitter systems (dopamine, GABA, glutamate, norepinephrine, opioid). Abrupt cessation removes the drug effect, unmasking the compensatory changes causing rebound hyperactivity or hypoactivity depending on the substance.

Etiology

- Alcohol withdrawal (potentially severe — DTs, seizures, death)
- Opioid withdrawal (flu-like symptoms, cravings, not life-threatening)
- Benzodiazepine withdrawal (prolonged, can cause seizures)
- Nicotine withdrawal (cigarettes, vaping)
- Stimulant withdrawal (amphetamine, cocaine — fatigue, depression)
- Cannabis withdrawal (irritability, insomnia, anxiety)
- Antidepressant discontinuation syndrome
- Caffeine withdrawal (headache, fatigue, irritability)
- Antipsychotic withdrawal
- Barbiturate withdrawal (severe, can be life-threatening)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe withdrawal symptoms, history of heavy alcohol/benzodiazepine use, suicidal thoughts, or hallucinations
- Accompanied by other concerning signs

Related Symptoms

- Anxiety
- Insomnia
- Agitation
- Sweating
- Palpitations

Note: Always consider serious underlying conditions when evaluating persistent addiction withdrawal.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 6

Allergic Reaction

6.1 Definition

An allergic reaction is an immune system response to a normally harmless substance (allergen). Reactions range from mild (sneezing, hives) to severe and life-threatening (anaphylaxis). Immediate hypersensitivity (type I) is IgE-mediated.

6.2 Pathophysiology

Allergic reactions involve sensitization and re-exposure. During first exposure, allergen-specific IgE antibodies bind to mast cells and basophils. On re-exposure, the allergen cross-links IgE, triggering mast cell degranulation and release of histamine, leukotrienes, and prostaglandins. These mediators cause vasodilation, increased vascular permeability, bronchoconstriction, and mucus secretion.

6.3 Etiology

- Foods: peanuts, tree nuts, shellfish, eggs, milk, wheat, soy
- Medications: penicillins, cephalosporins, sulfonamides, NSAIDs, aspirin
- Insect stings: bees, wasps, hornets, fire ants
- Environmental: pollen, dust mites, mold, pet dander
- Latex
- Contact: poison ivy, nickel, fragrances

6.4 Indications for Evaluation

- Skin: hives (urticaria), redness, itching, angioedema (swelling of lips, tongue, eyelids)
- Respiratory: sneezing, nasal congestion, wheezing, shortness of breath, throat tightness
- Gastrointestinal: nausea, vomiting, abdominal pain, diarrhea
- Cardiovascular: dizziness, hypotension, tachycardia, syncope
- Severe: difficulty breathing, stridor, hoarseness, loss of consciousness (anaphylaxis)

6.5 Related Symptoms

- Anaphylaxis (life-threatening)
- Urticaria (hives)
- Angioedema
- Allergic rhinitis (hay fever)
- Asthma
- Atopic dermatitis (eczema)
- Contact dermatitis

6.6 Self-Care/Home Management

- Identify and avoid known allergens
- Carry an epinephrine auto-injector if at risk for anaphylaxis
- Take OTC antihistamines for mild reactions
- Wear medical alert identification
- Inform healthcare providers of all allergies

6.7 Diagnostic Approach

- Clinical history of reaction to specific triggers
- Skin prick testing
- Specific IgE blood tests (ImmunoCAP)
- Oral food challenge in controlled setting

6.8 Treatment/Management Overview

- Epinephrine intramuscular is first-line for anaphylaxis
- Antihistamines (diphenhydramine, cetirizine) for mild reactions
- Corticosteroids for delayed reactions
- Bronchodilators for respiratory symptoms
- Allergen immunotherapy (allergy shots) for selected cases

Chapter 7

Amenorrhea (Absence of Menstruation)

7.1 Definition

Amenorrhea is the absence of menstrual bleeding in a female of reproductive age. It is classified as primary (failure to begin menstruation by age 15) or secondary (cessation of menstruation for three or more cycles or six months in a woman who previously menstruated).

7.2 Pathophysiology

Amenorrhea results from disruption of the hypothalamic-pituitary-ovarian (HPO) axis. In primary amenorrhea, causes include anatomical abnormalities (imperforate hymen, Mullerian agenesis), gonadal dysgenesis, or hypothalamic/pituitary disorders. Secondary amenorrhea typically involves anovulation due to hormonal imbalances, including hypothalamic suppression (stress, excessive exercise, low body weight), hyperprolactinemia, thyroid disorders, polycystic ovary syndrome (PCOS), premature ovarian failure, or intrauterine adhesions (Asherman syndrome).

7.3 Etiology

- Pregnancy (most common cause of secondary amenorrhea)
- Hypothalamic amenorrhea related to stress, excessive exercise, or low body weight
- Polycystic ovary syndrome (PCOS)
- Premature ovarian failure (primary ovarian insufficiency)
- Hyperprolactinemia
- Thyroid disorders (hypothyroidism or hyperthyroidism)
- Asherman syndrome (intrauterine adhesions)
- Mullerian agenesis (Mayer-Rokitansky-Kuster-Hauser syndrome)
- Gonadal dysgenesis (Turner syndrome)
- Medications (contraceptives, antipsychotics, chemotherapy)

7.4 Indications for Evaluation

- No menstruation by age 15 (primary amenorrhea)
- Absence of menstruation for three or more cycles in a previously menstruating woman
- Signs of pregnancy (nausea, breast tenderness, fatigue)
- Galactorrhea (nipple discharge suggesting hyperprolactinemia)
- Severe headaches or visual changes (possible pituitary tumor)
- Signs of androgen excess (hirsutism, acne, male-pattern balding)
- Hot flashes or vaginal dryness suggesting premature ovarian failure
- Rapid weight loss or excessive exercise history

7.5 Related Symptoms

- Irregular menstruation (oligomenorrhea)
- Heavy menstrual bleeding (menorrhagia)
- Infertility
- Hirsutism (excessive hair growth)
- Galactorrhea
- Hot flashes
- Vaginal dryness
- Mood changes
- Weight gain or weight loss

7.6 Self-Care/Home Management

- Maintain a healthy body weight through balanced nutrition
- Reduce excessive exercise if exercise-induced amenorrhea is suspected
- Manage stress through relaxation techniques and adequate sleep
- Track menstrual cycles with a calendar or app
- Discontinue medications that may contribute to amenorrhea under medical guidance
- Consider dietary adjustments if underweight or malnourished

7.7 Diagnostic Approach

- Detailed menstrual, sexual, and medication history
- Pregnancy test (serum hCG)
- Serum TSH, prolactin, FSH, LH, and estradiol levels
- Androgen profile (testosterone, DHEA-S) if signs of hyperandrogenism
- Pelvic ultrasound to evaluate uterine and ovarian anatomy
- MRI of the brain if pituitary tumor is suspected
- Hysteroscopy if Asherman syndrome is suspected

7.8 Treatment/Management Overview

- Address underlying cause (weight restoration, stress reduction, medication adjustment)
- Hormonal therapy with estrogen-progestin replacement for hypothalamic amenorrhea
- Dopamine agonists (cabergoline, bromocriptine) for hyperprolactinemia
- Thyroid hormone replacement for hypothyroidism
- Clomiphene citrate or letrozole for ovulation induction in PCOS
- Surgical correction for anatomical abnormalities or intrauterine adhesions
- Counseling regarding fertility options if premature ovarian failure is diagnosed

Chapter 8

Anal Itching (Pruritus Ani)

8.1 Definition

Anal itching (pruritus ani) is an intense, chronic itching of the skin around the anus. It affects up to 5 percent of the population and is more common in men. The condition can significantly impair quality of life due to discomfort, sleep disturbance, and social embarrassment.

8.2 Pathophysiology

Pruritus ani involves irritation of the perianal skin, which triggers the itch-scratch cycle. Scratching causes further skin damage and inflammation, perpetuating the itch. The perianal area has a thin, moist stratum corneum that is vulnerable to irritants. Fecal moisture, enzymes, and bile salts can damage the skin barrier. Secondary infection with *Candida albicans* or bacteria can exacerbate inflammation.

8.3 Etiology

- Poor hygiene or excessive cleaning (soap residue, vigorous wiping)
- Fecal soiling/incontinence leading to perianal moisture
- Hemorrhoids or anal fissures
- Pinworm infection (*Enterobius vermicularis*), especially in children
- *Candida* (yeast) infection, particularly in diabetics
- Contact dermatitis from soaps, lotions, toilet paper, or laundry detergents
- Psoriasis or seborrheic dermatitis involving the perianal area
- Food irritants (caffeine, spicy foods, citrus, chocolate, beer)
- Antibiotic use leading to secondary yeast overgrowth
- Anal fistula or other anorectal conditions

8.4 Indications for Evaluation

- Persistent perianal itching lasting more than one to two weeks
- Itching accompanied by rectal bleeding or discharge
- Visible worms in stool (suggesting pinworm infection)
- Perianal pain, swelling, or tenderness
- Skin changes such as redness, fissures, or lichenification
- Nocturnal itching (common with pinworms)
- History of diabetes, immunosuppression, or inflammatory bowel disease

8.5 Related Symptoms

- Hemorrhoids
- Anal fissure
- Rectal discharge
- Rectal pain
- Perianal rash
- Diarrhea or fecal incontinence
- Vaginal itching (*pruritus vulvae*)
- Pinworm infection

8.6 Self-Care/Home Management

- Clean the perianal area gently with plain warm water after bowel movements
- Pat dry (do not rub) and avoid harsh soaps or fragranced wipes
- Apply a thin layer of petroleum jelly or zinc oxide paste as a barrier
- Wear loose-fitting, breathable cotton underwear
- Avoid scratching; keep fingernails short
- Eliminate potential dietary triggers (caffeine, spicy foods, citrus)
- Use over-the-counter hydrocortisone cream (1

8.7 Diagnostic Approach

- Visual inspection of the perianal area
- Cellophane tape test for pinworm eggs (collected in the morning)
- Stool examination for ova and parasites if pinworm is suspected
- Skin scraping for fungal culture if Candida is suspected
- Anoscopy or sigmoidoscopy if underlying anorectal pathology is suspected
- Patch testing if contact dermatitis is suspected

8.8 Treatment/Management Overview

- Treat underlying cause (pinworm eradication, antifungal for Candida)
- Short course of topical corticosteroids for inflammation
- Topical antihistamines or pramoxine for symptomatic relief
- Behavioral modification to break the itch-scratch cycle
- Sitz baths with plain warm water after bowel movements
- Dietary modification to eliminate identified triggers
- Referral to a dermatologist or colorectal specialist for refractory cases

Chapter 9

Anal Pain

Definition

Pain or discomfort localized to the anal canal and perianal region, often sharp or throbbing, aggravated by sitting and bowel movements.

Pathophysiology

The anal canal's rich nerve supply (somatic innervation below the dentate line) makes it highly sensitive to pain. Acute anal pain commonly results from anal fissure (tear in the anal mucosa) or thrombosed external hemorrhoid. Chronic pain suggests fissure, fistula, abscess, or levator ani syndrome.

Etiology

- Anal fissure (acute or chronic)
- Thrombosed external hemorrhoid
- Perianal abscess
- Anal fistula
- Proctalgia fugax (brief sudden rectal spasm)
- Levator ani syndrome (chronic pelvic floor spasm)
- Inflammatory bowel disease (Crohn disease, UC)
- Proctitis (infectious or radiation-induced)
- Pruritus ani with skin breakdown
- Anal malignancy (rare)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe anal pain preventing bowel movements, fever, purulent drainage, or bleeding with pain
- Accompanied by other concerning signs

Related Symptoms

- Rectal bleeding
- Constipation
- Itching
- Swelling
- Fever

Note: Always consider serious underlying conditions when evaluating persistent anal pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 10

Anemia

10.1 Definition

Anemia is a condition characterized by a deficiency in the number or quality of red blood cells, leading to reduced oxygen-carrying capacity of the blood. It affects approximately one-third of the global population and manifests with fatigue, pallor, and weakness.

10.2 Pathophysiology

Anemia results from one of three mechanisms: decreased red blood cell production, increased red blood cell destruction (hemolysis), or blood loss. Decreased production can result from nutritional deficiencies (iron, vitamin B12, folate), bone marrow disorders, or chronic disease. Hemolytic anemias involve premature destruction of red blood cells due to intrinsic (sickle cell disease, thalassemia) or extrinsic (autoimmune, drug-induced) factors. Blood loss anemia follows acute hemorrhage or chronic low-grade bleeding.

10.3 Etiology

- Iron deficiency (most common cause worldwide)
- Vitamin B12 deficiency (pernicious anemia, malabsorption)
- Folate deficiency
- Chronic disease/inflammation (anemia of chronic disease)
- Acute or chronic blood loss (GI bleeding, heavy menstruation, trauma)
- Hemolytic anemias (sickle cell disease, thalassemia, autoimmune hemolytic anemia)
- Bone marrow disorders (aplastic anemia, myelodysplasia, leukemia)
- Kidney disease (decreased erythropoietin production)
- Medications (chemotherapy, antiretrovirals, some antibiotics)
- Genetic disorders (G6PD deficiency, hereditary spherocytosis)

10.4 Indications for Evaluation

- Unexplained fatigue, weakness, or lethargy
- Pallor (pale skin, conjunctiva, nail beds)
- Shortness of breath on exertion
- Dizziness or lightheadedness
- Rapid heart rate or palpitations
- Cold hands and feet
- Brittle nails or spoon-shaped nails (koilonychia) suggesting iron deficiency
- Glossitis (smooth, red tongue) suggesting B12 or folate deficiency
- Pica (craving ice, dirt, or starch) suggesting iron deficiency

10.5 Related Symptoms

- Fatigue
- Shortness of breath
- Pallor
- Dizziness
- Palpitations
- Cold intolerance
- Brittle nails
- Headache
- Restless legs syndrome (associated with iron deficiency)

10.6 Self-Care/Home Management

- Consume iron-rich foods (red meat, spinach, lentils, fortified cereals)
- Pair iron-rich foods with vitamin C (citrus, bell peppers) to enhance absorption
- Avoid tea or coffee with meals, as tannins inhibit iron absorption
- Eat foods rich in B12 (meat, fish, eggs, dairy) and folate (leafy greens, beans)
- Stay well-hydrated and rest when fatigued
- Avoid over-the-counter iron supplements without medical guidance
- Limit alcohol consumption, which can interfere with folate metabolism

10.7 Diagnostic Approach

- Complete blood count (CBC) with red cell indices
- Serum ferritin, iron, total iron-binding capacity (TIBC), transferrin saturation
- Serum vitamin B12 and folate levels
- Reticulocyte count to assess bone marrow response
- Peripheral blood smear for cell morphology
- Hemoglobin electrophoresis if hemoglobinopathy is suspected
- Bone marrow biopsy if bone marrow disorder is suspected

10.8 Treatment/Management Overview

- Oral or intravenous iron supplementation for iron deficiency
- Vitamin B12 injections or high-dose oral B12 for B12 deficiency
- Folic acid supplementation for folate deficiency
- Erythropoiesis-stimulating agents (ESA) for anemia of chronic kidney disease
- Treatment of underlying chronic condition (infection, inflammation, malignancy)
- Blood transfusion for severe symptomatic anemia
- Splenectomy for selected hemolytic anemias

Chapter 11

Anhedonia (Loss of Interest)

Definition

The reduced ability to experience pleasure or interest in activities that were previously enjoyable, a core symptom of depression that may also occur independently during hormonal transitions.

Pathophysiology

Anhedonia involves dysfunction of the brain reward circuitry, particularly the mesolimbic dopamine pathway projecting from the ventral tegmental area to the nucleus accumbens and prefrontal cortex. Oestrogen modulates dopamine receptor density and sensitivity; declining oestrogen during perimenopause reduces dopaminergic tone, blunting reward perception. Serotonergic and noradrenergic systems are also affected, contributing to diminished anticipation and experience of pleasure.

Etiology

- Major depressive disorder (most common association)
- Perimenopausal and menopausal hormonal changes
- Chronic stress (elevated cortisol suppressing dopamine)
- Schizophrenia or schizoaffective disorder
- Substance use disorders (dopamine receptor downregulation)
- Parkinson disease (dopamine depletion)
- Post-traumatic stress disorder
- Medication side effects (SSRIs, antipsychotics)
- Hypothyroidism
- Vitamin D deficiency

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent loss of interest lasting more than 2 weeks, especially with depressed mood, sleep or appetite changes, or suicidal thoughts
- Accompanied by other concerning signs

Related Symptoms

- Depression
- Fatigue
- Low mood
- Mood swings
- Difficulty concentrating

Note: Always consider serious underlying conditions when evaluating persistent anhedonia.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 12

Animal Bites

12.1 Definition

Animal bites are wounds caused by the teeth of animals. Dog bites are most common, followed by cat bites, which have a higher infection risk. Bites can cause puncture wounds, lacerations, crush injuries, and infections.

12.2 Pathophysiology

Animal teeth introduce bacteria deep into tissues. Cat bites have small, deep puncture wounds that inoculate bacteria into joint capsules and tendon sheaths. Dog bites cause more crush and tissue damage. Common pathogens include *Pasteurella multocida* (cats and dogs), *Staphylococcus*, *Streptococcus*, and anaerobes. Rabies is a rare but fatal viral infection transmitted through saliva.

12.3 Etiology

- Dog bites (most common, 80-90 percent of animal bites)
- Cat bites (higher infection risk)
- Human bites (highest infection risk, risk of hepatitis B/C, HIV)
- Rodent bites (rats, mice, hamsters)
- Wild animal bites (raccoons, bats, foxes - rabies risk)

12.4 Indications for Evaluation

- Puncture wound, laceration, or crush injury
- Pain, swelling, and redness at the bite site
- Signs of infection: purulent drainage, warmth, spreading erythema
- Fever
- Loss of function or sensation

- Deep wounds involving joints, tendons, or bones
- History of the biting animal (unknown, stray, or wild)

12.5 Related Symptoms

- Cellulitis (wound infection)
- Abscess formation
- Tetanus
- Rabies
- Osteomyelitis (bone infection)
- Septic arthritis

12.6 Self-Care/Home Management

- Wash the wound thoroughly with soap and water for 15 minutes
- Apply direct pressure if bleeding
- Elevate the wound
- Apply antibiotic ointment and cover with a clean bandage
- Seek medical evaluation for all but the most trivial bites

12.7 Diagnostic Approach

- Clinical assessment of wound depth, location, and contamination
- Wound culture if infection is present
- X-ray to rule out foreign body or fracture
- Blood tests for systemic infection

12.8 Treatment/Management Overview

- Wound irrigation and debridement
- Primary closure for cosmetic areas (face) or delayed closure for contaminated wounds
- Prophylactic antibiotics for high-risk bites (cat, human, hand wounds)
- Rabies postexposure prophylaxis if indicated
- Tetanus prophylaxis
- Elevation and wound care follow-up

Chapter 13

Ankle Pain

Definition

Pain or discomfort localized to the ankle joint and surrounding structures, including the medial and lateral malleoli, talocrural joint, and supporting ligaments and tendons.

Pathophysiology

Ankle pain most commonly results from ligamentous injury (ankle sprain) involving the lateral ligament complex (anterior talofibular, calcaneofibular, posterior talofibular). The ankle's weight-bearing role and its reliance on ligamentous stability make it vulnerable to inversion injuries. Chronic pain may arise from post-traumatic arthritis, tendinopathy (peroneal, Achilles, posterior tibial), or osteochondral defects of the talar dome.

Etiology

- Acute ankle sprain (lateral, medial, or syndesmotic)
- Ankle fracture (malleolar, bimalleolar, trimalleolar)
- Achilles tendinopathy or rupture
- Peroneal tendinopathy or subluxation
- Posterior tibial tendinopathy
- Osteochondritis dissecans of the talus
- Ankle osteoarthritis (post-traumatic)
- Gout or pseudogout
- Cellulitis or septic arthritis
- Tarsal coalition (congenital, presenting in adolescence)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Inability to bear weight after injury, severe swelling or deformity, fever with ankle pain, or chronic pain lasting more than 6 weeks
- Accompanied by other concerning signs

Related Symptoms

- Foot pain
- Leg pain
- Swelling
- Bruising
- Instability

Note: Always consider serious underlying conditions when evaluating persistent ankle pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 14

Anxiety

Definition

Anxiety represents excessive worry, nervousness, or unease about imminent events or uncertainties, often disproportionate to actual threat level, potentially interfering with daily functioning distinguishing transient fear states from generalized anxiety disorder panic disorder specific phobias clinical entities.

Pathophysiology

Anxiety involves complex neurocircuitry primarily centered on amygdala hypothalamic-pituitary-adrenal axis limbic system interactions Amygdala processes threat signals projecting to hypothalamus activating sympathetic nervous system releasing catecholamines cortisol HPA axis dysregulation observed chronic anxiety disorders Prefrontal cortex normally exerts top-down inhibition over amygdala diminished in GABAergic dysfunction serotonin norepinephrine dopamine imbalances contribute pathological anxiety states Central sensitization lowers threshold perceiving threats triggering exaggerated physiological responses including tachycardia hypertension sweating tremor respiratory changes gastrointestinal disturbances.

Etiology

- Generalized Anxiety Disorder persistent excessive worry uncontrollable symptoms >6 months Panic Disorder recurrent unexpected panic attacks agoraphobia Specific Phobias intense irrational fear objects/situations Social Anxiety Disorder fear scrutiny embarrassment Separation Anxiety Disorder excessive separation distress Obsessive Compulsive Disorder obsessions compulsions Post-Traumatic Stress Disorder trauma-related re-experiencing avoidance nightmares Acute Stress Disorder trauma within days-months Adjustment Disorder stress reaction medical illness/loss
- Medical causes: Hyperthyroidism pheochromocytoma cardiac arrhythmias hypoglycemia anemia drug withdrawal alcohol caffeine stimulants anticholinergics corticosteroids bronchodilators
- Genetic predisposition family history anxiety disorders twin studies show moderate heritability

- Environmental factors childhood trauma chronic stress adverse life events

Indications for Evaluation

Seek care if:

- Worry persisting beyond several weeks uncontrollable by reassurance
- Sleep disturbance difficulty falling/staying asleep frequent awakening
- Physical symptoms tachypnea palpitations muscle tension GI upset tremor paresthesia
- Functional impairment work school social relationships daily activities
- Safety checking behaviors reassurance seeking avoidance patterns escalating
- Suicidal ideation hopelessness accompanying severe depression anxiety comorbidity
- Panic attack frequency increasing or severity intensifying requiring emergency evaluation

Related Symptoms

- Restlessness fatigue concentration difficulties irritability muscle tension sleep disturbance palpitations diaphoresis tremor paresthesia abdominal discomfort urinary frequency fear of impending doom dread anticipation negative rumination catastrophizing hypervigilance startle response

Note: Comprehensive assessment differentiates primary psychiatric anxiety from medical etiologies thyroid function basic metabolic panel CBC toxicology screen when substance use suspected GAD-7 PHQ-9 scale screening validated instruments guide treatment Psychotherapy CBT first-line SSRIs/SNRIs pharmacotherapy benzodiazepines short-term use monitor dependence risk Referral psychiatry treatment resistant severe suicidal occupational dysfunction.

Self-Care / Home Management

- Establish a consistent daily routine with regular sleep, meals, and physical activity to support mental health stability.
- Practice stress management techniques including mindfulness, deep breathing exercises, and progressive muscle relaxation.
- Maintain social connections and avoid isolation by reaching out to trusted friends, family, or support groups.
- Avoid alcohol, caffeine, and recreational drugs which can destabilize mood and exacerbate psychiatric symptoms.
- If experiencing suicidal thoughts or crisis, contact emergency services or a crisis helpline immediately.

Diagnostic Approach

- Comprehensive psychiatric assessment includes history of present illness, past psychiatric history, substance use, medical history, and collateral information from family.
- Standardized screening tools (GAD-7, PHQ-9, PCL-5) help quantify symptom severity and track treatment response.
- Medical evaluation (thyroid function, basic metabolic panel, CBC, toxicology screen) rules out organic causes of psychiatric symptoms.
- Risk assessment for self-harm, suicide, and danger to others is an essential component of every psychiatric evaluation.

Treatment / Management Overview

- Psychotherapy (CBT, DBT, interpersonal therapy) is a first-line treatment for most psychiatric disorders.
- Pharmacotherapy with antidepressants, anxiolytics, mood stabilizers, or antipsychotics is tailored to the specific diagnosis and individual patient factors.
- Combined treatment (medication plus psychotherapy) generally yields superior outcomes compared to either modality alone.
- Referral to psychiatry is indicated for severe symptoms, treatment resistance, suicidal ideation, or diagnostic uncertainty.

Chapter 15

Arm Pain

Definition

Pain or discomfort anywhere in the upper extremity from the shoulder to the hand, arising from musculoskeletal, vascular, neurological, or referred sources.

Pathophysiology

Arm pain mechanisms include nociceptive (musculoskeletal strain, fracture, arthritis), neuropathic (cervical radiculopathy, peripheral nerve entrapment), ischemic (peripheral arterial disease, thoracic outlet syndrome), and referred (cardiac ischemia presenting as left arm pain). The brachial plexus distribution and shared cervical nerve roots mean arm pain frequently originates from the neck.

Etiology

- Cervical radiculopathy (herniated disc, foraminal stenosis)
- Rotator cuff tendinopathy or tear
- Fracture (humerus, radius, ulna)
- Lateral epicondylitis (tennis elbow)
- Medial epicondylitis (golfer's elbow)
- Carpal tunnel syndrome
- Thoracic outlet syndrome
- Bursitis (olecranon, subacromial)
- Referred cardiac pain (left arm — angina or MI)
- Peripheral neuropathy (diabetic, alcoholic)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Arm pain with chest pain, shortness of breath, or diaphoresis (possible cardiac), acute severe pain after injury, or arm pain with weakness or numbness
- Accompanied by other concerning signs

Related Symptoms

- Shoulder pain
- Neck pain
- Hand numbness
- Weakness
- Swelling

Note: Always consider serious underlying conditions when evaluating persistent arm pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 16

Ascites (Abdominal Fluid Accumulation)

16.1 Definition

Ascites is the pathological accumulation of fluid in the peritoneal cavity. It is most commonly caused by liver cirrhosis and portal hypertension. The presence of ascites indicates significant underlying disease and carries a poor prognosis in cirrhotic patients.

16.2 Pathophysiology

Ascites formation in cirrhosis results from two main mechanisms: portal hypertension and sodium retention. Increased pressure in the portal venous system forces fluid out of the hepatic sinusoids into the peritoneal space. Hypoalbuminemia (reduced oncotic pressure) exacerbates fluid leakage. Other causes involve inflammation (peritonitis), lymphatic obstruction, or increased vascular permeability.

16.3 Etiology

- Liver cirrhosis (most common cause)
- Portal hypertension (non-cirrhotic causes such as portal vein thrombosis)
- Congestive heart failure (right-sided heart failure)
- Nephrotic syndrome
- Peritoneal carcinomatosis (malignant ascites)
- Pancreatitis (pancreatic ascites)
- Tuberculous peritonitis
- Budd-Chiari syndrome (hepatic vein thrombosis)
- Myxedema (severe hypothyroidism)
- Serositis due to connective tissue diseases (lupus)

16.4 Indications for Evaluation

- Progressive abdominal distension and increased abdominal girth
- Rapid weight gain over days to weeks
- Abdominal discomfort, fullness, or pain
- Shortness of breath due to diaphragmatic compression
- Nausea and early satiety
- Umbilical hernia (from increased intra-abdominal pressure)
- Lower extremity edema
- Fever or abdominal tenderness (suggesting spontaneous bacterial peritonitis)

16.5 Related Symptoms

- Abdominal distension
- Edema (peripheral swelling)
- Shortness of breath
- Jaundice
- Abdominal pain
- Nausea and vomiting
- Umbilical hernia
- Weight gain
- Hepatomegaly or splenomegaly

16.6 Self-Care/Home Management

- Restrict dietary sodium to less than 2 g per day
- Monitor daily weight and abdominal girth
- Elevate the head of the bed for comfort
- Avoid alcohol completely if liver disease is present
- Take prescribed diuretics as directed
- Avoid NSAIDs which worsen sodium retention
- Seek medical attention if fever, worsening pain, or confusion develops

16.7 Diagnostic Approach

- Physical examination (shifting dullness, fluid wave test)
- Abdominal ultrasound to confirm ascites and assess liver morphology
- Diagnostic paracentesis with ascitic fluid analysis
- Serum-ascites albumin gradient (SAAG) to classify ascites
- Liver function tests, coagulation profile, serum albumin, renal function
- CT scan or MRI if malignancy is suspected

16.8 Treatment/Management Overview

- Sodium restriction (less than 2 g/day)
- Diuretic therapy (spironolactone with or without furosemide)
- Large-volume paracentesis for symptomatic relief
- Transjugular intrahepatic portosystemic shunt (TIPS) for refractory ascites
- Liver transplantation for end-stage liver disease
- Antibiotics for spontaneous bacterial peritonitis

Chapter 17

Back Pain

Definition

Pain along spine from neck to lower back leading cause of disability and missed work worldwide.

Pathophysiology

Originates from muscles, nerves, bones or joints; classified acute/subacute/chronic with mechanical vs inflammatory patterns.

Etiology

- Muscle strain
- Herniated disc
- Degenerative disc disease
- Osteoarthritis
- Osteoporotic fracture

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 7 days
- Accompanied by other concerning signs

Related Symptoms

- Stiffness
- Limited mobility

- Radiating leg pain
- Muscle spasms
- Reduced flexibility

Note: Always consider serious underlying conditions when evaluating persistent back pain.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.
- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 18

Bad Breath (Halitosis)

Definition

Unpleasant odor of the breath, caused primarily by volatile sulfur compounds produced by bacterial metabolism in the oral cavity, and occasionally from systemic conditions.

Pathophysiology

Anaerobic gram-negative bacteria in the oral biofilm (especially on the posterior tongue dorsum, periodontal pockets, and tonsillar crypts) degrade sulfur-containing amino acids producing hydrogen sulfide, methyl mercaptan, and dimethyl sulfide. Extrinsic sources include sinonasal, respiratory, gastrointestinal, and metabolic causes.

Etiology

- Poor oral hygiene (tongue coating, food debris)
- Periodontal disease (gingivitis, periodontitis)
- Tonsillitis or tonsil stones (tonsilloliths)
- Dry mouth (xerostomia)
- Dental caries or abscesses
- Sinusitis or postnasal drip
- Gastroesophageal reflux disease (GERD)
- Strong foods (garlic, onions, spices)
- Smoking or tobacco use
- Systemic diseases (diabetes, liver failure, kidney failure)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent halitosis despite good oral hygiene, associated with dental pain, fever, or other systemic symptoms
- Accompanied by other concerning signs

Related Symptoms

- Dry mouth
- Dental pain
- Postnasal drip
- Sore throat
- Acid reflux

Note: Always consider serious underlying conditions when evaluating persistent bad breath (halitosis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 19

Bad Taste in the Mouth (Dysgeusia)

Definition

A persistent abnormal or distorted sense of taste, ranging from a metallic, salty, bitter, or sour taste in the mouth without an external stimulus, caused by disorders of the taste buds, olfactory system, or systemic conditions.

Pathophysiology

Taste is mediated by taste buds on the tongue, palate, pharynx, and epiglottis, innervated by cranial nerves VII (chorda tympani), IX (glossopharyngeal), and X (vagus). Dysgeusia results from damage to taste receptors, alteration of saliva composition, or disruption of central taste pathways. Olfactory dysfunction (as in sinusitis or COVID-19) significantly alters flavor perception. Systemic conditions change saliva composition affecting taste receptor microenvironment.

Etiology

- Oral infections (gingivitis, periodontitis, oral thrush, sinusitis)
- Medications (metronidazole, clarithromycin, ACE inhibitors, metformin, lithium, chemotherapy)
- Dental problems (poor oral hygiene, dental caries, abscesses, amalgam fillings)
- GERD (regurgitation of acidic stomach contents)
- Sjogren syndrome or xerostomia (reduced saliva alters taste)
- Neurological disorders (Bell palsy, multiple sclerosis, stroke affecting taste pathways)
- COVID-19 infection (taste and smell dysfunction)
- Nutritional deficiencies (zinc, vitamin B12, copper)
- Smoking and tobacco use
- Radiation therapy to head and neck (damage to taste buds and salivary glands)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent bad taste with weight loss, difficulty swallowing, or any neurological symptoms (facial droop, weakness, speech changes)
- Accompanied by other concerning signs

Related Symptoms

- Bad breath
- Dry mouth
- Loss of taste
- Oral thrush
- Sinus pressure

Note: Always consider serious underlying conditions when evaluating persistent bad taste in the mouth (dysgeusia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 20

Bedwetting (Nocturnal Enuresis)

20.1 Definition

Bedwetting (nocturnal enuresis) is involuntary urination during sleep in a child aged five years or older. It affects approximately 15-20 percent of five-year-olds and resolves spontaneously in most cases, with an annual remission rate of about 15 percent.

20.2 Pathophysiology

Bedwetting involves a mismatch between the bladder's capacity to store urine overnight and the body's ability to either produce less urine or awaken in response to a full bladder. Three main mechanisms contribute: (1) nocturnal polyuria from insufficient antidiuretic hormone (ADH) secretion at night; (2) reduced functional bladder capacity; (3) impaired arousal from deep sleep in response to bladder distention.

20.3 Etiology

- Delayed maturation of the neural control of urination
- Insufficient nocturnal ADH secretion (nocturnal polyuria)
- Small functional bladder capacity
- Deep sleep pattern with impaired arousal
- Genetic predisposition (family history)
- Constipation pressing on the bladder
- Urinary tract infection
- Type 1 diabetes (polydipsia and polyuria)
- Psychological stress
- Obstructive sleep apnea

20.4 Indications for Evaluation

- Involuntary urination during sleep in a child aged five years or older
- Occurs at least twice per week for three or more months
- Daytime wetting (suggests more complex voiding dysfunction)
- Painful urination, excessive thirst, or weight loss
- Constipation or soiling (encopresis)
- Snoring or breathing difficulties during sleep
- New-onset bedwetting after six months of dryness (secondary enuresis)
- History of urinary tract infections

20.5 Related Symptoms

- Frequent urination (urinary frequency)
- Urinary urgency
- Urinary tract infection
- Constipation
- Daytime wetting (diurnal enuresis)
- Sleep apnea or snoring
- Nocturnal polyuria
- Diabetes mellitus

20.6 Self-Care/Home Management

- Limit fluids in the hour before bedtime
- Encourage regular toileting before sleep (double voiding)
- Use a moisture alarm to condition arousal to bladder fullness
- Protect the mattress with a waterproof cover
- Reward dry nights with praise
- Ensure regular bowel movements to prevent constipation
- Avoid caffeine and carbonated drinks in the evening

20.7 Diagnostic Approach

- Complete history and voiding diary
- Urinalysis and urine culture if infection is suspected
- Assessment of bowel function (constipation evaluation)
- Bladder ultrasound to assess post-void residual and bladder capacity
- Flow rate measurement if voiding dysfunction is suspected
- Sleep study if obstructive sleep apnea is suspected

20.8 Treatment/Management Overview

- Behavioral therapy: moisture alarms (first-line treatment)
- Desmopressin (synthetic ADH) for short-term dryness
- Oxybutynin if overactive bladder is suspected
- Treat underlying constipation with stool softeners and dietary fiber
- Address any underlying psychological or sleep disorder
- Combination therapy (alarm plus desmopressin) for refractory cases
- Reassurance and patience, as most cases resolve spontaneously

Chapter 21

Bent Penis (Peyronie Disease)

Definition

Acquired curvature of the erect penis caused by fibrous plaque formation in the tunica albuginea of the corpora cavernosa, often associated with pain and erectile dysfunction.

Pathophysiology

Peyronie disease results from repetitive microvascular trauma or buckling injury to the erect penis, triggering an abnormal wound-healing response. Fibrin deposition leads to chronic inflammation, dysregulated collagen production, and plaque formation in the tunica albuginea. The inelastic plaque prevents uniform expansion of the tunica during erection, causing curvature toward the side of the plaque.

Etiology

- Peyronie disease (idiopathic, most common)
- Penile trauma or fracture during intercourse
- Dupuytren contracture (associated condition)
- Genetic predisposition (family history)
- Connective tissue disorders
- Diabetes mellitus
- Autoimmune conditions
- Radiation therapy to the pelvis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms

- Sudden onset of penile curvature with severe pain, inability to have intercourse, or associated erectile dysfunction
- Accompanied by other concerning signs

Related Symptoms

- Erectile dysfunction
- Penile pain
- Penile shortening
- Dupuytren contracture
- Penile plaque

Note: Always consider serious underlying conditions when evaluating persistent bent penis.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 22

Black Eye (Periorbital Ecchymosis)

Definition

Bruising around the eye caused by blunt trauma to the periorbital region, resulting in blood accumulation in the loose subcutaneous connective tissue of the eyelids.

Pathophysiology

Blunt force fractures small blood vessels in the periorbital tissues, causing extravasation of blood into the subcutaneous space. The loose eyelid skin allows significant swelling. Blood tracks along tissue planes, often spreading to the cheek and contralateral eye. The classic black-blue-purple-yellow color progression reflects hemoglobin breakdown over 7-14 days.

Etiology

- Blunt trauma (sports injury, assault, fall)
- Nasal fracture with periorbital tracking
- Basilar skull fracture (raccoon eyes — bilateral, delayed)
- Orbital blowout fracture
- Post-surgical (eyelid surgery, rhinoplasty, sinus surgery)
- Coagulation disorders or anticoagulant medications
- Subgaleal hematoma tracking from scalp injury
- Insect bite or allergic reaction (mimics black eye)
- Periorbital cellulitis (not traumatic, requires urgent care)
- Child abuse (suspicious pattern or mechanism)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Blurry or double vision, eye pain with movement, inability to move the eye, blood in the eye, or trauma from high-velocity object
- Accompanied by other concerning signs

Related Symptoms

- Periorbital swelling
- Vision changes
- Headache
- Nasal pain
- Nausea

Note: Always consider serious underlying conditions when evaluating persistent black eye (periorbital ecchymosis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 23

Bleeding After Vaginal Sex (Postcoital Bleeding)

Definition

Vaginal bleeding or spotting occurring after sexual intercourse, ranging from light spotting to heavier flow, originating from the cervix, vagina, or uterus.

Pathophysiology

Postcoital bleeding most commonly arises from cervical pathology: cervicitis (infectious or inflammatory), cervical ectropion (exposure of columnar epithelium on the ectocervix), cervical polyps, or cervical dysplasia/neoplasia. Vaginal atrophy (especially postmenopausal) and vaginal lacerations from insufficient lubrication or trauma may also cause bleeding. Uterine sources include endometrial polyps, fibroids, or pregnancy-related conditions.

Etiology

- Cervicitis (chlamydia, gonorrhea, trichomonas, HSV)
- Cervical ectropion (ectopy)
- Cervical or endometrial polyps
- Cervical dysplasia or carcinoma
- Vaginal atrophy (genitourinary syndrome of menopause)
- Insufficient lubrication
- Vaginal or cervical trauma/laceration
- Endometritis or pelvic inflammatory disease
- Pregnancy (cervical friability)
- Coagulation disorders or anticoagulant therapy

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent or heavy postcoital bleeding, pain during intercourse, abnormal Pap smear history, or postmenopausal bleeding
- Accompanied by other concerning signs

Related Symptoms

- Vaginal bleeding between periods
- Pelvic pain
- Vaginal discharge
- Painful sex
- Abnormal Pap smear

Note: Always consider serious underlying conditions when evaluating persistent bleeding after vaginal sex.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 24

Bleeding Between Periods (Spotting)

Definition

Vaginal bleeding occurring between regular menstrual periods, ranging from light spotting to heavy flow, and including any bleeding after menopause.

Pathophysiology

Intermenstrual bleeding arises from hormonal imbalance (anovulation, estrogen breakthrough), uterine pathology (polyps, fibroids, hyperplasia), cervical lesions (polyps, cervicitis, dysplasia), or implantation bleeding in pregnancy. Postmenopausal bleeding always requires evaluation to exclude endometrial cancer.

Etiology

- Hormonal imbalance (anovulatory cycles, perimenopause)
- Uterine fibroids or polyps
- Cervical polyps, cervicitis, or ectropion
- Endometrial hyperplasia or carcinoma
- Hormonal contraception (IUD, pills, implant) breakthrough
- Pregnancy complications (ectopic pregnancy, threatened miscarriage)
- Infection (pelvic inflammatory disease, sexually transmitted infections)
- Thyroid disorders
- Coagulation disorders
- Cervical or vaginal trauma

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any postmenopausal bleeding, heavy intermenstrual bleeding, or bleeding with pain, fever, or pregnancy concerns
- Accompanied by other concerning signs

Related Symptoms

- Abnormal menstrual cycles
- Pelvic pain
- Fatigue
- Heavy periods
- Vaginal discharge

Note: Always consider serious underlying conditions when evaluating persistent bleeding between periods (spotting).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 25

Bleeding During Pregnancy

Definition

Any vaginal bleeding occurring during pregnancy, which requires immediate evaluation to exclude potentially life-threatening causes for both mother and fetus.

Pathophysiology

First-trimester bleeding (most common) results from implantation bleeding, threatened miscarriage, or ectopic pregnancy. Second- and third-trimester bleeding raises concern for placenta previa (placenta covering the cervical os), placental abruption (premature separation of the placenta from the uterine wall), or vasa previa (fetal vessels crossing the membranes over the cervical os). Uterine rupture is a rare but catastrophic cause.

Etiology

- Implantation bleeding (early, light, self-limited)
- Threatened, inevitable, or complete miscarriage
- Ectopic pregnancy (fallopian tube)
- Molar pregnancy (gestational trophoblastic disease)
- Placenta previa
- Placental abruption
- Cervical insufficiency or incompetent cervix
- Cervical or vaginal infection (cervicitis, vaginitis)
- Vasa previa
- Uterine rupture (rare, usually scar-related)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any vaginal bleeding during pregnancy requires immediate medical evaluation, especially if accompanied by pain, fever, or decreased fetal movement
- Accompanied by other concerning signs

Related Symptoms

- Abdominal pain
- Back pain
- Contractions
- Fever
- Decreased fetal movement

Note: Always consider serious underlying conditions when evaluating persistent bleeding during pregnancy.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 26

Bloating

Definition

Abdominal distension, fullness, or tightness sensation caused by excess gas accumulation in the gastrointestinal tract, often accompanied by visible swelling of the abdomen.

Pathophysiology

Gas accumulation results from swallowed air (aerophagia), bacterial fermentation of undigested food in the colon, or impaired gas transit/evacuation. Visceral hypersensitivity amplifies the perception of normal gas volumes in patients with functional bowel disorders.

Etiology

- Irritable bowel syndrome (IBS)
- Small intestinal bacterial overgrowth (SIBO)
- Constipation
- Food intolerances (lactose, fructose, gluten)
- Swallowing air (aerophagia)
- Gastroparesis or delayed gastric emptying
- Ovarian pathology (cysts, masses)
- Celiac disease
- Gastrointestinal obstruction
- Ascites from liver disease or malignancy

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent bloating lasting more than 2 weeks, bloating with abdominal pain, unexplained weight loss, or changes in bowel habits
- Accompanied by other concerning signs

Related Symptoms

- Abdominal pain
- Flatulence
- Constipation
- Diarrhea
- Nausea

Note: Always consider serious underlying conditions when evaluating persistent bloating.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 27

Blood Clots (Venous Thromboembolism)

Definition

Formation of a blood clot (thrombus) within a deep vein, most commonly in the lower extremities (deep vein thrombosis), which may dislodge and travel to the lungs (pulmonary embolism).

Pathophysiology

Virchow triad describes predisposing factors: venous stasis (immobility, surgery, long travel), endothelial injury (trauma, surgery, catheterization), and hypercoagulability (inherited thrombophilia, malignancy, pregnancy, oral contraceptives). DVT typically starts in the calf veins and may propagate proximally. Embolization to pulmonary arteries causes ventilation-perfusion mismatch, increased pulmonary vascular resistance, and potentially hemodynamic collapse.

Etiology

- Prolonged immobility (surgery, hospitalization, long-haul travel)
- Malignancy (cancer-associated thrombosis)
- Oral contraceptives or hormone replacement therapy
- Pregnancy and postpartum period
- Inherited thrombophilia (Factor V Leiden, prothrombin mutation)
- Antiphospholipid syndrome
- Trauma or major surgery (especially orthopedic)
- Central venous catheters
- Obesity
- Advanced age

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Unilateral leg swelling, warmth, and redness (possible DVT), or sudden chest pain, shortness of breath, and hemoptysis (possible PE) — seek emergency care
- Accompanied by other concerning signs

Related Symptoms

- Leg swelling
- Chest pain
- Shortness of breath
- Calf pain
- Hemoptysis

Note: Always consider serious underlying conditions when evaluating persistent blood clots.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 28

Blood in Semen (Hematospermia)

Definition

Presence of blood in the ejaculate, presenting as pink, red, or brown-tinged semen, most often benign and self-limiting but requiring careful evaluation.

Pathophysiology

Bleeding typically arises from the prostate (most common), seminal vesicles, vas deferens, or urethra. In men under 40, infection or iatrogenic causes predominate; in men over 40, prostate pathology including malignancy must be excluded. Most cases are idiopathic with spontaneous resolution.

Etiology

- Seminal vesicle or prostate infection (prostatitis, seminal vesiculitis)
- Post-prostate biopsy (most common iatrogenic cause)
- Urethral stricture or trauma
- Prostate calculi
- Sexually transmitted infections
- Prostate or seminal vesicle cysts
- Hypertension or coagulopathy
- Prostate cancer (especially in men over 40)
- Testicular or prostate trauma
- Schistosomiasis (in endemic areas)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent blood in semen beyond 3-4 weeks, recurrent episodes, or men over 40 with associated urinary symptoms
- Accompanied by other concerning signs

Related Symptoms

- Dysuria
- Urinary frequency
- Perineal pain
- Fever
- Lower back pain

Note: Always consider serious underlying conditions when evaluating persistent blood in semen (hematospermia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 29

Blood in Stool

Definition

Presence of blood appearing bright red (lower GI) or black/tarry (upper GI).

Pathophysiology

Indicates gastrointestinal bleeding requiring localization; ranges from benign hemorrhoids to serious malignancy.

Etiology

- Hemorrhoids
- Anal fissure
- Inflammatory bowel disease
- Diverticulosis
- Colorectal cancer

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Prompt evaluation
- Accompanied by other concerning signs

Related Symptoms

- Abdominal pain
- Change in bowel habits

- Weight loss
- Anemia symptoms
- Mucus/discharge

Note: Always consider serious underlying conditions when evaluating persistent blood in stool.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 30

Boils (Furuncles and Carbuncles)

30.1 Definition

A boil (furuncle) is a deep folliculitis that forms a tender, pus-filled nodule in the skin. A carbuncle is a cluster of interconnected furuncles. Boils are most commonly caused by *Staphylococcus aureus*.

30.2 Pathophysiology

Boils develop when *Staphylococcus aureus* enters the skin through a hair follicle or minor break in the skin. The body's immune response recruits neutrophils to the site, leading to pus formation. The infection extends into the deep dermis and subcutaneous tissue, forming an abscess cavity. A carbuncle occurs when multiple adjacent follicles are infected and coalesce.

30.3 Etiology

- *Staphylococcus aureus* infection (most common pathogen)
- Methicillin-resistant *Staphylococcus aureus* (MRSA)
- Streptococcal species (less common)
- Diabetes mellitus (increased susceptibility)
- Immunocompromised states (HIV, chemotherapy, organ transplantation)
- Obesity (increased skin folds and friction)
- Poor hygiene or close living conditions
- Recurrent skin trauma or friction (shaving, tight clothing)
- Iron deficiency anemia
- Contact sports with skin-to-skin transmission

30.4 Indications for Evaluation

- Tender, red, swollen nodule with a central pustule
- Enlarging, painful skin lesion over several days
- Fever, chills, or malaise accompanying a skin lesion
- Multiple or recurrent lesions
- Lesions in immunocompromised or diabetic patients
- Lesions on the face, spine, or groin
- Red streaks extending from the lesion (lymphangitis)
- Lesion that fails to drain or heal within two weeks

30.5 Related Symptoms

- Cellulitis (surrounding skin infection)
- Abscess (larger pus collection)
- Folliculitis (superficial hair follicle infection)
- Impetigo (superficial skin infection)
- Lymphangitis (red streaks from infection)
- Fever and chills (systemic signs of infection)
- Skin abscess
- Hidradenitis suppurativa

30.6 Self-Care/Home Management

- Apply warm compresses to promote drainage and relieve pain
- Keep the area clean with antibacterial soap
- Cover the boil with a clean, dry dressing once it drains
- Do NOT squeeze or lance the boil (risk of spreading infection)
- Wash hands thoroughly after touching the lesion
- Change towels, washcloths, and bed linens daily during treatment
- Avoid sharing personal items (towels, razors, clothing)

30.7 Diagnostic Approach

- Clinical diagnosis based on physical examination
- Wound culture and sensitivity if recurrent or suspected MRSA
- Blood glucose testing to screen for diabetes
- Complete blood count if systemic infection is suspected
- Incision and drainage by a healthcare professional if large or painful

30.8 Treatment/Management Overview

- Incision and drainage for fluctuant abscesses (definitive treatment)
- Warm compresses to promote spontaneous drainage
- Oral antibiotics if cellulitis, fever, or immunosuppression
- MRSA coverage (trimethoprim-sulfamethoxazole, doxycycline) if suspected
- Mupirocin topical for small, early lesions
- Chlorhexidine washes and intranasal mupirocin for carrier eradication
- Diabetes management and glycemic control to prevent recurrence

Chapter 31

Bowel Incontinence

Definition

Involuntary loss of stool varying from occasional minor leakage to complete inability control bowel movements representing significant impacting quality life social isolation depression.

Pathophysiology

Distinction between fecal incontinence urgency urge incontinence passive soiling overflow encopresis; pathophysiology involves sphincter weakness rectal hypersensitivity sensory loss reduced compliance neurological impairment obstetric trauma pelvic floor dysfunction common elderly postpartum women multiple pregnancies constipation diarrhea rectal prolapse rectocele rectoscele anal fistula.

Etiology

- Obstetric injury anal sphincter damage
- Neurological conditions stroke MS Parkinson's
- Chronic constipation rectal impaction overflow
- Pelvic floor muscle weakness
- Diarrhea inflammatory bowel disease

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than new onset progressive associated neurological deficits weight loss rectal bleeding requires specialist assessment colorectal examination imaging manometry ultrasound defecography considered referral gastroenterology coloproctology
- Accompanied by other concerning signs

Related Symptoms

- Urgency Constipation abdominal pain hemorrhoids skin irritation perianal rash urinary frequency

Note: Always consider serious underlying conditions when evaluating persistent bowel incontinence.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 32

Brain Fog

Definition

A subjective sensation of mental clouding or cognitive dysfunction characterized by difficulty concentrating, forgetfulness, slowed thinking, and reduced mental clarity.

Pathophysiology

Multifactorial disruption of cognitive processing involving neuroinflammation, altered neurotransmitter balance, impaired cerebral blood flow, hormonal dysregulation, and sleep disruption. Often associated with chronic illness, stress, hormonal changes, or medication effects.

Etiology

- Chronic fatigue syndrome (myalgic encephalomyelitis)
- Fibromyalgia
- Autoimmune diseases (lupus, multiple sclerosis, Sjogren syndrome)
- Thyroid dysfunction (hypothyroidism)
- Sleep disorders (insomnia, sleep apnea)
- Depression and anxiety
- Nutritional deficiencies (B12, vitamin D, iron)
- Medication side effects (antihistamines, benzodiazepines, opioids)
- Long COVID or post-viral syndrome
- Chronic stress or burnout

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent cognitive symptoms affecting daily function, progressive worsening, or accompanied by other neurological symptoms
- Accompanied by other concerning signs

Related Symptoms

- Fatigue
- Memory loss
- Headache
- Sleep disturbance
- Irritability

Note: Always consider serious underlying conditions when evaluating persistent brain fog.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 33

Breast Lump

Definition

A palpable mass, nodule, or thickening in the breast tissue, which may be benign (cyst, fibroadenoma, fibrocystic change) or malignant, requiring systematic evaluation to determine etiology.

Pathophysiology

Breast lumps arise from proliferation of breast tissue components: epithelial (fibroadenoma, carcinoma), connective tissue (phyllodes tumor), or cystic dilation of ducts (simple cysts). Malignant transformation involves genetic mutations (BRCA1/2, HER2, ER/PR) leading to uncontrolled cellular proliferation, angiogenesis, and potential metastasis. Hormonal influences (estrogen, progesterone) drive growth of many benign and malignant breast lesions.

Etiology

- Fibroadenoma (most common benign solid breast mass in young women)
- Simple breast cyst (fluid-filled sac, common in premenopausal women)
- Fibrocystic changes (nodularity, tenderness with menstrual cycle)
- Breast cancer (invasive ductal carcinoma most common)
- Intraductal papilloma (nipple discharge, often bloody)
- Mastitis or breast abscess (infection, often during lactation)
- Lipoma or hamartoma (benign fatty tumors)
- Fat necrosis (post-traumatic or post-surgical)
- Phyllodes tumor (rare, can be benign or malignant)
- Gynecomastia (male breast enlargement from hormonal imbalance)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any new breast lump, particularly in women over 30; lump with skin dimpling, nipple retraction, bloody discharge, or axillary lymphadenopathy
- Accompanied by other concerning signs

Related Symptoms

- Breast pain
- Nipple discharge
- Axillary lump
- Skin changes
- Swelling

Note: Always consider serious underlying conditions when evaluating persistent breast lump.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 34

Breast Tenderness (Mastalgia)

Definition

Pain, discomfort, or tenderness in one or both breasts, ranging from mild soreness to severe pain that interferes with daily activities, often cyclical in relation to hormonal fluctuations.

Pathophysiology

Cyclical mastalgia correlates with the luteal phase of the menstrual cycle when progesterone and oestrogen peak, causing breast tissue oedema, ductal proliferation, and stromal expansion. Hormone-sensitive breast tissue contains oestrogen and progesterone receptors; hormonal fluctuations during perimenopause alter the normal cyclical pattern. Non-cyclical mastalgia arises from extramammary structures (chest wall, costochondral junctions) or focal breast pathology (cysts, fibroadenomas, duct ectasia).

Etiology

- Cyclical hormonal fluctuations (perimenopause, menstrual cycle)
- Hormone replacement therapy (HRT) or oral contraceptives
- Breast cysts (fibrocystic breast changes)
- Fibroadenomas
- Mastitis (infectious or non-infectious inflammation)
- Duct ectasia
- Costochondritis (chest wall pain referred to breast)
- Trauma or prior breast surgery
- Large or heavy breasts (mechanical strain)
- Breast cancer (rare, usually painless but possible)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Breast pain with a new palpable lump, skin changes (dimpling, redness, peau d'orange), nipple discharge, or persistent focal pain lasting more than one menstrual cycle
- Accompanied by other concerning signs

Related Symptoms

- Breast lump
- Nipple discharge
- Breast swelling
- Premenstrual syndrome
- Painful sex

Note: Always consider serious underlying conditions when evaluating persistent breast tenderness.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 35

Brittle Nails (Nail Changes)

Definition

Nails that are weak, thin, split, peel, or break easily, reflecting impaired keratinisation or nutritional deficiencies affecting the nail matrix and plate.

Pathophysiology

Nail plate formation occurs in the nail matrix through keratinocyte differentiation and keratinisation. Oestrogen receptors in the nail matrix regulate keratin production; declining oestrogen during menopause reduces nail thickness and integrity. Reduced oil production (sebum) also affects the nail cuticle and hydration. Nutritional deficiencies, repeated wet-dry cycles (housework, swimming), and trauma from manicure techniques further compromise nail structure.

Etiology

- Hormonal changes of perimenopause and menopause
- Iron deficiency anaemia
- Zinc deficiency
- Biotin deficiency
- Hypothyroidism or hyperthyroidism
- Repeated exposure to water and detergents
- Fungal nail infection (onychomycosis)
- Psoriasis (nail pitting, onycholysis)
- Trauma or aggressive manicure techniques
- Raynaud phenomenon or peripheral vascular disease

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Brittle nails with nail discolouration, thickening, separation from nail bed, or associated with fatigue, pallor, or weight changes
- Accompanied by other concerning signs

Related Symptoms

- Hair loss
- Dry skin
- Fatigue
- Anaemia
- Hypothyroidism

Note: Always consider serious underlying conditions when evaluating persistent brittle nails.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 36

Bruises (Contusions and Ecchymosis)

36.1 Definition

A bruise (contusion or ecchymosis) is a discoloration of the skin caused by bleeding into the subcutaneous tissue from ruptured blood vessels following blunt trauma. Bruises evolve through characteristic color changes as hemoglobin is broken down and reabsorbed.

36.2 Pathophysiology

Bruising occurs when blunt force damages capillaries and small venules, allowing blood to extravasate into the surrounding tissue. Initially, the bruise appears red-purple due to deoxygenated hemoglobin. Over time, hemoglobin is converted to biliverdin (green color), then to bilirubin (yellow-brown). Macrophages phagocytose the breakdown products, and the bruise gradually fades over one to three weeks.

36.3 Etiology

- Accidental trauma (most common cause)
- Anticoagulant therapy (warfarin, heparin, DOACs)
- Antiplatelet therapy (aspirin, clopidogrel)
- Coagulation disorders (hemophilia, von Willebrand disease)
- Thrombocytopenia (low platelet count)
- Corticosteroid use (skin thinning and vascular fragility)
- Vitamin C deficiency (scurvy)
- Vitamin K deficiency
- Liver disease
- Aging (increased vascular fragility)
- Physical abuse

36.4 Indications for Evaluation

- Large or painful bruises without significant trauma
- Frequent or recurrent bruising with minimal provocation
- Bruises accompanied by prolonged bleeding from cuts or gums
- Spontaneous bruising (no identifiable cause)
- Bruising with nosebleeds, blood in urine or stool
- Petechiae (tiny red spots indicating platelet disorders)
- Bruising in unusual locations
- Bruising in a child with an inconsistent explanation (possible abuse)

36.5 Related Symptoms

- Petechiae (tiny capillary hemorrhages)
- Hematoma (larger collection of blood in tissue)
- Purpura (bleeding into skin)
- Epistaxis (nosebleeds)
- Gingival bleeding (bleeding gums)
- Hemarthrosis (bleeding into joints)
- Coagulopathy (bleeding tendency)
- Thrombocytopenia

36.6 Self-Care/Home Management

- Apply ice wrapped in a cloth for 15-20 minutes to reduce swelling
- Elevate the injured area if possible
- Rest the injured area to prevent further trauma
- Apply a compression bandage to minimize swelling
- Avoid heat for the first 48 hours
- Over-the-counter arnica or bromelain may reduce bruise appearance
- Avoid aspirin or NSAIDs if bruising is frequent

36.7 Diagnostic Approach

- Complete blood count (platelet count)
- Coagulation studies (PT, aPTT, INR)
- von Willebrand factor and factor VIII assays if hemophilia suspected
- Liver function tests if liver disease is suspected
- Vitamin C and vitamin K levels if deficiency suspected
- Imaging if deep hematoma or underlying fracture is suspected

36.8 Treatment/Management Overview

- Treat underlying coagulopathy (vitamin K, fresh frozen plasma, factor replacement)
- Adjust or temporarily hold anticoagulant/antiplatelet therapy if excessive
- Discontinue causative medications if possible
- Platelet transfusion for severe thrombocytopenia
- Surgical evacuation for large, tense hematomas
- Counseling and social services if abuse is suspected

Chapter 37

Bunions (Hallux Valgus)

37.1 Definition

A bunion (hallux valgus) is a bony deformity at the base of the great toe, characterized by lateral deviation of the big toe and medial prominence of the first metatarsal head. It affects approximately 23 percent of adults and is more common in women.

37.2 Pathophysiology

Bunions develop when the first metatarsal head deviates medially while the proximal phalanx deviates laterally, creating an increased hallux valgus angle (greater than 15 degrees). This misalignment causes the distal end of the first metatarsal to become prominent medially, where it rubs against footwear. Chronic pressure leads to bursal inflammation, bony proliferation, and pain.

37.3 Etiology

- Genetic predisposition (inherited foot structure)
- Tight, narrow, or high-heeled footwear
- Flat feet (pes planus) with excessive pronation
- Ligamentous laxity or hypermobility
- Rheumatoid arthritis
- Neuromuscular disorders affecting foot muscle balance
- Pes cavus (high-arched feet)
- Occupational factors (prolonged standing)
- Female sex (four times more common than in men)

37.4 Indications for Evaluation

- Visible bump on the inner side of the foot at the base of the big toe
- Pain over the bunion that worsens with walking or tight shoes
- Redness, swelling, or warmth over the joint (bursitis)
- Numbness or tingling in the big toe
- Difficulty fitting into regular shoes
- Corns or calluses on the bunion or overlapping second toe
- Restricted range of motion at the great toe joint
- Progressive worsening of the deformity

37.5 Related Symptoms

- Foot pain (metatarsalgia)
- Corns and calluses
- Hammer toe deformity
- Flat feet (pes planus)
- Arthritis of the great toe joint (hallux rigidus)
- Bursitis
- Sesamoiditis

37.6 Self-Care/Home Management

- Wear wide-toed shoes with low heels
- Use bunion pads (gel or foam) to cushion the prominence
- Apply ice to reduce acute inflammation after prolonged walking
- Take over-the-counter pain relievers as needed
- Perform toe stretches and strengthening exercises
- Use orthotic inserts to support the arch and redistribute pressure
- Avoid activities that exacerbate pain until inflammation subsides

37.7 Diagnostic Approach

- Clinical examination (inspect deformity, assess range of motion)
- Weight-bearing X-rays of the foot to measure hallux valgus angle
- Assess for associated deformities (hammer toes, metatarsalgia, flat feet)
- Evaluate vascular status and sensation of the foot

37.8 Treatment/Management Overview

- Conservative: footwear modification, orthotics, padding, ice, NSAIDs
- Physical therapy for strengthening intrinsic foot muscles
- Corticosteroid injection into the bursa if bursitis is present
- Surgical correction (bunionectomy) if conservative measures fail
- Various surgical techniques (osteotomy, arthrodesis, exostectomy)

Chapter 38

Burning Feet

Definition

A sensation of heat, burning, or warmth in the feet, often accompanied by tingling, numbness, or sharp pain, indicating underlying peripheral nerve dysfunction.

Pathophysiology

Burning feet most commonly result from peripheral neuropathy, particularly length-dependent axonal neuropathy where distal nerve endings are affected first. Metabolic (diabetes), toxic (alcohol), nutritional (B vitamin deficiencies), and inflammatory (autoimmune) causes damage to small nerve fibers (A-delta and C fibers) that transmit pain and temperature sensation, leading to aberrant nociceptive signaling.

Etiology

- Diabetic peripheral neuropathy (most common)
- Alcoholic neuropathy
- Vitamin B deficiency (B1, B6, B12)
- Peripheral arterial disease (critical limb ischemia)
- Tarsal tunnel syndrome
- Hypothyroidism
- Chemotherapy-induced peripheral neuropathy
- Chronic kidney disease (uremic neuropathy)
- Erythromelalgia (rare vascular disorder)
- Heavy metal toxicity (lead, mercury, arsenic)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Burning feet with uncontrolled diabetes, progressive weakness, skin breakdown, or weight loss
- Accompanied by other concerning signs

Related Symptoms

- Foot numbness
- Tingling
- Leg pain
- Skin changes
- Foot ulcers

Note: Always consider serious underlying conditions when evaluating persistent burning feet.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 39

Burning Mouth Syndrome

Definition

A chronic or recurrent burning sensation in the oral cavity without clinically identifiable mucosal abnormalities, commonly affecting the tongue, lips, palate, or entire mouth.

Pathophysiology

Burning mouth syndrome (BMS) involves dysfunction of the trigeminal nerve (CN V) and gustatory pathways, likely due to central and peripheral neuropathic mechanisms. Declining oestrogen alters oral mucosal sensitivity, salivary composition, and neurosensory processing. Reduced salivary flow (xerostomia) contributes to mucosal dryness and irritation. Cranial small fibre neuropathy, possibly autoimmune or hormonal in origin, is implicated in many cases. Subtypes include: Type 1 (worsens during the day), Type 2 (constant), Type 3 (intermittent with symptom-free periods).

Etiology

- Perimenopausal and menopausal hormonal changes (most common)
- Xerostomia (dry mouth from medications, autoimmune, or radiation)
- Nutritional deficiencies (iron, zinc, B vitamins, folate)
- Oral candidiasis (thrush)
- Gastro-oesophageal reflux disease (acid reaching the mouth)
- Medication side effects (ACE inhibitors, antiretrovirals, chemotherapy)
- Anxiety and depression (psychogenic BMS)
- Allergic contact stomatitis (dental materials, flavourings)
- Diabetes mellitus (peripheral neuropathy)
- Geographic tongue or oral lichen planus

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Burning mouth lasting more than 2 weeks with weight loss, difficulty eating, mucosal lesions, or associated systemic symptoms
- Accompanied by other concerning signs

Related Symptoms

- Dry mouth
- Altered taste
- Oral thrush
- Mouth ulcers
- Bad breath

Note: Always consider serious underlying conditions when evaluating persistent burning mouth syndrome.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 40

Burping (Eructation)

Definition

The involuntary or voluntary expulsion of gas from the upper gastrointestinal tract through the mouth, caused by swallowed air (aerophagia) or intraluminal gas production, often accompanied by abdominal distension and discomfort.

Pathophysiology

Burping occurs when swallowed air accumulates in the stomach and is expelled via relaxation of the upper esophageal sphincter. Aerophagia (excessive air swallowing) occurs during eating, drinking, chewing gum, smoking, or anxiety. Supragastric belching involves voluntary air aspiration into the esophagus followed by immediate expulsion without gastric involvement. Gastric fermentation from certain foods or bacterial overgrowth can increase intraluminal gas production.

Etiology

- Aerophagia (eating too quickly, talking while eating, chewing gum, drinking carbonated beverages)
- Gastroesophageal reflux disease (GERD)
- Functional dyspepsia
- Gastroparesis (delayed gastric emptying)
- *Helicobacter pylori* infection
- Chronic pancreatitis or pancreatic insufficiency
- Celiac disease or lactose intolerance
- Small intestinal bacterial overgrowth (SIBO)
- Hiatal hernia
- Anxiety, stress, or behavioral factors (supragastric belching)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Burping accompanied by severe abdominal pain, vomiting, weight loss, or signs of GI bleeding; inability to pass gas with abdominal distension (possible obstruction)
- Accompanied by other concerning signs

Related Symptoms

- Bloating
- Abdominal pain
- Heartburn
- Nausea
- Indigestion

Note: Always consider serious underlying conditions when evaluating persistent burping (eructation).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 41

Calf Pain

Definition

Pain localized to the posterior lower leg (gastrocnemius and soleus muscles), ranging from mild soreness to severe debilitating pain.

Pathophysiology

Calf pain most commonly arises from musculoskeletal strain (gastrocnemius or soleus tears), but critical vascular causes (DVT, compartment syndrome, arterial occlusion) must be excluded. The calf muscles are particularly vulnerable to strain during activities involving sudden push-off (sprinting, jumping).

Etiology

- Gastrocnemius or soleus strain (tennis leg)
- Deep vein thrombosis (DVT)
- Muscle contusion or hematoma
- Achilles tendinopathy or rupture
- Baker's cyst rupture (pseudothrombophlebitis)
- Peripheral arterial disease (claudication)
- Compartment syndrome
- Cellulitis or soft tissue infection
- Electric contusion (charlie horse)
- Ruptured popliteal aneurysm

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden severe calf pain with swelling, redness, and warmth (DVT concern), or pain after trauma with inability to bear weight
- Accompanied by other concerning signs

Related Symptoms

- Leg swelling
- Knee pain
- Ankle pain
- Numbness or tingling
- Redness or warmth

Note: Always consider serious underlying conditions when evaluating persistent calf pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 42

Canker Sores (Aphthous Ulcers)

42.1 Definition

Canker sores (aphthous ulcers) are shallow, painful ulcers that develop on the oral mucosa. They affect approximately 20 percent of the population and are classified as minor (less than 1 cm), major (greater than 1 cm), or herpetiform.

42.2 Pathophysiology

The exact pathogenesis involves an immune-mediated attack on oral epithelial cells, driven by T-cell activation and cytokine release. Focal necrosis of the epithelium results in a painful, crater-like defect covered by a grayish fibrinous membrane surrounded by an erythematous halo. The ulcers heal spontaneously by epithelial regeneration over one to three weeks.

42.3 Etiology

- Minor trauma (biting the cheek, sharp foods, dental appliances)
- Stress and anxiety
- Hormonal changes (menstruation, pregnancy)
- Food sensitivities (spicy, acidic foods; chocolate, nuts, cheese)
- Nutritional deficiencies (vitamin B12, folate, iron, zinc)
- Gluten sensitivity/celiac disease
- Immunodeficiency states
- Medications (NSAIDs, beta-blockers)
- Family history (genetic predisposition)
- Autoimmune conditions (Behcet disease, IBD)

42.4 Indications for Evaluation

- Painful, shallow ulcers on non-keratinized oral mucosa
- Ulcers with a grayish-white center and red border
- Recurrent episodes lasting 7-14 days
- Difficulty eating, drinking, or speaking due to pain
- Prodromal burning or tingling sensation
- Ulcers with fever, rash, or genital ulcers (possible Behcet disease)
- Ulcers persisting longer than two weeks

42.5 Related Symptoms

- Oral pain (mouth pain)
- Difficulty swallowing (dysphagia)
- Mouth sores (oral ulcerations)
- Cold sores (herpes labialis)
- Glossitis (tongue inflammation)
- Gingivitis (gum inflammation)
- Behcet disease

42.6 Self-Care/Home Management

- Avoid spicy, acidic, or rough foods during outbreaks
- Use over-the-counter topical preparations (benzocaine)
- Apply milk of magnesia or baking soda paste
- Rinse with salt water several times daily
- Use a soft-bristled toothbrush
- Identify and avoid personal dietary triggers
- Consider vitamin B12, folate, or iron supplements if deficient

42.7 Diagnostic Approach

- Clinical diagnosis based on appearance and history
- Biopsy for atypical, persistent, or unusually large ulcers
- Blood tests for nutritional deficiencies
- Tests for celiac disease if gastrointestinal symptoms present
- Evaluation for Behcet disease if oral and genital ulcers coexist

42.8 Treatment/Management Overview

- Topical corticosteroids (triamcinolone acetonide)
- Topical analgesics (lidocaine, benzocaine)
- Oral corticosteroid short course for severe cases
- Immunomodulatory agents (colchicine) for refractory cases
- Antimicrobial mouth rinses (chlorhexidine)
- Chemical cautery (silver nitrate) for rapid pain relief
- Nutritional supplementation if deficiencies identified

Chapter 43

Catarrh

Definition

Inflammatory exudate accumulating mucous membranes sinuses nasal passages throat causing congestion postnasal drip difficulty breathing through nose cough hoarseness throat clearing sensation.

Pathophysiology

Common upper respiratory infection allergic rhinitis sinusitis occupational irritant exposure dry air pollution smoke environmental allergens seasonal variations temperature humidity changes chronic conditions gastroesophageal reflux disease contributing mechanism increased mucosal blood vessel permeability glandular secretion edema obstruction impaired drainage pathways recurrent infections polyps anatomical deviations deviated septum concha bullosa nasal tumors.

Etiology

- Allergic rhinitis hay fever
- Upper respiratory viral infections
- Sinusitis acute chronic
- Gastroesophageal reflux disease
- Environmental irritants pollutants smoke

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent beyond typical cold duration one-sided unilateral obstruction blood-tinged discharge severe facial pain visual changes orbital swelling periorbital tenderness ophthalmologic emergency suspected orbital cellulitis intracranial complications require urgent imaging surgical consultation otolaryngology
- Accompanied by other concerning signs

Related Symptoms

- Sneezing runny nose sore throat cough
- Postnasal drip headache facial pressure
- Reduced sense of smell sleep disturbance snoring daytime fatigue
- Cognitive impairment concentration difficulty
- Voice changes hoarseness laryngitis

Note: Always consider serious underlying conditions when evaluating persistent catarrh.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 44

Cellulite

44.1 Definition

Cellulite is a common cosmetic condition characterized by dimpled, uneven skin surface, typically on the thighs, buttocks, and abdomen. It affects 80-90 percent of post-adolescent women.

44.2 Pathophysiology

Cellulite results from structural changes in the subcutaneous fat layer and its fibrous connective tissue. In women, the fibrous septae connecting the dermis to the deep fascia run vertically, creating compartments that allow fat protrusion into the dermis when fat hypertrophies. This creates the characteristic cottage cheese or orange peel appearance.

44.3 Etiology

- Female sex (female-pattern fat distribution)
- Genetic predisposition
- Hormonal factors (estrogen, insulin, prolactin)
- Age-related skin thinning and collagen loss
- Increased body fat percentage
- Sedentary lifestyle
- Diet high in processed foods, salt, and carbohydrates
- Dehydration (reduced skin elasticity)
- Pregnancy (hormonal and weight changes)

44.4 Indications for Evaluation

- Dimpled, uneven skin on the thighs, buttocks, hips, or abdomen
- No associated pain, redness, or systemic symptoms

- Skin may feel cooler or have reduced sensation
- No ulceration, discharge, or signs of infection

44.5 Related Symptoms

- Obesity or overweight
- Skin laxity (loss of skin elasticity)
- Poor circulation (venous or lymphatic insufficiency)
- Stretch marks (striae)
- Lymphedema (different condition)

44.6 Self-Care/Home Management

- Maintain a healthy weight through balanced diet and regular exercise
- Strength training to build muscle and improve skin tone
- Stay well-hydrated to maintain skin elasticity
- Dry brushing to stimulate circulation
- Topical creams containing caffeine or retinol (temporary improvement)
- Massage therapy to improve local circulation
- Accept that cellulite is a normal variation

44.7 Diagnostic Approach

- Clinical diagnosis based on visual inspection
- No laboratory or imaging tests are necessary

44.8 Treatment/Management Overview

- Lifestyle modification: weight loss, exercise, healthy diet
- Topical treatments (retinol, caffeine) minimal temporary benefit
- Laser and radiofrequency therapies for mild improvement
- Subcision (Cellfina) to release fibrous septae
- Acoustic wave therapy
- Endermologie (mechanical massage) for temporary reduction

Chapter 45

Chest Pain

Definition

Chest pain is discomfort, pressure, squeezing, burning, sharpness, or heaviness in the anterior chest wall, potentially originating from cardiac, pulmonary, gastrointestinal, musculoskeletal, or psychological structures, ranging from benign to life-threatening etiologies requiring prompt assessment.

Pathophysiology

Cardiac ischemia activates myocardial sympathetic afferent fibers releasing substance P and CGRP producing retrograde somatic convergence in spinal cord (T1-T4) interpreted as chest pain pericarditis involves parietal pericardium innervated by phrenic nerves producing sharp positional pain pleuritic chest pain arises from parietal pleura innervated by intercostal nerves aggravated by respiration Esophageal pain follows vagal/sympathetic pathways mimicking angina Musculoskeletal pain derives from costosternal junctions, ribs, muscles with reproducible tenderness Anxiety/panic disorders produce chest tightness through hyperventilation and heightened thoracic awareness

Etiology

- Cardiac: Acute coronary syndrome/unstable angina/myocardial infarction, pericarditis, myocarditis, aortic dissection, hypertrophic cardiomyopathy, mitral valve prolapse
- Pulmonary: Pulmonary embolism, pneumothorax, pneumonia, pleurisy, COPD exacerbation, pulmonary hypertension
- Gastroesophageal: GERD, esophageal spasm, peptic ulcer disease, pancreatitis, cholecystitis
- Musculoskeletal: Costochondritis/Tietze syndrome, muscle strain, rib fracture, fibromyalgia
- Psychiatric: Panic attack, anxiety, somatization
- Other: Herpes zoster (prodromal phase), anxiety-induced chest tightness, factitious pain

Indications for Evaluation

Seek care if:

- Chest pain with diaphoresis, nausea, vomiting, syncope suggesting ACS
- Chest pain with sudden onset/dyspnea/hemoptysis suggesting PE
- Sudden "tearing" chest pain suggesting dissection
- Fever/respiratory symptoms suggesting pneumonia
- Positional pain relieved/sit forward suggesting pericarditis
- Episodic substernal pressure with exertion suggesting angina
- New onset in patient >40 years with CV risk factors

Related Symptoms

- Dyspnea
- Orthopnea
- Paroxysmal nocturnal dyspnea
- Palpitations
- Dizziness
- Presyncope
- Syncope
- Cough
- Hemoptysis
- Heartburn
- Regurgitation
- Odynophagia
- Arm/jaw/back radiation
- Sweating
- Nausea/vomiting
- Anxiety
- Fear of impending doom

Note: Immediate assessment prioritizes ruling out life-threatening causes (ACS, PE, dissection, tension pneumothorax, cardiac tamponade). ECG, troponin, D-dimer, CXR typically first-line. History includes character, location, duration, radiation, aggravating/alleviating factors, associated symptoms, CV risk factors. Never dismiss chest pain without adequate evaluation regardless of apparent benign explanation.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.
- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 46

Chest Tightness and Discomfort

Definition

A subjective sensation of pressure, squeezing, constriction, or fullness in the chest area, distinct from sharp or stabbing chest pain but equally requiring careful evaluation.

Pathophysiology

Chest tightness may reflect bronchoconstriction (asthma, COPD), esophageal spasm (GERD, motility disorder), anxiety/hyperventilation (muscle tension), or cardiac ischemia (angina presenting as tightness without frank pain). Differentiating cardiac from non-cardiac causes is critical.

Etiology

- Asthma or reactive airway disease
- Gastroesophageal reflux disease (GERD) with esophageal spasm
- Anxiety, panic attacks, or hyperventilation syndrome
- Angina pectoris or coronary artery disease
- Chronic obstructive pulmonary disease (COPD) exacerbation
- Costochondritis (chest wall tenderness)
- Pulmonary embolism
- Pneumonia or pleurisy
- Laryngeal or tracheal irritation
- Heart failure (especially with orthopnea)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Chest tightness with shortness of breath, radiating pain, sweating, nausea, or lightheadedness (possible cardiac cause — emergency)
- Accompanied by other concerning signs

Related Symptoms

- Shortness of breath
- Wheezing
- Cough
- Anxiety
- Palpitations

Note: Always consider serious underlying conditions when evaluating persistent chest tightness and discomfort.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 47

Chills

Definition

Shivering sensation often accompanying fever as body tries to raise temperature set point.

Pathophysiology

Physiological response during febrile illness characterized by peripheral vasoconstriction and shivering thermogenesis.

Etiology

- Bacterial infection
- Malaria
- Drug reaction
- Post-operative state
- Heat stroke recovery

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 24 hours With fever or persistent
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Sweating

- Malaise
- Headache
- Muscle aches

Note: Always consider serious underlying conditions when evaluating persistent chills.

Self-Care / Home Management

- Rest and maintain adequate hydration with water, clear fluids, or oral rehydration solutions.
- Use acetaminophen or ibuprofen to reduce fever and body aches (follow label dosing instructions).
- Dress lightly and keep the room at a comfortable temperature — avoid bundling excessively.
- Monitor temperature at regular intervals and track other symptoms to identify worsening trends.
- Seek medical attention for fever above 40 °C (104 °F), fever lasting more than 3 days, or fever with stiff neck, severe headache, rash, or difficulty breathing.

Diagnostic Approach

- History includes fever duration, pattern (intermittent, remittent, continuous), associated symptoms (chills, rigors, sweats), and exposure history.
- Vital signs assess fever severity and hemodynamic stability; complete physical examination seeks a source.
- Laboratory evaluation includes CBC with differential, blood cultures, urinalysis, inflammatory markers (CRP, ESR), and targeted testing based on exposure history.
- Imaging (chest X-ray, abdominal ultrasound, CT) is guided by clinical findings and suspected source.

Treatment / Management Overview

- Antipyretics (acetaminophen, NSAIDs) provide symptomatic relief but do not treat the underlying cause.
- Empiric antibiotics may be indicated for suspected bacterial infection after appropriate cultures are obtained.
- Antiviral, antifungal, or antimalarial therapy is directed at specific pathogens based on exposure and testing.
- Source control (drainage of abscess, removal of infected device) is essential for focal infections.

Chapter 48

Choking (Airway Obstruction)

Definition

Partial or complete blockage of the airway by a foreign object, food, or anatomical structure, preventing adequate breathing and requiring immediate intervention.

Pathophysiology

Foreign body lodges in the pharynx, larynx, or trachea causing mechanical obstruction. Partial obstruction allows some air exchange (coughing, wheezing, stridor). Complete obstruction causes silent distress with inability to cough, speak, or breathe, leading to hypoxia, loss of consciousness, and cardiac arrest within minutes without intervention.

Etiology

- Food aspiration (meat, nuts, seeds, hot dogs, grapes)
- Small objects (coins, marbles, toy parts, buttons)
- Inadequate chewing or eating too fast
- Alcohol intoxication (impaired swallowing reflex)
- Dental issues or poorly fitting dentures
- Neurological conditions (dysphagia from stroke, Parkinson, dementia)
- Anaphylaxis (laryngeal edema causing airway closure)
- Croup or epiglottitis (infectious airway narrowing)
- Trauma causing laryngeal fracture or airway compression
- Tongue obstruction in unconscious patient

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Complete airway obstruction (unable to cough, speak, or breathe) requires immediate emergency response and abdominal thrusts (Heimlich maneuver)
- Accompanied by other concerning signs

Related Symptoms

- Shortness of breath
- Stridor
- Cough
- Cyanosis
- Loss of consciousness

Note: Always consider serious underlying conditions when evaluating persistent choking (airway obstruction).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 49

Chronic Wounds

Definition

Wounds that fail to proceed through orderly healing stages within 4-6 weeks, including pressure ulcers, venous stasis ulcers, diabetic foot ulcers, and arterial insufficiency ulcers.

Pathophysiology

Healing failure results from prolonged inflammation, impaired angiogenesis, bacterial biofilm formation, and matrix metalloproteinase imbalance. Underlying causes include ischemia (arterial disease), venous hypertension (venous insufficiency), neuropathy and pressure (diabetes), or prolonged pressure (immobility). Biofilm bacteria evade immune clearance and resist antibiotics.

Etiology

- Venous insufficiency (venous stasis ulcers)
- Peripheral arterial disease (ischemic ulcers)
- Diabetes mellitus (diabetic foot ulcers)
- Prolonged pressure (pressure ulcers/bedsores)
- Chronic infection or osteomyelitis
- Vasculitis or autoimmune disease
- Malignancy (basal cell carcinoma, squamous cell carcinoma)
- Radiation therapy damage
- Lymphedema
- Pyoderma gangrenosum

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any non-healing wound persisting more than 4 weeks, wound with increasing pain, redness, swelling, or purulent drainage
- Accompanied by other concerning signs

Related Symptoms

- Leg swelling
- Skin discoloration
- Pain
- Fever
- Reduced mobility

Note: Always consider serious underlying conditions when evaluating persistent chronic wounds.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 50

Clitoral Pain (Clitorodynia)

Definition

Pain, discomfort, hypersensitivity, or altered sensation localised to the clitoris, which may be burning, stabbing, or aching, and can occur spontaneously or with stimulation.

Pathophysiology

Clitoral tissue is densely innervated with sensory nerve endings and contains abundant oestrogen and androgen receptors. Declining oestrogen and testosterone during menopause cause clitoral tissue atrophy, reduced blood flow, and altered nerve sensitivity. The clitoral nerve is a terminal branch of the pudendal nerve (S2-S4); entrapment or irritation may cause referred or localised pain. Vulvodynia and clitorodynia share overlapping neuropathic and musculoskeletal mechanisms.

Etiology

- Genitourinary syndrome of menopause (clitoral atrophy)
- Vulvodynia (generalised or localised vulvar pain)
- Pudendal nerve entrapment or neuralgia
- Trauma (childbirth, surgery, sexual)
- Lichen sclerosus or lichen planus
- Recurrent candidiasis or fungal infection
- Contact dermatitis (soaps, lubricants, laundry products)
- Peyronie disease (affecting clitoral suspensory ligament, rare)
- Endometriosis of the clitoris (rare)
- Psychosexual factors (anxiety, history of abuse)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Clitoral pain with visible lesions, discharge, fever, or associated with bladder/bowel symptoms, or pain that interferes with daily function
- Accompanied by other concerning signs

Related Symptoms

- Vulvovaginal irritation
- Vaginal dryness
- Painful sex
- Pelvic pain
- Labia changes

Note: Always consider serious underlying conditions when evaluating persistent clitoral pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 51

Cognitive Deficits (Brain Fog and Memory Lapses)

51.1 Definition

Cognitive deficits refer to impairments in mental processes such as memory, attention, executive function, language, and visuospatial skills. These deficits can range from mild subjective complaints to severe functional impairment as in dementia.

51.2 Pathophysiology

Cognitive function depends on complex neural networks distributed across the brain. Deficits arise when these networks are disrupted by structural damage (stroke, trauma), neurodegenerative processes (Alzheimer disease), metabolic derangements, neuroinflammation, or psychiatric conditions. Cholinergic circuits in the basal forebrain and hippocampus are particularly vulnerable.

51.3 Etiology

- Neurodegenerative diseases (Alzheimer disease, frontotemporal dementia)
- Cerebrovascular disease (stroke, vascular dementia)
- Traumatic brain injury
- Mild cognitive impairment (MCI)
- Depression, anxiety, or chronic stress (pseudodementia)
- Sleep disorders (sleep apnea, insomnia)
- Vitamin B12 deficiency
- Thyroid dysfunction
- Medication side effects (anticholinergics, benzodiazepines, opioids)
- Alcohol or substance abuse
- Chronic medical conditions (renal failure, hepatic encephalopathy)

51.4 Indications for Evaluation

- Memory loss that disrupts daily life
- Difficulty with planning, problem-solving, or multitasking
- Word-finding difficulty or trouble following conversations
- Getting lost in familiar places
- Changes in judgment or decision-making
- Personality changes, apathy, or social withdrawal
- Difficulty with familiar tasks (cooking, finances)
- Onset of confusion or delirium

51.5 Related Symptoms

- Memory loss
- Confusion
- Disorientation
- Difficulty concentrating
- Fatigue
- Mood changes (depression, anxiety, irritability)
- Sleep disturbances
- Headache

51.6 Self-Care/Home Management

- Engage in regular mental stimulation (reading, puzzles)
- Maintain a consistent daily routine and use memory aids
- Get adequate sleep (7-9 hours per night)
- Exercise regularly to improve cerebral blood flow
- Eat a Mediterranean-style diet rich in omega-3 fatty acids
- Reduce alcohol consumption
- Stay socially connected
- Manage chronic conditions (hypertension, diabetes, depression)

51.7 Diagnostic Approach

- Detailed history from patient and family member
- Cognitive screening (Montreal Cognitive Assessment)
- Neuropsychological testing
- Brain imaging (MRI or CT) to identify structural causes
- Blood tests: vitamin B12, TSH, complete blood count
- Assessment for depression
- Lumbar puncture if early-onset dementia suspected

51.8 Treatment/Management Overview

- Treat reversible causes: replace B12, correct thyroid function
- Cholinesterase inhibitors (donepezil, rivastigmine) for Alzheimer disease
- Memantine for moderate-to-severe Alzheimer disease
- Antidepressants for depression-related cognitive impairment
- Cognitive rehabilitation and occupational therapy
- Management of vascular risk factors
- Referral to a memory clinic or neurologist

Chapter 52

Cold Hands and Feet (Raynaud Phenomenon)

Definition

A vasospastic disorder characterized by episodic digital ischemia triggered by cold or stress, causing well-demarcated color changes (white, blue, red) in the fingers and/or toes, often with numbness, tingling, and pain during rewarming.

Pathophysiology

Raynaud phenomenon involves exaggerated vasoconstriction of digital arteries and arterioles in response to cold or emotional stress. Primary Raynaud is idiopathic, benign, and associated with hyperactivation of alpha-2 adrenergic receptors on digital vessels. Secondary Raynaud involves underlying pathology (scleroderma, lupus) causing structural vessel damage, intimal hyperplasia, and obliterative microvascular disease, leading to more severe ischemia and potential digital ulceration.

Etiology

- Primary Raynaud disease (benign, no underlying cause, onset before age 30)
- Connective tissue diseases (scleroderma, systemic lupus, mixed connective tissue disease)
- Cryoglobulinemia (abnormal cold-precipitating proteins)
- Buerger disease (thromboangiitis obliterans, linked to smoking)
- Atherosclerosis or peripheral artery disease
- Thoracic outlet syndrome (neurovascular compression)
- Medications (beta-blockers, ergotamines, chemotherapy bleomycin, cisplatin)
- Repetitive vibration injury (hand-arm vibration syndrome from power tools)
- Hypothyroidism (reduced metabolic rate, cold intolerance)
- Carpal tunnel syndrome (associated with vasomotor symptoms)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Unilateral symptoms, skin ulceration or gangrene, symptoms beginning after age 30, or associated with rash, joint pain, or systemic symptoms
- Accompanied by other concerning signs

Related Symptoms

- Numbness
- Tingling
- Finger swelling
- Joint pain
- Digital ulcers

Note: Always consider serious underlying conditions when evaluating persistent cold hands and feet (raynaud phenomenon).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 53

Cold Intolerance (Feeling Cold)

Definition

An exaggerated sensitivity to cold temperatures or a persistent sensation of feeling cold even in normal environmental conditions, often with visible vasoconstriction of the extremities.

Pathophysiology

Cold intolerance results from altered thermoregulation, reduced metabolic heat production, or impaired vasomotor control. Oestrogen influences hypothalamic thermoregulatory set-point; declining oestrogen destabilises the thermoregulatory centre, causing both hot flushes and cold spells. Reduced basal metabolic rate (age-related and thyroid-related) decreases heat production. Vasomotor symptoms alternate between vasodilation (hot flushes) and vasoconstriction (cold flushes) as the hypothalamus attempts to regulate temperature.

Etiology

- Perimenopausal vasomotor instability (cold flushes)
- Hypothyroidism (reduced metabolic rate)
- Iron deficiency anaemia (impaired oxygen delivery)
- Peripheral arterial disease
- Raynaud phenomenon or disease
- Low body weight or inadequate caloric intake
- Anorexia nervosa (loss of insulating fat)
- Vitamin B12 deficiency (impaired myelination)
- Diabetes mellitus (peripheral neuropathy)
- Medications (beta-blockers, ergot derivatives)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent cold intolerance with fatigue, weight gain, constipation, dry skin, or colour changes in fingers/toes (blue/white)
- Accompanied by other concerning signs

Related Symptoms

- Fatigue
- Cold hands and feet
- Weight gain
- Dry skin
- Hot flushes

Note: Always consider serious underlying conditions when evaluating persistent cold intolerance.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 54

Cold Sores (Herpes Labialis)

Definition

Recurrent, painful, vesicular eruptions on the lips and perioral skin caused by herpes simplex virus (typically HSV-1), characterized by prodromal tingling followed by clustered blisters that crust and heal within 7-14 days.

Pathophysiology

Primary HSV-1 infection typically occurs in childhood (gingivostomatitis). The virus then establishes lifelong latency in the trigeminal ganglion. Reactivation triggers include UV radiation, fever, stress, immunosuppression, trauma, and hormonal changes. Reactivated virus travels along sensory nerve axons to the skin, causing epithelial cell lysis, vesicle formation, and local inflammation. Healing involves cell-mediated immune clearance.

Etiology

- HSV-1 reactivation (most common, triggered by UV light, stress, illness)
- Fever or viral illness (hence "cold sores")
- Ultraviolet radiation (sun exposure, tanning beds)
- Physical or emotional stress
- Immunosuppression (HIV, chemotherapy, transplant)
- Hormonal changes (menstruation)
- Facial trauma or dental procedures
- Fatigue and sleep deprivation
- Cold weather or wind exposure
- Close contact with active lesions (autoinoculation or transmission)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Frequent recurrent episodes (more than 6 per year), lesions spreading to eyes, or lesions lasting more than 2 weeks
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Swollen lymph nodes
- Sore throat
- Mouth ulcers
- Facial pain

Note: Always consider serious underlying conditions when evaluating persistent cold sores (herpes labialis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 55

Coma

55.1 Definition

Coma is a state of profound unconsciousness in which the patient cannot be aroused and lacks normal responses to external stimuli. It represents a medical emergency requiring immediate evaluation. Coma results from bilateral cerebral hemisphere dysfunction or brainstem reticular activating system disruption.

55.2 Pathophysiology

Consciousness requires an intact ascending reticular activating system (ARAS) in the brainstem and functional connections to the cerebral cortex. Coma results from either diffuse bilateral cortical injury (hypoxia, toxins, metabolic derangement) or direct damage to the ARAS (stroke, hemorrhage, compression). Metabolic causes interfere with neuronal energy metabolism. Structural causes disrupt neural pathways directly.

55.3 Etiology

- Traumatic brain injury (most common cause in young adults)
- Stroke (ischemic or hemorrhagic)
- Hypoxic-ischemic encephalopathy (cardiac arrest, near-drowning)
- Drug overdose (opioids, benzodiazepines, barbiturates)
- Metabolic disorders (hypoglycemia, hepatic encephalopathy, uremia)
- CNS infection (meningitis, encephalitis)
- Brain tumor with herniation
- Seizure (status epilepticus, post-ictal state)
- Carbon monoxide poisoning
- Hypothermia or hyperthermia
- Electrolyte disturbances

55.4 Indications for Evaluation

- Unresponsiveness to verbal or painful stimuli
- Glasgow Coma Scale score of 8 or less
- Bilateral pupillary abnormalities
- Abnormal breathing patterns
- Posturing (decorticate or decerebrate rigidity)
- Loss of corneal, oculocephalic, or gag reflexes
- Asymmetric motor responses or abnormal eye movements
- Seizure activity or posturing

55.5 Related Symptoms

- Loss of consciousness
- Seizures
- Syncope (transient loss of consciousness)
- Delirium (acute confusion)
- Brain herniation
- Respiratory failure
- Hypotension or shock

55.6 Self-Care/Home Management

- This condition requires emergency medical care, not self-management.
- If a person suddenly becomes unresponsive, call emergency services immediately.
- Do not give food, drink, or medications to an unconscious person.
- Place the person on their side (recovery position) if no spinal injury.
- Check for breathing and pulse; begin CPR if absent.

55.7 Diagnostic Approach

- Glasgow Coma Scale (GCS) assessment
- Rapid blood glucose measurement
- Complete blood count, electrolytes, renal and liver function
- Arterial blood gas analysis
- Toxicology screen for drug overdose
- Non-contrast CT head for structural causes
- MRI brain for more detailed assessment
- Electroencephalogram (EEG) to detect seizures

55.8 Treatment/Management Overview

- Airway, breathing, and circulation management (intubation if GCS less than 8)
- Correct metabolic derangements (dextrose, naloxone for opioids)
- Control intracranial pressure if elevated
- Treat underlying cause (antibiotics, surgery)
- Antiepileptic drugs for status epilepticus
- Therapeutic hypothermia for post-cardiac arrest
- Intensive care monitoring and supportive care

Chapter 56

Concussion (Mild Traumatic Brain Injury)

56.1 Definition

A concussion is a mild traumatic brain injury (mTBI) caused by a blow to the head or body that causes the brain to move rapidly within the skull. It temporarily impairs brain function and is not typically life-threatening but requires careful management.

56.2 Pathophysiology

The sudden acceleration-deceleration force causes shearing of axons, disruption of neuronal membranes, and a complex neurometabolic cascade. There is an influx of calcium and efflux of potassium, impaired oxidative metabolism, decreased cerebral blood flow, and altered neurotransmitter release. These changes produce the clinical symptoms of concussion, which are functional rather than structural.

56.3 Etiology

- Sports injuries (football, soccer, hockey, boxing, rugby)
- Falls (most common cause in children and elderly)
- Motor vehicle accidents
- Assault or physical violence
- Workplace injuries
- Blast injuries in military settings

56.4 Indications for Evaluation

- Headache or pressure in the head
- Dizziness or balance problems

- Nausea or vomiting
- Blurred or double vision
- Sensitivity to light or noise
- Confusion, disorientation, or feeling foggy
- Difficulty concentrating or remembering
- Slowed reaction times
- Mood changes (irritability, sadness, anxiety)
- Sleep disturbances (insomnia, drowsiness)
- Loss of consciousness (occurs in less than 10 percent of concussions)

56.5 Related Symptoms

- Head injury
- Headache (post-concussive)
- Dizziness
- Memory loss
- Neck pain (whiplash-associated)
- Post-concussion syndrome

56.6 Self-Care/Home Management

- Physical and cognitive rest for 24-48 hours
- Avoid screen time (phones, computers, TV)
- Avoid sports, exercise, and strenuous activity
- Get adequate sleep
- Avoid alcohol and recreational drugs
- Use acetaminophen for headache (avoid NSAIDs/aspirin due to bleeding risk)
- Return to activities gradually under medical supervision

56.7 Diagnostic Approach

- Clinical evaluation using standardized assessment tools (SCAT5, ACE)
- Neurological examination
- CT scan of the head if indicated (loss of consciousness, focal deficit, worsening symptoms, older age, anticoagulant use)
- MRI for persistent symptoms
- Neuropsychological testing for return-to-play decisions

56.8 Treatment/Management Overview

- Physical and cognitive rest initially
- Gradual return to activity protocol (school, work, sport)
- Symptom management (headache, sleep, mood)
- Referral to concussion specialist for persistent symptoms
- Vestibular therapy for balance issues
- Avoid repeat concussion risk during recovery

Chapter 57

Confusion

Definition

Altered mental state characterized by reduced awareness of surroundings, disorganized thinking, and impaired attention.

Pathophysiology

Ranges from mild disorientation to delirium; acute onset suggests metabolic toxic infectious or structural causes while chronic progressive indicates neurodegenerative processes.

Etiology

- Infection/sepsis
- Metabolic disorders (electrolyte imbalances)
- Medication effects/toxicity
- Stroke/TIA
- Head injury

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than acute confusion requires immediate evaluation; new or progressive warrants prompt assessment
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Headache
- Weakness
- Seizures
- Memory loss

Note: Always consider serious underlying conditions when evaluating persistent confusion.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 58

Constipation

Definition

Infrequent spontaneous bowel movements less than three per week accompanied by straining hard stool sensation incomplete evacuation manual maneuvers Rome IV criteria

Pathophysiology

Caused slow colonic transit dyssynergic defecation functional disorders medication induced opioids anticholinergics calcium channel blockers iron supplements hypothyroidism Parkinson disease IBS-C

Etiology

- Low fiber fluid diet lack exercise
- Opioid medication use
- Hypothyroidism
- Parkinson's disease
- IBS-C constipation predominant

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than New rectal bleeding change habitual pattern family colon cancer history investigate exclude malignancy lower GI workup
- Accompanied by other concerning signs

Related Symptoms

- Abdominal bloating straining feeling rectal blockage hard stools incomplete emptying

Note: Always consider serious underlying conditions when evaluating persistent constipation

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 59

Corns and Calluses

59.1 Definition

Corns and calluses are areas of thickened, hardened skin that develop in response to repeated friction or pressure. Calluses are diffuse areas of hyperkeratosis on weight-bearing surfaces, while corns are smaller, well-circumscribed lesions with a central core over bony prominences.

59.2 Pathophysiology

Repeated mechanical trauma stimulates an adaptive thickening of the stratum corneum (hyperkeratosis) as the skin attempts to protect itself. Calluses form on pressure points and are generally painless. Corns form over bony prominences and have a central keratinous core that presses on underlying dermal nerve endings, causing pain.

59.3 Etiology

- Ill-fitting footwear (tight, narrow, or high-heeled shoes)
- Foot deformities (hammertoes, bunions, bone spurs)
- Abnormal gait or biomechanics
- Repetitive manual labor (calluses on palms)
- Walking barefoot on hard surfaces
- Age-related loss of cushioning fat pads
- Arthritis causing bony prominences

59.4 Indications for Evaluation

- Thickened, hardened patches of skin on feet or hands
- Pain or tenderness when walking or applying pressure
- Yellowish or grayish discoloration

- Central hard core (nucleus) characteristic of corns
- Pain between toes (soft corn)
- Difficulty finding comfortable shoes
- Redness or signs of inflammation (possible infection)

59.5 Related Symptoms

- Bunions (hallux valgus)
- Hammer toe deformity
- Flat feet (pes planus)
- Foot pain (metatarsalgia)
- Plantar warts (may be confused with calluses)
- Athlete's foot (tinea pedis)
- Skin fissures (cracks in thickened skin)

59.6 Self-Care/Home Management

- Soak feet in warm water and gently file with a pumice stone
- Apply moisturizing cream containing urea or salicylic acid
- Use corn pads or donut-shaped felt pads to relieve pressure
- Wear properly fitted shoes with a wide toe box
- Avoid cutting corns or calluses with sharp instruments
- Use OTC salicylic acid patches for corns
- Wear padded gloves for hand calluses

59.7 Diagnostic Approach

- Clinical examination and inspection of the lesion
- Distinguish from plantar warts
- Evaluate footwear for proper fit
- Assess foot biomechanics and gait
- X-ray if underlying bone spur suspected

59.8 Treatment/Management Overview

- Pumice stone debridement after soaking
- Salicylic acid 40 percent patches for corns
- Paramedical or podiatric debridement
- Custom orthotics to redistribute pressure
- Padding and cushioning (felt pads, silicone toe separators)
- Surgical correction of underlying foot deformity if needed
- Proper footwear modification

Chapter 60

Cough

Definition

Reflex action clearing airways of mucus or irritants; classified acute, subacute, chronic by duration.

Pathophysiology

Productive vs dry, timing, triggers help differentiate etiologies; upper airway cough syndrome most common cause of chronic cough.

Etiology

- Viral URI
- Asthma/COPD
- GERD
- Postnasal drip
- Smoking/smoke irritation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 3 weeks (persistent)
- Accompanied by other concerning signs

Related Symptoms

- Sore throat
- Congestion

- Wheezing
- Chest tightness
- Phlegm production

Note: Always consider serious underlying conditions when evaluating persistent cough.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 61

Crying Spells

Definition

Episodes of uncontrollable or excessive tearfulness that occur unexpectedly or with minimal provocation, often unrelated to the severity of the triggering event.

Pathophysiology

Crying spells during perimenopause result from hormonal fluctuations affecting neurotransmitter systems that regulate emotional control. Oestrogen modulates serotonin synthesis, receptor sensitivity, and reuptake; declining and fluctuating oestrogen reduces serotonergic tone, impairing emotional regulation. The limbic system, particularly the amygdala and anterior cingulate cortex, becomes more reactive to emotional stimuli. Reduced GABAergic inhibition further disinhibits emotional expression.

Etiology

- Perimenopausal and menopausal hormonal fluctuations
- Premenstrual syndrome or premenstrual dysphoric disorder
- Major depressive disorder
- Anxiety disorders (generalised, panic)
- Adjustment disorder with depressed mood
- Post-traumatic stress disorder
- Pseudobulbar affect (neurological condition)
- Chronic stress and burnout
- Bereavement or grief
- Medication side effects (hormonal contraceptives, corticosteroids)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Crying spells with suicidal thoughts, hopelessness, sleep or appetite changes, or lasting more than 2 weeks with functional impairment
- Accompanied by other concerning signs

Related Symptoms

- Depression
- Anxiety
- Mood swings
- Irritability
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent crying spells.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 62

Cyst

62.1 Definition

A cyst is a closed capsule or sac-like structure filled with liquid, semisolid, or gaseous material. Cysts can develop in any tissue and vary widely in size. Most cysts are benign, though some require evaluation to exclude malignancy.

62.2 Pathophysiology

Cysts form when fluid-producing cells become enclosed within a membranous sac, or when a duct becomes blocked and secretions accumulate behind the obstruction. Common types include epidermal (sebaceous) cysts from obstructed hair follicles, ganglion cysts from synovial fluid herniation, ovarian cysts, Baker's cysts, and breast cysts.

62.3 Etiology

- Blocked ducts or glands (sebaceous gland obstruction)
- Trauma (inclusion cysts)
- Congenital remnants (branchial cleft cyst, thyroglossal duct cyst)
- Infection (pilonidal cyst)
- Hormonal factors (ovarian follicles, breast cysts)
- Joint or tendon pathology (ganglion cyst, Baker's cyst)
- Tumors (cystic degeneration of neoplasms)
- Inflammatory conditions (ovarian endometrioma)

62.4 Indications for Evaluation

- Palpable lump under the skin or in deeper tissue
- Smooth, round, mobile mass

- Slow or no growth over time
- Pain, tenderness, or redness if inflamed or infected
- Rapid growth or change in character
- Cyst in unusual location
- Cyst causing functional impairment (pain, pressure on nerves)

62.5 Related Symptoms

- Abscess (infected, pus-filled cavity)
- Lipoma (benign fatty tumor)
- Tumor (solid neoplasm versus cystic structure)
- Boil (furuncle, infected hair follicle)
- Lymphedema (swelling from lymphatic obstruction)
- Hematoma (blood collection)

62.6 Self-Care/Home Management

- Do NOT attempt to squeeze, puncture, or drain a cyst at home
- Apply warm compresses if the cyst is inflamed
- Keep the area clean to prevent infection
- Monitor for changes in size, color, or pain level
- Avoid friction or pressure over the cyst
- Seek medical evaluation if the cyst becomes painful, red, or grows rapidly

62.7 Diagnostic Approach

- Clinical examination (palpation, transillumination)
- Ultrasound to determine cystic vs. solid nature
- CT or MRI for deeper cysts
- Fine-needle aspiration for cytology if malignancy suspected
- Excisional biopsy for definitive diagnosis

62.8 Treatment/Management Overview

- Observation if asymptomatic and benign-appearing
- Incision and drainage for infected cysts
- Excision (complete removal of cyst and lining) to prevent recurrence
- Aspiration for simple cysts (higher recurrence rate)
- Sclerotherapy for some types (ganglion, ovarian)
- Antibiotics for secondary infection
- Treat underlying condition

Chapter 63

Dandruff (Seborrheic Dermatitis of the Scalp)

63.1 Definition

Dandruff is a common, chronic scalp condition characterized by flaking of the scalp skin. It affects approximately 50 percent of the population worldwide. It is a mild form of seborrheic dermatitis and is not contagious.

63.2 Pathophysiology

Dandruff results from a combination of factors: colonization by the yeast *Malassezia globosa* which breaks down sebum into irritating fatty acids; individual susceptibility to *Malassezia* metabolites; and increased sebum production. The inflammatory response accelerates epidermal cell turnover, causing immature keratinocytes to clump together and shed as visible flakes.

63.3 Etiology

- *Malassezia* yeast colonization
- Increased sebum production (seborrhea)
- Genetic predisposition
- Hormonal fluctuations
- Stress and fatigue
- Cold, dry weather
- Infrequent shampooing
- Neurological conditions (Parkinson disease, HIV)
- Use of harsh hair products

63.4 Indications for Evaluation

- White or yellowish flakes on the scalp, hair, and shoulders
- Itchy scalp
- Dry or greasy scales on the scalp
- Red patches on the scalp
- Flaking on other sebaceous areas (eyebrows, nose, ears, chest)
- No significant hair loss or scarring
- Flakes worsen in winter and improve in summer

63.5 Related Symptoms

- Seborrheic dermatitis (more severe, inflammatory form)
- Scalp itching (pruritus)
- Scalp redness or irritation
- Dry skin (xerosis)
- Psoriasis of the scalp (thicker, silvery plaques)
- Contact dermatitis
- Eczema (atopic dermatitis)

63.6 Self-Care/Home Management

- Shampoo regularly (daily or every other day)
- Use OTC dandruff shampoos (zinc pyrithione, selenium sulfide, ketoconazole)
- Alternate dandruff shampoos if one becomes less effective
- Leave shampoo on the scalp for 5 minutes before rinsing
- Apply coal tar shampoo for stubborn cases
- Avoid excessive use of styling products
- Gentle brushing to remove loose flakes

63.7 Diagnostic Approach

- Clinical diagnosis based on history and scalp examination
- Rule out other scalp conditions (psoriasis, tinea capitis)
- Scalp scraping with KOH preparation to exclude fungal infection
- Skin biopsy in atypical or treatment-resistant cases

63.8 Treatment/Management Overview

- Medicated shampoos: ketoconazole, zinc pyrithione, selenium sulfide, coal tar
- Topical corticosteroids for inflammation
- Topical antifungal creams for non-scalp areas
- Calcineurin inhibitors for refractory seborrheic dermatitis
- Salicylic acid shampoos to loosen thick scales
- UV phototherapy for widespread or severe cases

Chapter 64

Dark Circles Under Eyes

Definition

Dark, discolored appearance of the skin beneath the eyes, which may be black, blue, brown, or purple, resulting from vascular, pigmentary, or structural causes.

Pathophysiology

Dark circles arise from multiple mechanisms: vascular congestion (thin periorbital skin reveals underlying blood vessels, exacerbated by fatigue, allergies, or aging), melanin deposition (constitutional/genetic hyperpigmentation, more common in darker skin types), post-inflammatory hyperpigmentation (from atopic or allergic dermatitis), and shadowing from infraorbital fat herniation or tear trough hollowing with aging.

Etiology

- Fatigue and sleep deprivation (vascular stasis)
- Allergic rhinitis and atopic dermatitis (venous congestion)
- Genetic/constitutional periorbital hyperpigmentation
- Aging (skin thinning, fat herniation, tear trough hollowing)
- Dehydration
- Iron deficiency anemia
- Sun exposure (increased melanin production)
- Eye strain or prolonged screen time
- Periorbital edema (fluid retention)
- Medications (prostaglandin analogs for glaucoma)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Dark circles with persistent fatigue, pallor, or other signs of anemia, or if accompanied by periorbital swelling or pain
- Accompanied by other concerning signs

Related Symptoms

- Eye puffiness
- Fatigue
- Nasal congestion
- Headache
- Pale skin

Note: Always consider serious underlying conditions when evaluating persistent dark circles under eyes.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 65

Dehydration

Definition

A state of negative fluid balance resulting from excessive water loss (through skin, lungs, kidneys, or GI tract) or inadequate intake, leading to intracellular and extracellular volume depletion with potential electrolyte disturbances.

Pathophysiology

Water loss exceeds intake, causing reduced total body water, increased plasma osmolality, and stimulation of osmoreceptors in the hypothalamus. Antidiuretic hormone (ADH) is released, increasing water reabsorption in renal collecting ducts. Thirst is stimulated. Continued loss without replacement leads to hypovolemia, reduced cardiac output, compensatory tachycardia, vasoconstriction, and eventually hemodynamic instability and shock. Cellular dehydration impairs enzyme function and organ performance.

Etiology

- Inadequate fluid intake (common in elderly, infants, dementia, during illness)
- Vomiting and diarrhea (gastroenteritis most common cause in children)
- Excessive sweating (exercise, fever, heat exposure)
- Polyuria (diabetes mellitus, diabetes insipidus, diuretic medications)
- Blood loss (hemorrhage)
- Burns (loss of skin barrier leading to fluid evaporation)
- Fever (increased insensible water loss)
- Alcohol intake (diuretic effect)
- Gastrointestinal losses (NG suction, fistulas, ostomy output)
- Reduced thirst sensation in elderly (presbyphagia, cognitive impairment)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe dehydration signs: no urine output for 8 hours, lethargy or confusion, rapid heart rate, low blood pressure, sunken eyes, or inability to keep fluids down
- Accompanied by other concerning signs

Related Symptoms

- Thirst
- Dark urine
- Fatigue
- Dizziness
- Dry mouth

Note: Always consider serious underlying conditions when evaluating persistent dehydration.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
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Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
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- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 66

Delirium

Definition

Acute confusional state characterized by fluctuating consciousness impaired attention disorientation altered perception cognitive dysfunction.

Pathophysiology

Medical emergency requiring rapid identification of underlying cause; differs from chronic confusion/dementia by abrupt onset fluctuating course reversible etiologies hyperactive hypoactive mixed subtypes associated with increased mortality morbidity prolonged hospitalization long-term cognitive decline especially elderly population.

Etiology

- Infection/sepsis
- Medication toxicity/polypharmacy
- Metabolic abnormalities
- Alcohol withdrawal
- Postoperative state

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than acute onset rapid fluctuation requires immediate investigation identifying treatable underlying cause emergent evaluation essential
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Agitation
- Hallucinations
- Disorganized thinking
- Sleep-wake cycle disruption

Note: Always consider serious underlying conditions when evaluating persistent delirium.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 67

Delusions

Definition

Fixed, false beliefs that persist despite contradictory evidence, inconsistent with the individual's cultural or religious background, and representing a disturbance in thought content.

Pathophysiology

Dysfunction in prefrontal cortex (reality testing and belief evaluation) and limbic system (salience attribution) leads to formation and maintenance of delusional beliefs. Dopamine hyperactivity in mesolimbic pathways contributes to aberrant salience where neutral stimuli acquire false meaning. Delusions are categorized by content: persecutory, grandiose, referential, somatic, erotomanic, and nihilistic.

Etiology

- Schizophrenia or schizoaffective disorder
- Bipolar disorder (manic phase)
- Major depressive disorder with psychotic features
- Delirium (acute confusional state)
- Dementia (Alzheimer, Lewy body)
- Substance use or withdrawal (amphetamine, cocaine, alcohol withdrawal)
- Medication side effects (corticosteroids, anticholinergics, dopaminergic drugs)
- Metabolic or endocrine disorders (thyroid disease, B12 deficiency)
- Parkinson disease (medication-induced)
- Brain tumors or traumatic brain injury

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- New-onset delusions, especially with agitation, self-harm risk, or inability to care for oneself (immediate psychiatric evaluation)
- Accompanied by other concerning signs

Related Symptoms

- Hallucinations
- Agitation
- Confusion
- Suspiciousness
- Sleep disturbance

Note: Always consider serious underlying conditions when evaluating persistent delusions.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 68

Depression

Definition

Persistent sad empty irritable mood plus loss of interest pleasure lasting minimum two weeks affecting daily function.

Pathophysiology

Major depressive disorder involves multiple symptoms beyond low mood including sleep appetite energy concentration suicidal ideation; differential distinguishes from bipolar grief adjustment reactions.

Etiology

- Chemical imbalance in brain
- Life events trauma genetic predisposition
- Chronic medical conditions
- Medication side effects
- Substance use disorders

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent mood symptoms beyond two weeks or suicidal ideation require professional assessment
- Accompanied by other concerning signs

Related Symptoms

- Sleep disturbance
- Appetite weight changes
- Fatigue/loss of energy
- Feelings worthlessness guilt
- Thoughts death suicide

Note: Always consider serious underlying conditions when evaluating persistent depression.

Self-Care / Home Management

- Establish a consistent daily routine with regular sleep, meals, and physical activity to support mental health stability.
- Practice stress management techniques including mindfulness, deep breathing exercises, and progressive muscle relaxation.
- Maintain social connections and avoid isolation by reaching out to trusted friends, family, or support groups.
- Avoid alcohol, caffeine, and recreational drugs which can destabilize mood and exacerbate psychiatric symptoms.
- If experiencing suicidal thoughts or crisis, contact emergency services or a crisis helpline immediately.

Diagnostic Approach

- Comprehensive psychiatric assessment includes history of present illness, past psychiatric history, substance use, medical history, and collateral information from family.
- Standardized screening tools (GAD-7, PHQ-9, PCL-5) help quantify symptom severity and track treatment response.
- Medical evaluation (thyroid function, basic metabolic panel, CBC, toxicology screen) rules out organic causes of psychiatric symptoms.
- Risk assessment for self-harm, suicide, and danger to others is an essential component of every psychiatric evaluation.

Treatment / Management Overview

- Psychotherapy (CBT, DBT, interpersonal therapy) is a first-line treatment for most psychiatric disorders.
- Pharmacotherapy with antidepressants, anxiolytics, mood stabilizers, or antipsychotics is tailored to the specific diagnosis and individual patient factors.

- Combined treatment (medication plus psychotherapy) generally yields superior outcomes compared to either modality alone.
- Referral to psychiatry is indicated for severe symptoms, treatment resistance, suicidal ideation, or diagnostic uncertainty.

Chapter 69

Diarrhea

Definition

Frequent passage of loose or watery stools exceeding usual frequency and consistency.

Pathophysiology

Caused by increased secretion, decreased absorption, altered transit time, or osmotic load; acute vs chronic guides evaluation.

Etiology

- Viral gastroenteritis
- Bacterial infection
- Antibiotic-associated diarrhea
- IBS-D
- Inflammatory bowel disease

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 2-3 days (persistent)
- Accompanied by other concerning signs

Related Symptoms

- Abdominal cramps
- Urgency

- Dehydration
- Electrolyte imbalance
- Weight loss

Note: Always consider serious underlying conditions when evaluating persistent diarrhea.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 70

Disorientation

Definition

Impairment of awareness of time, place, or person, often accompanied by confusion, difficulty concentrating, and inability to perform familiar tasks.

Pathophysiology

Disorientation results from dysfunction in cortical and subcortical networks supporting orientation and attention. Acute disorientation (delirium) involves global cognitive impairment from metabolic, toxic, infectious, or structural causes. Chronic disorientation (dementia) reflects progressive neurodegenerative damage. The reticular activating system and prefrontal cortex are key affected regions.

Etiology

- Delirium (metabolic, infectious, medication-induced)
- Alzheimer disease and other dementias
- Head trauma or traumatic brain injury
- Stroke or transient ischemic attack
- Seizure (especially complex partial or post-ictal)
- Hypoglycemia or hyperglycemia
- Electrolyte imbalances
- Medication side effects (anticholinergics, benzodiazepines, opioids)
- Alcohol intoxication or withdrawal
- Severe dehydration or malnutrition

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden-onset disorientation (possible stroke, hypoglycemia, or delirium — emergency evaluation)
- Accompanied by other concerning signs

Related Symptoms

- Memory loss
- Confusion
- Agitation
- Dizziness
- Speech difficulty

Note: Always consider serious underlying conditions when evaluating persistent disorientation.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 71

Dizziness

Definition

Feeling of lightheadedness, unsteadiness, or vertigo (spinning sensation); distinct sensations have different etiologies.

Pathophysiology

Encompasses vertigo, presyncope, disequilibrium, lightheadedness - each with distinct differential diagnoses.

Etiology

- BPPV
- Orthostatic hypotension
- Dehydration
- Medication side effects
- Anxiety/panic

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluation needed if persistent/recurring
- Accompanied by other concerning signs

Related Symptoms

- Nausea
- Imbalance
- Hearing changes
- Visual disturbances
- Tinnitus

Note: Always consider serious underlying conditions when evaluating persistent dizziness.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 72

Double Vision (Diplopia)

Definition

The perception of two images of a single object, classified as monocular (persists when one eye is covered, originates from ocular media) or binocular (resolves with covering either eye, originates from ocular misalignment).

Pathophysiology

Binocular diplopia results from misalignment of the visual axes (strabismus) due to cranial nerve palsy (CN III, IV, VI), neuromuscular junction disorder (myasthenia gravis), restrictive orbitopathy (thyroid eye disease), or brainstem/cerebellar lesions affecting ocular motor control. Monocular diplopia arises from refractive errors, cataract, corneal irregularity, or retinal problems causing light scattering within one eye.

Etiology

- Cranial nerve palsy (CN III, IV, VI from microvascular disease, aneurysm, trauma, tumor)
- Myasthenia gravis (fatigable diplopia, worse with sustained gaze)
- Thyroid eye disease (Graves orbitopathy, restrictive myopathy)
- Strabismus (childhood squint, decompensating in adulthood)
- Multiple sclerosis (demyelinating brainstem lesion)
- Stroke or transient ischemic attack (brainstem, cerebellum)
- Orbital trauma or fracture
- Cataract (monocular diplopia)
- Corneal irregularity (dry eye, keratoconus, scar)
- Botulism or Guillain-Barre syndrome (Miller Fisher variant)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden-onset diplopia (possible stroke, aneurysm, or brainstem lesion), diplopia with headache, ptosis, or pupil changes
- Accompanied by other concerning signs

Related Symptoms

- Headache
- Ptosis (drooping eyelid)
- Blurred vision
- Dizziness
- Strabismus

Note: Always consider serious underlying conditions when evaluating persistent double vision (diplopia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 73

Drooling (Sialorrhea)

Definition

The unintentional loss of saliva from the mouth, resulting from excessive saliva production (hyper-salivation) or impaired oral motor control causing difficulty swallowing saliva, which can lead to skin irritation, aspiration, and social embarrassment.

Pathophysiology

Saliva is produced by three major paired salivary glands (parotid, submandibular, sublingual) and minor glands, with daily production of 0.5-1.5 L. Drooling occurs due to excess production (rare) or, more commonly, from oropharyngeal dysphagia, impaired lip seal, poor head control, or reduced swallowing frequency. Neurological conditions disrupt the coordinated neuromuscular sequence of saliva clearance.

Etiology

- Neurological disorders (cerebral palsy, Parkinson disease, ALS, stroke)
- Developmental delay (children with poor oral motor control)
- Oropharyngeal dysphagia (difficulty swallowing)
- Dental problems (teething in infants, ill-fitting dentures)
- GERD (increased salivation as compensatory response)
- Medications (clozapine, risperidone, cholinesterase inhibitors, ketamine)
- Toxins (mercury, organophosphates, arsenic)
- Pregnancy (hormonal changes causing hypersalivation)
- Mouth breathing (nasal obstruction, enlarged tonsils/adenoids)
- Parkinson disease (stooped posture, reduced swallowing frequency, rigidity)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden onset of drooling (possible stroke, especially with facial droop or speech difficulty), drooling with fever and difficulty swallowing (possible epiglottitis or retropharyngeal abscess)
- Accompanied by other concerning signs

Related Symptoms

- Dysphagia
- Speech difficulty
- Facial droop
- Bad breath
- Choking

Note: Always consider serious underlying conditions when evaluating persistent drooling (sialorrhea).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 74

Dry Eye Disease (Dry Eyes)

Definition

A multifactorial disease of the ocular surface characterized by loss of homeostasis of the tear film, accompanied by ocular discomfort symptoms and visual disturbance.

Pathophysiology

Tear film instability from either reduced tear production (aqueous-deficient dry eye — Sjogren, age, medications) or excessive tear evaporation (meibomian gland dysfunction leading to lipid layer deficiency). Ocular surface inflammation creates a vicious cycle of tear dysfunction, hyperosmolarity, and epithelial damage.

Etiology

- Aging (reduced tear production with age)
- Meibomian gland dysfunction (most common cause)
- Environmental factors (dry climate, AC, wind, screen time)
- Contact lens wear
- Sjogren syndrome (autoimmune)
- Medication side effects (antihistamines, decongestants, diuretics, antidepressants)
- Hormonal changes (menopause)
- Blepharitis (eyelid inflammation)
- Refractive surgery (LASIK)
- Systemic diseases (rheumatoid arthritis, lupus, diabetes)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent dry eye symptoms despite artificial tears, eye pain, photophobia, or vision changes
- Accompanied by other concerning signs

Related Symptoms

- Eye redness
- Blurred vision
- Photophobia
- Eye fatigue
- Foreign body sensation

Note: Always consider serious underlying conditions when evaluating persistent dry eye disease (dry eyes).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 75

Dry Mouth

Definition

Reduced salivary flow causing oral mucosa dryness difficulty swallowing tasting speaking altering normal mouth sensation.

Pathophysiology

Xerostomia can lead dental caries periodontal disease taste disturbances oral infections halitosis speaking eating difficulties; distinguished from normal temporary dryness persists despite hydration medication induced systemic disease radiation therapy Sjogren syndrome autoimmune diabetes aging dehydration chemotherapy opioids antihistamines antidepressants.

Etiology

- Medications (anticholinergics antihistamines)
- Autoimmune disorders (Sjogren syndrome)
- Radiation therapy head/neck
- Diabetes mellitus
- Dehydration/insufficient fluid intake

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent beyond simple dehydration unexplained accompanied other systemic symptoms warrants dental medical evaluation
- Accompanied by other concerning signs

Related Symptoms

- Difficulty swallowing dry throat tooth decay bad breath altered taste frequent thirst

Note: Always consider serious underlying conditions when evaluating persistent dry mouth.

Self-Care / Home Management

- Maintain good oral hygiene: brush twice daily, floss daily, and rinse with warm salt water (1/2 tsp salt in 8 oz water).
- Avoid tobacco, alcohol, and spicy or acidic foods that may irritate oral tissues.
- Stay hydrated by sipping water throughout the day to combat dry mouth.
- Over-the-counter pain relievers and topical oral gels can provide temporary relief for toothache or gum discomfort.
- See a dentist promptly for persistent oral symptoms, as early intervention improves outcomes.

Diagnostic Approach

- History covers onset, character of pain, aggravating/relieving factors, oral hygiene habits, and dental care history.
- Oral examination inspects teeth, gums, mucosa, tongue, and oropharynx for caries, inflammation, lesions, or masses.
- Dental X-rays (panoramic, periapical) identify caries, abscesses, impactions, and bone pathology.
- Referral to a dentist or oral surgeon is appropriate for definitive diagnosis and treatment of dental pathology.

Treatment / Management Overview

- Dental caries require restorative treatment (fillings, crowns, root canal) by a dentist.
- Oral infections (abscess, pericoronitis) may require drainage, antibiotics, and definitive dental treatment.
- Dry mouth management includes hydration, sugar-free lozenges, saliva substitutes, and addressing underlying causes.
- Regular dental check-ups every 6 months are essential for prevention and early detection of oral pathology.

Chapter 76

Dry Orgasm

Definition

Orgasm achieved without ejaculation of semen (anejaculation) or with significantly reduced or absent seminal fluid, occurring retrograde into the bladder or due to failure of seminal emission.

Pathophysiology

Dry orgasm results from either retrograde ejaculation (semen flows backward into the bladder due to failure of the bladder neck to close during emission) or anejaculation (failure of seminal emission from the seminal vesicles, prostate, and vas deferens). The internal bladder sphincter normally contracts during orgasm to prevent retrograde flow; disruption by surgery, medication, or neuropathy causes retrograde ejaculation.

Etiology

- Transurethral resection of the prostate (TURP)
- Spinal cord injury or neurological disease
- Diabetes mellitus with autonomic neuropathy
- Medication side effects (alpha-blockers, antipsychotics, antidepressants)
- Retroperitoneal lymph node dissection (testicular cancer surgery)
- Multiple sclerosis
- Spinal stenosis or cauda equina syndrome
- Bladder neck surgery
- Severe hypospadias or congenital abnormalities
- Psychogenic anejaculation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent dry orgasm with fertility concerns, associated with back pain or neurological symptoms, or after pelvic surgery
- Accompanied by other concerning signs

Related Symptoms

- Erectile dysfunction
- Infertility
- Reduced libido
- Testicular pain
- Urinary symptoms

Note: Always consider serious underlying conditions when evaluating persistent dry orgasm.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 77

Dry Skin (Xerosis)

Definition

A common skin condition characterized by rough, scaling, flaking, and sometimes pruritic skin due to reduced water content in the stratum corneum, often exacerbated by cold weather, low humidity, and frequent washing.

Pathophysiology

Dry skin results from disruption of the skin barrier function, particularly the stratum corneum lipid matrix (ceramides, cholesterol, free fatty acids) that normally retains water. Decreased natural moisturizing factors (amino acids, urea, lactate) and impaired desquamation lead to accumulation of corneocytes on the surface. Environmental factors (low humidity, cold, wind, excessive bathing) accelerate transepidermal water loss. Aging reduces sebaceous and sweat gland activity.

Etiology

- Cold, dry weather (winter xerosis, most common)
- Excessive bathing or hand washing (strips natural oils)
- Harsh soaps and detergents (disrupt skin barrier)
- Aging (decreased sebum production, thinner skin)
- Central heating or air conditioning (low indoor humidity)
- Atopic dermatitis (eczema, genetic barrier defect)
- Psoriasis (increased epidermal turnover with scaling)
- Hypothyroidism (reduced sweating and sebum production)
- Diabetes mellitus (dehydration, autonomic dysfunction)
- Medications (diuretics, retinoids, antihistamines, statins)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Dry skin with severe itching, signs of infection (increasing redness, warmth, swelling, pus), extensive cracking or fissures, or rash unresponsive to moisturizers
- Accompanied by other concerning signs

Related Symptoms

- Itching (pruritus)
- Rash
- Skin flaking
- Erythema
- Skin fissures

Note: Always consider serious underlying conditions when evaluating persistent dry skin (xerosis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 78

Dysphagia (Difficulty Swallowing)

78.1 Definition

Dysphagia is the subjective sensation of difficulty or abnormality in swallowing. It can involve the oral, pharyngeal, or esophageal phase of swallowing. Dysphagia affects approximately 15 percent of older adults and can lead to malnutrition, dehydration, and aspiration pneumonia.

78.2 Pathophysiology

Swallowing is a complex coordinated process involving cranial nerves V, VII, IX, X, and XII, as well as over 30 muscles. The oral phase involves mastication and bolus formation. The pharyngeal phase triggers the swallowing reflex, closing the airway. The esophageal phase uses peristalsis to move the bolus into the stomach. Dysphagia results from disruption at any phase.

78.3 Etiology

- Stroke (most common neurological cause)
- Gastroesophageal reflux disease (GERD) with esophagitis
- Esophageal stricture
- Esophageal cancer
- Neuromuscular disorders (Parkinson disease, multiple sclerosis, ALS)
- Zenker diverticulum
- Cricopharyngeal muscle dysfunction
- Eosinophilic esophagitis
- Achalasia (failure of LES relaxation)
- Head and neck cancer or its treatment
- Dementia or cognitive impairment

78.4 Indications for Evaluation

- Sensation of food sticking in the throat or chest
- Pain with swallowing (odynophagia)
- Coughing or choking during meals
- Regurgitation of food or liquids
- Recurrent pneumonia (from aspiration)
- Unexplained weight loss or dehydration
- Changes in voice quality after swallowing
- Prolonged meal times or avoidance of certain foods

78.5 Related Symptoms

- Gastroesophageal reflux disease (GERD)
- Heartburn
- Regurgitation
- Weight loss
- Cough
- Aspiration pneumonia
- Sialorrhea (drooling)
- Sore throat

78.6 Self-Care/Home Management

- Eat slowly and take small bites
- Chew thoroughly before swallowing
- Sit upright during and for 30 minutes after meals
- Consider thickened liquids if thin liquids cause coughing
- Moisten dry foods with sauce, gravy, or broth
- Alternate bites of food with sips of liquid
- Avoid talking while eating

78.7 Diagnostic Approach

- Bedside swallowing assessment by speech-language pathologist
- Videofluoroscopic swallow study (modified barium swallow)
- Fiberoptic endoscopic evaluation of swallowing (FEES)
- Esophagogastroduodenoscopy (EGD) with biopsy
- Esophageal manometry to assess motility
- Barium swallow (esophagram)

78.8 Treatment/Management Overview

- Swallowing therapy by speech-language pathologist
- Dietary modifications (texture-modified foods, thickened liquids)
- Treat underlying cause (PPI for reflux, dilation for stricture)
- Esophageal dilation for strictures
- Botulinum toxin for cricopharyngeal dysfunction or achalasia
- Surgical myotomy for achalasia
- Percutaneous endoscopic gastrostomy (PEG) for severe dysphagia

Chapter 79

Earache

Definition

Pain or discomfort localized to the ear, ranging from dull ache to sharp stabbing sensation, originating from the external, middle, or inner ear structures or referred from adjacent regions.

Pathophysiology

Ear pain arises from inflammation, infection, pressure changes, or irritation of the auricle, ear canal, tympanic membrane, or middle ear structures. Referred pain occurs via shared innervation with the temporomandibular joint, teeth, sinuses, and pharynx through cranial nerves V, VII, IX, and X.

Etiology

- Otitis media (middle ear infection)
- Otitis externa (swimmer's ear)
- Eustachian tube dysfunction
- Temporomandibular joint (TMJ) disorders
- Dental infections or abscesses
- Foreign body in ear canal
- Earwax impaction
- Perforated tympanic membrane
- Mastoiditis
- Referred pain from tonsillitis or pharyngitis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe ear pain, fever, hearing loss, ear discharge, or pain persisting more than 2-3 days
- Accompanied by other concerning signs

Related Symptoms

- Hearing loss
- Fever
- Ear discharge
- Dizziness
- Headache

Note: Always consider serious underlying conditions when evaluating persistent earache.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 80

Earwax (Cerumen Impaction)

80.1 Definition

Earwax (cerumen) is a natural substance produced by ceruminous glands in the outer ear canal that protects the ear by trapping debris and repelling insects. Cerumen impaction occurs when wax accumulates to the point of causing symptoms, affecting approximately 10 percent of children and 5 percent of adults.

80.2 Pathophysiology

Cerumen is composed of shed skin cells, hair, and secretions from ceruminous and sebaceous glands. It normally migrates outward from the tympanic membrane along the ear canal, aided by jaw movement. Impaction occurs when this migration fails due to narrow canals, excessive hair, dry cerumen type, or use of cotton swabs which push wax deeper.

80.3 Etiology

- Cotton swab use (pushing wax deeper)
- Anatomical factors (narrow, tortuous ear canals)
- Excessive ear hair
- Dry cerumen phenotype (genetic variation)
- Hearing aid or earplug use
- Advanced age (drier cerumen)
- Occupational exposure to dusty environments
- Excessive swimming or water exposure

80.4 Indications for Evaluation

- Decreased hearing or feeling of fullness in the ear

- Earache or discomfort
- Tinnitus (ringing in the ear)
- Dizziness or vertigo
- Itching in the ear canal
- Cough (auricular branch of vagus nerve stimulation)
- Visible wax blocking the ear canal

80.5 Related Symptoms

- Hearing loss (conductive)
- Earache (otalgia)
- Tinnitus
- Dizziness or vertigo
- Otitis externa (swimmer's ear)
- Ear fullness or pressure

80.6 Self-Care/Home Management

- Do NOT use cotton swabs or other objects to clean the ear canal
- Use OTC cerumenolytic drops (carbamide peroxide, mineral oil)
- Tilt head and allow drops to remain for several minutes
- Irrigate gently with warm water using a bulb syringe (if no eardrum perforation)
- Dry the ear canal afterward
- Never attempt to remove wax if eardrum is perforated

80.7 Diagnostic Approach

- Otoscopic examination to visualize cerumen and tympanic membrane
- Tympanometry if hearing loss is present
- Audiometry if needed
- Rule out other causes of hearing loss

80.8 Treatment/Management Overview

- Cerumenolytic agents to soften wax
- Ear irrigation with warm water (after ensuring intact tympanic membrane)
- Microsuction or manual removal under otoscopic guidance
- Referral to otolaryngologist for complicated or recurrent impaction

Chapter 81

Edema (Fluid Retention)

81.1 Definition

Edema is the accumulation of excess fluid in the interstitial space of tissues, causing visible swelling. It can be localized (venous insufficiency, lymphatic obstruction) or generalized (heart failure, kidney disease, liver disease).

81.2 Pathophysiology

Edema arises from an imbalance in Starling forces: (1) increased capillary hydrostatic pressure (heart failure, venous insufficiency); (2) decreased plasma oncotic pressure (hypoalbuminemia); (3) increased capillary permeability (inflammation, infection, allergic reaction); (4) lymphatic obstruction (lymphedema).

81.3 Etiology

- Chronic venous insufficiency (most common cause of bilateral leg edema)
- Congestive heart failure
- Cirrhosis with hypoalbuminemia
- Nephrotic syndrome
- Acute or chronic kidney disease
- Deep vein thrombosis (unilateral edema)
- Lymphedema (cancer treatment, infection, congenital)
- Medications (calcium channel blockers, NSAIDs, corticosteroids)
- Pregnancy (physiologic edema)
- Idiopathic edema

81.4 Indications for Evaluation

- Visible swelling of one or both legs, ankles, feet
- Pitting edema on examination
- Rapid weight gain over days
- Swelling accompanied by shortness of breath (possible heart failure)
- Unilateral leg swelling with pain (possible DVT)
- Periorbital edema (suggesting kidney disease)
- Ascites or abdominal distension
- Skin changes: thinning, dry, ulcerated skin

81.5 Related Symptoms

- Shortness of breath (if pulmonary edema)
- Ascites (abdominal fluid)
- Weight gain (fluid retention)
- Orthopnea (difficulty breathing when lying flat)
- Paroxysmal nocturnal dyspnea
- Skin ulcers (from chronic venous stasis)
- Cellulitis (infection of edematous skin)

81.6 Self-Care/Home Management

- Elevate the legs above heart level several times daily
- Reduce dietary sodium to less than 2 g per day
- Wear compression stockings if venous insufficiency
- Avoid prolonged sitting or standing
- Stay hydrated
- Limit alcohol consumption
- Monitor daily weight and ankle circumference

81.7 Diagnostic Approach

- History and physical examination
- Basic metabolic panel and serum albumin
- Urinalysis for proteinuria
- Brain natriuretic peptide (BNP) if heart failure suspected
- Liver function tests if cirrhosis suspected
- Venous duplex ultrasound if DVT suspected
- Chest X-ray if pulmonary edema suspected

81.8 Treatment/Management Overview

- Treat underlying cause (heart failure, liver disease, kidney disease)
- Diuretic therapy (loop diuretics: furosemide)
- Sodium restriction
- Compression therapy for venous insufficiency or lymphedema
- ACE inhibitors for nephrotic syndrome
- Discontinue contributing medications
- Leg elevation and mobilization

Chapter 82

Elbow Pain

Definition

Pain or discomfort localized to the elbow joint and surrounding soft tissues, commonly arising from tendinopathy, arthritis, trauma, or nerve entrapment.

Pathophysiology

Elbow pain localizes based on anatomy: lateral epicondylitis (tennis elbow) involves degenerative changes in the extensor carpi radialis brevis origin; medial epicondylitis (golfer's elbow) affects the common flexor origin; olecranon bursitis involves inflammation of the bursa overlying the olecranon process. Articular pain suggests osteoarthritis, inflammatory arthritis, or intra-articular loose bodies.

Etiology

- Lateral epicondylitis (tennis elbow)
- Medial epicondylitis (golfer's elbow)
- Olecranon bursitis
- Elbow osteoarthritis
- Rheumatoid arthritis
- Cubital tunnel syndrome (ulnar nerve entrapment)
- Distal biceps or triceps tendinopathy
- Elbow dislocation or fracture (radial head, olecranon)
- Ligamentous injury (UCL tear — Tommy John lesion)
- Septic arthritis or gout

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Elbow pain after acute injury with deformity or inability to move, fever, or severe pain with swelling and warmth
- Accompanied by other concerning signs

Related Symptoms

- Arm pain
- Forearm pain
- Numbness in fingers
- Swelling
- Decreased range of motion

Note: Always consider serious underlying conditions when evaluating persistent elbow pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 83

Electric Shock Sensation (Formication)

Definition

A sudden, brief sensation resembling an electric shock, tingling, or crawling feeling on or under the skin, typically lasting seconds, and often occurring without an external stimulus.

Pathophysiology

Electric shock sensations, medically termed paresthesias, arise from ectopic impulse generation in peripheral sensory nerves or central sensory pathways. Oestrogen supports myelin integrity and nerve conduction; declining oestrogen during perimenopause may increase nerve irritability. Reduced progesterone (which has a membrane-stabilising effect) further lowers the nerve excitation threshold. Hormonal fluctuations alter ion channel function (sodium, potassium, calcium) in sensory neurons, leading to spontaneous depolarisation.

Etiology

- Perimenopausal and menopausal hormonal fluctuations
- Cervical or lumbar nerve root compression (radiculopathy)
- Peripheral neuropathy (diabetic, alcoholic, nutritional)
- Multiple sclerosis (central demyelination)
- Migraine aura (sensory variant)
- Anxiety and hyperventilation syndrome
- Medication side effects (antidepressants, anticonvulsants)
- Vitamin B12 deficiency
- Hypocalcaemia or hypomagnesaemia
- Transient ischaemic attack (sensory cortex involvement)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Frequent or persistent electric shock sensations with weakness, numbness, vision changes, or other neurological symptoms
- Accompanied by other concerning signs

Related Symptoms

- Tingling numbness
- Nerve pain
- Dizziness
- Anxiety
- Headache

Note: Always consider serious underlying conditions when evaluating persistent electric shock sensation.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 84

Enlarged Prostate (Benign Prostatic Hyperplasia)

84.1 Definition

An enlarged prostate (benign prostatic hyperplasia, BPH) is a noncancerous increase in prostate size that commonly affects older men. BPH can compress the urethra and cause bothersome lower urinary tract symptoms.

84.2 Pathophysiology

BPH involves hyperplasia of stromal and glandular elements in the transition zone of the prostate. As the prostate enlarges, it compresses the prostatic urethra, increasing resistance to urine flow. Bladder detrusor muscle hypertrophies to overcome the obstruction, leading to instability, reduced compliance, and eventual decompensation with incomplete emptying.

84.3 Etiology

- Aging (BPH affects over 50 percent of men by age 60 and 90 percent by age 85)
- Androgenic hormones (dihydrotestosterone drives prostatic growth)
- Genetic and familial factors
- Ethnic variation (higher in African-American men)
- Obesity
- Lack of physical activity
- Erectile dysfunction medications (may worsen symptoms)

84.4 Indications for Evaluation

- Frequent urination (urinary frequency, especially at night)
- Urgency to urinate

- Weak urine stream or difficulty starting urination
- Dribbling at the end of urination
- Feeling of incomplete bladder emptying
- Straining to urinate
- Sudden inability to urinate (acute urinary retention)

84.5 Related Symptoms

- Frequent urination
- Nocturia (nighttime urination)
- Urinary urgency
- Overactive bladder
- Urinary retention (inability to urinate)
- Urinary tract infection (from incomplete emptying)
- Hematuria (blood in urine)
- Prostate cancer

84.6 Self-Care/Home Management

- Limit fluid intake before bedtime
- Reduce caffeine and alcohol consumption
- Avoid decongestants and antihistamines
- Empty the bladder completely when urinating
- Double-void (wait a moment then try again)
- Maintain a healthy weight
- Stay physically active
- Practice timed voiding (urinate on schedule)

84.7 Diagnostic Approach

- Digital rectal examination
- International Prostate Symptom Score questionnaire
- Urinalysis and urine culture
- Serum PSA (prostate-specific antigen)
- Urinary flow rate measurement (uroflowmetry)
- Post-void residual volume measurement
- Transrectal ultrasound of the prostate

84.8 Treatment/Management Overview

- Watchful waiting for mild symptoms
- Alpha-blockers (tamsulosin, terazosin) for symptom relief
- 5-alpha reductase inhibitors (finasteride, dutasteride) to shrink the prostate
- Combination therapy for moderate-to-severe symptoms
- TURP (transurethral resection of the prostate)
- Minimally invasive procedures (Rezum, UroLift)
- Surgery for severe cases or complications

Chapter 85

Enlarged Spleen (Splenomegaly)

85.1 Definition

Splenomegaly is an enlargement of the spleen beyond its normal size (approximately 12 cm in length, 150-200 g in weight). It is a sign of underlying disease rather than a primary condition.

85.2 Pathophysiology

The spleen serves as a filter for blood, removing old or abnormal red blood cells and pathogens. Splenomegaly occurs through: (1) increased workload (hemolytic anemias, infections); (2) congestion from portal hypertension; (3) infiltration by abnormal cells (leukemia, lymphoma); (4) extramedullary hematopoiesis.

85.3 Etiology

- Portal hypertension (cirrhosis, portal vein thrombosis)
- Infections (mononucleosis, malaria, endocarditis, tuberculosis, HIV)
- Hematologic disorders (hemolytic anemias, sickle cell disease)
- Myeloproliferative neoplasms (myelofibrosis, CML)
- Lymphoproliferative disorders (lymphoma, CLL)
- Infiltrative diseases (sarcoidosis, amyloidosis, Gaucher disease)
- Autoimmune diseases (lupus, rheumatoid arthritis)
- Splenic cysts or abscesses
- Splenic infarction

85.4 Indications for Evaluation

- Left upper quadrant abdominal pain or fullness
- Early satiety (spleen compressing the stomach)

- Palpable spleen on abdominal examination
- Unexplained weight loss, fever, or night sweats
- Fatigue, pallor, or easy bruising
- Signs of portal hypertension (ascites, jaundice)
- Signs of infection (fever, lymphadenopathy)

85.5 Related Symptoms

- Left upper quadrant pain
- Abdominal fullness or bloating
- Early satiety
- Fatigue
- Jaundice
- Lymphadenopathy
- Ascites
- Fever

85.6 Self-Care/Home Management

- Avoid contact sports or abdominal trauma
- Avoid heavy lifting or straining
- Use a seatbelt to protect the abdomen
- Take acetaminophen for mild discomfort (avoid NSAIDs)
- Monitor for signs of splenic rupture

85.7 Diagnostic Approach

- Abdominal ultrasound to confirm splenomegaly
- Complete blood count with differential
- Liver function tests
- Serologic tests for infectious causes
- CT or MRI of the abdomen
- Bone marrow biopsy if hematologic malignancy suspected

85.8 Treatment/Management Overview

- Treat the underlying cause
- Splenectomy for massive splenomegaly causing symptoms
- Splenic embolization (less invasive alternative)
- Vaccination before splenectomy
- Avoid contact sports to prevent splenic rupture
- Folic acid supplementation in hemolytic anemias

Chapter 86

Erectile Dysfunction (Impotence)

Definition

The persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance, lasting at least 3 months, with significant distress or interpersonal difficulty.

Pathophysiology

Erection requires intact neurological, vascular, endocrine, and psychological systems. Nitric oxide released from cavernous nerve terminals and endothelium activates guanylate cyclase, increasing cGMP, causing smooth muscle relaxation and arterial dilation. ED results from disruption at any level: endothelial dysfunction (atherosclerosis, diabetes) impairs NO production; venous leak prevents trapping of blood; neurological damage disrupts signaling; low testosterone reduces libido and erectile function.

Etiology

- Vascular disease (atherosclerosis, hypertension, diabetes mellitus)
- Neurological disorders (spinal cord injury, multiple sclerosis, Parkinson, stroke)
- Endocrine causes (hypogonadism, hyperprolactinemia, thyroid disorders)
- Psychological factors (anxiety, depression, performance anxiety, stress)
- Medications (antidepressants SSRIs, antihypertensives beta-blockers, diuretics, antipsychotics)
- Lifestyle factors (smoking, alcohol, obesity, sedentary lifestyle, recreational drugs)
- Pelvic surgery or trauma (prostatectomy, cystectomy, pelvic radiation)
- Peyronie disease (penile curvature with plaques)
- Renal failure or liver cirrhosis
- Aging-related decline in testosterone and vascular function

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden onset of ED after pelvic trauma or surgery, ED with loss of morning erections (possible hypogonadism), or ED causing significant relationship distress
- Accompanied by other concerning signs

Related Symptoms

- Low libido
- Premature ejaculation
- Difficulty achieving orgasm
- Penile curvature
- Urinary symptoms

Note: Always consider serious underlying conditions when evaluating persistent erectile dysfunction (impotence).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 87

Excessive Sweating

Definition

Profuse sweating beyond thermoregulatory needs occurring minimal heat exertion emotional stress significant impairment daily activities social functioning occupational performance.

Pathophysiology

Primary focal hyperhidrosis excessive sweating localized palms soles axillae face cranium without identifiable cause onset childhood adolescence autonomic dysfunction idiopathic secondary generalized hyperhidrosis whole body caused medications fever thyroid disease malignancy menopause diabetes hypoglycemia neurological conditions drugs antipyretics opioids insulin sulfonylureas cholinergic agents antidepressants withdrawal states alcohol consumption.

Etiology

- Primary hyperhidrosis focal excessive sweating
- Medications antidepressants opioids
- Hyperthyroidism diabetes mellitus
- Malignancy lymphoma infections
- Menopausal hot flashes withdrawal

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than progressive unexplained recent onset accompanying weight loss fever palpitations night sweats requires endocrine metabolic infectious neoplastic workup dermatology referral consideration topical antiperspirants iontophoresis botulinum toxin sympathectomy options evaluated specialist assessment beneficial
- Accompanied by other concerning signs

Related Symptoms

- Night sweats chills anxiety weight loss
- Fever palpitations orthostatic intolerance
- Dehydration electrolyte imbalance social isolation embarrassment
- Clothing stains occupational limitation reduced quality of life
- Social avoidance psychological impact depression anxiety

Note: Always consider serious underlying conditions when evaluating persistent excessive sweating.

Self-Care / Home Management

- Rest and maintain adequate hydration with water, clear fluids, or oral rehydration solutions.
- Use acetaminophen or ibuprofen to reduce fever and body aches (follow label dosing instructions).
- Dress lightly and keep the room at a comfortable temperature — avoid bundling excessively.
- Monitor temperature at regular intervals and track other symptoms to identify worsening trends.
- Seek medical attention for fever above 40 extdegreeC (104 extdegreeF), fever lasting more than 3 days, or fever with stiff neck, severe headache, rash, or difficulty breathing.

Diagnostic Approach

- History includes fever duration, pattern (intermittent, remittent, continuous), associated symptoms (chills, rigors, sweats), and exposure history.
- Vital signs assess fever severity and hemodynamic stability; complete physical examination seeks a source.
- Laboratory evaluation includes CBC with differential, blood cultures, urinalysis, inflammatory markers (CRP, ESR), and targeted testing based on exposure history.
- Imaging (chest X-ray, abdominal ultrasound, CT) is guided by clinical findings and suspected source.

Treatment / Management Overview

- Antipyretics (acetaminophen, NSAIDs) provide symptomatic relief but do not treat the underlying cause.
- Empiric antibiotics may be indicated for suspected bacterial infection after appropriate cultures are obtained.

- Antiviral, antifungal, or antimalarial therapy is directed at specific pathogens based on exposure and testing.
- Source control (drainage of abscess, removal of infected device) is essential for focal infections.

Chapter 88

Eye Allergies (Allergic Conjunctivitis)

88.1 Definition

Eye allergies (allergic conjunctivitis) are inflammation of the conjunctiva triggered by exposure to allergens such as pollen, dust mites, pet dander, or mold. It is the most common cause of eye allergy symptoms and is estimated to affect 15-20 percent of the population.

88.2 Pathophysiology

When an allergen contacts the conjunctiva, it cross-links IgE antibodies on the surface of mast cells, triggering degranulation and release of histamine and other inflammatory mediators. Histamine causes vasodilation (redness), increased vascular permeability (swelling and tearing), and nerve stimulation (itching).

88.3 Etiology

- Seasonal pollens (ragweed, grass, tree pollen)
- House dust mites
- Pet dander (cats, dogs)
- Mold spores
- Cockroach allergens
- Contact lens solutions or preservatives
- Air pollution or smoke
- Cosmetics or eye makeup

88.4 Indications for Evaluation

- Eye itching (most characteristic symptom)
- Redness of the eyes

- Watery or stringy discharge
- Swollen eyelids
- Burning sensation
- Sensitivity to light (photophobia)
- Nasal congestion and sneezing (often concurrent)

88.5 Related Symptoms

- Itchy eyes
- Red eyes
- Watery eyes
- Swollen eyelids
- Allergic rhinitis (hay fever)
- Conjunctivitis (infectious)
- Contact dermatitis of the eyelids

88.6 Self-Care/Home Management

- Avoid known allergens
- Wear wrap-around sunglasses outdoors
- Use artificial tears (preservative-free) to flush allergens
- Apply cold compresses to itchy eyes
- Remove contact lenses and switch to glasses
- Wash hands after touching pets
- Shower and change clothes after outdoor exposure

88.7 Diagnostic Approach

- Clinical history and eye examination
- Slit-lamp examination of the conjunctiva
- Allergy skin testing or specific IgE blood tests
- Conjunctival scraping for eosinophils in chronic cases

88.8 Treatment/Management Overview

- Antihistamine eye drops (olopatadine, ketotifen)
- Mast cell stabilizer eye drops (cromolyn sodium)
- Combination drops (antihistamine + mast cell stabilizer)
- Topical NSAID eye drops
- Lubricating eye drops (artificial tears)
- Oral antihistamines
- Topical corticosteroid eye drops for severe cases

Chapter 89

Eye Discharge

Definition

Abnormal secretion from the eye ranging from clear and watery to thick, purulent, or mucoid, indicating underlying ocular surface or lacrimal system pathology.

Pathophysiology

Watery discharge typically accompanies viral conjunctivitis or allergic reaction with increased lacrimation. Purulent (yellow/green) discharge indicates bacterial infection with neutrophil accumulation. Mucoid discharge (stringy, white) suggests allergic conjunctivitis. Chronic discharge may indicate lacrimal duct obstruction or dacryocystitis.

Etiology

- Conjunctivitis (viral, bacterial, or allergic)
- Dacryocystitis (lacrimal sac infection)
- Nasolacrimal duct obstruction
- Keratitis (corneal infection or inflammation)
- Stye (hordeolum) or chalazion
- Blepharitis (eyelid margin inflammation)
- Entropion or ectropion (eyelid malposition)
- Foreign body
- Contact lens-related infection
- Herpes simplex or herpes zoster ophthalmicus

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Eye discharge with vision changes, severe eye pain, photophobia, or in contact lens wearers (risk of corneal ulcer)
- Accompanied by other concerning signs

Related Symptoms

- Eye redness
- Itchy eyes
- Blurred vision
- Eye pain
- Photophobia

Note: Always consider serious underlying conditions when evaluating persistent eye discharge.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 90

Eye Floaters (Vitreous Floaters)

90.1 Definition

Eye floaters are small specks, spots, or thread-like shapes that drift across the field of vision. They are most noticeable when looking at a plain bright background. While usually benign, the sudden onset of floaters, especially with flashes, requires urgent evaluation.

90.2 Pathophysiology

Floaters are shadows cast on the retina by clumps within the vitreous humor of the eye. With age, the vitreous humor undergoes syneresis (liquefaction) and contraction, causing collagen fibers to aggregate. Posterior vitreous detachment (PVD) can pull on the retina, causing the sudden appearance of floaters and flashes.

90.3 Etiology

- Age-related vitreous degeneration (most common)
- Posterior vitreous detachment
- Retinal tear
- Retinal detachment
- Vitreous hemorrhage (diabetic retinopathy, trauma)
- Inflammatory cells (uveitis)
- Myopia (nearsightedness)
- Eye trauma or surgery
- Migraine (visual aura with scintillating scotomas)

90.4 Indications for Evaluation

- Small spots, specks, or cobweb-like shapes in vision
- Floaters that drift slowly across the visual field
- Shadows or dark shapes that move with eye movement
- Sudden increase in number or size of floaters
- Flashes of light in peripheral vision
- Curtain-like shadow or loss of vision

90.5 Related Symptoms

- Flashing lights (photopsia)
- Vision loss or visual field defect
- Blurred vision
- Retinal detachment
- Posterior vitreous detachment
- Migraine with aura

90.6 Self-Care/Home Management

- Move the eyes up and down to shift the floater out of the visual axis
- Try to ignore benign floaters (they often become less noticeable over time)
- Seek immediate evaluation if new floaters or flashes appear suddenly
- Do not rub the eyes
- Keep a healthy diet rich in antioxidants

90.7 Diagnostic Approach

- Comprehensive eye examination with pupillary dilation
- Slit-lamp biomicroscopy of the vitreous
- Indirect ophthalmoscopy of the retina
- B-scan ultrasound if view is obstructed
- Optical coherence tomography (OCT) of the retina

90.8 Treatment/Management Overview

- Observation and reassurance for benign floaters
- Treatment of underlying cause (diabetic retinopathy, uveitis)
- Laser vitreolysis (YAG laser for symptomatic floaters)
- Vitrectomy for severe, disabling floaters (rarely needed)
- Surgical repair for retinal tear or detachment

Chapter 91

Eye Pain (Ophthalmalgia)

Definition

Pain or discomfort localized to the eye or orbital region, which may be sharp, dull, burning, or throbbing, originating from ocular or periocular structures.

Pathophysiology

Eye pain is categorized as ocular (surface: cornea, conjunctiva, sclera) or orbital (deeper: optic nerve, extraocular muscles, vascular structures). Surface pain (foreign body sensation, burning) arises from corneal epithelial disruption (abrasion, dry eye, keratitis) or conjunctival inflammation. Deep, aching pain suggests glaucoma, optic neuritis, uveitis, or orbital pathology. Referred pain from trigeminal nerve branches may mimic eye pain.

Etiology

- Corneal abrasion or foreign body
- Dry eye disease (keratoconjunctivitis sicca)
- Conjunctivitis (viral, bacterial, allergic)
- Keratitis (infectious, contact lens-related)
- Acute angle-closure glaucoma (emergency)
- Uveitis or iritis
- Optic neuritis (associated with multiple sclerosis)
- Sinusitis (referred pain to eye)
- Orbital cellulitis (emergency)
- Cluster headache or migraine (ocular pain)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden severe eye pain with vision loss, redness, and halos (possible acute glaucoma), eye pain after trauma, or eye pain with fever and proptosis
- Accompanied by other concerning signs

Related Symptoms

- Headache
- Vision blurred
- Red eye
- Photophobia
- Nausea

Note: Always consider serious underlying conditions when evaluating persistent eye pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 92

Eye Strain (Asthenopia)

Definition

Ocular discomfort, fatigue, or pain associated with prolonged visual tasks, eye muscle imbalance, or uncorrected refractive errors.

Pathophysiology

Prolonged near-work (digital screens, reading) causes sustained contraction of ciliary muscle (accommodation) and medial rectus muscles (convergence), leading to muscle fatigue, reduced blink rate, and ocular surface drying. Accommodative spasm may follow prolonged near-focus, causing transient distance blur.

Etiology

- Prolonged screen time (computer vision syndrome / digital eye strain)
- Uncorrected refractive errors (myopia, hyperopia, astigmatism, presbyopia)
- Poor lighting or glare
- Decreased blink rate while concentrating
- Dry eye disease
- Accommodative or convergence insufficiency
- Small font sizes or poor screen contrast
- Postural issues (looking up vs. down at screens)
- Reading in dim light
- Eye muscle imbalance

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent eye strain despite taking breaks and correcting ergonomics, or eye strain with headaches, double vision, or vision changes
- Accompanied by other concerning signs

Related Symptoms

- Headache
- Blurred vision
- Dry eyes
- Neck pain
- Photophobia

Note: Always consider serious underlying conditions when evaluating persistent eye strain (asthenopia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 93

Eye Twitching (Blepharospasm and Myokymia)

Definition

Intermittent, involuntary, repetitive contractions of the orbicularis oculi muscle. Benign eyelid myokymia involves fine, rippling twitching of a single eyelid (usually lower), while blepharospasm involves sustained, bilateral, forceful eyelid closure.

Pathophysiology

Eyelid myokymia results from spontaneous depolarization of motor units in the orbicularis oculi muscle, often triggered by fatigue, stress, or caffeine. The pathophysiology involves increased excitability of the facial nerve nucleus or motor cortex. Blepharospasm is a focal dystonia involving involuntary contraction of the orbicularis oculi due to basal ganglia dysfunction and altered inhibition in the brainstem blink reflex circuits.

Etiology

- Fatigue and sleep deprivation (most common cause of myokymia)
- Stress and anxiety
- Caffeine or stimulant intake
- Eye strain (prolonged screen time, uncorrected vision)
- Dry eyes (ocular surface irritation triggers reflex blinking)
- Nutritional deficiencies (magnesium, potassium, calcium)
- Alcohol or tobacco use
- Benign essential blepharospasm (focal dystonia, typically bilateral)
- Hemifacial spasm (unilateral, involving entire side of face, vascular compression of CN VII)
- Medications (antipsychotics, antihistamines, SSRIs)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Twitching spreading to other parts of the face, both eyes closing involuntarily, eye redness or discharge, or twitching lasting more than 2 weeks
- Accompanied by other concerning signs

Related Symptoms

- Eye strain
- Dry eyes
- Facial twitching
- Blepharospasm
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent eye twitching (blepharospasm and myokymia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 94

Eyelid Bump (Stye and Chalazion)

Definition

A localized, tender, erythematous nodule on the eyelid margin caused by acute bacterial infection of eyelash follicle (external hordeolum/style) or meibomian gland (internal hordeolum); a chalazion is a chronic, painless, inflammatory granuloma of the meibomian gland.

Pathophysiology

External styes involve infection of the Zeis or Moll glands (eyelash follicles) by *Staphylococcus aureus*, causing acute suppurative inflammation. Internal styes infect meibomian glands with similar pathophysiology. Chalazion results from obstruction of the meibomian gland duct without acute infection, leading to lipogranulomatous inflammation with lipid-laden macrophages, giant cells, and a firm, nontender nodule.

Etiology

- *Staphylococcus aureus* infection (most common cause of stye)
- Poor eyelid hygiene
- Blepharitis (chronic eyelid inflammation predisposing to styes)
- Contact lens use (improper hygiene)
- Rosacea (associated with meibomian gland dysfunction)
- Seborrheic dermatitis
- Hormonal changes
- Stress and fatigue
- Use of expired or contaminated eye makeup
- Duct obstruction from thickened meibomian secretions (chalazion)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Recurrent styes, lesion not resolving within 2 weeks, vision changes, eyelid swelling spreading to entire eye or face, or fever
- Accompanied by other concerning signs

Related Symptoms

- Eyelid swelling
- Eye redness
- Eye discharge
- Eye pain
- Blepharitis

Note: Always consider serious underlying conditions when evaluating persistent eyelid bump (stye and chalazion).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 95

Eyelid Inflammation (Blepharitis)

95.1 Definition

Blepharitis is a common, chronic inflammation of the eyelid margins. It affects people of all ages and is frequently associated with dry eye syndrome and meibomian gland dysfunction.

95.2 Pathophysiology

Blepharitis is classified as anterior (affecting the eyelash follicles) or posterior (affecting the meibomian glands). Anterior blepharitis is often caused by staphylococcal colonization or seborrheic dermatitis. Posterior blepharitis involves meibomian gland dysfunction with abnormal lipid secretion, leading to tear film instability and ocular surface inflammation.

95.3 Etiology

- Staphylococcal bacterial colonization of lid margins
- Seborrheic dermatitis
- Meibomian gland dysfunction
- Rosacea (ocular rosacea)
- Demodex mite infestation
- Contact lens wear
- Allergies
- Dry eye syndrome

95.4 Indications for Evaluation

- Red, swollen, and irritated eyelid margins
- Crusting and flaking of the eyelids (especially on waking)
- Burning or gritty sensation in the eyes

- Itching of the eyelids
- Watery eyes
- Crusty eyelashes or loss of eyelashes
- Sensitivity to light
- Chalazion or hordeolum (stye) formation

95.5 Related Symptoms

- Dry eyes (dry eye syndrome)
- Conjunctivitis (pink eye)
- Stye (hordeolum)
- Chalazion
- Eyelash loss (madarosis)
- Ocular rosacea

95.6 Self-Care/Home Management

- Apply warm compresses for 5-10 minutes daily
- Gently scrub the eyelid margins with a clean washcloth or lid scrub
- Use OTC artificial tears for dry eye symptoms
- Avoid eye makeup during flare-ups
- Clean the eyelids with baby shampoo diluted in warm water
- Replace eye makeup every 3 months

95.7 Diagnostic Approach

- Slit-lamp examination of the eyelid margins
- Evaluation of meibomian gland function
- Eyelash microscopy for Demodex
- Assessment of tear film break-up time
- Culture of lid margins if severe or treatment-resistant

95.8 Treatment/Management Overview

- Warm compresses and eyelid hygiene (cornerstone of treatment)
- Antibiotic ointments (erythromycin, bacitracin) for bacterial blepharitis
- Topical cyclosporine for ocular rosacea
- Oral doxycycline or minocycline for meibomian gland dysfunction
- Topical corticosteroid drops for acute inflammation
- Management of associated conditions (dry eye, rosacea)

Chapter 96

Facial Droop

Definition

Unilateral or bilateral weakness of facial muscles resulting in asymmetry of expression, inability to close the eye or raise the eyebrow, and flattening of the nasolabial fold.

Pathophysiology

Facial droop results from dysfunction of the facial nerve (cranial nerve VII) at any point from its nucleus in the pons to its peripheral branches. Lower motor neuron lesions (Bell palsy, Ramsay Hunt, Lyme) affect entire ipsilateral face; upper motor neuron lesions (stroke, tumor) spare the forehead due to bilateral cortical innervation.

Etiology

- Bell palsy (idiopathic facial nerve palsy)
- Stroke or transient ischemic attack (upper motor neuron)
- Ramsay Hunt syndrome (herpes zoster oticus)
- Lyme disease (borreliosis)
- Brain tumor or metastatic lesion
- Traumatic facial nerve injury
- Otitis media or mastoiditis
- Guillain-Barre syndrome (bilateral facial weakness)
- Sarcoidosis (Heerfordt syndrome)
- Myasthenia gravis (fatigable weakness)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden facial droop requires immediate medical evaluation to differentiate stroke from Bell palsy (FAST assessment)
- Accompanied by other concerning signs

Related Symptoms

- Arm or leg weakness
- Speech difficulty
- Ear pain
- Eye closure difficulty
- Drooling

Note: Always consider serious underlying conditions when evaluating persistent facial droop.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 97

Facial Numbness

Definition

Loss or alteration sensation face involving trigeminal nerve distribution may unilateral bilateral partial complete sensory changes paresthesia dysesthesia anesthesia hypoesthesia affecting dermatomes ophthalmic maxillary mandibular divisions trigeminal nerve fifth cranial nerve CN V

Pathophysiology

Results disruption transmission sensory information facial skin mucosa oral cavity nasal cavity teeth gums tongue anterior two-thirds via trigeminal pathways central nervous system brainstem thalamus somatosensory cortex compression ischemia inflammation demyelination infiltration destruction afferent fibers peripheral ganglia nuclei tracts ascending pathways suprasynaptic processing interruption various etiologies spanning structural inflammatory infectious neoplastic vascular metabolic toxic degenerative autoimmune traumatic iatrogenic congenital developmental anomalies differentiate paresthesia tingling prickling abnormal positive sensations numbness represents reduced diminished absent perception negative sensory phenomena clinical distinction critical localization lesion site differential diagnostic consideration management implications prognosis outcomes

Etiology

- Bell palsy acute idiopathic facial paralysis most common cause isolated facial sensory deficit transient recovery weeks months
- Multiple sclerosis demyelinating trigeminal pathway young adults relapsing remitting course MRI findings oligoclonal bands CSF analysis confirmatory immune-modulating therapy
- Trigeminal neuralgia paroxysmal sharp lancinating electric shock-like trigger zone light touch cold wind chewing talking episodic seconds minutes responds carbamazepine oxcarbazepine surgery refractory gamma knife radiosurgery
- Acoustic neuroma vestibular schwannoma benign tumor cerebellopontine angle compress trigeminal labyrinthine nerves progressive hearing loss tinnitus balance issues facial weakness develop slowly years surgical radiation treatment options depend size age comorbidities

functional outcomes quality of life rehabilitation support multidisciplinary team otolaryngologists neurosurgeons radiation oncologists audiologists speech therapists psychologists social workers coordinate holistic care individualized strategies recovery adaptation coping community resources ongoing follow-up surveillance monitoring recurrence complications proactive intervention prevent worsening preserve function maximize independence autonomy dignity ethical considerations informed consent shared responsibility collaborative partnership patient family healthcare providers best wellbeing thrive adversity challenges journey recovery healing restoration wholeness harmony peace tranquility fulfillment happiness love compassion empathy gratitude acceptance belonging connection trust security stability dependability commitment loyalty devotion patience endurance strength courage honor integrity justice

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Persistent any duration warrant neurological evaluation especially onset sudden progressive association other neurological deficits speech difficulty facial droop limb weakness visual disturbance gait imbalance double vision slurred speech confusion memory problems mood changes behavioral alterations cognitive decline executive dysfunction apraxia agnosia aphasia dysarthria dysphagia vertigo ataxia tremor rigidity bradykinesia hyperreflexia hyporeflexia Babinski clonus myoclonus ptosis miosis mydriasis anisocoria oculomotor palsy abducens trochlear diplopia nystagmus strabismus exophthalmos enophthalmos conjunctivitis uveitis scleritis retinitis choroiditis vitreoretinopathy optic neuritis pseudotumor intracranial hypertension hydrocephalus syrinx cyst tumor abscess hemorrhage infarction embolism stroke TIA cerebral venous sinus thrombosis arteriovenous malformation aneurysm cavernous angioma capillary telangiectasia watershed borderzone territory ischemic hemorrhagic stroke intracerebral subarachnoid hematoma contusion concussion fracture skull base temporal bone orbitofrontal ethmoid sphenoid occipital parietal frontal sinus mastoid petrous jugular canal cochlear vestibule semicircular ampulla crista utricle saccule Meniere's disease vestibular migraine bilateral vestibulopathy superior semicircular canal dehiscence fistula glioblastoma medulloblastoma ependymoma oligodendrocytoma urgent assessment indicated emergent evaluation required immediate attention warranted concerning presentation demands professional medical intervention consultation evaluation diagnostic workup therapeutic management follow-up care coordination monitoring assessment of progression resolution outcome prognosis risk factors preventive measures lifestyle modifications supportive care rehabilitation services psychosocial support educational resources community access equitable healthcare delivery systemic integration collaborative effort interdisciplinary teamwork comprehensive approach holistic patient-centered care continuous quality improvement evidence-based practice guidelines recommendations standards protocols benchmarks indicators metrics measurements evaluations assessments reviews audits feedback mechanisms processes procedures systems infrastructure technology platforms tools resources supplies equipment facilities personnel training education professional development competency certification accreditation regulatory compliance legal ethical considerations policy frameworks governance structures accountability transparency communication engagement stakeholder partnerships collaborations alliances networks consortia organizations associations societies leagues federations unions groups clubs bands syndicates cartels trusts cooperatives joint-stock corporations

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 UNESCO world heritage listings national parks state parks provincial parks municipal parks
 county parks city parks township parks neighborhood parks pocket parks green spaces lawns
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 acrobats magicians illusionists escape artists clowns jesters fools jesting humor wit comedy
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 reliefs paintings drawings sketches photographs images illustrations graphics cartoons comics
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Related Symptoms

- Sensory loss pattern division involvement unilateral bilateral progression associated facial pain weakness drooping eye closure difficulties corneal reflex changes taste alterations saliva production dry mouth drooling difficulty speaking swallowing food spillage emotional expressions flat affect reduced facial mobility social interaction embarrassment psychological impact quality of life deterioration depression anxiety isolation withdrawal

Note: Always consider serious underlying conditions when evaluating persistent facial numbness

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 98

Fainting (Syncope)

98.1 Definition

Syncope (fainting) is a transient loss of consciousness due to global cerebral hypoperfusion, characterized by rapid onset, brief duration, and spontaneous complete recovery. It is a common symptom with many potential causes.

98.2 Pathophysiology

Cerebral blood flow requires adequate cardiac output, systemic vascular resistance, and cerebral perfusion pressure. Syncope occurs when there is a sudden reduction in one or more of these factors: reflex-mediated bradycardia and vasodilation (reflex syncope), orthostatic hypotension with inadequate compensation, or cardiac causes with reduced pump function or arrhythmias.

98.3 Etiology

- Vasovagal syncope (emotional stress, pain, fear, prolonged standing)
- Orthostatic hypotension (dehydration, blood loss, medications)
- Situational syncope (micturition, defecation, coughing, swallowing)
- Cardiac arrhythmias (bradycardia, heart block, ventricular tachycardia)
- Structural heart disease (aortic stenosis, hypertrophic cardiomyopathy)
- Pulmonary embolism
- Medication-induced (vasodilators, diuretics, beta-blockers)

98.4 Indications for Evaluation

- Sudden, brief loss of consciousness
- Prodrome of lightheadedness, nausea, sweating, pallor
- Rapid return to full consciousness without confusion

- Precipitating factors (prolonged standing, warm environment, emotional triggers)
- Palpitations or chest pain preceding syncope
- Syncope during exercise
- Family history of sudden cardiac death

98.5 Related Symptoms

- Dizziness or lightheadedness
- Presyncope (near-fainting)
- Palpitations
- Seizures (may be confused with syncope)
- Fall or injury from loss of consciousness
- Sudden cardiac arrest

98.6 Self-Care/Home Management

- If feeling faint, lie down or sit with head between knees
- Cross legs and tense muscles to increase blood pressure
- Stay well hydrated
- Avoid prolonged standing in hot environments
- Stand up slowly from sitting or lying position
- Avoid alcohol and heavy meals before activities
- Recognize and avoid individual triggers

98.7 Diagnostic Approach

- Detailed history of the event and triggers
- Orthostatic vital signs (blood pressure and heart rate supine and standing)
- 12-lead electrocardiogram to exclude arrhythmia
- Echocardiogram to evaluate structural heart disease
- Holter or event monitor for suspected arrhythmia
- Tilt-table test for recurrent vasovagal syncope

98.8 Treatment/Management Overview

- Reassurance and education for reflex syncope
- Increase fluid and salt intake (if not contraindicated)
- Compression stockings for orthostatic hypotension
- Discontinue or adjust offending medications
- Pacemaker for bradycardia or heart block
- ICD for ventricular arrhythmias
- Specific therapy for underlying cardiac or pulmonary cause

Chapter 99

Fatigue

Definition

Extreme tiredness, lack of energy, distinct from ordinary physical tiredness; persists despite rest.

Pathophysiology

Multifactorial symptom representing potential combination of peripheral muscle fatigue and central nervous system mechanisms across numerous conditions.

Etiology

- Depression/anxiety disorders
- Anemia
- Obstructive sleep apnea
- Hypothyroidism
- Medication side effects

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Persistent interfering with daily life (>4 weeks) warrants investigation
- Accompanied by other concerning signs

Related Symptoms

- Weakness/concentration difficulty
- Sleep disturbance
- Mood changes
- Malaise
- Reduced exercise tolerance

Note: Always consider serious underlying conditions when evaluating persistent fatigue.

Self-Care / Home Management

- Prioritize adequate rest and sleep to support the body's healing and recovery processes.
- Maintain a balanced diet with regular meals and stay well-hydrated.
- Monitor symptoms and track any changes in severity, frequency, or associated features.
- Avoid overexertion and gradually increase activity levels as tolerated.
- Seek medical evaluation if symptoms persist, worsen, or interfere significantly with daily function.

Diagnostic Approach

- Comprehensive history and physical examination guide the diagnostic workup based on the specific symptom presentation.
- Laboratory testing (CBC, metabolic panel, inflammatory markers) helps screen for underlying systemic conditions.
- Imaging studies are tailored to the suspected anatomic region and pathology.
- Referral to appropriate specialists is guided by the differential diagnosis established through initial evaluation.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic management may include pharmacotherapy, lifestyle modifications, and supportive care.
- Multidisciplinary care is often beneficial for complex or chronic symptoms.
- Close follow-up ensures treatment effectiveness and allows adjustment of the management plan as needed.

Chapter 100

Febrile Seizure

Definition

A seizure occurring in children aged 6 months to 5 years, triggered by fever (temperature above 38 extdegreeC) in the absence of intracranial infection, metabolic disturbance, or history of afebrile seizures. Simple febrile seizures last less than 15 minutes and are generalized; complex febrile seizures are focal, prolonged, or recurrent within 24 hours.

Pathophysiology

Fever causes rapid temperature elevation that lowers the seizure threshold in the developing brain through cytokine-mediated effects (IL-1, IL-6) on neuronal excitability and GABAergic inhibition. Genetic susceptibility involving sodium channel and GABA receptor genes increases risk. The immature blood-brain barrier and developing myelination make young children particularly vulnerable. Simple febrile seizures do not cause brain damage.

Etiology

- Upper respiratory tract infections (otitis media, pharyngitis, tonsillitis)
- Viral infections (influenza, adenovirus, RSV, HHV-6 roseola, enterovirus)
- Pneumonia or bronchiolitis
- Urinary tract infection (especially in young children)
- Gastroenteritis
- Bacterial infections (pneumonia, occult bacteremia)
- Post-immunization fever (MMR, DTaP, pneumococcal, influenza)
- Roseola infantum (HHV-6) — classic cause of febrile seizures
- Dental abscess or other focal infection
- Genetic predisposition (family history of febrile seizures)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- First febrile seizure always requires emergency evaluation to exclude meningitis, encephalitis, or electrolyte disturbance; seizure lasting more than 5 minutes, focal seizure, or multiple seizures within 24 hours
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Seizures
- Loss of consciousness
- Confusion
- Shaking

Note: Always consider serious underlying conditions when evaluating persistent febrile seizure.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 101

Fever

Definition

Fever is defined as an elevation of core body temperature above the normal diurnal range, typically exceeding 38.0°C (100.4°F) when measured orally or 38.3°C (100.9°F) rectally. It represents a regulated increase in the hypothalamic set-point mediated by endogenous pyrogens during immune activation, distinguishing it from hyperthermia where thermoregulation fails.

Pathophysiology

Fever results from the action of exogenous and endogenous pyrogens on the hypothalamus. Exogenous pyrogens stimulate immune cells to release endogenous pyrogens including interleukin-1 (IL-1), IL-6, tumor necrosis factor-alpha (TNF-*alpha*), and interferon-alpha. These cytokines induce cyclooxygenase-2 expression leading to increased prostaglandin E2 synthesis which raises the thermoregulatory set-point. Heat-conservation mechanisms (vasoconstriction, shivering) activate until temperature reaches new set-point. Resolution involves active heat-loss (vasodilation, sweating).

Etiology

- Infectious causes (most common): Viral infections (URI, influenza, enteroviruses, hepatitis, HIV), Bacterial infections (pneumonia, UTI, cellulitis, endocarditis, abscesses, meningitis), Fungal infections (histoplasmosis, coccidioidomycosis), Parasitic infections (malaria, toxoplasmosis)
- Inflammatory/autoimmune conditions: Rheumatoid arthritis, systemic lupus erythematosus, vasculitides (giant cell arteritis, Takayasu arteritis), inflammatory bowel disease (Crohn disease, ulcerative colitis), sarcoidosis
- Neoplastic causes: Lymphomas (especially Hodgkin lymphoma), leukemias, renal cell carcinoma, hepatocellular carcinoma, melanoma, multiple myeloma
- Drug-induced fever: Antibiotics (sulfonaminals, beta-lactams), anticonvulsants (phenytoin, carbamazepine), antiarrhythmics (quinidine), biologics, vaccines
- Thromboembolic events: Pulmonary embolism, deep vein thrombophlebitis

- Other causes: Heat stroke, CNS hemorrhage, post-operative state, factitious disorder

Indications for Evaluation

Seek care if:

- Fever persisting beyond 3 days without identified source
- Temperature $\geq 40^{\circ}\text{C}$ (104°F) regardless of duration
- Fever accompanied by neck stiffness, photophobia, altered mental status (meningeal irritation)
- Immunocompromised patient with any fever (temperature $\geq 38.3^{\circ}\text{C}$)
- Fever with rash that does not blanch (possible meningococemia)
- Severe headache, abdominal pain, flank pain with fever

Related Symptoms

- Chills
- Rigors
- Diaphoresis
- Headache
- Myalgia
- Arthralgia
- Fatigue
- Anorexia
- Weight loss
- Night sweats
- Dehydration
- Tachycardia
- Tachypnea

Note: Persistent fever warrants systematic investigation including blood cultures, CBC with differential, inflammatory markers (ESR, CRP), urinalysis, chest imaging, and consideration of autoimmune/neoplastic etiologies when initial workup negative.

Self-Care / Home Management

- Rest and maintain adequate hydration with water, clear fluids, or oral rehydration solutions.
- Use acetaminophen or ibuprofen to reduce fever and body aches (follow label dosing instructions).
- Dress lightly and keep the room at a comfortable temperature — avoid bundling excessively.
- Monitor temperature at regular intervals and track other symptoms to identify worsening trends.
- Seek medical attention for fever above 40 °C (104 °F), fever lasting more than 3 days, or fever with stiff neck, severe headache, rash, or difficulty breathing.

Diagnostic Approach

- History includes fever duration, pattern (intermittent, remittent, continuous), associated symptoms (chills, rigors, sweats), and exposure history.
- Vital signs assess fever severity and hemodynamic stability; complete physical examination seeks a source.
- Laboratory evaluation includes CBC with differential, blood cultures, urinalysis, inflammatory markers (CRP, ESR), and targeted testing based on exposure history.
- Imaging (chest X-ray, abdominal ultrasound, CT) is guided by clinical findings and suspected source.

Treatment / Management Overview

- Antipyretics (acetaminophen, NSAIDs) provide symptomatic relief but do not treat the underlying cause.
- Empiric antibiotics may be indicated for suspected bacterial infection after appropriate cultures are obtained.
- Antiviral, antifungal, or antimalarial therapy is directed at specific pathogens based on exposure and testing.
- Source control (drainage of abscess, removal of infected device) is essential for focal infections.

Chapter 102

Fluid from the Ear (Otorrhea)

Definition

Any discharge from the ear canal, varying in character (clear, bloody, purulent, serous) depending on the underlying pathology of the external ear, middle ear, or mastoid.

Pathophysiology

Otorrhea results from external ear canal infection (otitis externa), middle ear infection with tympanic membrane perforation (otitis media suppurativa), cholesteatoma with secondary infection, or cerebrospinal fluid leakage from skull base fracture. Purulent discharge suggests bacterial infection; clear fluid suggests CSF leak (halo sign) or serous effusion.

Etiology

- Acute otitis externa (swimmer's ear)
- Acute otitis media with tympanic membrane perforation
- Chronic suppurative otitis media
- Cholesteatoma (keratinizing squamous epithelial cyst)
- Foreign body in ear canal with secondary infection
- Traumatic tympanic membrane perforation
- CSF otorrhea from temporal bone fracture (clear fluid)
- Ear canal eczema or seborrheic dermatitis
- Furuncle or abscess of ear canal
- Post-surgical (after ear tube placement, mastoidectomy)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Ear discharge with severe pain, fever, hearing loss, or head trauma; clear watery discharge after head injury
- Accompanied by other concerning signs

Related Symptoms

- Earache
- Hearing loss
- Fever
- Dizziness
- Ear fullness

Note: Always consider serious underlying conditions when evaluating persistent fluid from the ear (otorrhea).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 103

Foot Pain

Definition

Pain or discomfort localized anywhere in the foot, from the hindfoot (heel, ankle) through the midfoot (arch) to the forefoot (toes, metatarsals), with causes ranging from mechanical to systemic.

Pathophysiology

Foot pain etiology varies by location: heel pain most commonly from plantar fasciitis; midfoot pain from degenerative arthritis or stress fracture; metatarsalgia (forefoot pain) from weight-bearing overload, Morton neuroma, or Freiberg infarction; great toe pain from hallux valgus (bunion), hallux rigidus, or gout. The foot's 26 bones, 33 joints, and complex ligamentous/tendinous architecture create numerous pain generators.

Etiology

- Plantar fasciitis
- Metatarsalgia
- Morton neuroma (interdigital neuroma)
- Hallux valgus (bunion)
- Hallux rigidus (great toe arthritis)
- Hammer toe or claw toe deformity
- Stress fracture (metatarsal, navicular)
- Tarsal tunnel syndrome
- Gout (usually first metatarsophalangeal joint)
- Flatfoot (pes planus) or cavus foot deformity

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Foot pain with inability to bear weight, open wound or ulcer, fever, redness spreading up the leg, or sudden severe pain with swelling
- Accompanied by other concerning signs

Related Symptoms

- Heel pain
- Ankle pain
- Toe pain
- Swelling
- Numbness or tingling

Note: Always consider serious underlying conditions when evaluating persistent foot pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 104

Frequent Bowel Movements

Definition

Increased frequency of defecation exceeding three bowel movements per day, often with loose or urgent stools, reflecting altered gastrointestinal motility or secretion.

Pathophysiology

Increased stool frequency results from reduced colonic transit time (rapid transit prevents water absorption leading to loose stools), increased secretion (enterotoxins, bile acid malabsorption), or increased smooth muscle contractility (IBS, hyperthyroidism). Acute onset suggests infection (gastroenteritis) while chronic suggests functional IBS, IBD, malabsorption, or medication effect.

Etiology

- Acute gastroenteritis (viral, bacterial, parasitic)
- Irritable bowel syndrome (IBS-D)
- Inflammatory bowel disease (Crohn disease, ulcerative colitis)
- Food intolerances (lactose, gluten, FODMAPs)
- Hyperthyroidism
- Bile acid malabsorption
- Medication side effects (metformin, antibiotics, laxatives, SSRIs)
- Anxiety or stress-related
- Celiac disease
- Colon cancer or carcinoid syndrome (rare)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Frequent bowel movements with blood, weight loss, fever, nocturnal symptoms, or dehydration
- Accompanied by other concerning signs

Related Symptoms

- Diarrhea
- Abdominal pain
- Nausea
- Urgency
- Bloating

Note: Always consider serious underlying conditions when evaluating persistent frequent bowel movements.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 105

Frequent Urination (Urinary Frequency)

105.1 Definition

Frequent urination is the need to urinate more often than normal. The typical adult urinates 4-8 times per day. Causes range from benign lifestyle factors to serious medical conditions.

105.2 Pathophysiology

Urinary frequency occurs when the bladder cannot hold the normal volume of urine (decreased functional capacity), the body produces excess urine (polyuria), or the bladder is hypersensitive. Mechanisms include: detrusor overactivity, bladder outlet obstruction, increased fluid intake, diabetes (polydipsia), urinary tract infection (irritation), or bladder compression.

105.3 Etiology

- Urinary tract infection (most common cause of acute frequency)
- Overactive bladder (detrusor overactivity)
- Benign prostatic hyperplasia (BPH) in men
- Diabetes mellitus (polyuria from hyperglycemia)
- Diabetes insipidus (excessive dilute urine)
- Diuretic use (medications, caffeine, alcohol)
- Excessive fluid intake (polydipsia)
- Pregnancy (compression from uterus)
- Pelvic organ prolapse
- Interstitial cystitis (painful bladder syndrome)
- Bladder stones or tumors

- Radiation cystitis
- Anxiety or habit

105.4 Indications for Evaluation

- Urinating more than 8 times in 24 hours
- Waking up to urinate (nocturia)
- Sudden onset of frequency (think UTI or diabetes)
- Pain or burning with urination
- Excessive thirst and increased fluid intake
- Incomplete bladder emptying
- Change in urine color or odor
- Fever or back pain

105.5 Related Symptoms

- Urinary urgency
- Nocturia (nighttime urination)
- Dysuria (painful urination)
- Polyuria (excessive urine output)
- Urinary incontinence

105.6 Self-Care/Home Management

- Reduce caffeine and alcohol intake
- Limit fluids before bedtime
- Bladder training: gradually increase intervals between urination
- Pelvic floor exercises
- Keep a bladder diary to track patterns
- Avoid bladder irritants (spicy foods, citrus, artificial sweeteners)

105.7 Diagnostic Approach

- Urinalysis and urine culture
- Bladder diary
- Blood glucose and HbA1c
- Post-void residual measurement
- Urodynamic studies if indicated

105.8 Treatment/Management Overview

- Treat underlying cause (antibiotics for UTI, diabetes management, BPH treatment)
- Behavioral therapy (bladder training, pelvic floor therapy)
- Anticholinergic or beta-3 agonist medications for OAB
- Avoid excessive fluid intake

Chapter 106

Frostbite

106.1 Definition

Frostbite is a cold-induced injury to tissues caused by freezing of water in cells and extracellular spaces. It most commonly affects exposed extremities (fingers, toes, ears, nose, cheeks). Severity ranges from superficial (first-degree) to deep tissue necrosis (fourth-degree).

106.2 Pathophysiology

Cold exposure causes vasoconstriction to preserve core temperature, reducing blood flow to extremities. At temperatures below 0 C (32 F), ice crystals form in extracellular fluid, causing osmotic shifts and cellular dehydration. Subsequent thawing and reperfusion triggers inflammatory cascade with microvascular thrombosis, edema, and tissue necrosis.

106.3 Etiology

- Prolonged exposure to freezing temperatures
- Inadequate clothing or protective gear
- High wind chill factor
- Contact with cold surfaces or freezing metals
- High altitude with decreased oxygen
- Alcohol or drug intoxication (impaired judgment)
- Peripheral vascular disease
- Diabetes mellitus
- Smoking (vasoconstriction)

106.4 Indications for Evaluation

- Cold, pale, and waxy skin
- Numbness or loss of sensation in the affected area
- Skin feels firm or frozen to the touch
- After rewarming: pain, throbbing, burning sensation
- Blisters or skin discoloration (red, blue, black)
- Tissue necrosis or gangrene in severe cases

106.5 Related Symptoms

- Hypothermia (systemic cold exposure)
- Chilblains (pernio, non-freezing cold injury)
- Trench foot (immersion foot)
- Raynaud phenomenon
- Gangrene (tissue death)
- Peripheral neuropathy

106.6 Self-Care/Home Management

- Move to a warm environment immediately
- Remove wet or constrictive clothing
- Immerse affected areas in warm water (37-39 C, 100-102 F) for 30-60 minutes
- Do NOT use direct heat (heating pads, fire) - causes burns
- Do NOT rub or massage frostbitten tissue
- Do NOT break blisters
- Elevate affected extremities

106.7 Diagnostic Approach

- Clinical assessment of depth and extent of injury
- Rewarming in warm water and post-thaw assessment
- Technetium-99m bone scan or angiography for deep injury
- MRI or magnetic resonance angiography for tissue viability
- Doppler ultrasound for blood flow assessment

106.8 Treatment/Management Overview

- Rapid rewarming in warm water (37-39 C) with analgesia
- Tetanus prophylaxis if needed
- Ibuprofen for inflammation
- Topical aloe vera gel
- Debridement of clearly necrotic tissue (delayed, 4-6 weeks)
- Thrombolytic therapy with tPA for severe deep frostbite
- Amputation for non-viable tissue (delayed until demarcation)

Chapter 107

Ganglion Cyst

107.1 Definition

A ganglion cyst is a noncancerous, fluid-filled swelling that most commonly develops along the tendons or joints of the wrists, hands, and feet. It is the most common soft-tissue mass of the hand and wrist.

107.2 Pathophysiology

The exact mechanism is unclear, but ganglion cysts are believed to arise from myxoid degeneration of the connective tissue in joint capsules or tendon sheaths. The cyst fluid resembles synovial fluid but with higher levels of hyaluronic acid. A small stalk connecting to the joint cavity is often present, with a one-way valve mechanism thought to allow fluid accumulation.

107.3 Etiology

- Joint or tendon stress from repetitive use
- Joint or tendon injury
- Joint or tendon irritation
- Degenerative changes
- Connective tissue weakness (generally unknown cause)
- More common in women aged 20-40 and in gymnasts

107.4 Indications for Evaluation

- Visible or palpable lump under the skin
- Most commonly on the dorsum of the wrist
- Smooth, round, and firm or slightly fluctuant
- Transilluminates brightly

- Size may vary over time
- May or may not be painful
- Pain or weakness with joint movement
- Possible tingling or numbness if pressing on a nerve

107.5 Related Symptoms

- Wrist pain
- Hand or wrist swelling
- Carpal tunnel syndrome
- Arthritis of the wrist
- Tendonitis

107.6 Self-Care/Home Management

- Observe and monitor (many resolve spontaneously)
- Wear a wrist brace or splint to limit movement
- Avoid activities that aggravate the cyst
- Avoid attempting to pop or drain the cyst at home
- Apply over-the-counter pain relievers for discomfort

107.7 Diagnostic Approach

- Clinical examination
- Transillumination (confirms cystic nature)
- Ultrasound to confirm cystic nature and rule out solid mass
- MRI if diagnosis is uncertain or for deep cysts
- Aspiration reveals thick, clear, jelly-like fluid

107.8 Treatment/Management Overview

- Observation (up to 50 percent resolve spontaneously)
- Aspiration with a large-bore needle
- Aspiration followed by corticosteroid injection
- Surgical excision with ligation of the stalk
- Splinting to limit movement during recovery

Chapter 108

Green Stool

Definition

Green-colored stool resulting from the presence of bile pigments, rapid intestinal transit, or ingestion of green-colored foods or supplements.

Pathophysiology

Bilirubin produced from heme breakdown is conjugated in the liver, secreted into bile, and normally metabolized by gut bacteria to stercobilin (brown pigment). Green stool occurs when bile passes through the intestine too rapidly for bacterial conversion to stercobilin, or when biliverdin (green pigment) is not fully reduced. Dietary chlorophyll (green vegetables) and artificial food coloring (green dyes) are common benign causes.

Etiology

- Dietary: green leafy vegetables (spinach, kale)
- Food coloring (green dyes in beverages, candies, gelatin)
- Rapid intestinal transit (diarrhea, IBS-D)
- Bile acid malabsorption
- Antibiotic use (altered gut flora)
- Iron supplements
- Green-colored medications or supplements
- Infectious gastroenteritis (especially *Salmonella*)
- Crohn disease or ulcerative colitis
- Malabsorption syndromes

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent green stool with diarrhea, weight loss, abdominal pain, or other concerning GI symptoms
- Accompanied by other concerning signs

Related Symptoms

- Diarrhea
- Abdominal pain
- Bloating
- Weight loss
- Nausea

Note: Always consider serious underlying conditions when evaluating persistent green stool.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 109

Groin Pain or Swelling

Definition

Pain, discomfort, or visible enlargement in the inguinal or pubic region, arising from musculoskeletal, hernia-related, vascular, or genitourinary causes.

Pathophysiology

Groin pain in younger patients commonly results from adductor muscle strain (sports hernia or athletic pubalgia). Hernias present with bulging aggravated by increased intra-abdominal pressure. Lymphadenopathy suggests infection or malignancy. Testicular causes (torsion, epididymitis) refer pain to groin via shared innervation.

Etiology

- Inguinal or femoral hernia
- Adductor muscle strain (groin pull)
- Lymphadenopathy (inguinal lymph nodes)
- Epididymitis or orchitis
- Testicular torsion or tumor
- Osteitis pubis
- Hip joint pathology (labral tear, OA)
- Kidney stones (referred pain)
- Femoral artery pseudoaneurysm (post-catheterization)
- Psoas abscess

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe groin pain with nausea/vomiting (possible strangulated hernia), testicular pain with swelling, or fever with groin pain
- Accompanied by other concerning signs

Related Symptoms

- Hip pain
- Lower abdominal pain
- Testicular pain
- Nausea
- Lump in groin

Note: Always consider serious underlying conditions when evaluating persistent groin pain or swelling.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 110

Gum Swelling

Definition

Localized enlargement of gum tissue resulting from inflammation, edema, accumulation of fluid and cellular components, vascular changes at the gingival margin area surrounding the tooth neck; visible protrusion beyond normal contour, typically red, tender, bleeding upon brushing, eating, or spicy foods.

Pathophysiology

Gum swelling arises from inflammatory response triggered by bacterial plaque, biofilm, dental caries, calculus/tartar, foreign material, trauma, ill-fitting restorations, braces, retainers, dentures, implants, systemic medications, immunosuppressants, hormonal fluctuations, pregnancy, puberty, thyroid dysfunction, blood disorders, hematologic diseases, leukemias, lymphomas, autoimmune conditions, lupus, vasculitides, granulomatous diseases, Crohn's, sarcoidosis, Behçet's, drug eruptions, angioedema, allergic reactions, infections (viral: herpes simplex HSV, Epstein-Barr EBV, cytomegalovirus CMV, HIV), HIV-associated periodontitis, necrotizing ulcerative gum disease (NUG/ANUG), acute necrotizing ulcerative gingivitis, Vincent infection (spirochetes, *Fusobacterium necrophorum*), anaerobic bacteria overgrowth.

Etiology

- Gingivitis (early reversible stage of gum disease)
- Periodontal disease (gum disease progressing to bone loss)
- Dental abscess (infection at tooth root or between gums)
- Impacted wisdom teeth
- Trauma or injury to gums
- Poor oral hygiene
- Hormonal changes (pregnancy, puberty, menstruation)
- Certain medications (phenytoin, cyclosporine, calcium channel blockers)

- Vitamin deficiencies (especially vitamin C)
- Blood disorders (leukemia, anemia)

Indications for Evaluation

Seek care if:

- Severe or worsening swelling
- Difficulty breathing or swallowing
- Fever with gum swelling
- Persistent bleeding
- Swelling interfering with speech or eating
- Loose teeth without obvious cause

Related Symptoms

- Bad breath (halitosis)
- Tooth pain
- Mouth odor
- Difficulty chewing
- Gum tenderness
- Swollen lymph nodes in neck

Note: Always consider serious underlying conditions when evaluating persistent gum swelling.

Self-Care / Home Management

- Maintain good oral hygiene: brush twice daily, floss daily, and rinse with warm salt water (1/2 tsp salt in 8 oz water).
- Avoid tobacco, alcohol, and spicy or acidic foods that may irritate oral tissues.
- Stay hydrated by sipping water throughout the day to combat dry mouth.
- Over-the-counter pain relievers and topical oral gels can provide temporary relief for toothache or gum discomfort.
- See a dentist promptly for persistent oral symptoms, as early intervention improves outcomes.

Diagnostic Approach

- History covers onset, character of pain, aggravating/relieving factors, oral hygiene habits, and dental care history.
- Oral examination inspects teeth, gums, mucosa, tongue, and oropharynx for caries, inflammation, lesions, or masses.
- Dental X-rays (panoramic, periapical) identify caries, abscesses, impactions, and bone pathology.
- Referral to a dentist or oral surgeon is appropriate for definitive diagnosis and treatment of dental pathology.

Treatment / Management Overview

- Dental caries require restorative treatment (fillings, crowns, root canal) by a dentist.
- Oral infections (abscess, pericoronitis) may require drainage, antibiotics, and definitive dental treatment.
- Dry mouth management includes hydration, sugar-free lozenges, saliva substitutes, and addressing underlying causes.
- Regular dental check-ups every 6 months are essential for prevention and early detection of oral pathology.

Chapter 111

Gynecomastia (Enlarged Male Breast Tissue)

111.1 Definition

Gynecomastia is benign enlargement of male breast tissue caused by an imbalance of estrogen to androgen activity. It affects up to 65 percent of adolescent boys and 30-50 percent of older men at some point in their lives.

111.2 Pathophysiology

Normal male breast tissue is maintained by a balance between androgens and estrogens. Gynecomastia develops when there is an increased estrogen-to-androgen ratio, either from increased estrogen production, decreased androgen production, or increased peripheral conversion of androgens to estrogens by aromatase enzyme. Histologically, there is ductal and stromal hyperplasia.

111.3 Etiology

- Physiologic (neonatal, pubertal, aging)
- Medication-induced (spironolactone, cimetidine, calcium channel blockers, anabolic steroids, alcohol)
- Liver cirrhosis (impaired estrogen metabolism)
- Testosterone deficiency (primary or secondary hypogonadism)
- Testicular tumors (Leydig cell, Sertoli cell, germ cell)
- Adrenal tumors (estrogen-secreting)
- Hyperthyroidism
- Obesity (increased aromatase activity in fat)
- Chronic kidney disease or dialysis
- Malnutrition or refeeding

111.4 Indications for Evaluation

- Enlargement of one or both breasts in a male
- Palpable glandular breast tissue (firm, rubbery disc) centered under the nipple
- Tenderness or sensitivity of the breast tissue
- May be unilateral or bilateral
- No skin changes, retraction, or nipple discharge
- Slow or gradual growth

111.5 Related Symptoms

- Breast tenderness in males
- Testicular pain or swelling
- Hypogonadism (low testosterone)
- Breast cancer in males (rare)
- Pseudogynecomastia (enlargement from fat, not glandular tissue)

111.6 Self-Care/Home Management

- Reassurance for mild, physiologic cases
- Lose weight if overweight (reduces aromatase activity)
- Avoid anabolic steroids, alcohol, marijuana
- Discontinue offending medications if possible
- Wear loose-fitting clothing to reduce discomfort

111.7 Diagnostic Approach

- History and physical examination (glandular vs. fatty tissue)
- CBC, liver function, renal function, TSH
- Testosterone, estradiol, LH, FSH levels
- hCG and tumor markers if testicular tumor suspected
- Mammography or ultrasound to distinguish from breast cancer
- Testicular ultrasound if exam is abnormal

111.8 Treatment/Management Overview

- Discontinue or change offending medication
- Treat underlying cause (hypogonadism, hyperthyroidism)
- Selective estrogen receptor modulators (tamoxifen)
- Aromatase inhibitors (anastrozole)
- Surgery (subcutaneous mastectomy, liposuction) for persistent gynecomastia
- Observation for pubertal gynecomastia (spontaneous resolution in 90 percent)

Chapter 112

Haemoptysis (Coughing Up Blood)

Definition

Expectoration of blood or blood-tinged sputum from the respiratory tract below the larynx, distinguishing it from hematemesis (gastrointestinal) and epistaxis (nasal).

Pathophysiology

Bleeding originates from bronchial arteries (systemic circulation, 90

Etiology

- Acute bronchitis (most common cause)
- Bronchiectasis
- Tuberculosis
- Lung cancer or endobronchial tumor
- Pneumonia (including lung abscess)
- Pulmonary embolism with infarction
- Aspergilloma (fungus ball in pre-existing cavity)
- Bronchial adenoma or carcinoid tumor
- Pulmonary arteriovenous malformation
- Autoimmune disease (Wegener granulomatosis, Goodpasture syndrome)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms

- Any episode of hemoptysis requires medical evaluation; large volume (>100mL) or hemodynamic instability requires emergency care
- Accompanied by other concerning signs

Related Symptoms

- Cough
- Shortness of breath
- Chest pain
- Fever
- Weight loss

Note: Always consider serious underlying conditions when evaluating persistent haemoptysis (coughing up blood).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 113

Hair Loss

Definition

Excessive shedding thinning hair scalp body beyond normal turnover telogen effluvium alopecia areata androgenetic alopecia distinguishing diffuse patchy pattern gradual rapid onset location extent extent severity

Pathophysiology

Diverse etiologies disrupting hair growth cycle anagen (growth) phase prematurely terminated pushing hairs into telogen (resting) phase causing shedding later mechanical traction damage inflammation follicle destruction hormonal influences nutritional deficiencies medications stress genetics autoimmune phenomena disrupt follicular integrity regeneration mechanisms

Etiology

- Stress anemia nutritional deficiencies postpartum illness surgery weight crash dieting
- Medication side effects chemotherapy anticoagulants retinoids beta-blockers antidepressants
- Autoimmune disease alopecia areata vitiligo lupus thyroiditis
- Hormonal imbalance thyroid dysfunction polycystic ovary syndrome androgens male/female pattern baldness
- Genetic predisposition androgenetic alopecia familial hereditary pattern receding crown thinning

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Gradual months requires evaluation rapid patchy bald areas scalps tender painful inflamed pustular crusting scarring indicates scarring alopecia requiring

prompt intervention family history autoimmune thyroid disease rheumatologic conditions extensive weight loss fatigue fever night sweats suggestive systemic illness requiring workup

- Accompanied by other concerning signs

Related Symptoms

- Scalp tenderness nail changes brittle splitting clubbing hair shaft thinning inflammation redness swelling pus discharge facial hair excess facial scalp oily skin menstrual irregularities mood changes sleep disturbance appetite changes

Note: Always consider serious underlying conditions when evaluating persistent hair loss

Self-Care / Home Management

- Keep the affected area clean and dry; wash gently with mild, fragrance-free soap and lukewarm water.
- Avoid scratching — use cool compresses, calamine lotion, or OTC hydrocortisone cream to relieve itching.
- Apply moisturizer regularly to maintain skin barrier integrity, especially after bathing.
- Identify and avoid potential triggers (irritants, allergens, excessive heat, tight clothing).
- Seek medical evaluation if the rash spreads rapidly, develops blisters, shows signs of infection (pus, increasing redness, warmth), or is accompanied by fever.

Diagnostic Approach

- History includes onset, location, progression, associated symptoms (itching, pain, fever), exposure to triggers, and prior skin conditions.
- Physical examination assesses morphology (macules, papules, vesicles, etc.), distribution, and extent of skin involvement.
- Diagnostic tests may include skin scraping for fungal examination, patch testing for allergic contact dermatitis, or biopsy for suspicious lesions.
- Laboratory studies (CBC, ESR, autoimmune serologies) are obtained when systemic disease is suspected.

Treatment / Management Overview

- Topical therapies (corticosteroids, antifungals, antibiotics, emollients) are first-line for most localized skin conditions.
- Systemic therapy (oral antihistamines, antibiotics, antifungals, or immunosuppressants) is reserved for extensive or refractory cases.

- Phototherapy or biologic agents may be indicated for chronic inflammatory skin diseases (psoriasis, eczema).
- Referral to dermatology is appropriate for uncertain diagnoses, treatment failure, or skin lesions suspicious for malignancy.

Chapter 114

Hallucinations

Definition

Perception of sensory experiences (visual, auditory, tactile, olfactory, or gustatory) in the absence of external stimuli, arising from neurological or psychiatric dysfunction.

Pathophysiology

Hallucinations involve aberrant activation of sensory cortex without corresponding external input. Auditory hallucinations (most common in schizophrenia) arise from misattribution of inner speech to external sources via dysfunction in frontotemporal networks. Visual hallucinations suggest organic pathology (delirium, dementia, seizures, migraine).

Etiology

- Psychiatric disorders (schizophrenia, schizoaffective disorder, bipolar disorder)
- Delirium (especially in elderly or hospitalized patients)
- Dementia (Lewy body dementia, Alzheimer disease)
- Substance use (LSD, psilocybin, PCP, cannabis)
- Alcohol withdrawal (delirium tremens)
- Seizures (temporal lobe epilepsy)
- Migraine with aura (scintillating scotomas, fortification spectra)
- Sensory deprivation (Charles Bonnet syndrome in visually impaired)
- Brain tumors or strokes affecting sensory pathways
- Metabolic derangements (electrolyte imbalance, hepatic encephalopathy)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- New-onset hallucinations with confusion, agitation, head injury, or risk of harm to self or others (emergency evaluation)
- Accompanied by other concerning signs

Related Symptoms

- Delusions
- Confusion
- Agitation
- Sleep disturbance
- Anxiety

Note: Always consider serious underlying conditions when evaluating persistent hallucinations.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 115

Hand Tremor

Definition

Hand tremor is an involuntary, rhythmic, oscillatory movement of one or both hands, typically ranging from 4 to 12 Hz in frequency. It may occur at rest (resting tremor), during maintained posture (postural tremor), or with voluntary movement (kinetic/intention tremor), and can significantly impair fine motor tasks such as writing, eating, or dressing.

Pathophysiology

Tremor arises from dysfunction within the central nervous system circuits involving the cerebellum, thalamus, basal ganglia, and cortical areas. Essential tremor involves abnormal cerebellar-olivary-thalamic-cortical oscillations. Parkinsonian tremor results from degeneration of dopaminergic neurons in the substantia nigra pars compacta, causing abnormal basal ganglia-thalamic output. Cerebellar tremor results from lesions in the cerebellum or its pathways, disrupting coordination of voluntary movement.

Etiology

- Essential tremor (most common cause of action tremor)
- Parkinson disease (resting tremor, pill-rolling, typically asymmetric)
- Cerebellar tremor (intention tremor from stroke, MS, tumor, or degeneration)
- Drug-induced tremor (beta-agonists, antidepressants, lithium, valproate, amiodarone, corticosteroids)
- Metabolic causes (hyperthyroidism, hypoglycemia, hepatic encephalopathy, Wilson disease)
- Enhanced physiologic tremor (stress, fatigue, anxiety, caffeine, alcohol withdrawal)
- Dystonic tremor (associated with dystonic postures)
- Psychogenic tremor (sudden onset, variable, distractible, entrainable)
- Orthostatic tremor (leg tremor on standing, may involve hands)
- Fragile X-associated tremor/ataxia syndrome (FXTAS)

Indications for Evaluation

Seek care if:

- Tremor interfering with daily activities (writing, eating, drinking, dressing)
- Sudden onset or rapid progression of tremor
- Tremor accompanied by other neurological symptoms (rigidity, bradykinesia, gait disturbance, speech changes)
- Tremor beginning after starting a new medication
- Onset before age 40 (may suggest genetic or metabolic etiology)

Related Symptoms

- Rigidity (resistance to passive movement)
- Bradykinesia (slowness of movement)
- Gait disturbance or imbalance
- Changes in handwriting (micrographia in Parkinson, large shaky writing in cerebellar)
- Head tremor (yes-yes or no-no oscillation)
- Voice tremor (quavering speech)
- Fatigue or weakness in affected limb

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes; avoid driving or operating machinery if tremor is severe.
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates tremors.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many tremor conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.

- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include beta-blockers for essential tremor, levodopa for Parkinson disease, and antiepileptics for certain tremor types.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 116

Headache

Definition

Pain perceived anywhere in head region among most common human experiences encountered clinically.

Pathophysiology

Primary headaches lack identifiable structural cause; secondary headaches attributable to another condition with diverse mechanisms.

Etiology

- Tension-type headache (most common)
- Migraine
- Cluster headache
- Medication-overuse
- Secondary causes requiring urgent evaluation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Thunderclap onset, progressive worsening, new after age 50 require urgent evaluation
- Accompanied by other concerning signs

Related Symptoms

- Nausea/vomiting
- Photophobia/phonophobia
- Neck stiffness
- Visual aura changes
- Worsening with Valsalva

Note: Always consider serious underlying conditions when evaluating persistent headache.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 117

Hearing Loss

Definition

A partial or total inability to hear sound in one or both ears, ranging from mild (difficulty with soft speech) to profound (deafness), classified as conductive (outer/middle ear), sensorineural (inner ear/auditory nerve), or mixed type.

Pathophysiology

Conductive hearing loss results from obstruction or damage to the external ear canal, tympanic membrane, or ossicular chain impeding sound transmission to the cochlea. Sensorineural hearing loss involves damage to cochlear hair cells, the auditory nerve, or central auditory pathways. Noise exposure causes mechanical damage to hair cells and metabolic exhaustion. Presbycusis (age-related) involves cumulative hair cell loss, stria vascularis atrophy, and central processing decline.

Etiology

- Presbycusis (age-related hearing loss, most common)
- Noise-induced hearing loss (occupational, recreational, acute acoustic trauma)
- Otitis media with effusion (glue ear, most common in children)
- Otosclerosis (abnormal bone remodeling of stapes)
- Cerumen impaction (earwax blockage)
- Meniere disease (episodic vertigo, tinnitus, fluctuating hearing)
- Ototoxic medications (aminoglycosides, loop diuretics, cisplatin, salicylates)
- Sudden sensorineural hearing loss (viral, vascular, autoimmune)
- Acoustic neuroma (vestibular schwannoma)
- Trauma (temporal bone fracture, barotrauma, perforated TM)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden hearing loss (within 72 hours), hearing loss with vertigo, tinnitus, or ear pain, or hearing loss in one ear only
- Accompanied by other concerning signs

Related Symptoms

- Tinnitus
- Earache
- Vertigo
- Ear fullness
- Balance problems

Note: Always consider serious underlying conditions when evaluating persistent hearing loss.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 118

Heart Murmur

Definition

An extra or unusual heart sound heard during auscultation, caused by turbulent blood flow across cardiac valves, septal defects, or vascular structures.

Pathophysiology

Murmurs result from high-flow across normal valves, forward flow across stenotic valves, backward flow across regurgitant valves, or flow across septal defects. Graded by intensity (I-VI), timing (systolic, diastolic, continuous), configuration (crescendo, decrescendo), and quality (blowing, harsh, rumbling).

Etiology

- Innocent (functional) murmur — benign, no structural heart disease
- Aortic stenosis (degenerative, bicuspid valve)
- Aortic regurgitation
- Mitral regurgitation (mitral valve prolapse, papillary muscle dysfunction)
- Mitral stenosis (typically rheumatic heart disease)
- Pulmonic stenosis or regurgitation
- Hypertrophic cardiomyopathy (HOCM)
- Ventricular septal defect (congenital)
- Atrial septal defect (congenital)
- Infective endocarditis with valve vegetation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- New murmur with chest pain, shortness of breath, syncope, fever (possible endocarditis), or unexplained weight loss
- Accompanied by other concerning signs

Related Symptoms

- Shortness of breath
- Chest pain
- Palpitations
- Fatigue
- Leg swelling

Note: Always consider serious underlying conditions when evaluating persistent heart murmur.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
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Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 119

Heartburn

Definition

Heartburn is a burning retrosternal discomfort or pain that rises from the epigastrium toward the throat, typically occurring after meals or when lying down. It results from gastroesophageal reflux of acidic gastric contents into the esophagus and represents the most common symptom of gastroesophageal reflux disease (GERD).

Pathophysiology

Heartburn occurs when the lower esophageal sphincter (LES) is incompetent or relaxes inappropriately, allowing gastric acid, bile, and pepsin to reflux into the distal esophagus. The esophageal mucosa lacks the protective barrier of gastric epithelium, making it vulnerable to injury. Transient LES relaxations (tLESRs) are the primary mechanism in most patients. Risk factors include obesity, hiatal hernia, delayed gastric emptying, and increased intra-abdominal pressure. Chronic exposure leads to esophagitis, strictures, Barrett esophagus, and esophageal adenocarcinoma.

Etiology

- Gastroesophageal reflux disease (GERD) — most common cause
- Hiatal hernia (sliding or paraesophageal)
- Functional heartburn (no visible mucosal injury on endoscopy)
- Eosinophilic esophagitis (allergic inflammatory condition)
- Infectious esophagitis (Candida, HSV, CMV — typically in immunocompromised)
- Pill esophagitis (NSAIDs, bisphosphonates, potassium, tetracyclines, iron)
- Gastroparesis (delayed gastric emptying, common in diabetes)
- Pregnancy (hormonal relaxation of LES and increased abdominal pressure)
- Obesity (increased intra-abdominal pressure promoting reflux)
- Lifestyle triggers (large meals, fatty/spicy foods, caffeine, alcohol, tobacco, chocolate, mint)

Indications for Evaluation

Seek care if:

- Heartburn occurring more than twice per week
- Symptoms persisting despite over-the-counter antacids or lifestyle modifications
- Difficulty or pain with swallowing (dysphagia, odynophagia)
- Unexplained weight loss, persistent nausea/vomiting, or signs of GI bleeding
- Chest pain with exertion or pressure (must exclude cardiac causes first)

Related Symptoms

- Regurgitation of sour or bitter fluid into the mouth
- Dysphagia (difficulty swallowing) or odynophagia (painful swallowing)
- Chronic cough or hoarseness (laryngopharyngeal reflux)
- Globus sensation (lump in throat)
- Belching or bloating
- Asthma exacerbations (reflux-triggered bronchospasm)
- Nausea and epigastric fullness
- Dental erosions from acid exposure

Self-Care / Home Management

- Eat smaller, more frequent meals and avoid lying down for at least 2-3 hours after eating.
- Elevate the head of the bed by 6-8 inches using blocks or a wedge pillow.
- Avoid trigger foods: fatty/fried foods, chocolate, mint, caffeine, alcohol, spicy foods, citrus, and tomatoes.
- Maintain a healthy weight, stop smoking, and avoid tight clothing around the abdomen.
- Over-the-counter antacids (calcium carbonate, magnesium/aluminum hydroxide) or H₂ blockers provide short-term relief.

Diagnostic Approach

- History covers onset, duration, frequency, severity, aggravating/relieving factors, and red flag symptoms (dysphagia, weight loss, GI bleeding).
- Upper endoscopy (esophagogastroduodenoscopy, EGD) is the gold standard for evaluating GERD complications, esophagitis, Barrett esophagus, and excluding malignancy.
- Ambulatory pH monitoring measures esophageal acid exposure time and correlates symptoms with reflux episodes.
- Esophageal manometry evaluates LES pressure and esophageal peristalsis, particularly before antireflux surgery.

Treatment / Management Overview

- First-line therapy includes lifestyle modifications and acid suppression with proton pump inhibitors (PPIs) or H₂ receptor antagonists.
- Prokinetic agents may be added for patients with delayed gastric emptying or weak peristalsis.
- Antireflux surgery (Nissen fundoplication) is reserved for well-selected patients with confirmed GERD who are intolerant of or refractory to medical therapy.
- Endoscopic therapies (Stretta procedure, transoral incisionless fundoplication) are less invasive alternatives for appropriate candidates.

Chapter 120

Heat Cramps

120.1 Definition

Heat cramps are painful, involuntary muscle spasms that occur during or after intense exercise in hot conditions. They are caused by electrolyte depletion, particularly sodium, from heavy sweating.

120.2 Pathophysiology

Prolonged sweating during physical activity in heat leads to loss of sodium and water. When sweat losses are replaced with only water, dilutional hyponatremia can develop, disrupting neuromuscular junction function and causing involuntary muscle contractions. The large muscles used during exercise are most commonly affected.

120.3 Etiology

- Intense physical activity in hot, humid conditions
- Heavy sweating with inadequate salt replacement
- Drinking only water without electrolytes during prolonged exercise
- Sodium depletion
- Poor physical conditioning
- Inadequate acclimatization to heat
- Fatigue
- Dehydration

120.4 Indications for Evaluation

- Sudden onset of painful muscle cramps
- Most common in calves, quadriceps, hamstrings, and abdominal muscles
- Cramps during or after exercise

- Heavy sweating prior to cramp onset
- Muscles feel hard or knotted
- Normal body temperature (differentiating from heat exhaustion)

120.5 Related Symptoms

- Heat exhaustion
- Heat stroke
- Dehydration
- Muscle cramps (from other causes)
- Exercise-associated muscle cramping

120.6 Self-Care/Home Management

- Stop activity and rest in a cool environment
- Drink electrolyte-containing fluids (sports drinks, salt water)
- Gently stretch and massage the cramped muscle
- Apply firm pressure to the cramped muscle
- Eat salty foods if able
- Replace fluids containing electrolytes
- Resume activity slowly and cautiously

120.7 Diagnostic Approach

- Clinical diagnosis based on history and physical exam
- Electrolyte panel (may show hyponatremia or hypomagnesemia)
- Creatine kinase if muscle injury suspected

120.8 Treatment/Management Overview

- Rest, stretching, and massage of the affected muscle
- Oral electrolyte replacement
- IV normal saline for severe or persistent cases
- Prevention: proper hydration with electrolyte drinks
- Acclimatize slowly to hot working conditions

Chapter 121

Heat Exhaustion

121.1 Definition

Heat exhaustion is a moderate heat-related illness characterized by heavy sweating, weakness, dizziness, and nausea due to inadequate fluid and electrolyte replacement during heat exposure. It can progress to heat stroke if not treated promptly.

121.2 Pathophysiology

Prolonged exposure to high environmental temperatures causes excessive sweating with loss of water and electrolytes. The body attempts to maintain thermoregulation through vasodilation and increased cardiac output, but volume depletion leads to reduced stroke volume and cardiac output, causing fatigue, weakness, dizziness, and orthostatic hypotension.

121.3 Etiology

- Strenuous exercise in hot, humid conditions
- Prolonged exposure to high temperatures
- Inadequate fluid intake
- Excessive sweating with inadequate salt replacement
- Heat waves affecting vulnerable populations
- Medications that impair thermoregulation (diuretics, beta-blockers)
- Alcohol consumption
- Obesity
- Advancing age

121.4 Indications for Evaluation

- Heavy sweating, cool and clammy skin
- Weakness and fatigue
- Dizziness or lightheadedness
- Nausea and possible vomiting
- Headache
- Mild confusion or irritability
- Thirst
- Decreased urine output
- Body temperature normal to mildly elevated (37-40 C)
- Orthostatic hypotension, rapid pulse

121.5 Related Symptoms

- Heat stroke (severe progression)
- Heat cramps
- Heat syncope (fainting)
- Dehydration
- Hyponatremia (low sodium)
- Hypovolemia

121.6 Self-Care/Home Management

- Move to a cool, shaded or air-conditioned environment
- Lie down and elevate the feet
- Drink cool water or electrolyte-containing sports drinks
- Remove tight or excess clothing
- Apply cool, wet cloths or take a cool shower
- Rest for 24 hours after recovery
- Avoid strenuous activity and heat exposure for several days

121.7 Diagnostic Approach

- Clinical diagnosis based on history of heat exposure and symptoms
- Core body temperature measurement
- Electrolytes, renal function, BUN-to-creatinine ratio
- Urinalysis (specific gravity, color)

121.8 Treatment/Management Overview

- Rest in a cool environment
- Oral or IV fluid replacement with balanced electrolyte solutions
- Monitor for progression to heat stroke
- Avoid exertion and heat until fully recovered
- Education on prevention and hydration strategies

Chapter 122

Heat Rash (Miliaria)

Definition

A skin condition caused by blockage of eccrine sweat ducts during hot, humid weather, leading to sweat retention and superficial inflammation presenting as small red bumps or blisters.

Pathophysiology

Sweat duct obstruction from keratin plugs, bacteria, or inflammation leads to sweat retention and ductal rupture. Miliaria crystallina (superficial, clear vesicles) involves stratum corneum duct obstruction. Miliaria rubra (prickly heat, red papules) involves deeper epidermal duct rupture with periductal inflammation. Miliaria profunda (deep, flesh-colored papules) involves dermal duct rupture.

Etiology

- Hot, humid weather (most common)
- Excessive sweating (exercise, fever)
- Tight or non-breathable clothing
- Prolonged bed rest (friction and heat trapping)
- Occlusive dressings or ointments
- Neonatal (immature sweat ducts in infants)
- Fever with profuse sweating
- Intense physical activity in heat
- Overdressing in warm environments
- Tropical climate exposure

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Heat rash with fever, signs of heat exhaustion (nausea, dizziness, confusion), or rash with pus drainage suggesting secondary infection
- Accompanied by other concerning signs

Related Symptoms

- Itching (pruritus)
- Red bumps on skin
- Prickling sensation
- Heat exhaustion
- Dehydration

Note: Always consider serious underlying conditions when evaluating persistent heat rash (miliaria).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 123

Heat Stroke

123.1 Definition

Heat stroke is a life-threatening condition characterized by core body temperature above 40 C (104 F) and central nervous system dysfunction. It represents the most severe form of heat-related illness and requires emergency medical treatment.

123.2 Pathophysiology

Heat stroke occurs when thermoregulatory mechanisms fail under extreme heat stress. The body's core temperature rises uncontrollably, leading to denaturation of proteins, cellular damage, and a systemic inflammatory response. This triggers multi-organ dysfunction including encephalopathy, acute kidney injury, liver failure, disseminated intravascular coagulation, and cardiovascular collapse.

123.3 Etiology

- Exertional: strenuous physical activity in hot, humid conditions (common in athletes, military recruits, laborers)
- Non-exertional (classic): passive exposure to extreme heat in vulnerable populations (elderly, infants, chronic medical conditions)
- High ambient temperature and humidity
- Inadequate hydration
- Poor ventilation or lack of air conditioning
- Obesity
- Cardiovascular disease
- Medications: diuretics, anticholinergics, beta-blockers, antihistamines
- Alcohol use
- Drug use (ecstasy/MDMA, amphetamines, cocaine)

123.4 Indications for Evaluation

- Core body temperature above 40 C (104 F)
- Altered mental status: confusion, agitation, delirium, seizures, coma
- Hot, dry, red skin (non-exertional type) or profuse sweating (exertional type)
- Rapid, strong pulse
- Headache, dizziness, lightheadedness
- Nausea and vomiting
- Rapid, shallow breathing
- Hypotension or shock
- Multi-organ failure

123.5 Related Symptoms

- Heat exhaustion (precursor, milder form)
- Heat cramps
- Heat syncope (fainting from heat)
- Dehydration
- Rhabdomyolysis (muscle breakdown)
- Acute kidney injury

123.6 Self-Care/Home Management

- This is a medical emergency - CALL EMERGENCY SERVICES
- Move the person to a cool, shaded area
- Remove excess clothing
- Cool the body aggressively: immerse in cool water, apply ice packs to neck, armpits, and groin
- Fan the person while misting with cool water
- Do NOT give fluids if the person is confused or unconscious
- Monitor breathing and consciousness

123.7 Diagnostic Approach

- Core body temperature measurement (rectal thermometer)
- Clinical diagnosis based on hyperthermia and CNS dysfunction
- Complete blood count, electrolytes, renal and liver function
- Coagulation profile (disseminated intravascular coagulation)
- Arterial blood gas analysis
- Creatine kinase levels (rhabdomyolysis)
- Cardiac monitoring for arrhythmias

123.8 Treatment/Management Overview

- Rapid cooling to target temperature of 39 C (102 F) within 30 minutes
- Cold water immersion (most effective method)
- Evaporative cooling with mist and fans
- Ice packs to neck, axillae, groin
- Cold IV fluids (4 C normal saline)
- Benzodiazepines for shivering or seizures
- Supportive care for multi-organ dysfunction
- ICU admission for severe cases

Chapter 124

Heavy Periods (Menorrhagia)

Definition

Excessive or prolonged menstrual bleeding defined as soaking through one or more pads/tampons every hour for several consecutive hours, bleeding lasting more than 7 days, or passing large blood clots.

Pathophysiology

Menorrhagia results from hormonal imbalance (estrogen-progesterone ratio disruption), uterine structural abnormalities (fibroids, polyps, adenomyosis), coagulation disorders, or endometrial dysfunction. Anovulatory cycles common in perimenopause and adolescence lead to unopposed estrogen stimulation causing thickened endometrium and heavy bleeding.

Etiology

- Uterine fibroids (leiomyomas)
- Hormonal imbalance (anovulation, perimenopause)
- Adenomyosis
- Endometrial polyps
- Coagulation disorders (von Willebrand disease, platelet disorders)
- Pelvic inflammatory disease
- Endometriosis
- Thyroid dysfunction (hypothyroidism)
- Intrauterine device (IUD) side effects
- Endometrial hyperplasia or malignancy

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Profuse bleeding soaking more than 1 pad/tampon per hour, bleeding with dizziness or fatigue, or between-period bleeding
- Accompanied by other concerning signs

Related Symptoms

- Fatigue (anemia)
- Abdominal cramps
- Irregular periods
- Back pain
- Pale skin

Note: Always consider serious underlying conditions when evaluating persistent heavy periods (menorrhagia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 125

Heel Pain

Definition

Pain localized to the posterior or inferior aspect of the heel, commonly caused by mechanical stress on the plantar fascia or Achilles tendon insertion.

Pathophysiology

Plantar fasciitis results from repetitive microtrauma and degeneration of the plantar fascia origin at the medial calcaneal tuberosity. Achilles tendinopathy involves degeneration of the Achilles tendon fibers from overuse, poor footwear, or biomechanical abnormalities. Heel pain is characteristically worse with the first steps in the morning or after prolonged sitting.

Etiology

- Plantar fasciitis (most common cause)
- Achilles tendinopathy or insertional tendinitis
- Calcaneal stress fracture
- Retrocalcaneal bursitis
- Heel spur syndrome
- Sever disease (calcaneal apophysitis in children)
- Fat pad atrophy (especially in elderly)
- Tarsal tunnel syndrome
- Gout or pseudogout
- Seronegative spondyloarthropathy (ankylosing spondylitis, reactive arthritis)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe heel pain with inability to bear weight, fever, redness, or swelling over the heel
- Accompanied by other concerning signs

Related Symptoms

- Arch pain
- Achilles tendon pain
- Ankle pain
- Foot numbness
- Morning stiffness

Note: Always consider serious underlying conditions when evaluating persistent heel pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 126

Hematoma

126.1 Definition

A hematoma is a collection of blood outside blood vessels, usually in liquid form within the tissues. It results from trauma that causes blood vessel rupture. Hematomas can occur anywhere in the body and range from minor bruises to life-threatening intracranial or retroperitoneal bleeds.

126.2 Pathophysiology

When a blood vessel is damaged, blood escapes into the surrounding tissue. The blood clot forms a localized collection. Over time, the body breaks down the blood products, leading to color changes (red to purple to green to yellow). Large hematomas can cause compression of adjacent structures, organ dysfunction, or hypovolemia.

126.3 Etiology

- Trauma (most common): contusions, fractures, crush injuries
- Surgery (postoperative hematoma)
- Anticoagulant therapy (warfarin, heparin, DOACs)
- Coagulation disorders (hemophilia, thrombocytopenia, liver disease)
- Spontaneous: ruptured aneurysm, vascular malformation
- Repeated minor trauma (subungual hematoma)
- Venipuncture or injection

126.4 Indications for Evaluation

- Localized swelling and discoloration
- Pain and tenderness at the site
- Firm or fluctuant mass on palpation

- Ecchymosis (black-and-blue discoloration)
- Neurological deficits if intracranial
- Signs of hypovolemia if large or internal

126.5 Related Symptoms

- Bruise (ecchymosis, superficial hematoma)
- Subdural hematoma (intracranial)
- Epidural hematoma (intracranial)
- Subungual hematoma (under the nail)
- Retroperitoneal hematoma
- Hypovolemic shock

126.6 Self-Care/Home Management

- RICE protocol: Rest, Ice, Compression, Elevation
- Apply ice for the first 24-48 hours, then warm compresses
- Elevate the affected area
- Take acetaminophen for pain (avoid NSAIDs if bleeding risk)
- Monitor for signs of expansion or infection

126.7 Diagnostic Approach

- Clinical examination
- Ultrasound for superficial or deep hematomas
- CT scan for intracranial, thoracic, or abdominal hematomas
- MRI for better characterization
- Complete blood count and coagulation profile

126.8 Treatment/Management Overview

- Observation for small, stable hematomas
- Drainage for large, tense, or infected hematomas
- Surgical evacuation for intracranial hematomas causing mass effect
- Reverse anticoagulation if applicable
- Compression bandaging for extremity hematomas

Chapter 127

Hematuria

Definition

Hematuria is the presence of blood in the urine, classified as gross (visible to the naked eye, appearing pink, red, or cola-colored) or microscopic (detected only on urinalysis or microscopy, defined as three or more red blood cells per high-power field). It may be symptomatic or asymptomatic and can originate from any site along the urinary tract from the glomerulus to the urethra.

Pathophysiology

Hematuria results from disruption of vascular integrity within the urinary tract due to inflammation, infection, calculi, trauma, neoplasm, or glomerular basement membrane damage. Glomerular hematuria (dysmorphic RBCs, RBC casts) arises from immune complex deposition, basement membrane thinning, or podocyte injury causing filtration barrier disruption. Nonglomerular hematuria (eumorphic RBCs) results from mucosal bleeding due to stones, malignancy, infection, or trauma along the collecting system.

Etiology

- Urinary tract infection (cystitis, pyelonephritis, prostatitis, urethritis)
- Nephrolithiasis (kidney stones, ureteral stones, bladder stones)
- Benign prostatic hyperplasia (BPH) — common in older men
- Glomerulonephritis (IgA nephropathy, post-streptococcal, membranous, crescentic)
- Malignancy (renal cell carcinoma, urothelial carcinoma of bladder/ureter, prostate cancer)
- Trauma (straddle injury, flank trauma, catheter-related)
- Strenuous exercise (benign, resolves with rest)
- Anticoagulant use (warfarin, DOACs, heparin) — often exacerbates underlying pathology
- Polycystic kidney disease
- Sickle cell disease or trait (papillary necrosis, hematuria)

Indications for Evaluation

Seek care if:

- Gross hematuria (visible blood in urine) regardless of associated symptoms
- Microscopic hematuria persisting on repeat testing
- Hematuria accompanied by flank pain, fever, or difficulty urinating
- Painless hematuria in older adults (must exclude malignancy)
- Hematuria in patients with smoking history or occupational exposure to carcinogens

Related Symptoms

- Dysuria (painful urination) suggesting infection
- Flank pain radiating to groin suggesting ureteral stone
- Frequency, urgency, and suprapubic discomfort
- Passage of clots (irregular shapes suggest bladder source)
- Foamy urine (suggests proteinuria with glomerular disease)
- Lower extremity edema or hypertension (suggests glomerulonephritis)

Self-Care / Home Management

- Drink adequate water throughout the day unless fluid restriction is medically indicated.
- Avoid bladder irritants including caffeine, alcohol, acidic foods, and artificial sweeteners.
- Practice good hygiene and urinate promptly when the urge arises to prevent urinary stasis.
- Apply a heating pad to the lower abdomen or back for comfort during urinary discomfort.
- Seek medical attention for fever, flank pain, visible blood in urine, or difficulty urinating.

Diagnostic Approach

- History covers urinary symptoms (frequency, urgency, dysuria, hesitancy, stream changes), fever, and flank pain.
- Urinalysis with microscopy and culture is the first-line diagnostic test; urine cytology is obtained when malignancy is suspected.
- Imaging (CT urogram, renal ultrasound) evaluates for stones, obstruction, structural abnormalities, or masses.
- Cystoscopy is indicated for persistent hematuria or when bladder pathology is suspected, particularly in patients with risk factors for bladder cancer.

Treatment / Management Overview

- Urinary tract infections are treated with targeted antibiotics based on culture sensitivity results.
- Nephrolithiasis management includes hydration, analgesics, medical expulsive therapy (alpha-blockers), or urologic intervention (ESWL, ureteroscopy) for large stones.
- Glomerular disease requires nephrology consultation for immunosuppression, blood pressure control, and proteinuria reduction.
- Urology referral is indicated for recurrent hematuria, suspected malignancy, obstructive symptoms, or definitive management of stones.

Chapter 128

Hiccups (Singultus)

Definition

Involuntary, spasmodic contractions of the diaphragm followed by sudden glottic closure producing the characteristic "hic" sound, mediated by a reflex arc involving the phrenic nerve and vagus nerve.

Pathophysiology

The hiccup reflex arc involves afferent fibers (phrenic and vagus nerves, sympathetic chain from T6-T12), central coordinator in the brainstem (medulla, reticular formation), and efferent fibers (phrenic nerve to diaphragm, vagus nerve to glottis). Hiccups lasting >48h (persistent) or >1 month (intractable) require investigation for underlying pathology.

Etiology

- Gastric distension (overeating, carbonated drinks, swallowing air)
- Sudden temperature changes (hot then cold food/drink)
- Alcohol consumption
- Excitement or emotional stress
- Gastroesophageal reflux disease (GERD)
- Medication side effects (benzodiazepines, barbiturates, corticosteroids)
- Central nervous system disorders (stroke, tumor, multiple sclerosis)
- Diaphragmatic irritation (pneumonia, pleurisy, pericarditis)
- Metabolic disorders (uremia, diabetes, electrolyte imbalance)
- Post-operative (especially abdominal or thoracic surgery)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Hiccups persisting more than 48 hours, hiccups with severe pain, or associated with neurological symptoms
- Accompanied by other concerning signs

Related Symptoms

- GERD
- Abdominal distension
- Chest discomfort
- Nausea
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent hiccups (singultus).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 129

Hip Pain

Definition

Discomfort localized to the hip joint or surrounding structures (groin, lateral thigh, buttock), arising from intra-articular pathology, periarticular soft tissues, or referred sources.

Pathophysiology

Hip pain localizes differently based on origin: groin pain typically indicates intra-articular pathology (osteoarthritis, labral tear, AVN); lateral thigh pain suggests trochanteric bursitis; buttock pain may arise from the sacroiliac joint, lumbar spine, or posterior hip structures.

Etiology

- Hip osteoarthritis (degenerative joint disease)
- Trochanteric bursitis
- Hip labral tear
- Avascular necrosis (AVN) of femoral head
- Femoroacetabular impingement (FAI)
- Hip fracture (especially in elderly/osteoporotic)
- Iliopsoas tendinopathy or bursitis
- Referred low back pain or sacroiliac dysfunction
- Osteitis pubis
- Septic arthritis (urgent, with fever and severe pain)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe hip pain after fall or injury, inability to bear weight, fever with hip pain, or groin pain with fever
- Accompanied by other concerning signs

Related Symptoms

- Groin pain
- Limping
- Stiffness
- Lower back pain
- Knee pain

Note: Always consider serious underlying conditions when evaluating persistent hip pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 130

Hives and Angioedema (Urticaria)

130.1 Definition

Hives (urticaria) are raised, itchy, red or skin-colored welts that appear suddenly on the skin. Angioedema is deeper swelling of the subcutaneous and submucosal tissues, often affecting the face, lips, tongue, throat, and genitals. They can occur separately or together.

130.2 Pathophysiology

Both conditions result from mast cell degranulation and release of histamine and other inflammatory mediators. In urticaria, vasodilation and increased vascular permeability in the superficial dermis causes the characteristic wheals. In angioedema, the same process occurs in the deep dermis and subcutaneous tissues. Triggers can be allergic (IgE-mediated) or non-allergic (direct mast cell activation).

130.3 Etiology

- Allergic reactions: foods (nuts, shellfish, eggs, milk), medications (antibiotics, NSAIDs, aspirin), insect stings, latex
- Physical triggers: pressure, cold, heat, sunlight, exercise, water, vibration
- Infections: viral, bacterial, parasitic
- Autoimmune conditions: thyroid disease, lupus
- Complement-mediated: hereditary angioedema (C1 inhibitor deficiency)
- Idiopathic: no identifiable cause
- Chronic spontaneous urticaria (lasting more than 6 weeks)

130.4 Indications for Evaluation

- Raised, red, itchy welts on the skin
- Welts that change shape, size, and location within hours

- Swelling of the face, lips, tongue, throat, hands, feet, or genitals
- Stinging or burning sensation
- Individual welts lasting less than 24 hours
- Angioedema: pain rather than itching
- Difficulty breathing, throat tightness, or hoarseness (anaphylaxis warning)

130.5 Related Symptoms

- Anaphylaxis (severe, life-threatening allergic reaction)
- Allergic rhinitis
- Asthma
- Atopic dermatitis (eczema)
- Contact dermatitis

130.6 Self-Care/Home Management

- Identify and avoid known triggers
- Take OTC antihistamines (cetirizine, loratadine, diphenhydramine)
- Apply cool compresses to reduce itching
- Wear loose, comfortable clothing
- Use fragrance-free moisturizers
- Avoid hot showers, alcohol, and NSAIDs which can worsen symptoms
- Keep a symptom diary to identify triggers

130.7 Diagnostic Approach

- Clinical history and physical examination
- Skin prick testing or specific IgE tests for suspected allergens
- Physical challenge tests for physical urticarias
- Complete blood count, ESR, CRP
- Thyroid antibodies and thyroid function tests
- Complement levels (if hereditary angioedema suspected)
- Skin biopsy for chronic or atypical cases

130.8 Treatment/Management Overview

- Second-generation antihistamines as first-line therapy
- Dose escalation of antihistamines up to 4x standard dose
- Add H2 blocker (ranitidine, famotidine)
- Leukotriene receptor antagonists (montelukast)
- Cyclosporine for refractory chronic urticaria
- Omalizumab (anti-IgE) for chronic spontaneous urticaria
- Epinephrine auto-injector for angioedema with anaphylaxis
- Corticosteroids for acute severe flare-ups

Chapter 131

Hoarseness

Definition

Change voice quality making rough strained breathy nasal due laryngeal pathology

Pathophysiology

Vocal cord dysfunction inflammation mass effect neurological involvement vocal abuse common contributing factor altered phonatory mechanism air flow turbulence vibration

Etiology

- Acute laryngitis viral URI
- Acid reflux laryngopharyngeal reflux
- Vocal nodules polyps
- Vocal cord paralysis
- Laryngeal carcinoma

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Voice change persisting beyond two-three weeks warrants ENT evaluation especially smokers heavy voice users progressive hoarseness neck mass dysphagia stridor
- Accompanied by other concerning signs

Related Symptoms

- Sore throat cough difficulty swallowing breathing problems voice change neck lump ear pain

Note: Always consider serious underlying conditions when evaluating persistent hoarseness

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 132

Hot Flashes (Hot Flashes)

Definition

Sudden episodes of intense heat sensation spreading over the upper body and face, often accompanied by sweating, palpitations, and anxiety, primarily associated with hormonal changes.

Pathophysiology

Estrogen withdrawal (menopausal transition) alters hypothalamic thermoregulatory set-point, narrowing the thermoneutral zone so minor core temperature elevations trigger heat-loss responses (vasodilation, sweating). Fluctuations in luteinizing hormone and norepinephrine also contribute to vasomotor instability.

Etiology

- Menopause (natural or surgical)
- Perimenopause
- Medication side effects (SSRIs, tamoxifen, raloxifene)
- Thyroid disorders (hyperthyroidism)
- Cancer treatment (chemotherapy, radiation, hormone therapy)
- Orchiectomy (in men)
- Carcinoid syndrome
- Adrenal insufficiency
- Alcohol or caffeine triggers
- Anxiety or panic disorder

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Hot flushes with fever, night sweats with weight loss, or flushes interfering significantly with quality of life
- Accompanied by other concerning signs

Related Symptoms

- Night sweats
- Palpitations
- Anxiety
- Sleep disturbance
- Mood swings

Note: Always consider serious underlying conditions when evaluating persistent hot flushes (hot flashes).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 133

Hyperhidrosis (Excessive Sweating)

133.1 Definition

Hyperhidrosis is a condition of excessive sweating beyond the amount needed for thermoregulation. It can be primary (idiopathic, focal) or secondary (due to an underlying medical condition or medication). It affects approximately 3 percent of the population.

133.2 Pathophysiology

Primary hyperhidrosis involves overactivity of the sympathetic nervous system, specifically the cholinergic fibers that innervate eccrine sweat glands. The cause is unknown but is thought to involve a genetic component with autosomal dominant inheritance. Secondary hyperhidrosis results from conditions that stimulate the hypothalamus or sympathetic nervous system, including infections, endocrine disorders, and malignancies.

133.3 Etiology

- Primary focal hyperhidrosis (palms, soles, axillae, face - no identifiable cause)
- Genetics: family history in 30-50 percent of cases
- Endocrine disorders: hyperthyroidism, diabetes (autonomic dysfunction), hypoglycemia, pheochromocytoma
- Menopause (hot flashes and sweating)
- Infections: tuberculosis, HIV, chronic infections
- Malignancies: lymphoma, leukemia, carcinoid tumors
- Neurological conditions: Parkinson disease, spinal cord injury
- Medications: antidepressants (SSRIs, TCAs), antipyretics, opioids, corticosteroids
- Anxiety and stress (trigger exacerbations)

133.4 Indications for Evaluation

- Excessive sweating that interferes with daily activities
- Visible, profuse sweating at rest without heat or exertion
- Sweating from palms, soles, axillae, or face (primary focal)
- Generalized sweating (secondary causes)
- Night sweats (suggests secondary cause)
- Onset in childhood or adolescence (primary)
- Family history of hyperhidrosis

133.5 Related Symptoms

- Heat rash (miliaria)
- Body odor (bromhidrosis)
- Skin maceration and secondary fungal infections
- Anxiety and social phobia related to sweating

133.6 Self-Care/Home Management

- Use clinical-strength antiperspirants containing aluminum chloride
- Bathe daily and dry skin thoroughly
- Wear loose, breathable clothing made of natural fibers
- Change socks and shoes frequently
- Avoid triggers (spicy foods, caffeine, alcohol)
- Use absorbent powders or pads
- Keep a sweat diary to identify triggers

133.7 Diagnostic Approach

- Clinical history and physical examination
- Starch-iodine test (Minor test) to map areas of sweating
- Blood tests: TSH, fasting glucose, complete blood count
- Consider 24-hour urinary catecholamines if pheochromocytoma suspected
- Chest X-ray or CT if tuberculosis or lymphoma suspected

133.8 Treatment/Management Overview

- Prescription-strength aluminum chloride antiperspirants
- Iontophoresis (for palmoplantar hyperhidrosis)
- Botulinum toxin type A injections into affected areas
- Oral anticholinergics (glycopyrrolate, oxybutynin)
- Endoscopic thoracic sympathectomy for severe cases
- Laser therapy or microwave thermolysis
- Management of secondary cause if identified

Chapter 134

Hyperventilation

134.1 Definition

Hyperventilation is rapid or deep breathing that exceeds the body's metabolic demand, leading to excessive elimination of carbon dioxide (hypocapnia) and respiratory alkalosis. It is most commonly triggered by anxiety but can occur with various medical conditions.

134.2 Pathophysiology

Ventilation removes CO₂ from the blood. When ventilation exceeds metabolic CO₂ production, arterial CO₂ drops and pH rises (respiratory alkalosis). Alkalosis causes cerebral vasoconstriction, reduced cerebral blood flow, and altered neuronal excitability, producing lightheadedness, paresthesias, and muscle cramps. Alkalosis also increases binding of calcium to albumin, reducing ionized calcium and causing neuromuscular hyperexcitability (Chvostek and Trousseau signs).

134.3 Etiology

- Anxiety, panic attacks, or acute stress (most common cause)
- Pulmonary causes: asthma exacerbation, pulmonary embolism, pneumonia
- Cardiac causes: heart failure, myocardial infarction
- Metabolic acidosis compensation (diabetic ketoacidosis)
- Salicylate overdose (aspirin toxicity)
- Pregnancy (progesterone stimulates respiration)
- Liver cirrhosis
- High altitude
- Central nervous system disorders
- Fever or sepsis

134.4 Indications for Evaluation

- Rapid, deep breathing out of proportion to exertion
- Lightheadedness, dizziness, or feeling faint
- Tingling or numbness of the lips, fingers, and toes
- Carpopedal spasm: muscle cramping in hands and feet
- Chest tightness or palpitations
- Respiratory alkalosis on arterial blood gas
- Anxiety or impending sense of doom
- Difficulty speaking in full sentences

134.5 Related Symptoms

- Anxiety and panic attacks
- Shortness of breath (dyspnea)
- Chest pain
- Palpitations
- Dizziness and presyncope
- Tingling and numbness (paresthesias)
- Hypocalcemia symptoms (tetany)

134.6 Self-Care/Home Management

- Recognize hyperventilation and practice slow, controlled breathing
- Inhale deeply through the nose for 4 seconds, hold for 4 seconds, exhale slowly through pursed lips for 6 seconds
- Reassure yourself that the episode is temporary and not dangerous
- Do NOT breathe into a paper bag (can cause hypoxia)
- Remove yourself from stressful situations if possible
- Practice relaxation techniques and mindfulness

134.7 Diagnostic Approach

- Clinical history: acute onset with anxiety or stress
- Observation of breathing pattern
- Arterial blood gas shows respiratory alkalosis and low PaCO₂
- Electrolytes, calcium, magnesium
- Chest X-ray if pulmonary cause suspected
- Electrocardiogram to rule out cardiac ischemia
- Pulse oximetry to exclude hypoxia

134.8 Treatment/Management Overview

- Treat underlying cause (anxiety, pulmonary embolus, salicylate overdose)
- Reassurance and verbal coaching to slow breathing
- Benzodiazepines for acute panic-related hyperventilation
- Rebreathing into a mask or reservoir may be used in emergency
- Oxygen for hypoxic patients
- IV calcium gluconate for severe tetany
- Cognitive behavioral therapy for chronic anxiety

Chapter 135

Hyphema (Bleeding in the Eye)

135.1 Definition

Hyphema is the accumulation of blood in the anterior chamber of the eye between the cornea and the iris. It most commonly results from blunt trauma but can occur spontaneously in certain conditions. Hyphema is an ocular emergency.

135.2 Pathophysiology

Blunt trauma to the eye causes a sudden increase in intraocular pressure that tears the blood vessels of the ciliary body or iris root. Blood leaks into the anterior chamber, where it settles by gravity. The blood can obstruct the trabecular meshwork, impairing aqueous humor drainage and increasing intraocular pressure. Rebleeding can occur 3-5 days after the initial injury.

135.3 Etiology

- Blunt eye trauma (most common cause, especially in children)
- Penetrating eye injury
- Sports-related eye injuries (racquetball, baseball, paintball)
- Spontaneous hyphema: iris neovascularization (diabetes), intraocular tumors, bleeding disorders
- Anticoagulant or antiplatelet therapy (warfarin, aspirin, clopidogrel)
- Sickle cell disease (increases risk of elevated IOP)
- Post-surgical (cataract surgery, glaucoma surgery)
- Child abuse (shaken baby syndrome)

135.4 Indications for Evaluation

- Visible blood settling in the lower part of the eye (anterior chamber)

- Blurred or decreased vision
- Eye pain
- Sensitivity to light (photophobia)
- Elevated intraocular pressure
- Blood may layer (microhyphema) or fill the entire anterior chamber (total hyphema, eight-ball hyphema)
- History of trauma or bleeding risk factors

135.5 Related Symptoms

- Eye pain
- Blurred vision
- Traumatic iritis
- Glaucoma (elevated intraocular pressure)
- Corneal blood staining (long-term complication)
- Orbital fracture (if associated with trauma)

135.6 Self-Care/Home Management

- Seek emergency eye care immediately
- Keep the head elevated 30-45 degrees
- Wear an eye shield or patch
- Do NOT rub the eye
- Avoid strenuous activity and heavy lifting
- Do not take aspirin or NSAIDs (may increase bleeding)
- Use a protective shield during sleep

135.7 Diagnostic Approach

- Slit-lamp examination to visualize blood in the anterior chamber
- Grading of hyphema (grade 0 to grade 4 based on blood level)
- Intraocular pressure measurement (tonometry)
- Detailed eye examination with dilation (after ruling out globe rupture)
- CT scan if orbital fracture or intraocular foreign body suspected
- Blood coagulation profile if spontaneous or recurrent

135.8 Treatment/Management Overview

- Elevate head of bed 30-45 degrees to promote settling
- Topical corticosteroids to reduce inflammation
- Cycloplegics (atropine) for comfort and to prevent synechiae
- Anti-glaucoma medications for elevated IOP
- Aminocaproic acid (tranexamic acid) to prevent rebleeding
- Anterior chamber washout if IOP is uncontrolled or total hyphema
- Surgical evacuation for persistent clot or corneal blood staining
- Treat underlying cause (sickle cell, coagulation disorder)

Chapter 136

Hypothermia

136.1 Definition

Hypothermia is a core body temperature below 35 C (95 F). It is classified as mild (32-35 C), moderate (28-32 C), or severe (below 28 C). Hypothermia is a life-threatening emergency that requires immediate intervention.

136.2 Pathophysiology

When the body loses heat faster than it can produce it, thermoregulatory mechanisms are overwhelmed. Initial shivering generates heat but becomes ineffective as temperature drops. Below 32 C, shivering ceases and metabolic rate declines by about 5 percent per degree. Cardiac conduction becomes unstable, with risk of ventricular fibrillation. Coagulation is impaired. Cerebral blood flow decreases, causing progressive loss of consciousness.

136.3 Etiology

- Prolonged exposure to cold environments
- Cold water immersion (heat loss is 25x faster in water)
- Inadequate clothing in cold weather
- Homelessness or inadequate shelter
- Advanced age (decreased thermoregulation)
- Alcohol or drug intoxication (vasodilation, impaired judgment)
- Hypoglycemia
- Hypothyroidism or adrenal insufficiency
- Sepsis
- Burns or extensive skin disease
- Spinal cord injury or immobility

136.4 Indications for Evaluation

- Shivering (mild) or no shivering (severe)
- Cold skin to the touch
- Confusion, apathy, or drowsiness
- Slurred speech and poor coordination
- Weak pulse and slow, shallow breathing
- Hypotension
- Arrhythmias (atrial fibrillation, ventricular fibrillation)
- Coma or unresponsiveness in severe hypothermia

136.5 Related Symptoms

- Frostbite (local cold injury)
- Chilblains (pernio)
- Trench foot
- Raynaud phenomenon
- Bradycardia
- Hypoglycemia
- Alcohol intoxication

136.6 Self-Care/Home Management

- Call emergency services - hypothermia is a medical emergency
- Move the person to a warm environment
- Remove wet clothing and replace with dry, warm layers
- Apply warm compresses to the trunk (not extremities)
- Provide warm, non-alcoholic, non-caffeinated drinks if conscious
- Do NOT apply direct heat (heating pads, hot water)
- Do NOT rub or massage limbs
- Handle the person gently to avoid triggering arrhythmias

136.7 Diagnostic Approach

- Core body temperature measurement with esophageal or rectal thermometer
- Clinical evaluation of severity based on mental status and shivering
- Electrocardiogram (Osborn J wave is characteristic)
- Complete blood count and electrolytes
- Coagulation profile
- Blood glucose
- Arterial blood gas analysis
- Renal and liver function tests

136.8 Treatment/Management Overview

- Passive external rewarming (warm environment, blankets) for mild hypothermia
- Active external rewarming (forced warm air, heating pads) for moderate hypothermia
- Active internal rewarming (warmed IV fluids, warmed humidified oxygen, peritoneal lavage) for severe hypothermia
- Extracorporeal membrane oxygenation (ECMO) for refractory severe hypothermia
- Cardiac monitoring and careful handling to prevent arrhythmias
- Treat underlying cause (hypoglycemia, hypothyroidism)
- Avoid rewarming shock: slow, controlled rewarming

Chapter 137

Inability to Urinate (Urinary Retention)

137.1 Definition

Urinary retention is the inability to empty the bladder completely or at all. Acute urinary retention is sudden and painful, while chronic retention develops gradually with incomplete emptying. It can affect both men and women but is most common in older men.

137.2 Pathophysiology

Urinary retention can be obstructive (physical blockage of urine flow) or non-obstructive (failure of detrusor contraction or coordination). Bladder outlet obstruction: prostate enlargement, urethral stricture, pelvic organ prolapse. Detrusor underactivity: neurologic conditions, medications, aging, or damage to bladder innervation. In acute retention, the bladder overdistends, causing severe pain and triggering afferent nerve signals.

137.3 Etiology

- Benign prostatic hyperplasia (BPH) is the most common cause in men
- Urethral stricture
- Prostate cancer
- Pelvic organ prolapse in women
- Bladder neck contracture
- Neurogenic bladder: spinal cord injury, multiple sclerosis, Parkinson disease, diabetic neuropathy
- Medications: anticholinergics, antihistamines, decongestants, opioids, alpha-agonists
- Post-surgical: anesthesia, pelvic surgery, post-hip replacement
- Constipation (fecal impaction compressing urethra)

- Infection: prostatitis, cystitis, urethritis
- Anatomical: phimosis, paraphimosis

137.4 Indications for Evaluation

- Inability to urinate or difficulty starting urination
- Painful, urgent need to urinate but unable to pass urine
- Lower abdominal pain and distension
- Feeling of bladder fullness
- Overflow incontinence (dribbling)
- Frequency or urgency with small volumes
- Palpable distended bladder on examination

137.5 Related Symptoms

- Enlarged prostate (BPH)
- Frequent urination
- Overactive bladder
- Urinary tract infection
- Hematuria (blood in urine)
- Constipation

137.6 Self-Care/Home Management

- Do NOT delay seeking medical care for acute retention
- Try to relax and find a comfortable position to urinate
- Run water or place a hand in warm water to stimulate urination
- Gentle suprapubic pressure (Cred maneuver) may help
- Avoid alcohol, caffeine, and decongestants
- Maintain regular bowel movements to avoid constipation

137.7 Diagnostic Approach

- History and physical examination including digital rectal exam
- Bladder scan or ultrasound for post-void residual
- Urinalysis and urine culture
- Uroflowmetry and pressure-flow studies
- Cystoscopy to evaluate for obstruction
- Urodynamic studies for suspected neurogenic bladder

137.8 Treatment/Management Overview

- Immediate catheterization for acute retention (Foley or intermittent)
- Alpha-blockers (tamsulosin, terazosin) for BPH
- 5-alpha reductase inhibitors for BPH
- Clean intermittent catheterization for chronic retention
- Surgery: TURP, urethrotomy, bladder neck incision
- Treatment of underlying cause (medication adjustment, infection treatment)
- Neuromodulation for non-obstructive retention

Chapter 138

Indigestion (Dyspepsia)

138.1 Definition

Indigestion (dyspepsia) is a group of upper abdominal symptoms including pain, burning, fullness, and nausea after eating. It affects approximately 20 percent of the population and is often functional (no identifiable organic cause).

138.2 Pathophysiology

Dyspepsia results from a complex interaction of gastric acid, impaired gastric accommodation, delayed gastric emptying, visceral hypersensitivity, and psychological factors. *Helicobacter pylori* infection, NSAID use, and gastritis contribute in a subset of patients. Functional dyspepsia involves abnormal brain-gut signaling without structural abnormality.

138.3 Etiology

- Functional dyspepsia (most common, no organic cause found)
- Gastroesophageal reflux disease (GERD)
- Peptic ulcer disease (gastric or duodenal ulcer)
- *H. pylori* infection
- Gastritis (acute or chronic)
- Medications: NSAIDs, aspirin, corticosteroids, antibiotics
- Gallbladder disease (biliary colic)
- Pancreatitis
- Gastric cancer (rare, but consider in older patients with warning signs)
- Delayed gastric emptying (gastroparesis)
- Anxiety and stress
- Food intolerances

138.4 Indications for Evaluation

- Upper abdominal pain or burning
- Feeling of fullness after a small meal (early satiety)
- Uncomfortable fullness after eating
- Nausea, belching, bloating
- Heartburn or regurgitation
- Symptoms for 4 or more weeks
- Unexplained weight loss (alarm symptom)
- Vomiting, especially with blood
- Difficulty swallowing
- Iron deficiency anemia

138.5 Related Symptoms

- Heartburn (GERD)
- Bloating and gas
- Nausea
- Vomiting
- Abdominal pain
- Belching

138.6 Self-Care/Home Management

- Eat smaller, more frequent meals
- Avoid trigger foods (spicy, fatty, acidic)
- Eat slowly and chew thoroughly
- Avoid eating within 2-3 hours of bedtime
- Elevate the head of the bed
- Reduce or stop smoking
- Limit alcohol and caffeine
- Maintain a healthy weight
- Manage stress through relaxation techniques

138.7 Diagnostic Approach

- Clinical history and physical examination
- Test for *H. pylori* (stool antigen or breath test)
- Upper endoscopy (EGD) for patients over 60 or with alarm symptoms
- Abdominal ultrasound if biliary or pancreatic cause suspected
- Gastric emptying study if gastroparesis suspected
- Ambulatory pH monitoring if refractory GERD

138.8 Treatment/Management Overview

- Antacids for mild symptoms
- H₂ receptor antagonists (famotidine)
- Proton pump inhibitors (omeprazole, pantoprazole)
- *H. pylori* eradication therapy if positive
- Prokinetic agents (metoclopramide, domperidone) for gastroparesis
- Tricyclic antidepressants (amitriptyline) for functional dyspepsia
- Cognitive behavioral therapy for functional dyspepsia

Chapter 139

Ingrown Hair (Pseudofolliculitis Barbae)

139.1 Definition

An ingrown hair is a hair that curls back or grows sideways into the skin, causing inflammation. It most commonly affects the beard area, neck, legs, and bikini area. It is particularly common in people with curly or coarse hair.

139.2 Pathophysiology

When a hair is shaved, the sharp tip can re-enter the skin as it grows. Curly hair is more prone to curling back into the follicle. The body mounts an inflammatory response, causing redness, papules, pustules, and sometimes hyperpigmentation or scarring. It is a foreign-body inflammatory reaction.

139.3 Etiology

- Shaving (most common cause)
- Tweezing or waxing
- Tight clothing (bikini area)
- Curly or coarse hair type
- Close shaving (multi-blade razors)
- Shaving against the direction of hair growth
- Stretching the skin while shaving
- Dry shaving or inadequate lubrication

139.4 Indications for Evaluation

- Small, raised, red bumps (papules) in areas of hair removal
- Pustules that may look like acne
- Pain or tenderness
- Itching
- Hyperpigmentation in chronic cases
- Curved hair visible under the skin
- Razor bumps (pseudofolliculitis barbae) on the neck and beard area

139.5 Related Symptoms

- Folliculitis (infection of hair follicles)
- Acne vulgaris
- Razor burn
- Pseudofolliculitis (razor bumps)
- Hyperpigmentation (post-inflammatory)

139.6 Self-Care/Home Management

- Stop shaving the affected area until healed
- Use a warm compress to bring the hair to the surface
- Gently extract the embedded hair with a sterile needle
- Shave with the grain, not against it
- Use a single-blade razor
- Apply warm water and shaving cream before shaving
- Exfoliate gently with a soft brush or washcloth
- Use electric clippers instead of a razor

139.7 Diagnostic Approach

- Clinical examination of affected skin
- Rule out other causes of folliculitis (bacterial, fungal)

139.8 Treatment/Management Overview

- Discontinue hair removal temporarily
- Apply topical antibiotics if secondarily infected
- Topical retinoids (tretinoin) to normalize follicular keratinization
- Topical corticosteroids for inflammation
- Laser hair removal for chronic, recurrent cases
- Electrolysis as a permanent solution
- Consider chemical depilatories instead of shaving

Chapter 140

Ingrown Toenails (Onychocryptosis)

140.1 Definition

An ingrown toenail is a condition where the nail grows into the surrounding skin, causing pain, redness, swelling, and often infection. It most commonly affects the great toe.

140.2 Pathophysiology

The nail plate pierces the lateral nail fold, causing a foreign-body inflammatory response. This leads to pain, edema, and granulation tissue formation. If the skin barrier is compromised, secondary bacterial infection can occur. Improper nail trimming, tight footwear, and genetic factors contribute to ingrowth.

140.3 Etiology

- Improper nail trimming (rounded corners instead of straight across)
- Tight shoes that compress the toes
- Trauma: stubbing the toe, dropping objects on the foot
- Genetic predisposition: wide nail fold or curved nails
- Poor foot hygiene
- Excessive sweating
- Obesity (increased pressure on toes)
- Repetitive activities (kicking, running)

140.4 Indications for Evaluation

- Pain along the edge of the toenail
- Redness and swelling of the nail fold
- Tenderness to touch

- Pus or drainage from the nail edge
- Overgrowth of skin (granulation tissue) around the nail
- Difficulty wearing shoes
- Chronic or recurrent episodes

140.5 Related Symptoms

- Paronychia (infection of the nail fold)
- Cellulitis of the toe
- Osteomyelitis (bone infection, rare)
- Fungal nail infection
- Subungual hematoma

140.6 Self-Care/Home Management

- Soak the foot in warm, soapy water 3-4 times daily
- Gently lift the nail edge and place a small piece of cotton or dental floss under it
- Apply antibiotic ointment
- Wear open-toed shoes or sandals
- Take OTC pain relievers
- Do NOT cut the nail at the corner or dig into the nail fold
- Trim nails straight across

140.7 Diagnostic Approach

- Clinical examination of the toe and nail
- Assess for signs of infection (erythema, purulence, warmth)
- X-ray if osteomyelitis suspected in severe cases

140.8 Treatment/Management Overview

- Conservative: warm soaks, cotton wick placement, antibiotics
- Partial nail avulsion (removal of the ingrown nail edge)
- Matrixectomy (chemical ablation with phenol) to prevent recurrence
- Complete nail avulsion for recurrent or severe cases

- Oral antibiotics for cellulitis
- Surgical correction by a podiatrist

Chapter 141

Insect Bites (Arthropod Bites and Stings)

141.1 Definition

Insect bites and stings are common injuries caused by arthropods including mosquitoes, bees, wasps, ants, fleas, ticks, and spiders. Most reactions are mild and local, but some can cause severe allergic reactions or transmit diseases.

141.2 Pathophysiology

Most insects inject saliva or venom through the skin. The body mounts an immediate immune response mediated by IgE, histamine, and other inflammatory mediators, causing local redness, swelling, itching, and pain. Delayed hypersensitivity reactions can occur hours to days later. Systemic allergic reactions (anaphylaxis) involve widespread mast cell degranulation.

141.3 Etiology

- Mosquitoes
- Bees, wasps, hornets, yellow jackets (Hymenoptera)
- Fire ants
- Fleas
- Ticks
- Spiders (black widow, brown recluse)
- Bed bugs
- Chiggers
- Scabies mites

141.4 Indications for Evaluation

- Small, red, itchy bump at the bite site
- Pain or burning sensation
- Swelling around the bite
- Central punctum (visible bite mark)
- Multiple bites in clusters or lines
- Hive-like reaction distant from the bite
- Signs of infection: spreading redness, warmth, pus
- Signs of anaphylaxis: difficulty breathing, swelling of lips/tongue, hives all over, dizziness

141.5 Related Symptoms

- Allergic reaction
- Anaphylaxis
- Cellulitis (secondary infection)
- Lyme disease (tick-borne)
- West Nile virus (mosquito-borne)
- Scabies
- Urticaria (hives)

141.6 Self-Care/Home Management

- Clean the bite area with soap and water
- Apply a cold compress to reduce swelling
- Use OTC hydrocortisone cream or calamine lotion for itching
- Take oral antihistamines (diphenhydramine, cetirizine)
- Do not scratch (increases infection risk)
- Elevate the affected limb if swollen
- Remove ticks carefully with tweezers at the skin surface

141.7 Diagnostic Approach

- Clinical history and appearance of the bite
- Identify the insect if possible
- Allergy testing (skin prick or specific IgE) for suspected sting allergy
- Blood tests for tick-borne diseases if indicated

141.8 Treatment/Management Overview

- Topical corticosteroids for local inflammation
- Oral antihistamines for itching
- Epinephrine auto-injector for those with history of anaphylaxis
- Antibiotics for secondary infection
- Tetanus prophylaxis if needed
- Specific antivenom for venomous spider bites
- Remove stingers by scraping (not squeezing)

Chapter 142

Insomnia

Definition

Difficulty falling asleep staying asleep waking too early resulting in daytime impairment.

Pathophysiology

Classified as transient (days) acute (weeks) chronic (months); primary insomnia vs secondary to medical psychiatric environmental factors; hyperarousal model predominant mechanism.

Etiology

- Stress/anxiety
- Depression
- Circadian rhythm disruption
- Medical conditions (pain GERD sleep apnea)
- Medication/stimulant effects

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than chronic insomnia (>3 months) affecting daily function warrants treatment
- Accompanied by other concerning signs

Related Symptoms

- Daytime fatigue
- Irritability
- Concentration difficulty
- Mood changes
- Decreased performance

Note: Always consider serious underlying conditions when evaluating persistent insomnia.

Self-Care / Home Management

- Establish a consistent daily routine with regular sleep, meals, and physical activity to support mental health stability.
- Practice stress management techniques including mindfulness, deep breathing exercises, and progressive muscle relaxation.
- Maintain social connections and avoid isolation by reaching out to trusted friends, family, or support groups.
- Avoid alcohol, caffeine, and recreational drugs which can destabilize mood and exacerbate psychiatric symptoms.
- If experiencing suicidal thoughts or crisis, contact emergency services or a crisis helpline immediately.

Diagnostic Approach

- Comprehensive psychiatric assessment includes history of present illness, past psychiatric history, substance use, medical history, and collateral information from family.
- Standardized screening tools (GAD-7, PHQ-9, PCL-5) help quantify symptom severity and track treatment response.
- Medical evaluation (thyroid function, basic metabolic panel, CBC, toxicology screen) rules out organic causes of psychiatric symptoms.
- Risk assessment for self-harm, suicide, and danger to others is an essential component of every psychiatric evaluation.

Treatment / Management Overview

- Psychotherapy (CBT, DBT, interpersonal therapy) is a first-line treatment for most psychiatric disorders.
- Pharmacotherapy with antidepressants, anxiolytics, mood stabilizers, or antipsychotics is tailored to the specific diagnosis and individual patient factors.

- Combined treatment (medication plus psychotherapy) generally yields superior outcomes compared to either modality alone.
- Referral to psychiatry is indicated for severe symptoms, treatment resistance, suicidal ideation, or diagnostic uncertainty.

Chapter 143

Internal Bleeding

143.1 Definition

Internal bleeding is bleeding that occurs inside the body, within tissues, organs, or body cavities. It is not visible externally and can be life-threatening depending on the rate and location of blood loss.

143.2 Pathophysiology

Internal bleeding results from injury to blood vessels within the body. Trauma can lacerate or rupture vessels. Non-traumatic causes include erosion of vessels by ulcers or tumors, rupture of aneurysms, bleeding from varices, or coagulopathy. Blood loss can lead to hypovolemic shock, organ compression (tamponade), and impaired organ function.

143.3 Etiology

- Trauma (blunt or penetrating): motor vehicle accidents, falls, assault
- Ruptured abdominal aortic aneurysm
- Hemorrhagic stroke (intracerebral or subarachnoid hemorrhage)
- Peptic ulcer disease (gastrointestinal bleeding)
- Esophageal varices (liver cirrhosis)
- Ectopic pregnancy rupture (gynecologic emergency)
- Ovarian cyst rupture
- Pelvic fractures
- Anticoagulant therapy
- Coagulation disorders (hemophilia, liver disease)
- Ruptured spleen (trauma, mononucleosis)

143.4 Indications for Evaluation

- Unexplained hypotension or shock
- Tachycardia (rapid heart rate)
- Pale, cool, clammy skin
- Abdominal pain, distension, or rigidity
- Vomiting blood (hematemesis)
- Black, tarry stools (melena)
- Blood in urine (hematuria)
- Neurological deficits (if intracranial)
- Shortness of breath (if hemothorax)
- Confusion or altered mental status

143.5 Related Symptoms

- Shock (hypovolemic)
- Gastrointestinal bleeding
- Hemorrhage
- Anemia
- Hematuria (blood in urine)
- Hemoptysis (coughing blood)

143.6 Self-Care/Home Management

- Internal bleeding is a medical emergency - CALL EMERGENCY SERVICES
- Do NOT eat or drink anything
- Lie down and keep calm
- Apply direct pressure to any external bleeding sites
- Do not move a person with suspected spinal injury

143.7 Diagnostic Approach

- Clinical assessment of vital signs and hemodynamic status
- Complete blood count (hemoglobin and hematocrit)
- Coagulation profile (PT, PTT, INR)
- CT scan of the affected area (head, chest, abdomen)
- Focused Assessment with Sonography for Trauma (FAST) ultrasound
- Angiography to identify bleeding source
- Diagnostic peritoneal lavage or thoracentesis

143.8 Treatment/Management Overview

- Resuscitation: IV fluids, blood transfusion
- Reverse anticoagulation if applicable
- Surgical intervention to control bleeding source
- Endoscopic therapy for GI bleeding
- Angiographic embolization
- Treat the underlying cause
- ICU monitoring for hemorrhagic shock

Chapter 144

Irregular Periods

Definition

Menstrual cycles that vary significantly from the typical 21-35 day pattern, including absent periods (amenorrhea), infrequent periods (oligomenorrhea), or unpredictable cycle length.

Pathophysiology

Normal menstruation requires intact hypothalamic-pituitary-ovarian axis with appropriate hormone levels. Disruption at any level causes irregular bleeding: hypothalamic (stress, exercise, weight loss), pituitary (prolactinoma), ovarian (PCOS, perimenopause, premature ovarian failure), uterine (fibroids, polyps), or endocrine (thyroid, adrenal) causes.

Etiology

- Polycystic ovary syndrome (PCOS)
- Perimenopause (transition to menopause)
- Thyroid disorders (hyperthyroidism or hypothyroidism)
- Hyperprolactinemia
- Stress or significant weight changes
- Excessive exercise (athletic amenorrhea)
- Eating disorders (anorexia nervosa)
- Uterine fibroids or polyps
- Primary ovarian insufficiency (premature menopause)
- Hormonal contraception effects

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Missed periods for 3+ cycles, very heavy bleeding, bleeding between periods, or difficulty conceiving
- Accompanied by other concerning signs

Related Symptoms

- Heavy bleeding
- Pelvic pain
- Weight changes
- Hair growth changes
- Acne

Note: Always consider serious underlying conditions when evaluating persistent irregular periods.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 145

Irritability

Definition

An excessive or disproportionate response to stimuli characterised by anger, frustration, impatience, and a lowered threshold for annoyance, often with verbal or behavioural outbursts.

Pathophysiology

Irritability in perimenopause is driven by oestrogen fluctuation affecting neurotransmitter systems involved in impulse control and emotional regulation. Oestrogen modulates serotonin synthesis and receptor sensitivity in the prefrontal cortex, which governs executive control over emotions. Declining oestrogen reduces serotonergic tone, decreasing the brain's ability to inhibit anger and frustration. Progesterone fluctuations affect GABAergic inhibition, further lowering the irritability threshold. Sleep disruption (common in menopause) amplifies irritability through prefrontal cortex dysfunction.

Etiology

- Perimenopausal and menopausal hormonal changes
- Sleep deprivation or insomnia
- Premenstrual syndrome or PMDD
- Major depressive disorder or bipolar disorder
- Generalised anxiety disorder
- Chronic stress and burnout
- Hyperthyroidism (thyrotoxicosis)
- Chronic pain conditions
- Substance use or withdrawal (alcohol, caffeine, nicotine)
- Medication side effects (corticosteroids, stimulants)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Irritability with mood swings, depression, anxiety, sleep disturbance, or relationship difficulties, especially if affecting daily function
- Accompanied by other concerning signs

Related Symptoms

- Mood swings
- Depression
- Anxiety
- Insomnia
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent irritability.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
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- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 146

Itching Pruritus

Definition

Unpleasant sensation provokes desire to scratch may localized generalized acute chronic primary secondary cause diverse etiologies dermatological systemic neurological psychological vascular mechanical factors

Pathophysiology

Neurotransmitter release substance P histamine cytokines nerve fiber activation itch-specific C fibers pruriceptors central processing spinal cord brain perception scratching provides temporary relief but damages skin creates itch-scratch cycle perpetuating condition

Etiology

- Dry skin xerosis most common winter dry climates aging reduced sebum production
- Allergic contact dermatitis poison ivy nickel cosmetics chemicals urticaria hives angioedema
- Viral exanthems measles rose chickenpox HSV VZV
- Drug eruption Stevens Johnson DRESS syndrome serum sickness-like reaction
- Autoimmune cutaneous lupus vasculitis psoriasis pemphigus pemphigoid

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Systemic mucosal surfaces fever rapidly progressive blister mucosal erosions hospital rule out drug-induced life-threatening syndromes SJS TEN scalded skin syndrome widespread unexplained itching jaundice dark urine pale stool suggest cholestatic liver disease renal failure lymphoma thyroid dysfunction diabetes neuropathy psychiatric depression anxiety psychogenic pruritus
- Accompanied by other concerning signs

Related Symptoms

- Itching sore warmth scaling crust oozing blister rupture peeling thickening pigmentation pain bacterial superinfection fever joint eye involvement rash lesions bumps blisters

Note: Always consider serious underlying conditions when evaluating persistent itching pruritus

Self-Care / Home Management

- Keep the affected area clean and dry; wash gently with mild, fragrance-free soap and lukewarm water.
- Avoid scratching — use cool compresses, calamine lotion, or OTC hydrocortisone cream to relieve itching.
- Apply moisturizer regularly to maintain skin barrier integrity, especially after bathing.
- Identify and avoid potential triggers (irritants, allergens, excessive heat, tight clothing).
- Seek medical evaluation if the rash spreads rapidly, develops blisters, shows signs of infection (pus, increasing redness, warmth), or is accompanied by fever.

Diagnostic Approach

- History includes onset, location, progression, associated symptoms (itching, pain, fever), exposure to triggers, and prior skin conditions.
- Physical examination assesses morphology (macules, papules, vesicles, etc.), distribution, and extent of skin involvement.
- Diagnostic tests may include skin scraping for fungal examination, patch testing for allergic contact dermatitis, or biopsy for suspicious lesions.
- Laboratory studies (CBC, ESR, autoimmune serologies) are obtained when systemic disease is suspected.

Treatment / Management Overview

- Topical therapies (corticosteroids, antifungals, antibiotics, emollients) are first-line for most localized skin conditions.
- Systemic therapy (oral antihistamines, antibiotics, antifungals, or immunosuppressants) is reserved for extensive or refractory cases.
- Phototherapy or biologic agents may be indicated for chronic inflammatory skin diseases (psoriasis, eczema).
- Referral to dermatology is appropriate for uncertain diagnoses, treatment failure, or skin lesions suspicious for malignancy.

Chapter 147

Jaundice

Definition

Yellowish discoloration of the skin, sclerae, and mucous membranes due to elevated serum bilirubin levels, clinically detectable when bilirubin exceeds 2-3 mg/dL (34-51 $\mu\text{mol/L}$).

Pathophysiology

Bilirubin accumulation results from pre-hepatic (hemolysis, ineffective erythropoiesis), hepatic (impaired hepatocyte uptake, conjugation, or excretion), or post-hepatic (biliary obstruction preventing bilirubin excretion into the gut) causes. Unconjugated (indirect) vs conjugated (direct) hyperbilirubinemia distinguished by laboratory testing guides diagnostic workup.

Etiology

- Viral hepatitis (A, B, C, E)
- Alcoholic liver disease or cirrhosis
- Gallstones or bile duct obstruction
- Pancreatic cancer (head of pancreas)
- Hemolytic anemias (sickle cell, spherocytosis)
- Drug-induced liver injury (acetaminophen, statins, isoniazid)
- Primary biliary cholangitis
- Gilbert syndrome (benign unconjugated hyperbilirubinemia)
- Acute fatty liver of pregnancy
- Autoimmune hepatitis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any jaundice requires medical evaluation to identify underlying cause quickly
- Accompanied by other concerning signs

Related Symptoms

- Dark urine
- Pale stools
- Abdominal pain
- Nausea
- Itching (pruritus)

Note: Always consider serious underlying conditions when evaluating persistent jaundice.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 148

Jock Itch (Tinea Cruris)

148.1 Definition

Jock itch (tinea cruris) is a common fungal infection of the groin and inner thighs caused by dermatophytes. It is more common in men and in athletes, especially in warm, humid conditions.

148.2 Pathophysiology

Dermatophyte fungi (most commonly *Trichophyton rubrum*) invade the superficial keratin layer of the skin. The warm, moist environment of the groin area promotes fungal growth. The infection elicits an inflammatory response that causes the characteristic red, itchy, ring-shaped rash. The scrotum is typically spared, which helps distinguish it from other conditions.

148.3 Etiology

- Excessive sweating and moisture in the groin area
- Tight, non-breathable clothing
- Obesity (increased skin folds and moisture)
- Immunosuppression (diabetes, HIV, corticosteroid use)
- Sharing contaminated towels or clothing
- Poor hygiene
- Prior fungal infection of the feet (athlete's foot)
- Hot, humid climate

148.4 Indications for Evaluation

- Intense itching in the groin and inner thigh area
- Red, raised, scaly rash with a well-defined border
- Rash may have central clearing (ringworm appearance)

- Burning or stinging sensation
- Flaking, peeling, or cracking skin
- Often spares the scrotum
- May extend to the buttocks and upper thighs

148.5 Related Symptoms

- Athlete's foot (tinea pedis)
- Ringworm (tinea corporis)
- Intertrigo (moisture-associated rash)
- Candidal intertrigo (yeast infection)
- Contact dermatitis of the groin
- Psoriasis

148.6 Self-Care/Home Management

- Keep the groin area clean and dry
- Apply OTC antifungal creams (clotrimazole, miconazole, terbinafine)
- Wear loose-fitting, breathable cotton underwear
- Change clothes after exercise
- Use antifungal powder to reduce moisture
- Avoid sharing towels
- Wash underwear in hot water
- Dry the groin area thoroughly after bathing

148.7 Diagnostic Approach

- Clinical examination: characteristic appearance and location
- Skin scraping with KOH preparation for microscopic fungal elements
- Fungal culture if diagnosis is uncertain

148.8 Treatment/Management Overview

- Topical antifungal creams for 2-4 weeks (terbinafine, clotrimazole, miconazole)
- Oral antifungal medication (terbinafine, itraconazole, fluconazole) for extensive or resistant cases
- Topical corticosteroids for inflammation (short-term)
- Keep area dry and avoid recurrence triggers

Chapter 149

Joint Pain

Definition

Pain or discomfort localized to one or more joints varying from dull ache to sharp stabbing sensation.

Pathophysiology

Differentiated monoarticular vs polyarticular, symmetric vs asymmetric, inflammatory vs degenerative patterns guiding toward rheumatologic or orthopedic etiologies.

Etiology

- Osteoarthritis (degenerative)
- Rheumatoid arthritis (inflammatory)
- Trauma/injury
- Septic arthritis (infectious)
- Gout (crystal arthropathy)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluated clinically based on pattern and associated findings
- Accompanied by other concerning signs

Related Symptoms

- Swelling
- Warmth/redness

- Morning stiffness >30min
- Reduced movement
- Systemic symptoms like fever/rash

Note: Always consider serious underlying conditions when evaluating persistent joint pain.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.
- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 150

Kidney Pain

Definition

Pain originating from the kidneys, typically felt in the flank region (costovertebral angle) between the ribs and hip, often radiating to the lower abdomen or groin.

Pathophysiology

Kidney pain arises from distention of the renal capsule (stretching from inflammation, obstruction, or mass), ureteral colic (smooth muscle spasm from stone passage), or inflammation of renal parenchyma (pyelonephritis). Unlike musculoskeletal back pain, renal pain is typically constant, not positional, and associated with urinary symptoms.

Etiology

- Nephrolithiasis (kidney stones)
- Pyelonephritis (kidney infection)
- Ureteral obstruction
- Renal infarction or thrombosis
- Polycystic kidney disease
- Renal cell carcinoma
- Glomerulonephritis
- Hydronephrosis
- Papillary necrosis
- Perinephric abscess

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe flank pain with fever and chills, inability to urinate, blood in urine, or severe unremitting pain
- Accompanied by other concerning signs

Related Symptoms

- Flank pain
- Blood in urine (hematuria)
- Fever
- Nausea and vomiting
- Dysuria

Note: Always consider serious underlying conditions when evaluating persistent kidney pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 151

Knee Injury

151.1 Definition

Knee injuries involve damage to the bones, ligaments, tendons, cartilage, or menisci of the knee joint. They are common in athletes but can occur in anyone. Severity ranges from minor sprains to complex multi-ligament injuries.

151.2 Pathophysiology

The knee is a hinge joint stabilized by four major ligaments (ACL, PCL, MCL, LCL), two menisci (medial and lateral), and surrounding muscles and tendons. Injury mechanisms include twisting (non-contact ACL tear), direct impact (MCL tear, fracture), or overuse (patellar tendinitis, bursitis). Damage causes pain, swelling, instability, and impaired function.

151.3 Etiology

- Sports injuries: pivoting, cutting, landing from a jump, direct blows
- ACL tear: non-contact twisting injury common in soccer, basketball, skiing
- MCL tear: valgus stress (blow to the outer knee)
- Meniscus tear: twisting with the foot planted
- Patellar dislocation: twisting or direct blow
- Patellofemoral pain syndrome: overuse, muscle imbalance
- Fractures: patella, femoral condyles, tibial plateau
- Overuse: runner's knee, jumper's knee
- Osteoarthritis: degenerative changes
- Direct trauma: falls, motor vehicle accidents

151.4 Indications for Evaluation

- Knee pain with weight-bearing or range of motion
- Swelling (effusion) within hours of injury
- Bruising around the knee
- Instability or giving way of the knee
- Locking or catching sensation (meniscus tear)
- Inability to fully extend or flex the knee
- Tenderness over specific structures (joint line, ligaments)
- Audible pop at the time of injury (classic ACL tear)

151.5 Related Symptoms

- Knee swelling
- Knee locking
- Knee instability
- Leg pain
- Joint stiffness
- Osteoarthritis (post-traumatic)

151.6 Self-Care/Home Management

- Rest, Ice, Compression, Elevation (RICE protocol)
- Avoid weight-bearing on the injured knee
- Use crutches for support
- Take OTC pain relievers (ibuprofen, acetaminophen)
- Apply ice for 20 minutes every 2-3 hours
- Use an elastic bandage for compression
- Begin gentle range-of-motion exercises once pain subsides

151.7 Diagnostic Approach

- History of injury mechanism and symptoms
- Physical examination: inspection, palpation, range of motion, stability tests (Lachman, drawer, McMurray)
- X-ray (weight-bearing) to rule out fracture
- MRI for soft tissue evaluation (ligaments, menisci, cartilage)
- CT scan for complex fractures
- Diagnostic arthroscopy in selected cases

151.8 Treatment/Management Overview

- Conservative: RICE protocol, physical therapy, activity modification
- Bracing for ligament injuries
- Physical therapy for muscle strengthening and proprioception
- Arthroscopic surgery for meniscus repair or partial meniscectomy
- ACL reconstruction for active individuals
- MCL healing with bracing (most heal non-operatively)
- Knee replacement for severe arthritis

Chapter 152

Knee Pain

Definition

Pain localized to the knee joint and surrounding structures, one of the most common musculoskeletal complaints affecting mobility and quality of life.

Pathophysiology

The knee's weight-bearing role and complex ligamentous/cartilage structures make it vulnerable to acute injury (ligament tears, meniscal tears) and chronic degenerative changes (osteoarthritis). Anterior knee pain suggests patellofemoral pathology; medial/lateral pain suggests meniscal or collateral ligament involvement; diffuse pain suggests inflammatory arthritis.

Etiology

- Osteoarthritis (degenerative joint disease)
- Meniscal tear
- Anterior cruciate ligament (ACL) injury
- Patellofemoral pain syndrome
- Patellar tendinopathy (jumper's knee)
- Baker's cyst (popliteal cyst)
- Rheumatoid arthritis or gout
- Iliotibial band syndrome
- Prepatellar bursitis (housemaid's knee)
- Septic arthritis or osteomyelitis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Acute knee injury with inability to bear weight, significant swelling, fever, locking, or instability
- Accompanied by other concerning signs

Related Symptoms

- Swelling
- Stiffness
- Locking or catching
- Instability (giving way)
- Limping

Note: Always consider serious underlying conditions when evaluating persistent knee pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 153

Labia Changes (Atrophy)

Definition

Structural changes to the labia majora and minora including shrinkage, loss of elasticity, colour changes, and reduced tissue volume, resulting from hormonal insufficiency.

Pathophysiology

Labial tissues contain oestrogen and androgen receptors that maintain collagen density, vascularity, and elastic fibres. Declining oestrogen during menopause causes dermal thinning, reduced collagen synthesis, loss of subcutaneous fat, and decreased blood flow. The labia minora may become pale, thin, and gradually resorbed; the labia majora lose subcutaneous fat volume. These changes contribute to vulvovaginal atrophy (part of genitourinary syndrome of menopause) and can affect urinary and sexual function.

Etiology

- Genitourinary syndrome of menopause (oestrogen deficiency)
- Natural ageing (collagen loss independent of hormones)
- Lichen sclerosus (autoimmune, causes scarring and fusion)
- Lichen planus (inflammatory dermatosis)
- Crohn disease (vulvar involvement)
- Radiation therapy to the pelvic region
- Surgical menopause (bilateral oophorectomy)
- Chemotherapy-induced ovarian failure
- Autoimmune oophoritis
- Smoking (accelerated oestrogen metabolism)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Labial changes with itching, burning, pain, skin thickening or whitening, fissures, or difficulty with intercourse
- Accompanied by other concerning signs

Related Symptoms

- Vaginal dryness
- Vaginal atrophy
- Painful sex
- Clitoral pain
- Urinary symptoms

Note: Always consider serious underlying conditions when evaluating persistent labia changes.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 154

Labor Signs (Signs of Labor)

154.1 Definition

Labor signs are the indicators that a pregnant woman is entering the active phase of childbirth. Recognizing these signs helps determine when to seek medical care.

154.2 Pathophysiology

As pregnancy reaches term (37-42 weeks), hormonal changes trigger uterine contractions, cervical ripening, and effacement. Prostaglandins and oxytocin stimulate coordinated uterine contractions. The cervix softens, shortens (effaces), and opens (dilates). The presenting fetal part descends into the pelvis. The amniotic sac may rupture spontaneously.

154.3 Etiology

- Natural progression of full-term pregnancy
- Preterm labor occurs before 37 weeks
- Induced labor (medical or elective)
- Spontaneous rupture of membranes

154.4 Indications for Evaluation

- Regular, painful contractions that become stronger and closer together
- Contractions lasting 30-70 seconds and occurring every 5-10 minutes
- Lower back pain that does not go away
- Bloody show: pink or blood-tinged mucus discharge
- Rupture of membranes: gush or leak of fluid from the vagina
- Water breaking may feel like a slow trickle or sudden gush
- Cervical dilation and effacement on examination

154.5 Related Symptoms

- False labor (Braxton Hicks contractions, which are irregular and do not dilate the cervix)
- Preterm labor (before 37 weeks)
- Vaginal bleeding in pregnancy
- Premature rupture of membranes
- Back pain during pregnancy

154.6 Self-Care/Home Management

- Time contractions from the start of one to the start of the next
- Drink fluids and rest between contractions
- Use breathing and relaxation techniques
- Take a warm shower or bath for comfort
- Change positions frequently (walking, sitting, lying on side)
- Eat light snacks if hungry (avoid heavy meals)
- Call the healthcare provider or go to the hospital when contractions are 5 minutes apart lasting 1 minute for 1 hour

154.7 Diagnostic Approach

- Pelvic examination to check for cervical dilation and effacement
- Fetal monitoring for contraction pattern and fetal heart rate
- AmniSure or Nitrazine test to confirm rupture of membranes
- Ultrasound to assess fetal position, fluid volume, and estimated weight

154.8 Treatment/Management Overview

- Admit for active labor management
- Pain management: epidural, IV narcotics, nitrous oxide
- Continuous fetal monitoring
- Oxytocin augmentation if labor slows
- Assisted delivery (forceps, vacuum) if needed
- Cesarean section for complications
- Neonatal care support after delivery

Chapter 155

Lack of Motivation (Avolition)

Definition

A reduction or absence of the drive to initiate and persist in goal-directed activities, often experienced as apathy, indifference, or difficulty starting tasks despite a desire to complete them.

Pathophysiology

Motivation depends on intact dopaminergic signalling in the mesolimbic and mesocortical pathways connecting the ventral tegmental area to the nucleus accumbens (reward anticipation) and prefrontal cortex (executive planning). Oestrogen upregulates dopamine synthesis and receptor density; declining oestrogen reduces dopaminergic tone, impairing the brain's reward anticipation and motivational drive. Serotonin-dopamine interactions also influence motivation; low serotonin (from hormonal fluctuation) can inhibit dopamine function.

Etiology

- Perimenopausal and menopausal hormonal changes
- Major depressive disorder
- Chronic fatigue syndrome (myalgic encephalomyelitis)
- Hypothyroidism
- Vitamin D deficiency
- Sleep disorders (obstructive sleep apnoea, insomnia)
- Attention deficit hyperactivity disorder (executive dysfunction)
- Schizophrenia (negative symptoms)
- Frontal lobe disorders (dementia, traumatic brain injury)
- Medication side effects (SSRIs, antipsychotics, beta-blockers)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent lack of motivation with depressed mood, sleep changes, weight changes, or inability to perform daily activities
- Accompanied by other concerning signs

Related Symptoms

- Fatigue
- Anhedonia
- Depression
- Difficulty concentrating
- Insomnia

Note: Always consider serious underlying conditions when evaluating persistent lack of motivation.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 156

Laryngitis

156.1 Definition

Laryngitis is inflammation of the larynx (voice box), most commonly caused by viral upper respiratory infections. It causes hoarseness or loss of voice. Acute laryngitis lasts less than 3 weeks, while chronic laryngitis persists longer.

156.2 Pathophysiology

Viral infection of the upper respiratory tract causes edema and inflammation of the vocal cords. The vocal cords become red, swollen, and less pliable, disrupting normal vibration and producing hoarseness. Overuse of the voice during inflammation can exacerbate the condition. Chronic laryngitis involves persistent irritation from reflux, smoking, or environmental factors.

156.3 Etiology

- Viral upper respiratory infections (most common)
- Vocal strain from yelling, singing, or prolonged speaking
- Gastroesophageal reflux disease (laryngopharyngeal reflux)
- Bacterial infections (less common)
- Allergies and postnasal drip
- Inhaled irritants (cigarette smoke, pollution)
- Chronic sinusitis
- Fungal laryngitis (immunosuppressed patients)
- Autoimmune conditions
- Intubation or surgery involving the airway

156.4 Indications for Evaluation

- Hoarseness or raspy voice
- Weak voice or loss of voice (aphonia)
- Sore throat or throat discomfort
- Dry, barking cough
- Feeling of a lump in the throat
- Frequent throat clearing
- Pain with speaking or swallowing
- Low-grade fever
- Symptoms lasting more than 3 weeks (chronic)

156.5 Related Symptoms

- Sore throat
- Cough
- Hoarseness
- Postnasal drip
- GERD (heartburn)
- Upper respiratory infection (common cold)
- Tonsillitis
- Pharyngitis

156.6 Self-Care/Home Management

- Voice rest: avoid speaking, whispering, or shouting
- Drink warm, soothing fluids (honey and lemon tea, broth)
- Use a humidifier or inhale steam
- Avoid smoking and secondhand smoke
- Avoid alcohol and caffeine (can dehydrate)
- Gargle with warm salt water
- Use OTC pain relievers for discomfort
- Do NOT whisper (strains the vocal cords more than normal speech)

156.7 Diagnostic Approach

- Clinical history and symptom pattern
- Laryngoscopy (mirror or flexible endoscopy) to visualize vocal cords
- Referral to otolaryngologist for chronic or atypical cases
- Laryngeal stroboscopy for detailed vocal cord assessment
- pH monitoring if reflux laryngitis suspected

156.8 Treatment/Management Overview

- Voice rest and hydration
- Treat underlying cause (antibiotics for bacterial, PPI for reflux)
- Corticosteroids for severe inflammation or airway compromise
- Speech therapy for vocal cord dysfunction or chronic hoarseness
- Surgery for vocal cord lesions (polyps, nodules) if present

Chapter 157

Leg Pain

Definition

Pain or discomfort anywhere in the lower extremity from the hip to the foot, arising from musculoskeletal, vascular, neurological, or dermatological sources.

Pathophysiology

Leg pain mechanisms include nociceptive (musculoskeletal injury, inflammation), neuropathic (nerve compression or damage from disc herniation, peripheral neuropathy), ischemic (peripheral arterial disease, DVT), and referred (hip or spine pathology presenting as thigh or leg pain).

Etiology

- Peripheral arterial disease (claudication)
- Deep vein thrombosis (DVT)
- Sciatica or lumbar radiculopathy
- Muscle strain or tear (hamstring, quadriceps, calf)
- Peripheral neuropathy (diabetic, alcoholic)
- Varicose veins or chronic venous insufficiency
- Compartment syndrome (acute or chronic)
- Shin splints (medial tibial stress syndrome)
- Stress fracture
- Cellulitis or soft tissue infection

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden severe leg pain with swelling and redness (possible DVT), leg pain with chest pain or shortness of breath, or non-healing ulcers
- Accompanied by other concerning signs

Related Symptoms

- Swelling
- Numbness or tingling
- Skin changes
- Ulcers
- Calf tenderness

Note: Always consider serious underlying conditions when evaluating persistent leg pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 158

Loss of Appetite

Definition

Reduced desire to eat prolonged anorexia can lead unintended weight loss nutritional deficiencies.

Pathophysiology

Distinction from nausea (urge to vomit) and early satiety (feeling full quickly); central hypothalamic regulation peripheral gastrointestinal signals interact influencing appetite.

Etiology

- Depression/anxiety
- Chronic infections
- Cancer/malignancy
- Thyroid disorders
- Medication side effects

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than unintentional significant weight loss requiring medical evaluation
- Accompanied by other concerning signs

Related Symptoms

- Weight loss
- Fatigue

- Nausea
- Taste changes
- Fullness after small meals

Note: Always consider serious underlying conditions when evaluating persistent loss of appetite.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 159

Loss of Bone Density (Osteopenia/Osteoporosis)

Definition

Progressive reduction in bone mineral density, leading to weakened skeletal structure and increased fracture risk, most commonly resulting from oestrogen deficiency after menopause.

Pathophysiology

Oestrogen inhibits osteoclast activity through receptor activator of nuclear factor- κ B ligand (RANKL) regulation. Oestrogen binds to oestrogen receptors on osteoclasts, promoting apoptosis and reducing bone resorption. The rapid decline in oestrogen during menopause removes this inhibitory signal, uncoupling bone remodelling: osteoclast activity increases while osteoblast activity does not keep pace, leading to net bone loss. Trabecular bone (vertebrae, distal radius) is affected more rapidly than cortical bone. Peak bone loss occurs in the first 3-5 years after menopause.

Etiology

- Menopausal oestrogen deficiency (primary cause)
- Advanced age
- Low peak bone mass (genetic, nutritional, exercise-related)
- Calcium and vitamin D deficiency
- Glucocorticoid use (corticosteroid-induced osteoporosis)
- Smoking
- Excessive alcohol consumption
- Low body weight (BMI < 19)
- Family history of osteoporosis or hip fracture
- Secondary causes (hyperthyroidism, hyperparathyroidism, malabsorption, rheumatoid arthritis)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Bone density screening (DXA scan) is recommended for all women aged 65 and older, or younger with risk factors; seek advice after any low-trauma fracture
- Accompanied by other concerning signs

Related Symptoms

- Back pain
- Loss of height
- Joint pain
- Fractures
- Muscle weakness

Note: Always consider serious underlying conditions when evaluating persistent loss of bone density.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 160

Loss of Smell (Anosmia)

Definition

Complete or partial inability to perceive odors, which can be temporary or permanent, congenital or acquired.

Pathophysiology

Anosmia results from either conductive loss (nasal obstruction preventing odorants from reaching olfactory epithelium — sinusitis, polyps, URI) or sensorineural loss (damage to olfactory neuroepithelium, olfactory nerve, or central pathways). Viral-induced anosmia (including COVID-19) arises from inflammation and damage to sustentacular cells supporting olfactory neurons.

Etiology

- Viral upper respiratory infection (including COVID-19)
- Nasal polyps or chronic rhinosinusitis
- Allergic rhinitis
- Head trauma with cribriform plate fracture
- Aging (presbyosmia)
- Smoking or toxic chemical exposure
- Neurodegenerative diseases (Parkinson disease, Alzheimer disease)
- Nasal tumors or sinus neoplasms
- Congenital anosmia (Kallmann syndrome)
- Medication side effects (zinc nasal sprays, some antibiotics)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden anosmia after head injury, anosmia with neurological symptoms, or persistent loss of smell beyond 4 weeks
- Accompanied by other concerning signs

Related Symptoms

- Loss of taste (ageusia)
- Nasal congestion
- Runny nose
- Headache
- Neurological symptoms

Note: Always consider serious underlying conditions when evaluating persistent loss of smell (anosmia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 161

Low Blood Sugar (Hypoglycemia)

161.1 Definition

Hypoglycemia is a condition of abnormally low blood glucose levels, typically below 70 mg/dL (3.9 mmol/L). It is most commonly associated with diabetes treatment but can occur in various other conditions.

161.2 Pathophysiology

The brain relies primarily on glucose for energy and cannot store significant amounts. When blood glucose falls below normal, the body releases counter-regulatory hormones (glucagon, epinephrine, cortisol, growth hormone) to raise glucose levels. The autonomic response causes adrenergic symptoms, while neuroglycopenia causes cognitive dysfunction. If prolonged, severe hypoglycemia can lead to seizures, coma, and brain damage.

161.3 Etiology

- Diabetes medications: insulin, sulfonylureas, meglitinides
- Skipping or delaying meals
- Increased physical activity without adjusting medication
- Excessive alcohol intake (inhibits gluconeogenesis)
- Critical illness: sepsis, liver failure, kidney failure
- Insulinoma (insulin-secreting pancreatic tumor)
- Hormonal deficiencies: adrenal insufficiency, hypopituitarism
- Reactive hypoglycemia (after gastric surgery or in prediabetes)
- Medications: beta-blockers, salicylates, quinolones
- Factitious hypoglycemia (self-administration of insulin)

161.4 Indications for Evaluation

- Autonomic symptoms: sweating, tremor, palpitations, anxiety, hunger
- Neuroglycopenic symptoms: confusion, drowsiness, difficulty speaking, weakness
- Blurred vision or double vision
- Headache
- Irritability or mood changes
- Seizures
- Loss of consciousness or coma
- Whipple triad: low blood glucose, symptoms, relief with glucose

161.5 Related Symptoms

- Diabetes mellitus (type 1 and type 2)
- Fatigue
- Dizziness
- Palpitations
- Sweating
- Seizures
- Coma

161.6 Self-Care/Home Management

- Check blood glucose if possible
- Consume 15 grams of fast-acting glucose (glucose tablets, fruit juice, regular soda, candy)
- Recheck blood glucose after 15 minutes
- Repeat treatment if still below 70 mg/dL
- Eat a protein-containing snack after recovery to prevent recurrence
- If unconscious or unable to swallow, emergency glucagon is needed
- Carry medical identification indicating diabetes

161.7 Diagnostic Approach

- Capillary blood glucose measurement (fingerstick)
- Detailed history of medication timing, meals, and symptoms
- Fasting blood glucose, insulin, and C-peptide levels
- 72-hour fasting test for suspected insulinoma
- Mixed-meal tolerance test for reactive hypoglycemia
- HbA1c to assess overall glycemic control

161.8 Treatment/Management Overview

- Immediate glucose administration (oral for conscious, IV glucose for unconscious)
- Glucagon injection (intramuscular or subcutaneous) for severe hypoglycemia
- Adjust diabetes medications to prevent recurrence
- Treat underlying cause (insulinoma resection, adrenal replacement)
- Education on hypoglycemia recognition and management
- Continuous glucose monitor for high-risk patients

Chapter 162

Low Libido (Female Hypoactive Sexual Desire Disorder)

Definition

Persistent or recurrent absence or deficiency of sexual fantasies, thoughts, and desire for sexual activity, causing marked distress or interpersonal difficulty.

Pathophysiology

Female sexual desire involves a complex interplay of neurological, hormonal, and psychosocial factors. Testosterone (produced by ovaries and adrenals) is the primary hormone driving sexual desire in women; its levels decline with age and fall significantly after menopause. Oestrogen maintains vaginal health and lubrication; its decline causes dyspareunia, which secondarily reduces desire. Dopamine (reward/anticipation) and serotonin (inhibition) balance sexual motivation. Menopausal hormonal shifts tilt this balance toward inhibition, while sleep deprivation, fatigue, and mood changes further suppress desire.

Etiology

- Menopausal testosterone and oestrogen decline
- Psychosocial factors (relationship conflict, stress, body image)
- Depression and anxiety
- Medication side effects (SSRIs, SNRIs, antipsychotics, hormonal contraceptives)
- Vaginal dryness and dyspareunia (fear of pain)
- Fatigue and sleep disturbance
- Hypothyroidism
- Hyperprolactinaemia
- Bilateral oophorectomy (surgical menopause)
- Chronic illness (diabetes, cardiovascular disease, cancer)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Low libido causing personal distress, affecting relationships, or associated with depression, pain during sex, or other menopausal symptoms
- Accompanied by other concerning signs

Related Symptoms

- Vaginal dryness
- Painful sex
- Fatigue
- Depression
- Anxiety

Note: Always consider serious underlying conditions when evaluating persistent low libido.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 163

Low Self-Esteem (Loss of Confidence)

Definition

A negative perception of self-worth and abilities, often accompanied by self-doubt, social withdrawal, and difficulty asserting oneself in personal or professional settings.

Pathophysiology

Self-esteem is influenced by prefrontal cortical regulation of amygdala and limbic system responses to social evaluation. Oestrogen modulates serotonin and dopamine neurotransmission; declining oestrogen reduces serotonergic tone, increasing sensitivity to negative social feedback and self-criticism. Physical symptoms of menopause (hot flushes, weight gain, skin/hair changes) may alter body image and self-perception. Sleep disruption and fatigue impair cognitive function, further undermining confidence in work and social roles.

Etiology

- Perimenopausal and menopausal changes (physical and emotional)
- Major depressive disorder
- Generalised anxiety disorder
- Social anxiety disorder
- Body image disturbance (weight gain, skin/hair changes)
- Chronic illness or pain
- Workplace stress or discrimination
- Relationship difficulties or divorce
- History of trauma or abuse
- Perfectionism and high self-standards

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Low self-esteem with depressed mood, social withdrawal, work impairment, or thoughts of self-harm
- Accompanied by other concerning signs

Related Symptoms

- Depression
- Anxiety
- Mood swings
- Fatigue
- Body image concerns

Note: Always consider serious underlying conditions when evaluating persistent low self-esteem.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 164

Lump Under Armpit (Axillary Mass)

Definition

A palpable swelling or mass in the axillary region, most commonly representing enlarged lymph nodes but also arising from breast tissue, skin lesions, or subcutaneous structures.

Pathophysiology

Axillary lymph nodes drain the upper limb, breast, chest wall, and upper back. Lymphadenopathy results from infection (local or systemic), inflammatory conditions, or malignancy (lymphoma, breast cancer metastasis). Other causes include sebaceous cysts, lipomas, hidradenitis suppurativa, and accessory breast tissue.

Etiology

- Lymphadenopathy (reactive from infection of arm/hand)
- Breast cancer metastasis
- Lymphoma or leukemia
- Sebaceous cyst or epidermoid cyst
- Lipoma (benign fatty tumor)
- Hidradenitis suppurativa (inflamed sweat gland)
- Cat scratch disease (*Bartonella henselae*)
- Tuberculosis or mycobacterial infection
- Sarcoidosis
- Vaccination reaction (recent COVID-19, flu, or other vaccine)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Firm, fixed, non-tender lump, progressive enlargement, or lump with fever, night sweats, or unexplained weight loss
- Accompanied by other concerning signs

Related Symptoms

- Breast lump
- Arm swelling
- Fever
- Night sweats
- Weight loss

Note: Always consider serious underlying conditions when evaluating persistent lump under armpit (axillary mass).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 165

Lymphedema

165.1 Definition

Lymphedema is chronic, progressive swelling of a body part due to impaired lymphatic drainage. It most commonly affects the arms or legs but can involve other areas. Primary lymphedema is congenital or genetic; secondary lymphedema results from damage to the lymphatic system.

165.2 Pathophysiology

The lymphatic system returns interstitial fluid to the bloodstream. When lymph vessels or nodes are damaged or absent, protein-rich fluid accumulates in the interstitial space. This causes swelling, tissue fibrosis, chronic inflammation, and impaired immune function. The accumulated fluid provides a growth medium for bacteria, increasing infection risk.

165.3 Etiology

- Secondary (most common): cancer treatment (lymph node dissection, radiation)
- Breast cancer-related lymphedema (arm swelling after axillary dissection)
- Pelvic cancer treatment (leg swelling)
- Infection: filariasis (tropical, caused by *Wuchereria bancrofti*)
- Infection: recurrent cellulitis or lymphangitis
- Trauma or burns
- Venous insufficiency (chronic, with secondary lymphatic overload)
- Primary (idiopathic): congenital or pubertal onset (Milroy disease, Meige disease)
- Obesity-associated lymphedema

165.4 Indications for Evaluation

- Persistent swelling of an arm or leg
- Swelling that begins distally and progresses proximally
- Non-pitting edema in later stages (fibrosis)
- Heaviness, tightness, or fullness in the affected limb
- Decreased flexibility in the affected joint
- Aching or discomfort
- Skin changes: thickening, hyperkeratosis, papillomatosis
- Positive Stemmer sign (inability to pinch the skin on the dorsum of the second toe)

165.5 Related Symptoms

- Edema (generalized fluid retention)
- Venous insufficiency
- Cellulitis (recurrent infections)
- Lymphangitis
- Elephantiasis (severe filariasis)

165.6 Self-Care/Home Management

- Elevate the affected limb when possible
- Practice good skin hygiene and moisturize daily
- Protect the limb from cuts, burns, and insect bites
- Wear compression garments as prescribed
- Perform light exercise with the affected limb
- Avoid constrictive clothing or jewelry on the affected side
- Avoid prolonged standing or sitting
- Maintain a healthy body weight

165.7 Diagnostic Approach

- Clinical examination: pitting, Stemmer sign, skin changes
- Lymphoscintigraphy (nuclear medicine imaging of lymphatic flow)
- Bioimpedance spectroscopy for early detection
- MRI or CT for detailed anatomy
- Duplex ultrasound to rule out venous causes

165.8 Treatment/Management Overview

- Complete decongestive therapy: manual lymphatic drainage, compression bandaging, exercise, skin care
- Compression garments for maintenance
- Pneumatic compression devices
- Surgical options: lymphovenous anastomosis, vascularized lymph node transfer
- Liposuction for late-stage fibrotic lymphedema
- Antibiotics for cellulitis prophylaxis
- Avoid blood pressure measurements and venipuncture on the affected arm

Chapter 166

Malaise

Definition

General feeling discomfort unease illness often predates specific symptoms nonspecific constitutional symptom.

Pathophysiology

Vague sense that something is wrong not easily patient quantified accompanies infections inflammatory states malignancies autoimmune diseases psychiatric conditions.

Etiology

- Viral/bacterial infections
- Autoimmune disorders
- Chronic kidney/liver disease
- Malignancy
- Psychological stress/depression

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent malaise with weight loss or systemic signs warrants investigation
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Fatigue
- Loss of appetite
- Muscle aches
- Sweating

Note: Always consider serious underlying conditions when evaluating persistent malaise.

Self-Care / Home Management

- Prioritize adequate rest and sleep to support the body's healing and recovery processes.
- Maintain a balanced diet with regular meals and stay well-hydrated.
- Monitor symptoms and track any changes in severity, frequency, or associated features.
- Avoid overexertion and gradually increase activity levels as tolerated.
- Seek medical evaluation if symptoms persist, worsen, or interfere significantly with daily function.

Diagnostic Approach

- Comprehensive history and physical examination guide the diagnostic workup based on the specific symptom presentation.
- Laboratory testing (CBC, metabolic panel, inflammatory markers) helps screen for underlying systemic conditions.
- Imaging studies are tailored to the suspected anatomic region and pathology.
- Referral to appropriate specialists is guided by the differential diagnosis established through initial evaluation.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic management may include pharmacotherapy, lifestyle modifications, and supportive care.
- Multidisciplinary care is often beneficial for complex or chronic symptoms.
- Close follow-up ensures treatment effectiveness and allows adjustment of the management plan as needed.

Chapter 167

Memory Loss

Definition

Impairment in ability to acquire new information or recall previously learned information.

Pathophysiology

Distinguishes between normal age-related forgetting and pathological impairment such as amnesic syndromes dementia or transient global amnesia; episodic memory most commonly affected

Alzheimer's disease (p.170) progressive neurodegenerative disorder characterized by gradual onset memory impairment affecting recent recall first then extends to remote memory accompanied by cognitive decline in language orientation executive function visuospatial skills behavioral changes mood disturbances agitation apathy depression psychosis sundowning syndrome functional independence progressively impaired requiring increasing care support eventual complete dependence end-stage bedbound incontinence nonverbal pathological hallmarks amyloid beta plaques neurofibrillary tangles tau protein neuronal loss cerebral atrophy hippocampal entorhinal cortex medial temporal lobe involvement early stages frontal parietal association areas later widespread cortical atrophy diagnosis clinical evaluation supported neuropsychological testing biomarkers CSF amyloid tau PET imaging APOE genetic risk factor no cure management cholinesterase inhibitors memantine supportive care safety modifications environmental adaptations caregiver support advance directives planning

Vascular dementia stepwise progression correlated with cerebrovascular events strategic infarcts multi-infarct state white matter ischemic small vessel disease subcortical vascular cognitive impairment depression pseudo-differential features from Alzheimer typically earlier gait disturbance urinary incontinence focal neurological signs pseudobulbar affect

Depression (pseudodementia) subjective complaints of memory difficulties often exaggerated apparent slowing of response times on examination concentration effort poor motivation depressive symptoms prominent true objective memory testing reveals intact recall with prompting improves with encouragement treatment responsive reversible unlike organic dementia

Thyroid dysfunction hypothyroidism can cause cognitive slowing memory complaints fatigue apathy weight gain cold dry skin bradycardia deep tendon reflexes delayed relaxation phase reversible with thyroid hormone replacement

Medication side effects anticholinergics benzodiazepines opioids antihistamines anticonvulsants alcohol sedatives hypnotics may impair attention encoding retrieval processes polypharmacy elderly particularly vulnerable deprescribing review medication reconciliation essential component

assessment

Etiology

- Alzheimer disease progressive neurodegeneration memory decline language dysfunction executive impairment behavioral changes eventually complete dependence
- Vascular dementia stepwise cognitive decline associated with cerebrovascular events gait disturbance urinary incontinence focal signs
- Depression pseudodementia subjective memory complaints concentration effort poor motivation improves with treatment realistic recall with prompting
- Thyroid dysfunction hypothyroidism cognitive slowing fatigue weight gain cold intolerance dry skin bradycardia reversible with therapy
- Medication side effects anticholinergics benzodiazepines opioids sedatives hypnotics alcohol especially problematic polypharmacy elderly patients

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than progressive or sudden onset warrants neurological evaluation including imaging and neuropsychological testing
- Accompanied by other concerning signs personality change gait disturbance incontinence focal deficits seizures

Related Symptoms

- Getting lost easily Repetitive questions Difficulty learning new names Misplacing items frequently Confusion about dates/times Executive dysfunction Language problems Visuospatial difficulties Mood changes Behavioral changes Sleep disturbances Appetite changes

Note: Always consider serious underlying conditions when evaluating persistent memory loss.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 168

Menstrual Pain (Dysmenorrhea)

168.1 Definition

Dysmenorrhea is painful menstruation that affects 50-90 percent of menstruating women. Primary dysmenorrhea is menstrual pain without underlying pathology. Secondary dysmenorrhea is caused by an identifiable condition such as endometriosis or fibroids.

168.2 Pathophysiology

Primary dysmenorrhea results from prostaglandin F₂-alpha release from the endometrium during menstruation, causing intense uterine contractions, ischemia, and pain. Secondary dysmenorrhea involves underlying pathology: endometriosis (ectopic endometrial tissue), adenomyosis (endometrial tissue in myometrium), uterine fibroids, or pelvic inflammatory disease.

168.3 Etiology

- Primary dysmenorrhea: prostaglandin-mediated uterine cramping
- Endometriosis (ectopic endometrial implants)
- Adenomyosis (endometrial tissue within the uterine wall)
- Uterine fibroids (leiomyomas)
- Pelvic inflammatory disease
- Ovarian cysts
- Cervical stenosis
- Intrauterine device (IUD) use
- Psychological factors (stress, anxiety)

168.4 Indications for Evaluation

- Cramping pain in the lower abdomen just before or during menstruation
- Pain radiating to the lower back or thighs
- Nausea, vomiting, or diarrhea during menstruation
- Headache or fatigue
- Pain that improves with age or after childbirth (primary)
- Pain that worsens over time (secondary)
- Heavy menstrual bleeding (menorrhagia)
- Pain with intercourse (dyspareunia)

168.5 Related Symptoms

- Heavy menstrual bleeding (menorrhagia)
- Irregular menstruation
- Premenstrual syndrome (PMS)
- Endometriosis
- Uterine fibroids
- Pelvic pain
- Back pain

168.6 Self-Care/Home Management

- Apply a heating pad or warm bath for relief
- Take OTC NSAIDs (ibuprofen, naproxen) starting 1-2 days before menstruation
- Exercise regularly (releases endorphins)
- Practice relaxation techniques and stress management
- Reduce salt, caffeine, and sugar intake before menses
- Try dietary supplements (magnesium, vitamin B1, omega-3 fatty acids)
- Rest when needed

168.7 Diagnostic Approach

- Clinical history: timing, character, and severity of pain
- Pelvic examination
- Pelvic ultrasound to rule out secondary causes
- Laparoscopy for definitive diagnosis of endometriosis
- MRI for adenomyosis in selected cases

168.8 Treatment/Management Overview

- NSAIDs as first-line therapy
- Combined oral contraceptive pills (suppress ovulation)
- Progestin-only contraceptives (pills, IUD, implant)
- GnRH agonists for endometriosis
- Surgery: endometriosis excision, myomectomy, hysterectomy
- Complementary therapies: acupuncture, transcutaneous electrical nerve stimulation
- Cognitive behavioral therapy for chronic pain

Chapter 169

Migraine

Definition

A primary headache disorder characterised by recurrent attacks of moderate to severe unilateral pulsating headache lasting 4-72 hours, often accompanied by nausea, photophobia, phonophobia, and sometimes aura.

Pathophysiology

Migraine involves cortical spreading depression (CSD) activating the trigeminovascular system, releasing calcitonin gene-related peptide (CGRP) and substance P, causing neurogenic inflammation and vasodilation of meningeal blood vessels. Oestrogen modulates trigeminal nerve sensitivity and CGRP release; the natural perimenopausal oestrogen decline and fluctuation can destabilise migraine patterns. The classic oestrogen withdrawal trigger occurs during the late luteal phase when oestrogen falls sharply. Perimenopause typically worsens migraine frequency in women with menstrual-related migraine, while post-menopause may improve it in some.

Etiology

- Oestrogen fluctuation or withdrawal (perimenopause, menstrual cycle)
- Genetic predisposition (family history of migraine)
- Stress and anxiety
- Sleep disturbance (insufficient or excessive sleep)
- Dietary triggers (aged cheese, chocolate, alcohol, caffeine)
- Weather changes (barometric pressure, humidity)
- Hormonal medications (combined oral contraceptives, HRT)
- Dehydration or skipped meals
- Sensory triggers (bright lights, strong odours, loud noises)
- Medication overuse (analgesic rebound headache)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Migraine with aura lasting more than 60 minutes, new-onset headache after age 50, worst headache of life, or headache with fever, stiff neck, or neurological deficits
- Accompanied by other concerning signs

Related Symptoms

- Headache
- Nausea
- Vision blurred
- Photophobia
- Dizziness

Note: Always consider serious underlying conditions when evaluating persistent migraine.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 170

Mittelschmerz (Ovulation Pain)

170.1 Definition

Mittelschmerz is mid-cycle abdominal pain that occurs at the time of ovulation, approximately 14 days before the next menstrual period. It is a normal physiologic phenomenon that is typically mild and self-limiting.

170.2 Pathophysiology

At ovulation, the mature ovarian follicle ruptures to release the egg, accompanied by a small amount of blood and follicular fluid. This fluid irritates the peritoneum, causing localized lower abdominal pain. The pain typically alternates sides each month depending on which ovary ovulates.

170.3 Etiology

- Normal ovulation (physiologic)
- Ovarian follicle rupture releasing fluid and blood
- Peritoneal irritation from follicular contents

170.4 Indications for Evaluation

- Lower abdominal pain midway through the menstrual cycle
- Pain typically on one side (alternating each cycle)
- Dull, cramping, or sharp pain lasting minutes to hours
- May be accompanied by light spotting
- No other symptoms of infection or pathology
- Pain resolves within 24-48 hours

170.5 Related Symptoms

- Menstrual cramps (dysmenorrhea)
- Ovarian cyst rupture (more severe pain)
- Appendicitis (right-sided pain, must be distinguished)
- Ectopic pregnancy (must be ruled out if sexually active)

170.6 Self-Care/Home Management

- Apply a heating pad to the lower abdomen
- Take OTC pain relievers (ibuprofen, acetaminophen)
- Rest until the pain subsides
- Track ovulation and mittelschmerz on a calendar
- Pain relief with warm baths
- Practice relaxation breathing

170.7 Diagnostic Approach

- Clinical diagnosis based on timing and character of pain
- Consistent relationship to menstrual cycle
- Pelvic examination is normal
- Ultrasound if pain is severe or atypical to rule out ovarian cyst or other pathology
- Urine pregnancy test to rule out ectopic pregnancy

170.8 Treatment/Management Overview

- Reassurance that mittelschmerz is a normal, benign condition
- NSAIDs or acetaminophen for pain
- Combined oral contraceptives suppress ovulation and eliminate mittelschmerz
- Heating pad and rest

Chapter 171

Mood Swings

Definition

Rapid or extreme fluctuations in emotional state, shifting between euphoria, irritability, sadness, or anger over short periods without clear precipitants.

Pathophysiology

Mood regulation involves complex interplay between neurotransmitters (serotonin, dopamine, norepinephrine), hormones (estrogen, progesterone, cortisol), and neural circuits (prefrontal cortex, amygdala, hippocampus). Disruption from hormonal changes, stress, sleep deprivation, or psychiatric conditions produces emotional lability.

Etiology

- Premenstrual syndrome (PMS) or PMDD
- Perimenopause and menopause
- Pregnancy and postpartum period
- Bipolar disorder
- Depression or anxiety disorders
- Chronic stress
- Sleep deprivation
- Substance use or withdrawal
- Thyroid disorders (hyperthyroidism)
- Medication side effects (corticosteroids, antidepressants)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Mood swings with thoughts of self-harm, significant functional impairment, or accompanying severe depression or mania
- Accompanied by other concerning signs

Related Symptoms

- Irritability
- Fatigue
- Sleep changes
- Anxiety
- Appetite changes

Note: Always consider serious underlying conditions when evaluating persistent mood swings.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 172

Morning Sickness (Nausea Gravidarum)

Definition

Nausea and vomiting experienced during pregnancy, typically beginning around week 6, peaking around week 9, and resolving by week 14-16, though symptoms may begin earlier or persist longer.

Pathophysiology

Rising human chorionic gonadotropin (hCG) and estrogen levels during early pregnancy stimulate the brainstem vomiting center. Other contributing factors include enhanced olfactory sensitivity, delayed gastric emptying, and gastroesophageal reflux. Psychological factors and prior history of motion sickness may increase susceptibility.

Etiology

- Normal pregnancy-related hormonal changes
- Elevated hCG levels (more severe in multiple pregnancies or molar pregnancy)
- Estrogen surge in first trimester
- Enhanced sense of smell
- Delayed gastric emptying
- Gastroesophageal reflux
- Multiparous pregnancy history
- Personal or family history of hyperemesis gravidarum
- Migraine susceptibility
- Psychological factors (stress, anxiety)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe vomiting causing dehydration (inability to keep fluids down), weight loss, or ketosis (hyperemesis gravidarum)
- Accompanied by other concerning signs

Related Symptoms

- Fatigue
- Food aversions
- Weight loss
- Dehydration
- Dizziness

Note: Always consider serious underlying conditions when evaluating persistent morning sickness (nausea gravidarum).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 173

Motion Sickness

Definition

A syndrome of nausea, dizziness, and autonomic symptoms triggered by real or perceived motion, arising from sensory conflict between visual, vestibular, and proprioceptive inputs.

Pathophysiology

The sensory conflict theory posits motion sickness occurs when information from the vestibular system (detecting motion) does not match visual and proprioceptive inputs. Autonomic activation via the brainstem vomiting center leads to nausea, pallor, cold sweats, and vomiting. Central habituation reduces symptoms over repeated exposure.

Etiology

- Car travel (especially winding roads)
- Boat or sea travel (sea sickness)
- Airplane travel
- Virtual reality or video simulations
- Amusement park rides
- Migraine-associated vertigo
- Inner ear disorders (Meniere disease, labyrinthitis)
- Pregnancy (first trimester)
- Migraine headache
- Panic disorder or anxiety

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Motion sickness with severe or persistent vomiting causing dehydration, or with neurological symptoms
- Accompanied by other concerning signs

Related Symptoms

- Nausea
- Dizziness
- Sweating
- Headache
- Pallor

Note: Always consider serious underlying conditions when evaluating persistent motion sickness.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 174

Mouth Ulcers (Aphthous Ulcers)

Definition

Painful, shallow, round or oval sores on the oral mucosa, typically covered by a yellow-gray fibrinous membrane and surrounded by an erythematous halo, recurring at intervals (minor, major, or herpetiform types).

Pathophysiology

Aphthous ulcers involve a localized immune-mediated inflammatory response targeting the oral epithelium. T-cell activation, tumor necrosis factor-alpha (TNF-alpha), and other cytokines cause epithelial breakdown. Trigger factors include minor trauma, stress, nutritional deficiencies, food sensitivities, and hormonal changes. Genetic predisposition is strong with positive family history in about 40% of cases.

Etiology

- Minor aphthous ulcers (most common, recur every few months)
- Major aphthous ulcers (larger, deeper, heal with scarring)
- Herpetiform ulcers (multiple clustered small ulcers)
- Trauma (biting cheek, sharp tooth, dental appliances, burns)
- Nutritional deficiencies (iron, folate, vitamin B12, zinc)
- Stress and anxiety (common trigger)
- Food sensitivities (chocolate, coffee, strawberries, nuts, cheese, tomatoes)
- Hormonal changes (premenstrual period)
- Autoimmune conditions (Crohn disease, Behcet disease, celiac disease)
- Immunosuppression (HIV, chemotherapy)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Ulcers lasting more than 3 weeks, unusually large or painful ulcers, ulcers accompanied by systemic symptoms (fever, weight loss, rash, joint pain), or difficulty eating/drinking
- Accompanied by other concerning signs

Related Symptoms

- Oral pain
- Difficulty swallowing
- Bad breath
- Swollen gums
- Sore throat

Note: Always consider serious underlying conditions when evaluating persistent mouth ulcers (aphthous ulcers).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 175

Muscle Cramps

Definition

Sudden involuntary sustained contraction of muscle causing severe pain temporary incapacity; soreness persists after cramp ceases more likely tired muscles at night.

Pathophysiology

Acute benign self-limiting differentiates from sustained spasm or tetany; dehydration electrolyte imbalance overuse pregnancy medication side effects common triggers; nocturnal cramps very common in elderly.

Etiology

- Dehydration electrolyte loss sweating sports activity
- Muscle overuse exercise fatigue prolonged positioning
- Pregnancy hormonal changes fluid retention
- Medication side effects statins diuretics
- Peripheral artery disease circulation issues

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent beyond relief frequent recurrence unexplained absence trauma requires medical evaluation
- Accompanied by other concerning signs

Related Symptoms

- Muscle soreness after episode
- Localized swelling tenderness
- Visible muscle knotting
- Difficulty walking immediately after
- Recurrent pattern in same muscle

Note: Always consider serious underlying conditions when evaluating persistent muscle cramps.

Self-Care / Home Management

- Apply the RICE protocol: Rest, Ice (15-20 min every 2-3 hours), Compression with elastic bandage, Elevation above heart level.
- Gradual return to activity with gentle stretching once acute pain subsides.
- Over-the-counter anti-inflammatory medications (ibuprofen, naproxen) can reduce pain and swelling.
- Avoid activities that worsen symptoms until healing is well underway.
- Seek evaluation for severe pain, inability to bear weight, joint deformity, or symptoms lasting beyond 2 weeks.

Diagnostic Approach

- History focuses on mechanism of injury, onset (acute vs insidious), location, aggravating/relieving factors, and prior episodes.
- Physical examination includes inspection (swelling, bruising, deformity), palpation, range-of-motion, and strength testing.
- Imaging (X-ray for fracture, ultrasound for tendon/ligament injury, MRI for soft tissue detail) guides diagnosis.
- Functional assessment evaluates ability to perform daily activities and sports-specific movements.

Treatment / Management Overview

- Acute injuries benefit from RICE, activity modification, and gradual rehabilitation exercises.
- Physical therapy is central to recovery, focusing on range-of-motion, strengthening, and proprioceptive training.
- Corticosteroid injections may be considered for significant inflammatory conditions (tendinitis, bursitis).

- Orthopedic referral is indicated for complete tears, fractures, joint instability, or failure of conservative therapy for 6-8 weeks.

Chapter 176

Muscle Pain

Definition

Pain discomfort or tenderness in skeletal muscles often described aching soreness cramping or stiffness.

Pathophysiology

Can be acute from exertion injury infection or chronic from inflammatory autoimmune medication metabolic conditions; myalgia differs from myositis (muscle inflammation).

Etiology

- Viral infections (influenza COVID-19)
- Overuse/exercise
- Statin medications
- Polymyalgia rheumatica morning stiffness shoulder hips elderly
- Electrolyte abnormalities (hypokalemia hypocalcemia)
- Fibromyalgia

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than severe unexplained or worsening muscle pain warrants evaluation
- Accompanied by other concerning signs

Related Symptoms

- Muscle weakness
- Fatigue
- Stiffness
- Swelling at site
- Feeling of muscle nodules

Note: Always consider serious underlying conditions when evaluating persistent muscle pain.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.
- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 177

Nail Injuries

177.1 Definition

Nail injuries include a range of traumas to the fingernails or toenails, such as subungual hematoma (blood under the nail), nail avulsion (partial or complete detachment), nail bed laceration, and fractures of the fingertip involving the nail. These are common hand injuries seen in emergency departments.

177.2 Pathophysiology

The nail is produced by the nail matrix, located under the proximal nail fold. Trauma to the nail complex can cause bleeding under the nail (subungual hematoma), tearing of the nail bed, or separation of the nail plate from the nail bed. Injury to the nail matrix can lead to permanent nail deformity.

177.3 Etiology

- Crush injury: closing a door on a finger, dropping a heavy object
- Blunt trauma: hitting the finger with a hammer
- Catching or tearing the nail on an object
- Sports injury: jamming a finger
- Stubbing a toe
- Repetitive trauma: running or hiking (subungual hematoma)
- Fingertip amputation (severe lacerations)

177.4 Indications for Evaluation

- Pain and throbbing at the nail site
- Visible blood under the nail (subungual hematoma)

- Partial or complete detachment of the nail
- Laceration or irregularity of the nail bed
- Swelling and tenderness of the fingertip
- Blood accumulating under the nail causing pressure
- Fracture of the distal phalanx

177.5 Related Symptoms

- Subungual hematoma (blood under nail)
- Nail avulsion
- Fingertip injury
- Finger fracture
- Nail bed laceration
- Paronychia (infection after injury)

177.6 Self-Care/Home Management

- For subungual hematoma: apply ice and elevate
- Wash minor cuts with soap and water
- Apply antibiotic ointment and a bandage
- Take OTC pain relievers
- Keep the nail clean and dry
- Do not attempt to drain the hematoma at home
- Protect the nail from further injury

177.7 Diagnostic Approach

- Clinical examination of the nail and fingertip
- X-ray to rule out distal phalanx fracture
- Assessment of nail bed involvement for repair decisions

177.8 Treatment/Management Overview

- Subungual hematoma: trephination (drainage through the nail)
- Nail bed laceration: repair with fine absorbable sutures
- Nail avulsion: replace the nail under the nail fold
- Splint the fingertip if associated fracture
- Tetanus prophylaxis if needed
- Antibiotics for open fractures or dirty wounds

Chapter 178

Nail Psoriasis

178.1 Definition

Nail psoriasis is a manifestation of psoriasis affecting the fingernails or toenails. It occurs in up to 50 percent of people with cutaneous psoriasis and can occur in isolation. Nail psoriasis significantly impacts quality of life and is often difficult to treat.

178.2 Pathophysiology

Psoriasis involves immune-mediated inflammation driven by T-cells. In the nail, inflammation affects the nail matrix and nail bed. Matrix involvement causes pitting, leukonychia, and crumbling. Nail bed involvement causes onycholysis (separation from nail bed), oil-drop discoloration, splinter hemorrhages, and subungual hyperkeratosis.

178.3 Etiology

- Psoriasis vulgaris (chronic plaque psoriasis)
- Psoriatic arthritis (nail changes are more common in psoriatic arthritis)
- Genetic predisposition
- Trauma (Koebner phenomenon: psoriasis develops at sites of injury)
- Infection (streptococcal infection can trigger flares)
- Stress
- Medications (lithium, beta-blockers, antimalarials)

178.4 Indications for Evaluation

- Small pits or depressions in the nail surface
- Thickening and crumbling of the nail
- Yellowish-brown oil-drop discoloration under the nail

- Separation of the nail from the nail bed (onycholysis)
- White patches or streaks on the nail
- Subungual hyperkeratosis (debris under the nail)
- Redness and swelling of the nail fold
- Pain or discomfort in the nail area

178.5 Related Symptoms

- Psoriasis (skin plaques)
- Psoriatic arthritis
- Fungal nail infection (may coexist)
- Paronychia (inflammation of nail fold)
- Eczema of the hands

178.6 Self-Care/Home Management

- Keep nails short and trimmed
- Avoid trauma to the nails
- Wear gloves for household work and gardening
- Moisturize hands and nails regularly
- Use nail polish to cosmetically conceal changes
- Avoid picking or manipulating the nails
- Gently clean under the nails

178.7 Diagnostic Approach

- Clinical examination of the nails
- Examination of skin for psoriatic plaques
- Dermoscopy of the nail
- Nail clipping or biopsy if diagnosis is uncertain
- Fungal culture to rule out coexistent onychomycosis

178.8 Treatment/Management Overview

- Topical corticosteroids with occlusion
- Calcipotriol (vitamin D analogue)
- Topical calcineurin inhibitors (tacrolimus)
- Intralesional corticosteroid injection
- Systemic therapy: methotrexate, cyclosporine, apremilast, biologics (adalimumab, secukinumab, ixekizumab)
- Excimer laser or PUVA phototherapy
- Treatment of associated skin and joint psoriasis

Chapter 179

Narcolepsy

179.1 Definition

Narcolepsy is a chronic neurological disorder characterized by excessive daytime sleepiness and sudden, uncontrollable episodes of falling asleep. It is caused by loss of hypocretin (orexin) neurons in the hypothalamus.

179.2 Pathophysiology

Narcolepsy type 1 results from autoimmune destruction of hypocretin-producing neurons in the lateral hypothalamus. Hypocretin is a neurotransmitter that promotes wakefulness and stabilizes sleep-wake transitions. Without hypocretin, patients experience rapid transitions into REM sleep, causing cataplexy (sudden muscle weakness triggered by emotions), sleep paralysis, and hypnagogic hallucinations.

179.3 Etiology

- Autoimmune destruction of hypocretin neurons (type 1 narcolepsy)
- Genetic predisposition (HLA-DQB1*0602 allele)
- Environmental triggers: influenza A (H1N1) infection or vaccination
- Traumatic brain injury affecting the hypothalamus
- Brain tumors or strokes involving the hypothalamus
- Multiple sclerosis affecting the hypothalamus
- Type 2 narcolepsy (idiopathic, without cataplexy)

179.4 Indications for Evaluation

- Excessive daytime sleepiness with irresistible sleep attacks
- Cataplexy (sudden loss of muscle tone triggered by laughter, anger, or surprise)

- Sleep paralysis (inability to move upon falling asleep or waking)
- Hypnagogic hallucinations (vivid, dream-like experiences at sleep onset)
- Fragmented nighttime sleep
- Automatic behaviors (performing tasks without awareness)
- Weight gain and obesity

179.5 Related Symptoms

- Excessive daytime sleepiness
- Fatigue
- Insomnia
- Sleep apnea
- Depression
- Hallucinations (hypnagogic)
- Obesity

179.6 Self-Care/Home Management

- Schedule short (15-20 minute) naps at regular times
- Maintain a consistent sleep schedule
- Avoid alcohol and caffeine close to bedtime
- Exercise daily (but not near bedtime)
- Inform employers, teachers, and family about the condition
- Avoid driving or operating heavy machinery if untreated
- Join a support group

179.7 Diagnostic Approach

- Polysomnography (overnight sleep study)
- Multiple Sleep Latency Test (MSLT) to measure sleep propensity
- Mean sleep latency less than 8 minutes and 2 or more sleep-onset REM periods
- HLA typing (presence of DQB1*0602)
- Measurement of CSF hypocretin-1 levels (less than 110 pg/mL in type 1)

179.8 Treatment/Management Overview

- Stimulants: modafinil, armodafinil (first-line for daytime sleepiness)
- Sodium oxybate (gamma-hydroxybutyrate) for cataplexy and fragmented sleep
- Antidepressants: venlafaxine, fluoxetine for cataplexy
- Pitolisant (histamine H3 receptor inverse agonist)
- Solriamfetol (dopamine/norepinephrine reuptake inhibitor)
- Scheduled naps and good sleep hygiene

Chapter 180

Nasal Congestion (Blocked Nose)

Definition

Obstruction of nasal airflow due to inflammation, swelling of nasal turbinates, mucus accumulation, or structural abnormalities, causing difficulty breathing through the nose.

Pathophysiology

Nasal congestion results from vasodilation and engorgement of venous sinusoids in the nasal turbinates triggered by inflammation (allergic, infectious), autonomic dysregulation, or medication effects. Mucosal edema narrows nasal passages while increased secretions further obstruct airflow. Chronic causes include structural abnormalities (deviated septum, nasal polyps) or chronic rhinosinusitis.

Etiology

- Common cold (viral upper respiratory infection)
- Allergic rhinitis (hay fever)
- Sinusitis (acute or chronic)
- Deviated nasal septum
- Nasal polyps
- Vasomotor rhinitis
- Pregnancy rhinitis
- Rhinitis medicamentosa (decongestant spray overuse)
- Environmental irritants (smoke, pollution, dry air)
- Enlarged adenoids or tonsils (in children)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Nasal congestion with high fever, facial swelling, visual changes, or unilateral congestion with bloody discharge
- Accompanied by other concerning signs

Related Symptoms

- Runny nose
- Sneezing
- Sinus pressure
- Headache
- Reduced sense of smell

Note: Always consider serious underlying conditions when evaluating persistent nasal congestion (blocked nose).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 181

Nausea

Definition

Unease stomach vomit often precedes vomiting alone emesis.

Pathophysiology

Brainstem mediated reflex triggered GI irritation vestibular stimulation increased intracranial pressure metabolic disturbances psychogenic factors.

Etiology

- Viral gastroenteritis intoxication
- Pregnancy morning sickness
- Medication side effects opioids chemotherapy
- Gastric outlet obstruction migration
- Migraine headache associated

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Prolonged unable retain fluids dehydration blood vomit high fever suspected obstruction
- Accompanied by other concerning signs

Related Symptoms

- Abdominal cramps loss appetite dehydration vomiting headache

Note: Always consider serious underlying conditions when evaluating persistent nausea.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 182

Neck Pain

Definition

Pain or discomfort localized to the cervical spine region, from the base of the skull to the upper shoulders, arising from musculoskeletal, neurological, or referred sources.

Pathophysiology

Neck pain most commonly results from mechanical dysfunction: muscle strain, ligamentous sprain, or facet joint irritation. Degenerative changes (cervical spondylosis) involve disc desiccation, osteophyte formation, and facet hypertrophy, which may lead to foraminal stenosis and nerve root compression (cervical radiculopathy). Myelopathy from spinal cord compression is a serious but less common cause.

Etiology

- Mechanical neck pain (muscle strain, poor posture)
- Cervical spondylosis (degenerative disc disease)
- Cervical radiculopathy (herniated disc, foraminal stenosis)
- Whiplash injury (motor vehicle accident)
- Tension-type headache (referred to neck)
- Meningitis (with fever, photophobia, Kernig sign)
- Cervical myelopathy (spinal cord compression)
- Fibromyalgia or myofascial pain syndrome
- Rheumatoid arthritis (atlantoaxial instability)
- Vertebral artery dissection (rare, with neurological symptoms)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Neck pain with fever and stiff neck (possible meningitis), neck pain after trauma, or neck pain with arm weakness/numbness
- Accompanied by other concerning signs

Related Symptoms

- Headache
- Shoulder pain
- Arm pain or numbness
- Stiffness
- Limited range of motion

Note: Always consider serious underlying conditions when evaluating persistent neck pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 183

Nerve Pain (Neuropathic Pain)

Definition

Pain caused by damage or dysfunction of the somatosensory nervous system, characterized by burning, shooting, electric shock-like, or stabbing sensations.

Pathophysiology

Nerve injury leads to abnormal ectopic discharge from damaged axons, altered ion channel expression, central sensitization in the spinal cord dorsal horn, and reduced descending inhibitory control. Neuroplastic changes in the brain amplify pain perception, making neuropathic pain often chronic and difficult to treat.

Etiology

- Diabetic peripheral neuropathy
- Postherpetic neuralgia (shingles)
- Trigeminal neuralgia
- Sciatica or lumbar radiculopathy
- Carpal tunnel syndrome
- Multiple sclerosis
- Chemotherapy-induced peripheral neuropathy
- HIV-associated neuropathy
- Alcohol-related neuropathy
- Complex regional pain syndrome (CRPS)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- New or progressive burning or shooting pain, pain with weakness, numbness, or changes in bladder/bowel function
- Accompanied by other concerning signs

Related Symptoms

- Numbness
- Tingling
- Burning sensation
- Weakness
- Allodynia (pain to light touch)

Note: Always consider serious underlying conditions when evaluating persistent nerve pain (neuropathic pain).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 184

Night Sweats

Definition

Nocturnal sweating refers to excessive perspiration during sleep saturating bedclothes, representing abnormal thermoregulatory response indicating underlying physiological or pathological processes.

Pathophysiology

Night sweats result from hypothalamic thermoregulatory dysregulation during sleep when autonomic control predominates. Increased sympathetic outflow triggers eccrine gland activation. Malignancy-related sweats involve cytokine effects (TNF-*alpha*, IL-1) altering set-point. Hormonal fluctuations decrease thermoneutral zone causing vasomotor instability.

Etiology

- Lymphoma (Pel-Ebstein syndrome alternating fevers/night sweats)
- Tuberculosis (evening fevers and night sweats)
- HIV/AIDS
- Hyperthyroidism (Graves disease)
- Menopause/vasomotor symptoms
- Antidepressants (SSRIs, SNRIs, TCAs)

Indications for Evaluation

Seek care if:

- Drenching night sweats requiring bedclothes change
- Unexplained weight loss with night sweats
- Fever accompanying night sweats
- Night sweats in malignancy patient
- Night sweats with palpitations/tremor

Related Symptoms

- Weight loss
- Fever
- Fatigue
- Malaise
- Lymphadenopathy
- Hot flashes
- Anxiety
- Cough

Note: Night sweats with unexplained weight loss and fever constitute red flags for lymphoma or tuberculosis. Investigations include CBC, ESR/CRP, thyroid function, HIV test, chest radiograph.

Self-Care / Home Management

- Rest and maintain adequate hydration with water, clear fluids, or oral rehydration solutions.
- Use acetaminophen or ibuprofen to reduce fever and body aches (follow label dosing instructions).
- Dress lightly and keep the room at a comfortable temperature — avoid bundling excessively.
- Monitor temperature at regular intervals and track other symptoms to identify worsening trends.
- Seek medical attention for fever above 40 extdegreeC (104 extdegreeF), fever lasting more than 3 days, or fever with stiff neck, severe headache, rash, or difficulty breathing.

Diagnostic Approach

- History includes fever duration, pattern (intermittent, remittent, continuous), associated symptoms (chills, rigors, sweats), and exposure history.
- Vital signs assess fever severity and hemodynamic stability; complete physical examination seeks a source.
- Laboratory evaluation includes CBC with differential, blood cultures, urinalysis, inflammatory markers (CRP, ESR), and targeted testing based on exposure history.
- Imaging (chest X-ray, abdominal ultrasound, CT) is guided by clinical findings and suspected source.

Treatment / Management Overview

- Antipyretics (acetaminophen, NSAIDs) provide symptomatic relief but do not treat the underlying cause.
- Empiric antibiotics may be indicated for suspected bacterial infection after appropriate cultures are obtained.
- Antiviral, antifungal, or antimalarial therapy is directed at specific pathogens based on exposure and testing.
- Source control (drainage of abscess, removal of infected device) is essential for focal infections.

Chapter 185

Night Terrors (Sleep Terrors)

185.1 Definition

Night terrors are a parasomnia disorder characterized by abrupt arousal from deep non-REM sleep with intense fear, screaming, and autonomic activation. The person is inconsolable and has no memory of the event upon waking. Night terrors most commonly affect children aged 3-7 years.

185.2 Pathophysiology

Night terrors occur during arousal from slow-wave (stage N3) sleep, typically in the first third of the night. There is a partial arousal from deep sleep with activation of the sympathetic nervous system (tachycardia, tachypnea, sweating, mydriasis). The brain remains in a mixed sleep-wake state, with the person unable to fully awaken. Genetic factors play a role.

185.3 Etiology

- Inadequate sleep or sleep deprivation
- Fever or illness
- Stress or anxiety
- Irregular sleep schedule
- Sleeping in an unfamiliar environment
- Bladder distension
- Genetics: family history of parasomnias
- Underlying sleep disorders (obstructive sleep apnea, restless legs syndrome)
- Febrile illness
- Medications (sedatives, hypnotics, psychostimulants)

185.4 Indications for Evaluation

- Sudden sitting up in bed with a frightened expression
- Loud screaming or crying during sleep
- Intense autonomic arousal: rapid heart rate, sweating, dilated pupils
- Inconsolable during the episode
- No response to comforting or verbal communication
- Episode lasts 1-10 minutes
- Returns to sleep spontaneously without full awakening
- Complete amnesia for the event in the morning

185.5 Related Symptoms

- Sleepwalking (somnambulism)
- Sleep talking
- Confusional arousals
- Nightmare disorder
- Sleep apnea
- Restless legs syndrome

185.6 Self-Care/Home Management

- Ensure adequate sleep (children need 10-12 hours)
- Maintain a consistent bedtime routine
- Create a calm, relaxing sleep environment
- Do not wake the child during a night terror (prolongs the episode)
- Gently guide the child back to bed and stay nearby
- Protect the child from injury (lock doors, remove obstacles)
- Reduce stress and screen time before bed

185.7 Diagnostic Approach

- Clinical history from parents or bed partner
- Polysomnography for atypical, frequent, or injurious episodes
- Differentiate from nightmares (which occur in REM sleep and are recalled)
- EEG to rule out nocturnal seizures

185.8 Treatment/Management Overview

- Reassurance and education for parents (spontaneous resolution is common)
- Address underlying sleep deprivation and schedule irregularity
- Treat coexisting sleep disorders
- Scheduled awakening (waking the child 15-30 minutes before typical episode time)
- Benzodiazepines (clonazepam) or tricyclic antidepressants for severe, frequent episodes

Chapter 186

Nipple Discharge

Definition

Any fluid that escapes from the nipple, varying in character (milky, serous, purulent, bloody, green/black) and significance from benign physiological discharge to a marker of underlying breast pathology including infection, duct ectasia, papilloma, or malignancy.

Pathophysiology

Milk production is regulated by prolactin from the anterior pituitary, normally suppressed by dopamine from the hypothalamus. Pathologic discharge arises from spontaneous activation of breast epithelial cells (hyperprolactinemia), ductal inflammation (duct ectasia, periductal mastitis), intraductal growth (papilloma), or malignant cells eroding into ducts. Unilateral, spontaneous, bloody discharge from a single duct raises highest concern for malignancy.

Etiology

- Physiological discharge (bilateral, milky, expressed only, common in reproductive-age women)
- Intraductal papilloma (most common cause of bloody unilateral discharge)
- Duct ectasia (benign dilation of mammary ducts, thick green/black discharge)
- Mastitis or breast abscess (purulent discharge with pain and erythema)
- Hyperprolactinemia (pituitary adenoma, medications, hypothyroidism)
- Breast cancer (invasive or DCIS, often unilateral spontaneous bloody discharge)
- Pregnancy and lactation (colostrum, milk)
- Galactorrhea (medication-induced: antipsychotics, metoclopramide, SSRIs, oral contraceptives)
- Breast trauma or surgery
- Fibrocystic breast changes

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Spontaneous, unilateral, bloody, or clear watery discharge; discharge associated with a palpable breast lump; or persistent discharge requiring pad use
- Accompanied by other concerning signs

Related Symptoms

- Breast lump
- Breast pain
- Breast swelling
- Galactorrhea
- Nipple changes

Note: Always consider serious underlying conditions when evaluating persistent nipple discharge.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 187

Nosebleed (Epistaxis)

Definition

Bleeding from the nasal cavity, arising from the anterior septum (Kiesselbach plexus, 90

Pathophysiology

Nasal mucosal vessels are superficial and fragile; anterior epistaxis from Little's area (Kiesselbach plexus) where several arteries anastomose. Posterior epistaxis arises from Woodruff plexus in the inferior meatus. Bleeding may be spontaneous or triggered by trauma, dryness, inflammation, or systemic coagulopathy.

Etiology

- Nasal trauma or picking
- Dry air (low humidity, heating, air conditioning)
- Allergic rhinitis or sinusitis
- Anticoagulant medications (warfarin, aspirin, clopidogrel)
- Nasal tumors or polyps
- Coagulation disorders (hemophilia, von Willebrand disease)
- Hypertension (especially posterior bleeds)
- Pregnancy (hormonal vascular changes)
- Cocaine use or intranasal medications
- Hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Bleeding persisting more than 20 minutes despite direct pressure, recurrent episodes, heavy blood loss, or bleeding on anticoagulation therapy
- Accompanied by other concerning signs

Related Symptoms

- Nasal congestion
- Headache
- Dizziness
- Pale skin (anemia)
- Easy bruising

Note: Always consider serious underlying conditions when evaluating persistent nosebleed (epistaxis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 188

Nosebleeds (Epistaxis)

188.1 Definition

Nosebleeds (epistaxis) are bleeding from the nasal cavity. They are very common, with approximately 60 percent of people experiencing at least one episode in their lifetime. Most nosebleeds are anterior (from Kiesselbach plexus) and self-limiting.

188.2 Pathophysiology

The nasal mucosa has a rich blood supply. Anterior epistaxis (90 percent of cases) arises from the Kiesselbach plexus (Little area) on the anterior nasal septum, where several arteries anastomose. Posterior epistaxis arises from the sphenopalatine artery and is more common in older adults, more severe, and harder to control.

188.3 Etiology

- Nasal trauma: picking the nose, blowing too hard, foreign body
- Dry air (low humidity, winter heating, air conditioning)
- Nasal allergies and sinusitis
- Upper respiratory infections
- Nasal sprays (corticosteroids, decongestants with overuse)
- Coagulation disorders (hemophilia, von Willebrand disease, thrombocytopenia)
- Anticoagulant or antiplatelet medications (warfarin, aspirin, clopidogrel)
- Liver disease (impaired coagulation factor synthesis)
- Hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome)
- Nasal tumors (benign or malignant)

188.4 Indications for Evaluation

- Active bleeding from one or both nostrils
- Blood dripping down the back of the throat (posterior nosebleed)
- Frequent or recurrent episodes
- Bleeding that lasts more than 20 minutes
- Heavy blood loss causing dizziness or lightheadedness
- Bleeding associated with easy bruising or bleeding from other sites
- History of anticoagulant use

188.5 Related Symptoms

- Nasal congestion
- Nasal trauma
- Sinusitis
- Allergic rhinitis
- Coagulopathy
- Anemia (from chronic blood loss)

188.6 Self-Care/Home Management

- Sit upright and lean forward (do NOT tilt the head back)
- Pinch the soft part of the nose firmly for 10-15 minutes
- Breathe through the mouth
- Apply a cold compress or ice pack to the bridge of the nose
- Do not pick or blow the nose for 24 hours after bleeding stops
- Use saline nasal spray to keep the nasal passages moist
- Use a humidifier in dry environments

188.7 Diagnostic Approach

- Anterior rhinoscopy to identify the bleeding site
- Nasal endoscopy to evaluate posterior bleeding
- Complete blood count (check for anemia or thrombocytopenia)
- Coagulation profile (PT, PTT, INR)
- Blood pressure assessment (hypertension can worsen bleeding)

188.8 Treatment/Management Overview

- Direct pressure as first-line self-care
- Nasal packing (anterior or posterior) for refractory bleeding
- Cauterization (chemical with silver nitrate or electrical)
- Tranexamic acid topical or oral
- Embolization of the sphenopalatine artery for severe posterior epistaxis
- Ligation of the sphenopalatine or anterior ethmoid artery
- Treat underlying cause (discontinue anticoagulants, correct coagulopathy)

Chapter 189

Oral Thrush (Oropharyngeal Candidiasis)

Definition

A fungal infection of the oral mucosa caused by overgrowth of *Candida* species (most commonly *Candida albicans*), presenting as white, creamy, curd-like plaques on the tongue, buccal mucosa, palate, and oropharynx that can be scraped off revealing erythematous underlying tissue.

Pathophysiology

Candida albicans is a normal commensal of the oral cavity kept in check by competitive oral flora, saliva, and immune surveillance. Disruption of this balance by antibiotics, corticosteroids, immunosuppression, xerostomia, or denture use allows *Candida* to proliferate and adhere to epithelial cells, forming hyphae that invade the superficial mucosa and trigger an inflammatory response.

Etiology

- Antibiotic use (suppression of competing oral bacteria)
- Inhaled or systemic corticosteroid use
- Immunosuppression (HIV/AIDS, chemotherapy, organ transplantation)
- Diabetes mellitus (especially poorly controlled)
- Denture use (denture stomatitis)
- Xerostomia (dry mouth from medications, Sjogren syndrome, radiation)
- Smoking
- Infancy (immature immune system)
- Iron or folate deficiency
- Nutritional deficiencies and general debilitation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- White plaques that do not scrape off easily, difficulty swallowing, fever, or thrush in immunocompromised patients or those receiving chemotherapy
- Accompanied by other concerning signs

Related Symptoms

- Dry mouth
- Bad breath
- Difficulty swallowing
- Altered taste
- Sore throat

Note: Always consider serious underlying conditions when evaluating persistent oral thrush (oropharyngeal candidiasis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 190

Overactive Bladder

190.1 Definition

Overactive bladder (OAB) is a syndrome characterized by urinary urgency, usually with increased daytime frequency and nocturia, with or without urgency urinary incontinence. It affects approximately 12-17 percent of the adult population.

190.2 Pathophysiology

OAB results from involuntary contractions of the detrusor muscle during the bladder filling phase (detrusor overactivity). This can be neurogenic (arising from neurological disease) or idiopathic. The normal bladder stores urine at low pressure; in OAB, inappropriate detrusor contractions cause the sensation of urgency. Mechanisms include increased afferent signaling, decreased central inhibition, and changes in urothelial function.

190.3 Etiology

- Idiopathic (most common, no identifiable cause)
- Neurological conditions: stroke, Parkinson disease, multiple sclerosis, spinal cord injury
- Bladder outlet obstruction (BPH in men, pelvic organ prolapse in women)
- Age-related changes in bladder function
- Urinary tract infection (acute, reversible)
- Bladder stones or tumors
- Diuretic use or excessive caffeine/alcohol intake
- Cognitive impairment (impaired signaling)

190.4 Indications for Evaluation

- Urinary urgency: sudden, compelling desire to urinate that is difficult to defer

- Urinary frequency: urinating 8 or more times in 24 hours
- Nocturia: waking up 2 or more times at night to urinate
- Urgency urinary incontinence: leaking urine with urgency
- Small-volume voids
- Sensation of incomplete emptying

190.5 Related Symptoms

- Urinary urgency and frequency
- Nocturia (nighttime urination)
- Incontinence (urge urinary incontinence)
- Enlarged prostate (BPH)
- Urinary tract infection
- Interstitial cystitis (bladder pain syndrome)

190.6 Self-Care/Home Management

- Bladder training: scheduled voiding every 2-3 hours
- Pelvic floor exercises (Kegel exercises) to strengthen the pelvic floor
- Avoid bladder irritants: caffeine, alcohol, spicy foods, acidic fruits
- Maintain a healthy weight
- Reduce fluid intake in the evening
- Double voiding: empty the bladder, wait, and try again

190.7 Diagnostic Approach

- Bladder diary (frequency, volume, urgency episodes)
- Urinalysis and urine culture to rule out infection
- Post-void residual measurement
- Urodynamic studies for complex or refractory cases
- Cystoscopy if hematuria or bladder pathology suspected

190.8 Treatment/Management Overview

- Behavioral therapy: bladder training, pelvic floor exercises, fluid management
- Anticholinergic medications: oxybutynin, solifenacin, tolterodine, darifenacin
- Beta-3 adrenergic agonists: mirabegron, vibegron
- Percutaneous tibial nerve stimulation (PTNS)
- Sacral neuromodulation (InterStim therapy)
- Botulinum toxin type A injection into the detrusor
- Surgery: augmentation cystoplasty or urinary diversion (rare)

Chapter 191

Pain

Definition

Unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Pathophysiology

Diverse etiologies across all body systems; characterization essential for diagnosis including location, quality, radiation, severity, timing.

Etiology

- Various depending on location and etiology

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Persistent/severe or interfering with function requires medical evaluation
- Accompanied by other concerning signs

Related Symptoms

- Varies greatly based on type and location

Note: Always consider serious underlying conditions when evaluating persistent pain.

Self-Care / Home Management

- Rest the affected area and avoid activities that aggravate the pain.

- Apply ice packs for 15-20 minutes at a time during the first 48 hours, then switch to heat if preferred.
- Over-the-counter analgesics such as acetaminophen or ibuprofen may provide relief (follow label instructions).
- Gentle stretching or movement within pain tolerance helps prevent stiffness.
- Seek medical evaluation if pain persists beyond 7 days despite self-care measures.

Diagnostic Approach

- A focused history assesses onset, location, quality, severity, duration, radiation, aggravating/relieving factors, and prior treatments.
- Physical examination includes inspection for swelling/deformity, palpation for tenderness point, range-of-motion testing, and neurovascular assessment.
- Imaging (X-ray, ultrasound, MRI, CT) may be indicated based on suspected etiology and red flag findings.
- Laboratory studies (CBC, CRP, ESR) help evaluate for inflammatory or infectious causes.

Treatment / Management Overview

- First-line management includes activity modification, physical therapy, and analgesics (NSAIDs or acetaminophen).
- Specific diagnoses may require targeted therapy: antibiotics for infection, corticosteroids for inflammation, surgery for structural pathology.
- Referral to a specialist (orthopedics, rheumatology, pain medicine) is indicated for refractory cases or complex diagnoses.
- Multimodal approaches combining medication, manual therapy, exercise, and behavioral strategies yield best outcomes for chronic pain.

Chapter 192

Pain after Surgery (Postoperative Pain)

192.1 Definition

Postoperative pain is acute pain following a surgical procedure. Effective management of postoperative pain is essential for recovery, mobility, and prevention of complications such as atelectasis, thromboembolism, and chronic pain syndromes.

192.2 Pathophysiology

Surgical incision causes tissue damage that releases inflammatory mediators (prostaglandins, bradykinin, histamine, substance P), which activate nociceptors. Peripheral and central sensitization can amplify pain signals. Uncontrolled pain leads to sympathetic activation, increased metabolic demand, impaired wound healing, and delayed recovery.

192.3 Etiology

- Surgical incision and tissue trauma
- Muscle retraction and dissection
- Inflammatory response to surgery
- Nerve injury or irritation
- Edema and swelling
- Drain sites and tube placement
- Patient positioning during surgery
- Immobilization of joints or limbs

192.4 Indications for Evaluation

- Pain at the surgical site
- Pain rated on a numeric scale (0-10)
- Pain that limits movement, coughing, or deep breathing
- Need for pain medication
- Pain accompanied by nausea, anxiety, or insomnia
- Signs of complications: fever, wound drainage, erythema (infection)
- Pain not responding to prescribed analgesics

192.5 Related Symptoms

- Acute pain
- Nausea and vomiting (opioid side effects)
- Constipation (opioid-induced)
- Fatigue
- Anxiety
- Wound infection
- Chronic post-surgical pain

192.6 Self-Care/Home Management

- Take pain medications as prescribed, on schedule
- Apply ice packs to the surgical site for the first 24-48 hours
- Elevate the surgical site when possible
- Use relaxation techniques and deep breathing
- Move and ambulate as tolerated (promotes recovery)
- Do not drive or operate machinery while on opioids
- Report uncontrolled pain to the healthcare provider

192.7 Diagnostic Approach

- Pain assessment using numeric or visual analog scale
- Clinical evaluation of the surgical site
- Vital sign monitoring (pain can cause tachycardia and hypertension)

192.8 Treatment/Management Overview

- Multimodal analgesia: combination of acetaminophen, NSAIDs, and opioids
- Regional anesthesia: nerve blocks, epidural analgesia
- Patient-controlled analgesia (PCA) pumps
- Local anesthetic infiltration at the surgical site
- Non-pharmacologic: ice, elevation, relaxation, distraction
- Treat side effects (antiemetics for nausea, stool softeners for constipation)

Chapter 193

Pain during Intercourse (Dyspareunia)

193.1 Definition

Pain during intercourse (dyspareunia) is recurrent or persistent pain associated with sexual activity. It can affect men and women at any age. Causes range from physical to psychological and often involve multiple factors.

193.2 Pathophysiology

In women, dyspareunia can result from inadequate lubrication, vulvodynia, vaginismus, endometriosis, pelvic inflammatory disease, ovarian cysts, or uterine fibroids. In men, causes include prostatitis, Peyronie disease, phimosis, or urethritis. Emotional factors such as anxiety, relationship issues, or past trauma can contribute to or exacerbate pain.

193.3 Etiology

- Inadequate lubrication (most common cause in women)
- Vulvodynia or vestibulodynia
- Vaginismus (involuntary pelvic floor muscle spasm)
- Endometriosis
- Pelvic inflammatory disease
- Uterine fibroids or adenomyosis
- Ovarian cysts
- Pelvic organ prolapse
- Prostatitis or chronic pelvic pain syndrome (men)
- Peyronie disease (penile curvature)

- Urethritis or sexually transmitted infections
- Postsurgical changes (episiotomy, hysterectomy)
- Radiation therapy
- Psychological factors: anxiety, depression, relationship issues, past trauma

193.4 Indications for Evaluation

- Pain during or after sexual intercourse
- Superficial pain at the vaginal opening or penis tip
- Deep pelvic pain during thrusting
- Burning, aching, or sharp pain
- Pain that has persisted for more than 3 months
- Avoidance of sexual activity due to pain
- Associated vaginal dryness, itching, or discharge
- Relationship distress

193.5 Related Symptoms

- Vaginal dryness
- Vaginismus
- Endometriosis
- Pelvic inflammatory disease
- Prostatitis
- Sexually transmitted infections
- Menopause (vaginal atrophy)
- Chronic pelvic pain

193.6 Self-Care/Home Management

- Use water-based lubricants generously
- Practice relaxation techniques and deep breathing
- Communicate with your partner about comfort
- Try different positions to reduce pain

- Engage in extended foreplay to enhance natural lubrication
- Empty the bladder before intercourse
- Apply a warm pack to the pelvic area before activity

193.7 Diagnostic Approach

- Detailed sexual and medical history
- Pelvic examination (women)
- Genital examination (men)
- Testing for sexually transmitted infections
- Pelvic ultrasound if endometriosis or fibroids suspected
- Laparoscopy for suspected endometriosis
- Psychological evaluation if indicated

193.8 Treatment/Management Overview

- Treat underlying cause (infection, endometriosis, prostatitis)
- Pelvic floor physical therapy
- Topical estrogen for vaginal atrophy
- Lidocaine or other topical analgesics
- Cognitive behavioral therapy or sex therapy
- Vaginal dilators for vaginismus
- Couples counseling
- Medication adjustments (change contraceptives or antidepressants)

Chapter 194

Painful Sex (Dyspareunia)

Definition

Recurrent or persistent genital pain occurring before, during, or after sexual intercourse, classified as superficial (introital) or deep pelvic pain.

Pathophysiology

Superficial dyspareunia involves vulvar, vestibular, or vaginal pathology (lack of lubrication, vulvodynia, vaginismus, infection, atrophy). Deep dyspareunia (pelvic pain with thrusting) indicates upper reproductive tract pathology (endometriosis, PID, ovarian cyst, adenomyosis) or non-gynecologic causes (IBS, interstitial cystitis).

Etiology

- Vaginal dryness (especially menopause or breastfeeding)
- Vaginismus (involuntary pelvic floor muscle spasm)
- Vulvodynia or vestibulodynia (chronic vulvar pain)
- Endometriosis
- Pelvic inflammatory disease (PID)
- Ovarian cysts
- Interstitial cystitis (painful bladder syndrome)
- Uterine fibroids or adenomyosis
- Irritable bowel syndrome
- Postpartum perineal changes or scar tissue

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Painful sex with fever, bleeding, discharge, or lasting beyond initial penetration attempts
- Accompanied by other concerning signs

Related Symptoms

- Vaginal dryness
- Pelvic pain
- Irregular bleeding
- Urinary symptoms
- Lower back pain

Note: Always consider serious underlying conditions when evaluating persistent painful sex (dyspareunia).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 195

Painful Urination (Dysuria)

Definition

Pain, burning, or discomfort experienced during urination, typically indicating irritation or inflammation of the lower urinary tract (urethra, bladder, or prostate).

Pathophysiology

Dysuria results from inflammation of the urethral or bladder mucosa, stimulating nociceptive nerve endings during urine passage. In women, the short urethra facilitates ascending bacterial infection (cystitis). In men, urethral irritation from infection (STI, prostatitis) or obstruction (BPH) is more common. Non-infectious causes include chemical irritation, interstitial cystitis, and atrophic vaginitis.

Etiology

- Urinary tract infection (cystitis)
- Urethritis (gonorrhea, chlamydia, trichomonas)
- Prostatitis (acute or chronic bacterial)
- Interstitial cystitis (painful bladder syndrome)
- Atrophic vaginitis (postmenopausal)
- Chemical irritants (soaps, spermicides, hygiene products)
- Urethral stricture
- Bladder stones
- Medication side effects (cyclophosphamide)
- Sexually transmitted infections

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Dysuria with fever, flank pain, blood in urine, vaginal/penile discharge, or inability to urinate
- Accompanied by other concerning signs

Related Symptoms

- Urinary frequency
- Urinary urgency
- Blood in urine
- Flank pain
- Fever

Note: Always consider serious underlying conditions when evaluating persistent painful urination.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 196

Palpitations

Definition

Awareness of heartbeat perceived as rapid, pounding, fluttering, or skipped beats.

Pathophysiology

Often benign but can indicate arrhythmias, structural heart disease, metabolic abnormalities, anxiety, or stimulant effects.

Etiology

- Anxiety/stress
- Arrhythmias (PVCs, SVT)
- Electrolyte imbalance
- Hyperthyroidism
- Caffeine/medication

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluate if new, irregular, or accompanied by chest pain/dizziness/syncope
- Accompanied by other concerning signs

Related Symptoms

- Dizziness
- Chest pain

- Shortness of breath
- Sweating
- Syncope/fainting

Note: Always consider serious underlying conditions when evaluating persistent palpitations.

Self-Care / Home Management

- Monitor symptoms including frequency, duration, triggers, and associated features; keep a symptom log.
- Avoid stimulants (caffeine, nicotine, decongestants, energy drinks) which can exacerbate palpitations and tachycardia.
- Practice relaxation techniques and stress reduction to minimize sympathetic activation.
- Maintain adequate hydration and electrolyte balance, especially during illness or exercise.
- Seek immediate evaluation for chest pain, shortness of breath, fainting, or palpitations with dizziness.

Diagnostic Approach

- History characterizes the symptom (quality, onset, duration, triggers, relieving factors) and assesses cardiac risk factors.
- Physical examination includes vital signs, cardiac auscultation (rate, rhythm, murmurs), lung exam, and peripheral edema assessment.
- Diagnostic testing includes ECG, ambulatory cardiac monitoring (Holter or event monitor), echocardiogram, and stress testing.
- Laboratory studies (troponin, BNP, electrolytes, thyroid function) help identify acute or contributing pathology.

Treatment / Management Overview

- Benign palpitations require reassurance and trigger avoidance; persistent symptoms may benefit from beta-blockers.
- Arrhythmias are treated based on type: rate control, rhythm control, anticoagulation, or ablation procedures.
- Heart murmurs require echocardiographic evaluation; management depends on the specific valvular lesion and severity.
- Cardiology referral is indicated for concerning symptoms, abnormal ECG findings, or newly discovered murmurs.

Chapter 197

Panic Attacks

Definition

Sudden episodes of intense fear or discomfort that peak within minutes, accompanied by physical symptoms such as palpitations, sweating, trembling, shortness of breath, chest pain, nausea, dizziness, and a sense of impending doom.

Pathophysiology

Panic attacks involve acute activation of the amygdala and its projections to the hypothalamus, periaqueductal grey, and locus coeruleus, triggering the sympathetic nervous system (fight-or-flight response). Oestrogen modulates the serotonergic and GABAergic systems that inhibit amygdala reactivity; declining oestrogen reduces inhibitory tone, lowering the panic threshold. Reduced progesterone (with its anxiolytic GABA-agonist metabolites) further disinhibits the panic circuit. Respiratory and interoceptive hypersensitivity (catastrophic misinterpretation of bodily sensations) amplifies the panic response.

Etiology

- Panic disorder (recurrent unexpected panic attacks with fear of recurrence)
- Perimenopausal and menopausal hormonal fluctuations
- Generalised anxiety disorder
- Major depressive disorder
- Post-traumatic stress disorder
- Hyperthyroidism (thyrotoxicosis)
- Hypoglycaemia
- Mitral valve prolapse
- Substance use or withdrawal (caffeine, alcohol, cocaine, cannabis)
- Medication side effects (stimulants, thyroid hormone)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Recurrent panic attacks, panic attacks with suicidal thoughts, new onset after age 40 without clear trigger, or chest pain with shortness of breath (rule out cardiac)
- Accompanied by other concerning signs

Related Symptoms

- Anxiety
- Heart palpitations
- Shortness of breath
- Dizziness
- Chest pain

Note: Always consider serious underlying conditions when evaluating persistent panic attacks.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 198

Peeling Skin

Definition

Desquamation or shedding of the outer epidermal layer of skin, which may be localized or generalized, resulting from environmental damage, infection, inflammation, or systemic disease.

Pathophysiology

Peeling skin results from disruption of intercellular adhesion between keratinocytes or accelerated epidermal turnover. Sunburn causes direct keratinocyte DNA damage triggering apoptosis and detachment. In inflammatory conditions (eczema, psoriasis), dysregulated immune responses accelerate epidermal proliferation and shedding. Infectious causes (staphylococcal scalded skin syndrome, fungal infections) directly disrupt desmosomal adhesion.

Etiology

- Sunburn (UV-induced skin damage)
- Dry skin (xerosis) from low humidity or over-washing
- Allergic contact dermatitis (poison ivy, nickel, cosmetics)
- Atopic dermatitis (eczema)
- Psoriasis (plaque or guttate)
- Fungal infections (tinea pedis/athlete's foot)
- Kawasaki disease (in children, with fever and rash)
- Scarlet fever (desquamation after fever resolves)
- Staphylococcal scalded skin syndrome (emergency)
- Medication reactions (Stevens-Johnson syndrome — emergency)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Peeling skin with fever, blistering, mucosal involvement, or widespread skin detachment (possible SJS/TEN)
- Accompanied by other concerning signs

Related Symptoms

- Rash
- Itching
- Dry skin
- Redness
- Blisters

Note: Always consider serious underlying conditions when evaluating persistent peeling skin.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 199

Pelvic Organ Prolapse

Definition

Descent of one or more pelvic organs (uterus, bladder, rectum, bowel, or vaginal vault) into or beyond the vaginal canal due to weakened pelvic floor support structures.

Pathophysiology

Pelvic organ prolapse results from compromise of the pelvic floor musculature (levator ani complex) and connective tissue (endopelvic fascia, uterosacral ligaments). Oestrogen maintains collagen and elastin in pelvic support tissues; declining oestrogen after menopause reduces tissue strength and elasticity. Vaginal childbirth (particularly multiple or traumatic deliveries) stretches and damages the levator ani and pudendal nerve. Increased intra-abdominal pressure (obesity, chronic cough, constipation, heavy lifting) accelerates prolapse progression. Prolapse is graded I-IV based on descent relative to the hymen.

Etiology

- Menopause (oestrogen deficiency weakening connective tissue)
- Vaginal childbirth (especially multiple, instrumental, or macrosomic deliveries)
- Advanced age
- Obesity (increased intra-abdominal pressure)
- Chronic constipation or straining
- Chronic cough (smoking, COPD, asthma)
- Heavy lifting (occupational or recreational)
- Connective tissue disorders (Ehlers-Danlos, Marfan)
- Prior pelvic surgery (hysterectomy)
- Genetic predisposition (family history of prolapse)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sensation of a bulge or pressure in the vagina, pelvic heaviness, urinary or faecal incontinence, or difficulty emptying bladder/bowel
- Accompanied by other concerning signs

Related Symptoms

- Pelvic pain
- Urinary incontinence
- Urinary frequency
- Back pain
- Painful sex

Note: Always consider serious underlying conditions when evaluating persistent pelvic organ prolapse.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 200

Pelvic Pain

Definition

Chronic or acute pain localized to the lower abdominal region below the umbilicus, involving reproductive, urinary, gastrointestinal, or musculoskeletal structures.

Pathophysiology

Pelvic pain in women most commonly arises from gynecological sources (endometriosis, adenomyosis, pelvic inflammatory disease, ovarian cysts) but can originate from gastrointestinal (IBS, diverticulitis), urological (interstitial cystitis), or musculoskeletal (pelvic floor dysfunction) systems. Visceral pain is poorly localized and often referred.

Etiology

- Endometriosis
- Pelvic inflammatory disease (PID)
- Ovarian cysts or torsion
- Uterine fibroids (leiomyomas)
- Adenomyosis
- Irritable bowel syndrome (IBS)
- Interstitial cystitis (painful bladder syndrome)
- Pelvic floor dysfunction
- Ectopic pregnancy (emergent)
- Chronic prostatitis (in men)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden severe pelvic pain, fever, vaginal bleeding in pregnancy, or pain with dizziness/syncope
- Accompanied by other concerning signs

Related Symptoms

- Lower back pain
- Dysmenorrhea (painful periods)
- Dyspareunia (painful sex)
- Urinary frequency
- Constipation

Note: Always consider serious underlying conditions when evaluating persistent pelvic pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 201

Penis Irritation (Balanitis)

Definition

Inflammation or irritation of the glans penis and/or foreskin (balanoposthitis), characterized by redness, swelling, itching, discharge, and discomfort.

Pathophysiology

Irritation results from infectious (candidal, bacterial, viral, parasitic), inflammatory (lichen sclerosus, contact dermatitis, psoriasis), or traumatic (friction, chemical irritation) causes. Poor hygiene, diabetes, and immunosuppression predispose to infection. Phimosis (tight foreskin) traps secretions increasing irritation risk.

Etiology

- Candidal infection (yeast — especially in diabetes)
- Poor hygiene (smegma accumulation)
- Contact dermatitis (soaps, detergents, lubricants, latex condoms)
- Bacterial infection (streptococcal, staphylococcal, anaerobic)
- Sexually transmitted infections (HSV, gonorrhea, chlamydia, syphilis)
- Lichen sclerosus (chronic inflammatory skin condition)
- Psoriasis or reactive arthritis
- Trauma or friction (vigorous intercourse, masturbation)
- Phimosis or Tight foreskin
- Allergic reaction (medications, topical creams)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Penis irritation with discharge, fever, difficulty urinating, or inability to retract foreskin (paraphimosis)
- Accompanied by other concerning signs

Related Symptoms

- Penile discharge
- Dysuria
- Swelling
- Itching
- Painful urination

Note: Always consider serious underlying conditions when evaluating persistent penis irritation (balanitis).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 202

Penis Swelling and Injury

Definition

Trauma or swelling involving the penis, ranging from minor contusions to serious injuries including fracture (tunica albuginea rupture), amputation, or strangulation.

Pathophysiology

Penile fracture occurs when an erect penis is subjected to bending force, causing rupture of the corpus cavernosum tunica albuginea with characteristic popping sound, immediate detumescence, swelling, and ecchymosis. Strangulation (hair tourniquet, constricting bands) causes venous outflow obstruction leading to edema and ischemia. Zipper injuries are common superficial trauma.

Etiology

- Penile fracture (tunica albuginea rupture during intercourse)
- Zipper entrapment injury (common in children)
- Hair tourniquet or constricting band (strangulation)
- Blunt trauma (sports injury, straddle injury)
- Penile amputation (rare, surgical emergency)
- Iatrogenic (post-circumcision, catheterization)
- Insect sting or bite causing angioedema
- Paraphimosis (retracted foreskin cannot be reduced)
- Peyronie disease (palpable plaque with curvature)
- Balanitis or cellulitis with significant edema

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Penile trauma with rapid swelling, inability to urinate, severe pain, blood at urethral meatus, or penile fracture (audible pop during erection)
- Accompanied by other concerning signs

Related Symptoms

- Penis pain
- Blood in urine
- Difficulty urinating
- Swelling
- Bruising

Note: Always consider serious underlying conditions when evaluating persistent penis swelling and injury.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 203

Petechiae

Definition

Small (1-3 mm), non-blanching, red or purple spots on the skin caused by intradermal hemorrhage from ruptured capillaries, indicating platelet or vascular abnormalities.

Pathophysiology

Petechiae result from bleeding into the dermis due to thrombocytopenia (low platelet count), platelet dysfunction, capillary fragility, or increased intravascular pressure. Non-blanching with diascopy distinguishes petechiae from erythema. Distribution provides clues: dependent areas (lower extremities) suggest vasculitis or increased pressure; widespread suggests systemic thrombocytopenia or coagulopathy.

Etiology

- Thrombocytopenia (ITP, chemotherapy, leukemia, bone marrow failure)
- Platelet dysfunction (aspirin, NSAIDs, uremia)
- Vasculitis (Henoch-Schonlein purpura, microscopic polyangiitis)
- Infections (meningococcemia, endocarditis, RMSF, viral)
- Coagulation disorders (hemophilia, von Willebrand disease)
- Trauma or vigorous coughing/vomiting (Valsalva)
- Medication reactions
- Fat embolism syndrome
- Sepsis or disseminated intravascular coagulation (DIC)
- Senile purpura (fragile capillaries in elderly)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Petechiae with fever, bleeding from other sites, neurological symptoms, or widespread rapid onset (possible meningococemia or DIC)
- Accompanied by other concerning signs

Related Symptoms

- Easy bruising
- Bleeding gums
- Nosebleeds
- Fever
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent petechiae.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 204

Pinched Nerve (Nerve Compression)

204.1 Definition

A pinched nerve occurs when excessive pressure is applied to a nerve by surrounding tissues such as bones, muscles, tendons, or discs. It causes pain, tingling, numbness, and weakness in the distribution of the affected nerve.

204.2 Pathophysiology

Compression of a nerve disrupts its blood supply and axonal transport, leading to demyelination and impaired signal conduction. The perineurium and endoneurium swell with edema. If pressure is relieved quickly, recovery is usually complete; prolonged compression can lead to Wallerian degeneration and permanent nerve damage.

204.3 Etiology

- Herniated intervertebral disc (cervical or lumbar radiculopathy)
- Spinal stenosis (narrowing of the spinal canal)
- Carpal tunnel syndrome (median nerve at the wrist)
- Cubital tunnel syndrome (ulnar nerve at the elbow)
- Tarsal tunnel syndrome (tibial nerve at the ankle)
- Meralgia paresthetica (lateral femoral cutaneous nerve)
- Piriformis syndrome (sciatic nerve compression)
- Bone spurs (osteophytes)
- Arthritis of the spine
- Repetitive motion injuries
- Poor posture or prolonged sitting
- Obesity
- Pregnancy

204.4 Indications for Evaluation

- Radiating pain along a nerve pathway
- Tingling, pins-and-needles sensation (paresthesia)
- Numbness in a specific distribution
- Muscle weakness in the affected area
- Burning sensation
- Pain worsened by certain positions or movements
- Muscle atrophy in chronic cases
- Tinel sign (tingling when tapping over the nerve)

204.5 Related Symptoms

- Back pain
- Neck pain
- Sciatica
- Carpal tunnel syndrome
- Radiculopathy
- Peripheral neuropathy
- Muscle weakness
- Muscle atrophy

204.6 Self-Care/Home Management

- Rest the affected area and avoid aggravating activities
- Apply ice for the first 48 hours, then heat
- Take OTC anti-inflammatory medications (ibuprofen, naproxen)
- Practice good posture
- Use ergonomic equipment (wrist support, standing desk)
- Stretch gently when symptoms improve
- Sleep with proper neck and back support

204.7 Diagnostic Approach

- Detailed neurological examination
- Nerve conduction studies and electromyography
- MRI of the affected spine region
- CT scan with myelography if MRI is contraindicated
- Ultrasound for peripheral nerve entrapment
- X-ray to evaluate for bone spurs or arthritis

204.8 Treatment/Management Overview

- Conservative: rest, physical therapy, NSAIDs
- Corticosteroid injections (epidural, nerve sheath)
- Activity modification and ergonomic changes
- Surgical decompression for refractory cases (carpal tunnel release, microdiscectomy)
- Nerve gliding exercises
- Treatment of underlying cause

Chapter 205

Pleurisy (Pleuritis)

205.1 Definition

Pleurisy is inflammation of the pleura, the double-layered membrane that surrounds the lungs and lines the chest cavity. The hallmark symptom is sharp, stabbing chest pain that worsens with breathing, coughing, or sneezing.

205.2 Pathophysiology

The pleural layers normally glide smoothly during respiration. When inflamed, they become swollen and rough, causing friction pain with each breath. Pleurisy can be dry (fibrinous) with no significant fluid accumulation, or exudative with pleural effusion (fluid between the layers). The underlying cause determines the nature of the effusion.

205.3 Etiology

- Viral infections (most common cause, especially coxsackievirus)
- Bacterial pneumonia (parapneumonic effusion)
- Pulmonary embolism with infarction
- Tuberculosis
- Connective tissue diseases (lupus, rheumatoid arthritis)
- Malignancy (lung cancer, mesothelioma, metastatic disease)
- Chest trauma or rib fracture
- Post-cardiac injury syndrome (Dressler syndrome)
- Asbestos exposure
- Uremia (end-stage renal disease)
- Pancreatitis (can cause left-sided pleural effusion)

205.4 Indications for Evaluation

- Sharp, stabbing chest pain that worsens with deep breathing, coughing, or sneezing
- Pain that may radiate to the shoulder or abdomen
- Shortness of breath (due to splinting or effusion)
- Cough
- Fever and chills (if infectious cause)
- Rapid, shallow breathing
- Friction rub heard on auscultation

205.5 Related Symptoms

- Chest pain (pleuritic)
- Shortness of breath
- Cough
- Fever
- Pleural effusion
- Pneumonia
- Pulmonary embolism

205.6 Self-Care/Home Management

- Take NSAIDs (ibuprofen, naproxen) for pain relief
- Rest and avoid heavy activity
- Apply a heating pad or ice pack to the chest
- Use pillows to splint the chest when coughing
- Practice deep breathing exercises to prevent atelectasis
- Stay hydrated

205.7 Diagnostic Approach

- Chest X-ray to evaluate for pleural effusion or pneumonia
- Ultrasound of the chest for pleural fluid detection
- CT chest with contrast for detailed evaluation
- Pleural fluid analysis (thoracentesis) for exudative effusion
- Blood tests: CBC, ESR, CRP, D-dimer
- ECG to rule out pericarditis or myocardial infarction

205.8 Treatment/Management Overview

- Treat the underlying cause (antibiotics for pneumonia, anticoagulation for PE)
- NSAIDs or corticosteroids for pain and inflammation
- Thoracentesis for symptomatic pleural effusion
- Chest tube with drainage for empyema
- Pleurodesis for recurrent malignant effusion
- Observation for mild viral pleurisy

Chapter 206

Pressure Sores (Bedsores)

206.1 Definition

Pressure sores (bedsores, decubitus ulcers) are localized injuries to the skin and underlying tissue caused by prolonged pressure. They most commonly develop over bony prominences in immobile patients. They are staged from I (non-blanchable erythema) to IV (full-thickness tissue loss).

206.2 Pathophysiology

Sustained pressure exceeding capillary filling pressure (32 mmHg) compresses blood vessels, causing ischemia, hypoxia, and tissue necrosis. Shear and friction forces contribute by damaging superficial tissues. Moisture from incontinence or sweating macerates the skin, increasing vulnerability. Pressure sores are graded by depth of tissue involvement.

206.3 Etiology

- Prolonged immobility (bedridden, wheelchair-bound)
- Neurological impairment (spinal cord injury, stroke, dementia)
- Advanced age (thinner skin, decreased subcutaneous fat)
- Malnutrition and dehydration
- Incontinence (urinary or fecal)
- Sensory loss (inability to feel pressure or pain)
- Friction and shear forces from repositioning
- Moisture from sweat, incontinence, or wound drainage
- Comorbidities: diabetes, peripheral vascular disease, anemia

206.4 Indications for Evaluation

- Non-blanchable erythema of intact skin (Stage I)
- Partial-thickness skin loss with blister or abrasion (Stage II)
- Full-thickness skin loss with visible subcutaneous fat (Stage III)
- Full-thickness tissue loss with exposed bone, muscle, or tendon (Stage IV)
- Pain or tenderness over bony prominences
- Location over sacrum, heels, elbows, hips, shoulders
- Signs of infection: purulent drainage, foul odor, surrounding cellulitis

206.5 Related Symptoms

- Cellulitis and soft tissue infection
- Osteomyelitis (bone infection)
- Sepsis
- Wound infection
- Malnutrition
- Dehydration

206.6 Self-Care/Home Management

- Reposition every 2 hours in bed or every 15 minutes in a wheelchair
- Use pressure-relieving surfaces (specialty mattresses, cushions)
- Keep skin clean and dry
- Apply barrier creams for incontinence
- Maintain adequate nutrition and hydration
- Inspect skin daily for early signs of breakdown
- Avoid massaging bony prominences

206.7 Diagnostic Approach

- Clinical staging of the pressure sore
- Wound culture if signs of infection
- Complete blood count, albumin, prealbumin levels
- X-ray or MRI if osteomyelitis suspected
- Wound biopsy if malignancy suspected

206.8 Treatment/Management Overview

- Pressure offloading (repositioning, specialty surfaces)
- Wound debridement of necrotic tissue
- Appropriate wound dressings (foam, hydrocolloid, alginate)
- Negative pressure wound therapy for Stage III/IV
- Antibiotics for cellulitis or osteomyelitis
- Surgical closure (flap or graft) for severe wounds
- Nutritional support (high-protein diet, supplements)

Chapter 207

Rash

Definition

Abnormal change in skin color or texture possibly involving bumps, patches, blisters.

Pathophysiology

Numerous etiologies including allergic reactions, infections, autoimmune diseases, drug eruptions, mechanical factors.

Etiology

- Allergic contact dermatitis
- Viral exanthems
- Drug eruption
- Autoimmune diseases
- Fungal infections

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluate with systemic features such as fever joint pain mucosal involvement
- Accompanied by other concerning signs

Related Symptoms

- Itching
- Pain
- Blisters/scaling
- Redness
- Swelling

Note: Always consider serious underlying conditions when evaluating persistent rash.

Self-Care / Home Management

- Keep the affected area clean and dry; wash gently with mild, fragrance-free soap and lukewarm water.
- Avoid scratching — use cool compresses, calamine lotion, or OTC hydrocortisone cream to relieve itching.
- Apply moisturizer regularly to maintain skin barrier integrity, especially after bathing.
- Identify and avoid potential triggers (irritants, allergens, excessive heat, tight clothing).
- Seek medical evaluation if the rash spreads rapidly, develops blisters, shows signs of infection (pus, increasing redness, warmth), or is accompanied by fever.

Diagnostic Approach

- History includes onset, location, progression, associated symptoms (itching, pain, fever), exposure to triggers, and prior skin conditions.
- Physical examination assesses morphology (macules, papules, vesicles, etc.), distribution, and extent of skin involvement.
- Diagnostic tests may include skin scraping for fungal examination, patch testing for allergic contact dermatitis, or biopsy for suspicious lesions.
- Laboratory studies (CBC, ESR, autoimmune serologies) are obtained when systemic disease is suspected.

Treatment / Management Overview

- Topical therapies (corticosteroids, antifungals, antibiotics, emollients) are first-line for most localized skin conditions.
- Systemic therapy (oral antihistamines, antibiotics, antifungals, or immunosuppressants) is reserved for extensive or refractory cases.

- Phototherapy or biologic agents may be indicated for chronic inflammatory skin diseases (psoriasis, eczema).
- Referral to dermatology is appropriate for uncertain diagnoses, treatment failure, or skin lesions suspicious for malignancy.

Chapter 208

Rectal Bleeding

Definition

Passage of blood from the rectum, which may be bright red (hematochezia, indicating distal GI source) or dark/maroon (suggesting proximal source).

Pathophysiology

Bright red blood coating stool or on toilet paper typically originates from hemorrhoids (internal or external), anal fissures, or distal rectal pathology. Darker blood mixed with stool suggests a colonic source (diverticulosis, colitis, polyps, malignancy). Massive bright red bleeding with clots may indicate diverticular hemorrhage or angiodysplasia. Hemorrhoids are the most common cause but colorectal cancer must always be excluded.

Etiology

- Internal or external hemorrhoids
- Anal fissure
- Diverticulosis (diverticular bleeding)
- Colorectal polyps or carcinoma
- Inflammatory bowel disease (ulcerative colitis, Crohn disease)
- Angiodysplasia (vascular malformation)
- Infectious colitis (E. coli, Shigella, Campylobacter)
- Radiation proctitis
- Ischemic colitis
- Rectal trauma or foreign body

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Heavy rectal bleeding with dizziness or syncope, bleeding with abdominal pain, or any rectal bleeding in patients over 50
- Accompanied by other concerning signs

Related Symptoms

- Blood in stool
- Anal pain
- Abdominal pain
- Constipation
- Weight loss

Note: Always consider serious underlying conditions when evaluating persistent rectal bleeding.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 209

Rectal Pain (Proctalgia)

209.1 Definition

Rectal pain is discomfort in the anal canal or rectum. It can range from sharp, brief episodes (proctalgia fugax) to chronic pain associated with underlying conditions such as hemorrhoids, anal fissures, or abscesses.

209.2 Pathophysiology

The anal canal is rich in sensory nerve endings, making it highly sensitive to pain. Causes include: stretching of the internal anal sphincter (fissure), inflammation or thrombosis of hemorrhoids, infection (abscess), muscle spasm (proctalgia fugax), or inflammation of the rectal lining (proctitis). The pelvic floor muscles can also contribute to pain.

209.3 Etiology

- Anal fissure (tear in the anal mucosa)
- Hemorrhoids (thrombosed or inflamed)
- Anorectal abscess
- Anal fistula
- Proctalgia fugax (brief, idiopathic sphincter spasm)
- Levator ani syndrome (chronic pelvic floor tension)
- Proctitis (inflammatory bowel disease, infection, radiation)
- Constipation or diarrhea (irritation)
- Foreign body or trauma
- Coccydynia (tailbone pain)
- Pudendal neuralgia

209.4 Indications for Evaluation

- Sharp or burning pain in the anal or rectal area
- Pain during bowel movements (defecation)
- Pain that radiates to the lower back, buttocks, or perineum
- Bleeding with bowel movements
- Sensation of a lump or mass
- Discharge or pus from the anus
- Itching or irritation
- Pain that awakens from sleep (proctalgia fugax)

209.5 Related Symptoms

- Rectal bleeding
- Hemorrhoids
- Anal fissure
- Anal itching (pruritus ani)
- Constipation
- Inflammatory bowel disease
- Proctitis

209.6 Self-Care/Home Management

- Sitz baths (warm water soaks) for 15-20 minutes several times a day
- Apply OTC hemorrhoid creams or suppositories
- Take stool softeners to prevent straining
- Drink plenty of water and eat a high-fiber diet
- Avoid prolonged sitting on hard surfaces
- Apply cold packs for acute swelling
- Use a cushion or donut pillow when sitting

209.7 Diagnostic Approach

- Digital rectal examination
- Anoscopy
- Proctoscopy or sigmoidoscopy for higher lesions
- Colonoscopy if bleeding or inflammatory bowel disease suspected
- MRI or endoanal ultrasound for complex fistulas or abscesses

209.8 Treatment/Management Overview

- Treat underlying cause (fissure, hemorrhoids, abscess)
- Topical nitroglycerin or nifedipine for anal fissure
- Hemorrhoidectomy or rubber band ligation for hemorrhoids
- Incision and drainage for abscess
- Fistulotomy for anal fistula
- Muscle relaxants or biofeedback for levator ani syndrome
- Nerve blocks for pudendal neuralgia

Chapter 210

Red Eye

Definition

Redness of the eye due to dilation of superficial blood vessels of the conjunctiva, episclera, or sclera, ranging from benign conjunctival injection to sight-threatening ocular emergencies.

Pathophysiology

Red eye results from vasodilation of conjunctival or episcleral vessels triggered by inflammation, infection, allergy, trauma, or increased intraocular pressure. Pattern of injection helps localize pathology: diffuse conjunctival injection suggests conjunctivitis; circumcorneal (ciliary flush) suggests keratitis, iritis, or acute glaucoma; sectoral injection suggests episcleritis or scleritis.

Etiology

- Conjunctivitis (viral, bacterial, allergic) — most common
- Subconjunctival hemorrhage (benign, spontaneous)
- Corneal abrasion or foreign body
- Keratitis (infectious, contact lens-related)
- Acute angle-closure glaucoma (emergency)
- Uveitis or iritis
- Episcleritis or scleritis
- Dry eye disease (keratoconjunctivitis sicca)
- Trauma (corneal laceration, globe rupture)
- Orbital cellulitis (with proptosis, fever, pain with eye movement)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Red eye with severe pain, vision loss, photophobia, halos around lights, corneal opacity, or trauma
- Accompanied by other concerning signs

Related Symptoms

- Eye pain
- Vision blurred
- Photophobia
- Eye discharge
- Headache

Note: Always consider serious underlying conditions when evaluating persistent red eye.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 211

Restless Legs Syndrome (Willis-Ekbom Disease)

Definition

An urge to move the legs, usually accompanied by uncomfortable sensations, that begins or worsens during rest or inactivity, is partially relieved by movement, and occurs predominantly in the evening or at night.

Pathophysiology

Restless legs syndrome (RLS) involves dysfunction of the central dopaminergic system, particularly the A11 hypothalamospinal pathway that modulates sensorimotor integration in the spinal cord. Oestrogen modulates dopamine function; oestrogen fluctuation during perimenopause may unmask or worsen RLS. Iron is a cofactor for tyrosine hydroxylase (the rate-limiting enzyme in dopamine synthesis); iron deficiency impairs dopamine production. Circadian dopamine fluctuations explain the evening/nocturnal worsening of symptoms. Genetic variants in BTBD9, MEIS1, and PTPRD are associated with primary RLS.

Etiology

- Primary (idiopathic) restless legs syndrome (genetic)
- Iron deficiency (low ferritin < 75 mcg/L)
- Perimenopause and menopause (hormonal trigger)
- Pregnancy (especially third trimester, hormonal and iron-related)
- Chronic kidney disease (uraemia)
- Peripheral neuropathy (diabetic, alcoholic)
- Medication side effects (antidepressants SSRI/SNRI, antipsychotics, antihistamines)
- Caffeine, alcohol, or nicotine use (evening)
- Hypothyroidism

- Parkinson disease (dopamine deficiency)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- RLS severely disrupting sleep, causing daytime fatigue, or accompanied by leg swelling, pain, or redness (rule out vascular causes)
- Accompanied by other concerning signs

Related Symptoms

- Insomnia
- Fatigue
- Leg pain
- Tingling numbness
- Muscle cramps

Note: Always consider serious underlying conditions when evaluating persistent restless legs syndrome.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 212

Runny Nose (Rhinorrhea)

Definition

Excessive discharge of nasal mucus from the nostrils, varying in consistency from clear and watery to thick and purulent, reflecting underlying etiology.

Pathophysiology

Increased nasal secretion from goblet cells and submucosal glands triggered by viral infection, allergic response (IgE-mediated mast cell degranulation), vasomotor instability, or irritation. Clear rhinorrhea suggests allergy or viral etiology; purulent discharge suggests bacterial sinusitis.

Etiology

- Common cold (viral URI)
- Allergic rhinitis (hay fever)
- Sinusitis (acute or chronic)
- Vasomotor rhinitis
- Cold weather or temperature changes
- Spicy foods (gustatory rhinitis)
- Nasal polyps
- Cerebrospinal fluid leak (clear watery unilateral)
- Pregnancy rhinitis
- Medication-induced (rhinitis medicamentosa from decongestant overuse)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent symptoms beyond 10 days, thick yellow/green discharge with facial pain, fever, bloody discharge, or unilateral clear drainage after head trauma
- Accompanied by other concerning signs

Related Symptoms

- Sneezing
- Nasal congestion
- Sore throat
- Cough
- Itchy eyes

Note: Always consider serious underlying conditions when evaluating persistent runny nose (rhinorrhea).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 213

Scabies

213.1 Definition

Scabies is a contagious skin infestation caused by the mite *Sarcoptes scabiei* var. *hominis*. It causes intense itching, especially at night, and a characteristic rash. It spreads through close skin-to-skin contact and is common worldwide.

213.2 Pathophysiology

The female mite burrows into the stratum corneum, creating tunnels where she lays eggs. The burrows trigger an immune response (type IV hypersensitivity) to the mites, eggs, and feces. Sensitization takes 2-6 weeks after initial infestation, so symptoms are delayed. The intense itch is primarily driven by the immune response, not the mites themselves.

213.3 Etiology

- Close skin-to-skin contact with an infested person
- Sexual contact
- Sharing clothing, bedding, or towels
- Crowded living conditions (nursing homes, prisons, dormitories)
- Immunosuppression (HIV, corticosteroid use) leading to crusted scabies
- Poverty and poor hygiene

213.4 Indications for Evaluation

- Intense itching, worse at night
- Thin, grayish-white, S-shaped burrows (most common between fingers, wrists, elbows, axillae, waist, genitals)
- Papules, vesicles, and excoriations

- Red, raised bumps (nodules) in the axillae and groin
- In infants: rash may involve the face, scalp, palms, and soles
- Crusted scabies: thick, crusted plaques with minimal itching (in immunocompromised)
- Multiple household members affected

213.5 Related Symptoms

- Intense itching (pruritus)
- Eczema (can be secondarily eczematized)
- Contact dermatitis
- Folliculitis
- Impetigo (secondary bacterial infection)

213.6 Self-Care/Home Management

- Wash all clothing, bedding, and towels in hot water and dry on high heat
- Items that cannot be washed should be sealed in a plastic bag for 72 hours
- Notify close contacts and household members for treatment
- Do not use OTC scabicides (permethrin is prescription)
- Avoid scratching to prevent secondary infection

213.7 Diagnostic Approach

- Clinical diagnosis based on history of itching and characteristic burrows
- Microscopic examination of skin scraping from a burrow (mites, eggs, feces visible)
- Dermoscopy to visualize the mite at the end of a burrow

213.8 Treatment/Management Overview

- Permethrin 5 percent cream (first-line): apply from neck to toes, wash off after 8-14 hours
- Ivermectin (oral): for resistant cases, crusted scabies, or institutional outbreaks
- Sulfur ointment for infants and pregnant women
- Lindane (second-line, due to neurotoxicity risk)
- Treat all household members and close contacts simultaneously
- Antihistamines for itching
- Topical corticosteroids for persistent nodules

Chapter 214

Sciatica

Definition

Pain radiating along the path of the sciatic nerve from lower back through hips and legs down to the foot.

Pathophysiology

Caused by compression or irritation of the sciatic nerve typically at lumbar spine level; burning shooting quality distinguishes from mechanical back pain; may include numbness tingling in affected leg distribution.

Etiology

- Herniated lumbar disc compressing nerve root
- Spinal stenosis narrowing spinal canal
- Piriformis syndrome muscle entrapment
- Degenerative disc disease nerve compression
- Spondylolisthesis vertebral slippage

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than sudden severe pain loss bladder bowel control saddle anesthesia requires immediate evaluation
- Accompanied by other concerning signs

Related Symptoms

- Lower back pain
- Numbness in leg
- Muscle weakness
- Radiating pain pattern
- Burning/shooting quality

Note: Always consider serious underlying conditions when evaluating persistent sciatica.

Self-Care / Home Management

- Apply the RICE protocol: Rest, Ice (15-20 min every 2-3 hours), Compression with elastic bandage, Elevation above heart level.
- Gradual return to activity with gentle stretching once acute pain subsides.
- Over-the-counter anti-inflammatory medications (ibuprofen, naproxen) can reduce pain and swelling.
- Avoid activities that worsen symptoms until healing is well underway.
- Seek evaluation for severe pain, inability to bear weight, joint deformity, or symptoms lasting beyond 2 weeks.

Diagnostic Approach

- History focuses on mechanism of injury, onset (acute vs insidious), location, aggravating/relieving factors, and prior episodes.
- Physical examination includes inspection (swelling, bruising, deformity), palpation, range-of-motion, and strength testing.
- Imaging (X-ray for fracture, ultrasound for tendon/ligament injury, MRI for soft tissue detail) guides diagnosis.
- Functional assessment evaluates ability to perform daily activities and sports-specific movements.

Treatment / Management Overview

- Acute injuries benefit from RICE, activity modification, and gradual rehabilitation exercises.
- Physical therapy is central to recovery, focusing on range-of-motion, strengthening, and proprioceptive training.
- Corticosteroid injections may be considered for significant inflammatory conditions (tendinitis, bursitis).

- Orthopedic referral is indicated for complete tears, fractures, joint instability, or failure of conservative therapy for 6-8 weeks.

Chapter 215

Seizures

Definition

Uncontrolled electrical activity in brain causing abnormal movements behaviors sensations consciousness alterations.

Pathophysiology

Present convulsions staring spells sensory changes requires neurological evaluation for etiation including epilepsy injury infection metabolic imbalance tumor.

Etiology

- Epilepsy
- Head injury
- Infection
- Metabolic imbalance
- Brain tumor

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than First episode warrants immediate neurological evaluation
- Accompanied by other concerning signs

Related Symptoms

- Post-seizure confusion
- Muscle soreness
- Headache
- Amnesia of event
- Temporary weakness

Note: Always consider serious underlying conditions when evaluating persistent seizures.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 216

Shin Splints (Medial Tibial Stress Syndrome)

216.1 Definition

Shin splints (medial tibial stress syndrome) is exercise-induced pain along the shinbone (tibia). It is a common overuse injury in runners, dancers, and military recruits.

216.2 Pathophysiology

Repetitive stress on the tibia and its surrounding muscles causes periostitis (inflammation of the periosteum) at the insertion of the posterior tibialis and soleus muscles along the medial tibial border. As the stress continues, it can progress to a stress reaction and eventually a stress fracture of the tibia.

216.3 Etiology

- Sudden increase in exercise intensity, duration, or frequency
- Running on hard or uneven surfaces
- Improper footwear (worn-out shoes, inadequate support)
- Overpronation of the feet (flat feet)
- Tight calf muscles
- Poor running mechanics
- Inadequate warm-up
- Female sex and low bone density (risk factors for progression to stress fracture)

216.4 Indications for Evaluation

- Dull, aching pain along the inner edge of the shinbone (tibia)

- Pain during and after exercise
- Pain that improves with rest
- Tenderness to touch along the medial tibia
- Mild swelling in the lower leg
- Pain that persists at rest or at night (may indicate stress fracture)

216.5 Related Symptoms

- Stress fracture of the tibia
- Compartment syndrome (acute or chronic exertional)
- Tendonitis of the posterior tibialis
- Calf muscle strain

216.6 Self-Care/Home Management

- Rest from high-impact activities
- Apply ice packs for 20 minutes several times a day
- Take OTC anti-inflammatory medications (ibuprofen, naproxen)
- Switch to low-impact activities (swimming, cycling)
- Stretch the calf muscles before and after activity
- Massage the affected area gently
- Consider orthotics for overpronation

216.7 Diagnostic Approach

- Clinical diagnosis based on history and tenderness along the medial tibia
- X-ray may be normal early but can show stress fracture later
- MRI for early detection of stress fracture or periosteal edema
- Bone scan (scintigraphy) to differentiate shin splints from stress fracture

216.8 Treatment/Management Overview

- Relative rest: reduce intensity and frequency of activity
- Ice and NSAIDs for pain and inflammation
- Physical therapy focusing on stretching and strengthening
- Proper footwear with good shock absorption and arch support
- Custom orthotics for biomechanical issues
- Gradual return to activity once pain-free
- Cross-training with low-impact exercises

Chapter 217

Shock (Hemodynamic Instability)

217.1 Definition

Shock is a life-threatening condition of inadequate tissue perfusion and oxygen delivery, leading to cellular dysfunction and organ failure. It requires immediate recognition and treatment. There are several types: hypovolemic, cardiogenic, distributive, and obstructive.

217.2 Pathophysiology

Shock results from failure of the circulatory system to deliver sufficient oxygen and nutrients to meet tissue demands. Hypovolemic shock: loss of circulating volume (hemorrhage, dehydration). Cardiogenic shock: pump failure (myocardial infarction, arrhythmia). Distributive shock: severe vasodilation (sepsis, anaphylaxis, neurogenic). Obstructive shock: mechanical obstruction to flow (pulmonary embolism, cardiac tamponade).

217.3 Etiology

- Hypovolemic: hemorrhage (trauma, GI bleed), dehydration (vomiting, diarrhea, burns)
- Cardiogenic: acute myocardial infarction, myocarditis, arrhythmias, valvular disease
- Distributive: sepsis (most common), anaphylaxis, spinal cord injury
- Obstructive: pulmonary embolism, cardiac tamponade, tension pneumothorax
- Trauma: hemorrhagic shock with or without obstructive component

217.4 Indications for Evaluation

- Hypotension (systolic BP less than 90 mmHg or MAP less than 65 mmHg)
- Tachycardia (rapid heart rate)
- Altered mental status: confusion, agitation, lethargy, coma
- Cold, pale, clammy skin (hypovolemic/cardiogenic) or warm, flushed skin (septic)

- Decreased urine output (less than 0.5 mL/kg/hour)
- Delayed capillary refill (greater than 2 seconds)
- Weak or thready peripheral pulses
- Metabolic acidosis with elevated lactate

217.5 Related Symptoms

- Hypotension
- Cardiac arrest
- Sepsis
- Anaphylaxis
- Hemorrhage
- Dehydration
- Trauma

217.6 Self-Care/Home Management

- Shock is a medical emergency - CALL EMERGENCY SERVICES immediately
- Lay the person flat on their back and elevate the legs about 12 inches
- Keep the person warm with a blanket
- Do not give anything to eat or drink
- Do not move the person if spinal injury is suspected
- Begin CPR if the person stops breathing

217.7 Diagnostic Approach

- Vital signs and clinical assessment of perfusion
- Blood lactate level (elevated in tissue hypoperfusion)
- Arterial blood gas analysis
- Complete blood count, electrolytes, renal function
- Cardiac enzymes and ECG for cardiogenic shock
- Chest X-ray
- Echocardiogram
- Central venous pressure and ScvO₂ monitoring

217.8 Treatment/Management Overview

- Airway, breathing, circulation (ABCs)
- IV access and fluid resuscitation (crystalloids for hypovolemic/septic)
- Blood transfusion for hemorrhagic shock
- Vasopressors (norepinephrine) for distributive/cardiogenic shock
- Treat underlying cause: antibiotics for sepsis, thrombolysis for PE, pericardiocentesis for tamponade
- ICU monitoring and organ support

Chapter 218

Shortness of Breath

Definition

Subjective breathing discomfort ranging mild effort intolerance severe air hunger known as dyspnea.

Pathophysiology

Complex sensory perception mismatch respiratory drive ventilation originates cardiac pulmonary hematologic metabolic psychiatric sources.

Etiology

- Heart failure exertion orthopnea
- Asthma COPD exacerbation
- Pulmonary embolism acute onset
- Pneumonia infection
- Anxiety panic attack hyperventilation

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Immediate acute distress cyanosis altered mental status
- Accompanied by other concerning signs

Related Symptoms

- Chest tightness wheezing cough orthopnea paroxysmal nocturnal dyspnea

Note: Always consider serious underlying conditions when evaluating persistent shortness of breath.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 219

Shoulder Pain

Definition

Pain or discomfort localized to the shoulder joint and surrounding structures, arising from the glenohumeral joint, acromioclavicular joint, rotator cuff, or referred from cervical spine or viscera.

Pathophysiology

The shoulder's complex anatomy allows extensive range of motion but predisposes to instability and impingement. Rotator cuff muscles (supraspinatus, infraspinatus, teres minor, subscapularis) stabilize the glenohumeral joint; their dysfunction leads to impingement, tendinopathy, or tears. Referred pain from the diaphragm, gallbladder, or heart may present as shoulder pain via phrenic nerve connections.

Etiology

- Rotator cuff tendinopathy or tear
- Frozen shoulder (adhesive capsulitis)
- Shoulder impingement syndrome
- Acromioclavicular joint arthritis or separation
- Glenohumeral osteoarthritis
- Bicipital tendinopathy
- Labral tear (SLAP tear)
- Cervical radiculopathy (referred from neck)
- Fracture (proximal humerus, clavicle)
- Referred cardiac pain (left shoulder — angina or MI)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Acute shoulder pain after injury, inability to raise arm, fever, or chest pain radiating to shoulder
- Accompanied by other concerning signs

Related Symptoms

- Neck pain
- Arm weakness
- Limited range of motion
- Night pain
- Arm numbness

Note: Always consider serious underlying conditions when evaluating persistent shoulder pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 220

Skin Tags (Acrochordons)

220.1 Definition

Skin tags (acrochordons) are small, soft, benign skin growths that hang off the skin by a thin stalk (peduncle). They are very common, affecting up to 60 percent of adults, and increase in frequency with age.

220.2 Pathophysiology

Skin tags are composed of loose fibrous tissue and are covered by epidermis. They develop in areas of skin friction, such as the neck, armpits, groin, and eyelids. Hormonal changes (pregnancy, growth hormone), insulin resistance, and genetic factors contribute to their formation. Human papillomavirus (HPV) has been implicated in some cases.

220.3 Etiology

- Friction from skin rubbing against skin or clothing
- Obesity (more skin folds and friction)
- Aging (increased prevalence after age 40)
- Pregnancy (hormonal changes)
- Insulin resistance and type 2 diabetes
- Acromegaly
- Genetics: family history
- HPV infection (some evidence)

220.4 Indications for Evaluation

- Small, soft, flesh-colored or slightly darker growth
- Pedunculated (attached by a stalk)

- Typically 2-5 mm but can be larger
- Most common on the neck, axillae, groin, and eyelids
- Usually painless but can become irritated from friction
- May twist (torsion) causing pain and discoloration
- Cosmetic concern is the most common reason for removal

220.5 Related Symptoms

- Skin growths and lesions
- Moles (nevi)
- Warts (verrucae)
- Seborrheic keratosis
- Neurofibromas

220.6 Self-Care/Home Management

- Do NOT attempt to cut or tie off skin tags at home
- Keep the area clean and dry
- Avoid irritation from clothing or jewelry
- Monitor for changes in size, shape, or color
- Consult a healthcare provider for removal if bothersome

220.7 Diagnostic Approach

- Clinical examination is usually sufficient
- Dermoscopy for confirmation
- Biopsy if atypical features suggest other pathology

220.8 Treatment/Management Overview

- Snip excision with scissors (with or without local anesthesia)
- Cryotherapy (liquid nitrogen freezing)
- Electrocautery (burning)
- Ligation (tying off the base)
- No treatment needed for asymptomatic tags

Chapter 221

Sneezing

Definition

Forceful expulsion of air through nose/mouth via coordinated reflex to remove irritants.

Pathophysiology

Common response to allergens viral infections nasal irritation helps clear upper airways.

Etiology

- Allergic rhinitis
- Viral infection
- Irritants/smoke
- Vasomotor rhinitis
- Bright light reflex

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Episodic typically resolve within week with allergen avoidance treatment
- Accompanied by other concerning signs

Related Symptoms

- Runny nose
- Congestion

- Itchy eyes/nose
- Watery eyes
- Postnasal drip

Note: Always consider serious underlying conditions when evaluating persistent sneezing.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 222

Snoring

Definition

A coarse, harsh breathing sound produced during sleep by vibration of the soft palate, uvula, pharyngeal walls, and tongue base due to partial upper airway obstruction, ranging from benign primary snoring to a marker of obstructive sleep apnea (OSA).

Pathophysiology

During sleep, reduced muscle tone in the pharyngeal dilator muscles allows the soft palate, uvula, and tongue to collapse partially into the airway. Airflow through the narrowed oropharynx creates negative pressure causing vibration of floppy soft tissue structures. Factors increasing collapse risk include obesity (fat deposition around airway), supine position, alcohol/sedatives (muscle relaxation), and anatomical narrowing (large tonsils, retrognathia, macroglossia).

Etiology

- Obesity (most common modifiable risk factor)
- Supine sleeping position (gravity worsens airway collapse)
- Alcohol, sedatives, or muscle relaxants before sleep
- Nasal congestion or obstruction (allergies, deviated septum, polyps)
- Enlarged tonsils or adenoids (common in children)
- Retrognathia or micrognathia (small or recessed jaw)
- Macroglossia (large tongue, seen in Down syndrome, acromegaly)
- Hypothyroidism (myxedematous changes, obesity, muscle dysfunction)
- Menopause (hormonal changes affecting airway muscle tone)
- Aging (decreased pharyngeal muscle tone with age)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Loud snoring with witnessed breathing pauses, choking/gasping during sleep, excessive daytime sleepiness, morning headaches, or hypertension (possible sleep apnea)
- Accompanied by other concerning signs

Related Symptoms

- Daytime sleepiness
- Fatigue
- Morning headache
- Sleep apnea
- Restless sleep

Note: Always consider serious underlying conditions when evaluating persistent snoring.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 223

Sore Throat

Definition

Pain scratchiness irritation in throat worsened swallowing talking.

Pathophysiology

Most commonly bacterial causes require different treatment approach post-nasal drip vocal strain acidic reflux smoking dry air contributing factors.

Etiology

- Viral URI
- Streptococcal pharyngitis
- Allergies
- Dry air/smoking
- Acid reflux

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than 1-2 weeks normally resolve persistent evaluate for strep cancer thyroid mass
- Accompanied by other concerning signs

Related Symptoms

- Fever
- Swollen glands
- Tonsillar exudate
- Voice change
- Difficulty swallowing

Note: Always consider serious underlying conditions when evaluating persistent sore throat.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 224

Splinters

224.1 Definition

A splinter is a foreign object that becomes embedded in the skin. Splinters are typically sharp, thin pieces of wood, glass, metal, or plastic. Most are superficial and can be removed at home.

224.2 Pathophysiology

Splinters penetrate the epidermis and dermis, causing a breach in the skin barrier. The body mounts an inflammatory response, which can lead to pain, redness, and swelling. If the splinter is not removed, the body may wall it off with granulation tissue, or secondary bacterial infection (cellulitis) may develop.

224.3 Etiology

- Wood (most common: from handling lumber, gardening)
- Metal (wire, staples, metal shavings)
- Glass (broken glass, fiberglass)
- Plastic or rubber
- Thorns or plant spines
- Fish fins or bones
- Pencils (graphite can leave a tattoo)

224.4 Indications for Evaluation

- Visible foreign body embedded in the skin
- Pain at the entry site
- Redness and mild swelling
- Sensation of something stuck in the skin

- Bleeding at the entry point
- Signs of infection: increased redness, warmth, pus, red streaks

224.5 Related Symptoms

- Cellulitis (secondary infection)
- Foreign body granuloma
- Tetanus risk (especially with soil-contaminated wounds)
- Puncture wound

224.6 Self-Care/Home Management

- Wash the area with soap and water
- Use tweezers sterilized with alcohol to grasp the splinter close to the skin surface
- Pull the splinter out in the same direction it entered
- For splinters under the skin, use a sterile needle to gently lift the skin over the splinter
- After removal, wash again and apply antibiotic ointment
- Monitor for signs of infection

224.7 Diagnostic Approach

- Clinical examination and visualization
- X-ray for radiopaque foreign bodies (metal, glass)
- Ultrasound for non-radiopaque objects

224.8 Treatment/Management Overview

- Removal with tweezers or sterile needle
- Local anesthesia for deep or difficult splinters
- Small incision for deeply embedded splinters
- Wound irrigation after removal
- Tetanus prophylaxis if vaccination is not up to date
- Antibiotics if signs of infection develop

Chapter 225

Stool Color Changes

225.1 Definition

Stool color changes refer to variations in the color of bowel movements that can be normal or indicate underlying disease. Stool color is influenced by diet, medications, and the presence of blood or bile.

225.2 Pathophysiology

Normal brown stool color comes from stercobilin, a breakdown product of bilirubin from red blood cell metabolism. Changes in stool color can result from: absence of bile (pale/clay-colored suggesting biliary obstruction), rapid transit (green from bile not being metabolized), upper GI bleeding (black, tarry from digested blood), lower GI bleeding (bright red from fresh blood), or dietary pigments.

225.3 Etiology

- Black/tarry (melena): upper GI bleeding (ulcer, gastritis, varices), iron supplements, bismuth (Pepto-Bismol), black licorice, blueberries
- Bright red blood (hematochezia): hemorrhoids, anal fissure, diverticular bleeding, colorectal cancer, inflammatory bowel disease
- Pale/clay-colored: biliary obstruction (gallstone, pancreatic cancer, hepatitis), barium contrast
- Green: leafy green vegetables, green food coloring, rapid intestinal transit (diarrhea)
- Yellow/greasy: malabsorption (celiac disease, pancreatic insufficiency, giardiasis)
- Orange: foods high in beta-carotene (carrots, sweet potatoes), antacids, rifampin

225.4 Indications for Evaluation

- Black, tarry, or maroon stools (requires urgent evaluation)
- Bright red blood on or in stool

- Persistent pale or clay-colored stools
- Stool color change persisting more than a few days
- Accompanying symptoms: abdominal pain, weight loss, jaundice, fever
- Change in bowel habits or stool caliber

225.5 Related Symptoms

- Rectal bleeding
- Gastrointestinal bleeding
- Jaundice
- Malabsorption
- Constipation
- Diarrhea

225.6 Self-Care/Home Management

- Review dietary intake for possible food-related causes
- Review medications and supplements
- If black stools from iron or bismuth, discontinue and observe
- Increase fiber and water intake
- Keep a stool diary if changes are intermittent

225.7 Diagnostic Approach

- History and physical examination (including digital rectal exam)
- Fecal occult blood test
- Complete blood count (check for anemia)
- Liver function tests and bilirubin
- Colonoscopy for suspected lower GI source
- Upper endoscopy for suspected upper GI bleeding

225.8 Treatment/Management Overview

- Treat underlying cause
- For GI bleeding: endoscopic therapy, IV fluids, transfusions
- For biliary obstruction: ERCP with stenting, surgery
- For malabsorption: dietary changes, enzyme replacement
- Discontinue or adjust causative medications

Chapter 226

Stress Health (Stress Management)

226.1 Definition

Stress is a state of mental or emotional strain resulting from adverse or demanding circumstances. Chronic stress can have significant negative effects on physical and mental health, contributing to a wide range of medical conditions.

226.2 Pathophysiology

The stress response involves activation of the hypothalamic-pituitary-adrenal (HPA) axis and sympathetic nervous system. Cortisol and catecholamines (epinephrine, norepinephrine) are released, increasing heart rate, blood pressure, and blood glucose. Chronic stress leads to persistent elevation of these hormones, causing immune suppression, inflammation, hypertension, insulin resistance, and mental health disorders.

226.3 Etiology

- Work-related pressure and job insecurity
- Financial difficulties
- Relationship problems (marital conflict, divorce, caregiving)
- Major life changes (moving, job loss, death of a loved one)
- Academic pressure
- Health problems (personal or family illness)
- Trauma (physical, emotional, or sexual abuse)
- Social isolation
- Discrimination or minority stress

226.4 Indications for Evaluation

- Anxiety, irritability, or mood swings
- Fatigue and low energy
- Difficulty sleeping (insomnia)
- Difficulty concentrating or making decisions
- Headaches, muscle tension, or body aches
- Changes in appetite (overeating or loss of appetite)
- Increased illness (stress suppresses immune function)
- Withdrawal from social activities
- Increased use of alcohol, tobacco, or other substances

226.5 Related Symptoms

- Anxiety disorders
- Depression
- Insomnia
- Headache (tension-type)
- Hypertension
- Irritable bowel syndrome
- Chronic fatigue

226.6 Self-Care/Home Management

- Practice deep breathing, meditation, or mindfulness daily
- Exercise regularly (30 minutes most days)
- Maintain a healthy sleep schedule (7-9 hours)
- Eat a balanced diet and avoid excessive caffeine and alcohol
- Set boundaries and learn to say no
- Connect with supportive friends and family
- Engage in enjoyable hobbies
- Limit screen time and news consumption

226.7 Diagnostic Approach

- Clinical evaluation with stress assessment tools (Perceived Stress Scale)
- Screen for anxiety and depression
- Assess lifestyle factors: sleep, exercise, diet, substance use

226.8 Treatment/Management Overview

- Cognitive behavioral therapy
- Mindfulness-based stress reduction
- Physical activity and exercise programs
- Stress management classes or workshops
- Counseling or psychotherapy for underlying issues
- Medication for co-occurring anxiety or depression
- Referral to a mental health specialist

Chapter 227

Stretch Marks (Striae Distensae)

227.1 Definition

Stretch marks (striae) are linear scars that develop in the dermis due to rapid stretching of the skin and hormonal factors. They are common during pregnancy, puberty, rapid weight gain, and in conditions such as Cushing syndrome.

227.2 Pathophysiology

Stretch marks result from a combination of mechanical stretching and hormonal changes that disrupt dermal collagen and elastin fibers. Glucocorticoids inhibit fibroblast function, reducing collagen synthesis. The dermis thins and tears, allowing the underlying subcutaneous tissue to show through. Initially, they appear red or purple (striae rubra), then fade to white or silver (striae alba).

227.3 Etiology

- Pregnancy (striae gravidarum)
- Puberty (rapid growth)
- Rapid weight gain or obesity
- Corticosteroid use (topical or systemic)
- Cushing syndrome
- Marfan syndrome (connective tissue disorder)
- Ehlers-Danlos syndrome
- Bodybuilding (rapid muscle growth)
- Genetics: family history increases risk

227.4 Indications for Evaluation

- Linear, parallel streaks on the skin
- Initially red, purple, or pink (*striae rubra*)
- Later become white, silvery, or hypopigmented (*striae alba*)
- Most common on abdomen, breasts, thighs, hips, buttocks, and arms
- Slightly depressed or wrinkled appearance
- May be itchy when newly formed
- No pain or other symptoms

227.5 Related Symptoms

- Cushing syndrome signs (moon face, buffalo hump, central obesity)
- Pregnancy-related skin changes
- Weight gain
- Obesity

227.6 Self-Care/Home Management

- Keep skin well-moisturized (cocoa butter, shea butter, vitamin E oil)
- Maintain a stable, healthy weight
- Stay hydrated
- Use sun protection on affected areas
- Apply silicone gel or sheeting for newer stretch marks
- Avoid prolonged corticosteroid use

227.7 Diagnostic Approach

- Clinical examination is diagnostic
- Rarely, biopsy for atypical cases

227.8 Treatment/Management Overview

- Topical retinoids (tretinoin) for early striae rubra
- Pulsed dye laser for red stretch marks
- Fractional laser resurfacing
- Microneedling
- Microdermabrasion
- Chemical peels
- No treatment is medically necessary

Chapter 228

Stroke

Definition

Sudden interruption of blood supply to brain causing rapid onset of focal neurological deficits representing medical emergency.

Pathophysiology

Ischemic stroke (87

Etiology

- Carotid artery stenosis embolic source
- Cardiac atrial fibrillation cardioembolism
- Small vessel disease lacunar infarcts
- Hypertension chronic vascular damage
- Hypercoagulable states rare causes

Indications for Evaluation

Seek care immediately:

- Sudden onset of any neurological deficit
- Time is brain - call emergency services FAST (Face drooping Arm weakness Speech difficulty Time to call 911)
- Accompanied by other concerning signs

Related Symptoms

- Facial weakness drooping
- Arm/leg weakness one side
- Speech difficulty aphasia slurred
- Vision loss monocular or visual field
- Severe sudden headache

Note: Stroke is a medical emergency requiring immediate evaluation.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 229

Sty (Hordeolum)

229.1 Definition

A sty (hordeolum) is an acute, painful infection of the eyelid margin involving the sebaceous glands of Zeis (external hordeolum) or meibomian glands (internal hordeolum). It is most commonly caused by *Staphylococcus aureus*.

229.2 Pathophysiology

Bacterial infection (usually *S. aureus*) of the eyelid glands causes localized inflammation, pus formation, and swelling. An external hordeolum involves the glands of Zeis at the base of the eyelash follicle. An internal hordeolum involves the meibomian glands within the eyelid tarsal plate and is usually more painful. Inflammation leads to a tender, red, swollen nodule.

229.3 Etiology

- Bacterial infection: *Staphylococcus aureus* (most common)
- Blepharitis (chronic eyelid inflammation is a risk factor)
- Poor eyelid hygiene
- Contact lens wear (improper hygiene)
- Rosacea
- Diabetes mellitus
- Immunosuppression
- Stress and fatigue

229.4 Indications for Evaluation

- Red, swollen, painful lump on the eyelid margin
- Localized tenderness

- Swelling of the entire eyelid (if severe)
- Gritty or foreign body sensation
- Watering of the eye
- Crusting along the eyelashes
- Pus may drain from the sty

229.5 Related Symptoms

- Blepharitis (eyelid inflammation)
- Chalazion (chronic, non-infectious meibomian cyst)
- Conjunctivitis (pink eye)
- Periorbital cellulitis (severe infection)

229.6 Self-Care/Home Management

- Apply warm compresses for 10-15 minutes, 3-4 times daily
- Gently massage the area after warm compresses
- Do NOT squeeze or pop the sty
- Clean the eyelid with diluted baby shampoo
- Avoid eye makeup and contact lens wear until healed
- Wash hands before and after touching the area

229.7 Diagnostic Approach

- Clinical examination is diagnostic
- Slit-lamp examination of the eyelid

229.8 Treatment/Management Overview

- Warm compresses are the mainstay of treatment
- Topical antibiotic ointments (erythromycin, bacitracin)
- Topical antibiotic-steroid combination for inflammation
- Oral antibiotics for recurrent or severe cases
- Incision and drainage for large or persistent sties
- Treat underlying blepharitis or rosacea

Chapter 230

Sunburn

230.1 Definition

Sunburn is acute cutaneous inflammation caused by excessive exposure to ultraviolet (UV) radiation from the sun or artificial sources (tanning beds). It is characterized by erythema, pain, and in severe cases, blistering.

230.2 Pathophysiology

UV-B radiation (290-320 nm) damages DNA in keratinocytes, causing the formation of pyrimidine dimers. Damaged cells release inflammatory mediators (prostaglandins, histamine, cytokines), causing vasodilation (erythema), pain, and edema. Apoptosis of damaged keratinocytes leads to peeling 3-7 days after exposure. UV-A radiation (320-400 nm) penetrates deeper, contributing to photoaging and immunosuppression.

230.3 Etiology

- Prolonged sun exposure without protection
- Tanning bed use
- Fair skin (Fitzpatrick types I-III)
- Outdoor activities near water, sand, or snow (reflected UV)
- High altitude (increased UV intensity)
- Midday sun (10 AM to 4 PM)
- Photosensitizing medications: tetracyclines, sulfonamides, NSAIDs, retinoids
- Ozone depletion

230.4 Indications for Evaluation

- Red, warm, tender skin in sun-exposed areas

- Pain or burning sensation
- Swelling (edema)
- Blisters in severe burns
- Peeling skin after 3-7 days
- Headache, fever, chills, nausea (sun poisoning with extensive burns)
- Dehydration

230.5 Related Symptoms

- Heat exhaustion and heat stroke (if prolonged heat exposure)
- Skin cancer (long-term risk)
- Photoaging (wrinkles, hyperpigmentation)
- Photosensitivity reactions

230.6 Self-Care/Home Management

- Cool the skin with cool (not cold) water or cool compresses
- Apply moisturizing lotion or aloe vera gel
- Take OTC pain relievers (ibuprofen, aspirin)
- Drink extra water to prevent dehydration
- Do NOT pop blisters (they protect the skin)
- Apply OTC hydrocortisone cream for itching
- Wear loose, soft clothing over sunburned skin

230.7 Diagnostic Approach

- Clinical diagnosis based on history of UV exposure and skin examination
- Assess extent and depth of burn for severe cases

230.8 Treatment/Management Overview

- Cool compresses and soothing lotions
- NSAIDs for pain and inflammation
- Topical corticosteroids for moderate burns
- Oral corticosteroids for severe, extensive burns

- Burn wound care for severe blistering
- Hydration and electrolyte replacement
- Tetanus prophylaxis if large area of blistering

Chapter 231

Swelling

Definition

Abnormal fluid accumulation tissues visible enlargement peripheral limbs dependent areas pitting non-pitting distinguishes generalized cardiac hepatic renal venous obstructive lymphatic arterial etiologies local trauma infection inflammation allergic reaction neoplasm

Pathophysiology

Interstitial fluid exceeds drainage capacity capillary hydrostatic pressure increased oncotic pressure decreased decreased lymphatic return obstructed lymphatic vessels tissue inflammation increases vascular permeability protein exudate fluid transudate accumulates intercellular spaces palpable firm firm indurate edematous pitting leaves depression finger releases immediate rebound non-pitting firm woody characteristic lymphatic obstruction fibrosis scleroderma lipodermatosclerosis myxedema fat deposits tumors cellulitis deep vein thrombosis lymphedema varicose veins congestive heart failure cirrhosis nephrotic syndrome hypothyroidism

Etiology

- Heart failure right-sided peripheral edema dependent ascites hepatomegaly jugular venous distension
- Kidney disease nephrotic syndrome glomerulonephritis renal failure proteinuria hypoalbuminemia hypertension hematuria
- Liver disease cirrhosis portal hypertension ascites splenomegaly varices coagulopathy encephalopathy jaundice
- Venous insufficiency varicose veins stasis dermatitis post-thrombotic syndrome deep vein thrombosis
- Medication side effects calcium channel blockers NSAIDs steroids hormones insulin thiazolidinediones vasodilators

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Progressive worsening unilateral sudden swelling pain redness warmth suggests deep vein thrombosis pulmonary embolism risk bilateral symmetrical morning facial/periorbital then progressing downward indicates nephrotic syndrome or nephritic syndrome hepatic congestion right heart failure associated ascites hepatomegaly spider angiomas caput medusae gynecomastia testicular atrophy endocrine manifestations thyroid swelling goiter myxedema facial puffiness delayed relaxation of deep tendon reflexes cold intolerance weight gain dry skin bradycardia constipation depression cognitive slowing requires thyroid function testing evaluation
- Accompanied by other concerning signs including shortness of breath chest pain palpitations syncope oliguria anuria dark urine foamy urine bloody urine flank pain fever chills weight loss night sweats abdominal distension pain vomiting diarrhea melena hematemesis jaundice altered mental status confusion seizures limb weakness speech difficulties visual disturbances headache neck stiffness photophobia

Related Symptoms

- Skin changes stretching glossiness discoloration erythema cyanosis pallor telangiectasias atrophy ulceration wound healing impairment infection cellulitis abscess lymphangitis lymphadenopathy warm tenderness pulsatile masses arterial bruit venous hum thrill auscultation findings cardiovascular respiratory abdominal neurological constitutional dermatological ophthalmological ENT dental oral maxillofacial extremity musculoskeletal genitourinary reproductive endocrine hematologic oncologic rheumatologic immunologic gastroenterologic hepatobiliary nephrologic infectious disease psychiatric neurological degenerative inflammatory metabolic toxic traumatic congenital developmental idiopathic rare syndromic presentations require multidisciplinary specialist evaluation comprehensive diagnostic workup targeted laboratory investigations imaging modalities specialty consultations therapeutic interventions lifestyle modifications preventive measures follow-up monitoring treatment response adverse effects complications outcomes prognosis quality of life assessments patient education self-management strategies shared decision-making partnerships healthcare providers

Note: Always consider serious underlying conditions when evaluating persistent swelling

Self-Care / Home Management

- Prioritize adequate rest and sleep to support the body's healing and recovery processes.
- Maintain a balanced diet with regular meals and stay well-hydrated.
- Monitor symptoms and track any changes in severity, frequency, or associated features.
- Avoid overexertion and gradually increase activity levels as tolerated.
- Seek medical evaluation if symptoms persist, worsen, or interfere significantly with daily function.

Diagnostic Approach

- Comprehensive history and physical examination guide the diagnostic workup based on the specific symptom presentation.
- Laboratory testing (CBC, metabolic panel, inflammatory markers) helps screen for underlying systemic conditions.
- Imaging studies are tailored to the suspected anatomic region and pathology.
- Referral to appropriate specialists is guided by the differential diagnosis established through initial evaluation.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic management may include pharmacotherapy, lifestyle modifications, and supportive care.
- Multidisciplinary care is often beneficial for complex or chronic symptoms.
- Close follow-up ensures treatment effectiveness and allows adjustment of the management plan as needed.

Chapter 232

Swollen or Painful Testicle

Definition

Enlargement, pain, or tenderness of one or both testicles, ranging from mild discomfort to acute severe pain requiring emergency evaluation.

Pathophysiology

Testicular torsion (spermatic cord twisting) compromises blood flow causing ischemia within 4-6 hours — surgical emergency. Epididymitis results from infection ascending from urethra/bladder. Hydrocele, varicocele, and spermatocele cause painless swelling. Testicular tumors typically present as painless solid mass.

Etiology

- Testicular torsion (surgical emergency)
- Epididymitis (often bacterial or sexually transmitted)
- Orchitis (viral — mumps, or bacterial)
- Hydrocele (fluid collection around testicle)
- Varicocele (dilated scrotal veins)
- Spermatocele (epididymal cyst)
- Testicular tumor or cancer
- Testicular trauma
- Inguinal hernia (scrotal extension)
- Torsion of appendix testis (small testicular appendage)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden severe testicular pain with nausea/vomiting (possible torsion — immediate emergency), or painless testicular lump
- Accompanied by other concerning signs

Related Symptoms

- Groin pain
- Scrotal swelling
- Nausea and vomiting
- Fever
- Urinary symptoms

Note: Always consider serious underlying conditions when evaluating persistent swollen or painful testicle.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 233

Syncope

Definition

Transient loss consciousness temporary reduction cerebral perfusion spontaneous recovery.

Pathophysiology

Differentiate vasovagal cardiac neurologic metabolic causes syncope exertion particularly concerning requires cardiac evaluation.

Etiology

- Vasovagal episode
- Cardiac arrhythmia
- Orthostatic hypotension
- Carotid sinus hypersensitivity
- Seizure mimic

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than First episode or recurrent with injury risk warrants medical evaluation especially if exertional precipitated by position changes cardiac family history present
- Accompanied by other concerning signs

Related Symptoms

- Palpitations
- Chest pain
- Shortness of breath
- Postural dizziness
- Prodromal warning symptoms

Note: Always consider serious underlying conditions when evaluating persistent syncope.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 234

Tachycardia

Definition

Resting heart rate exceeding 100 beats per minute, classified by origin (sinus, supraventricular, ventricular), duration (paroxysmal, persistent, permanent), and electrocardiographic characteristics.

Pathophysiology

Tachycardia mechanisms include increased automaticity (enhanced pacemaker activity), triggered activity (afterdepolarizations), or reentry (abnormal conduction circuit). Sinus tachycardia is a physiologic response to increased demand; pathologic tachyarrhythmias (atrial fibrillation, SVT, VT) require specific diagnosis and management.

Etiology

- Sinus tachycardia (fever, dehydration, anxiety, exercise)
- Atrial fibrillation or atrial flutter
- Supraventricular tachycardia (SVT, AVNRT, WPW)
- Ventricular tachycardia (VT)
- Hyperthyroidism
- Anemia
- Pulmonary embolism
- Heart failure or cardiomyopathy
- Electrolyte imbalances
- Medication or substance effects (caffeine, alcohol, stimulants, thyroid medication)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Resting tachycardia with chest pain, shortness of breath, syncope, or dizziness (possible cardiac arrhythmia)
- Accompanied by other concerning signs

Related Symptoms

- Palpitations
- Shortness of breath
- Dizziness
- Chest pain
- Fatigue

Note: Always consider serious underlying conditions when evaluating persistent tachycardia.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 235

Teething

235.1 Definition

Teething is the process of primary (baby) teeth emerging through the gums. It typically begins around 6 months of age and continues until about 2-3 years. While teething can cause mild discomfort, it does not cause significant illness.

235.2 Pathophysiology

As a tooth moves through the jawbone and gum tissue, it causes localized inflammation and pressure. This triggers the release of inflammatory mediators that can cause pain, swelling, and tenderness of the gums. The cutting of the gum tissue is a mechanical process that naturally resolves once the tooth erupts.

235.3 Etiology

- Normal eruption of primary teeth
- Typically begins at 6 months of age
- Lower central incisors usually erupt first
- Molars (around 12-18 months) may cause more discomfort

235.4 Indications for Evaluation

- Irritability and fussiness
- Excessive drooling
- Chewing and biting on objects
- Swollen, tender gums
- Mild temperature elevation (less than 38.3 C or 101 F)
- Decreased appetite for solid foods

- Disturbed sleep
- Rubbing the cheek or pulling the ear on the affected side

235.5 Related Symptoms

- Fever (if temperature is high, consider infection, not teething)
- Diarrhea (not caused by teething)
- Rash on the face (from drooling)
- Ear infection (can be confused with teething pain)

235.6 Self-Care/Home Management

- Gently massage the gums with a clean finger
- Provide a chilled (not frozen) teething ring
- Offer cold, soft foods (yogurt, pureed fruit) for older infants
- Use a clean, wet washcloth that has been chilled
- Wipe drool frequently to prevent facial rash
- Give acetaminophen or ibuprofen for pain if needed

235.7 Diagnostic Approach

- Clinical diagnosis based on age and symptoms
- Visual examination of the gums and emerging tooth

235.8 Treatment/Management Overview

- Pain relief: acetaminophen or ibuprofen (age-appropriate dosing)
- Teething rings and other non-pharmacologic measures
- Topical anesthetics (benzocaine) have risks and are not recommended by the FDA for children under 2
- Avoid amber necklaces (no evidence of benefit and pose strangulation risk)

Chapter 236

Testicular Pain (Orchalgia)

236.1 Definition

Testicular pain is discomfort in one or both testicles. It can be acute or chronic and may arise from the testicle itself, the epididymis, or referred from other structures. Testicular torsion is a urologic emergency that requires prompt surgical intervention.

236.2 Pathophysiology

Testicular pain can be caused by inflammation (epididymitis, orchitis), torsion (twisting of the spermatic cord causing ischemia), trauma, tumors, or varicocele (dilated veins). Torsion cuts off blood supply, causing ischemia and infarction within 4-6 hours. Referred pain can occur from kidney stones, hernia, or prostatitis via shared nerve pathways.

236.3 Etiology

- Epididymitis (inflammation of the epididymis, often infectious)
- Orchitis (inflammation of the testicle, often viral: mumps)
- Testicular torsion (surgical emergency)
- Torsion of the appendix testis
- Varicocele (dilated pampiniform plexus)
- Spermatocele (cystic dilation of the epididymis)
- Trauma or blunt injury
- Testicular tumor (usually painless but can cause dull ache)
- Inguinal hernia
- Referred pain from kidney stones or prostatitis

236.4 Indications for Evaluation

- Sudden, severe testicular pain (emergency - rule out torsion)
- Dull ache or heavy sensation
- Swelling or enlargement of the testicle
- Redness or warmth of the scrotal skin
- Nausea and vomiting (with torsion)
- Fever (with epididymitis or orchitis)
- Pain that improves with elevation (epididymitis)
- Urethral discharge (with epididymitis)

236.5 Related Symptoms

- Testicular swelling
- Scrotal pain
- Groin pain
- Lower abdominal pain
- Urinary symptoms (frequency, urgency, dysuria)

236.6 Self-Care/Home Management

- Wear supportive underwear (briefs instead of boxers)
- Apply ice packs to the scrotum
- Elevate the scrotum
- Take NSAIDs for pain
- Avoid heavy lifting and strenuous activity
- Do NOT apply heat (can worsen inflammation)
- Seek immediate care for sudden, severe pain

236.7 Diagnostic Approach

- Ultrasound with Doppler (urgent for suspected torsion)
- Urinalysis and urine culture
- Complete blood count
- STI testing (chlamydia, gonorrhea) for epididymitis
- Testicular tumor markers (AFP, hCG) if mass found

236.8 Treatment/Management Overview

- Testicular torsion: immediate surgical exploration and orchiopexy
- Epididymitis: antibiotics (doxycycline, ceftriaxone) and rest
- Orchitis: supportive care, NSAIDs, scrotal elevation
- Varicocele: observation or surgical ligation if symptomatic
- Trauma: rest, ice, elevation, NSAIDs; surgery for rupture
- Testicular tumor: radical orchiectomy and oncology referral

Chapter 237

Tingling Numbness

Definition

Abnormal sensation termed paresthesia tingling prickling burning loss normal skin sensation affecting extremities face trunk

Pathophysiology

Results nerve compression ischemia demyelination peripheral neuropathy central nervous system lesions stroke TIA vitamin deficiency multiple sclerosis carpal tunnel cervical radiculopathy disrupted sensory pathway abnormal firing patterns nerve signaling

Diabetic neuropathy presents burning shooting pains tingling numbness loss coordination affecting feet legs sometimes hands arms more common with poor diabetes control

Peripheral neuropathy typically begins finger toe tips slowly progresses along limbs causes burning shooting pains tingling numbness

Nutritional neuropathy associated vitamin deficiencies causes burning tingling sensory symptoms plus sore red tongue fatigue anemia signs

Etiology

- Carpal tunnel wrist numbness nocturnal predominant
- Peripheral neuropathy diabetes alcohol vitamin B12 deficiency
- Cervical radiculopathy neck arm dermatomal distribution
- Transient ischemic attack stroke sudden focal deficits
- Multiple sclerosis young adults relapsing remitting episodic
- Vitamin B12 deficiency pernicious anemia folate deficiency
- Alcohol toxic neuropathy chronic heavy consumption

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Severe associated weakness speech difficulty coordination problem evaluate stroke spinal cord compression immediately
- Accompanied by other concerning signs

Related Symptoms

- Weakness muscle twitching burning sensation allodynia distribution patterns dermatome nerve territory localized affected body part

Note: Always consider serious underlying conditions when evaluating persistent tingling numbness

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 238

Tinnitus (Ringing in the Ears)

Definition

Perception of sound in the absence of an external acoustic stimulus, described as ringing, buzzing, hissing, humming, or clicking, which may be unilateral or bilateral, continuous or intermittent.

Pathophysiology

Abnormal neural activity in the auditory pathway, from cochlear hair cell damage to central auditory cortex reorganization. Subtypes include objective (arising from vascular or muscular sources near the ear) and subjective (much more common, arising from auditory system dysfunction).

Etiology

- Noise-induced hearing loss
- Age-related hearing loss (presbycusis)
- Meniere disease
- Otosclerosis
- Cerumen impaction (earwax)
- Ototoxic medications (aspirin, NSAIDs, aminoglycosides, loop diuretics)
- Vascular disorders (pulsatile tinnitus — AV malformations, carotid stenosis)
- Temporomandibular joint (TMJ) disorders
- Head or neck injury
- Acoustic neuroma (vestibular schwannoma)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Sudden onset tinnitus, pulsatile tinnitus, tinnitus with hearing loss, dizziness, or focal neurological symptoms
- Accompanied by other concerning signs

Related Symptoms

- Hearing loss
- Vertigo
- Ear fullness
- Headache
- Dizziness

Note: Always consider serious underlying conditions when evaluating persistent tinnitus (ringing in the ears).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 239

Toothache (Dental Pain)

Definition

Pain originating from a tooth or its supporting structures, ranging from mild sensitivity to severe throbbing pain, caused by dental caries, infection, trauma, or dental procedures.

Pathophysiology

Dental caries (decay) progresses through enamel into dentin, exposing sensitive nerve endings. Pulpitis (inflammation of dental pulp) results from deep decay, trauma, or thermal injury. Periapical abscess forms when pulp infection extends through the apical foramen into surrounding bone. Periodontal abscess arises from gum infection in periodontal pockets.

Etiology

- Dental caries (tooth decay) — most common cause
- Pulpitis (reversible or irreversible)
- Periapical abscess (dental abscess)
- Periodontal abscess (gum infection)
- Cracked tooth syndrome
- Dental trauma (fractured or avulsed tooth)
- Impacted wisdom teeth (pericoronitis)
- Temporomandibular joint (TMJ) disorders (referred pain)
- Sinusitis (referred pain to upper teeth)
- Dental procedures (recent filling, root canal, extraction)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Severe tooth pain with facial swelling, fever, difficulty swallowing or breathing (possible deep space neck infection), or persistent pain beyond 2 days
- Accompanied by other concerning signs

Related Symptoms

- Facial swelling
- Fever
- Headache
- Gum swelling
- Earache (referred)

Note: Always consider serious underlying conditions when evaluating persistent toothache (dental pain).

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 240

Tremors

240.1 Definition

Tremors are involuntary, rhythmic, oscillatory movements of a body part. They are the most common movement disorder. Tremors are classified by their appearance (resting, action, intention) and underlying cause.

240.2 Pathophysiology

Tremors arise from dysfunction in the central nervous system circuits that control movement. Resting tremor (Parkinson disease) involves degeneration of dopaminergic neurons in the substantia nigra. Essential tremor involves abnormal oscillations in the cerebello-thalamo-cortical circuit. Cerebellar tremor results from damage to the cerebellum or its connections. Enhanced physiologic tremor occurs when the normal tremor is amplified by metabolic or drug effects.

240.3 Etiology

- Essential tremor (most common movement disorder, often familial)
- Parkinson disease (resting tremor, bradykinesia, rigidity)
- Enhanced physiologic tremor: anxiety, stress, fatigue, caffeine, stimulants
- Medication-induced: beta-agonists, antidepressants, valproate, lithium, corticosteroids
- Metabolic: hyperthyroidism, hypoglycemia, electrolyte imbalances
- Alcohol withdrawal
- Cerebellar tremor: multiple sclerosis, stroke, tumor, trauma
- Drug-induced: alcohol, benzodiazepine withdrawal
- Wilson disease (young-onset, with other neurological signs)

240.4 Indications for Evaluation

- Rhythmic, involuntary shaking of a body part
- Resting tremor (present at rest, decreases with movement - Parkinson)
- Action tremor (occurs with voluntary movement - essential tremor)
- Intention tremor (worsens at the end of a goal-directed movement - cerebellar)
- Postural tremor (occurs when maintaining a position against gravity)
- Tremor that interferes with writing, eating, drinking, or other daily tasks
- Anxiety or social embarrassment due to visible tremor

240.5 Related Symptoms

- Parkinson disease
- Essential tremor
- Multiple sclerosis
- Hyperthyroidism
- Anxiety
- Alcohol withdrawal

240.6 Self-Care/Home Management

- Reduce or avoid caffeine, nicotine, and other stimulants
- Practice relaxation techniques to reduce stress
- Get adequate sleep
- Use weighted utensils, cups, and pens to dampen tremor
- Avoid alcohol (temporary improvement can lead to dependence)
- Use adaptive devices (tremor-canceling spoons, non-slip mats)

240.7 Diagnostic Approach

- Clinical history and physical examination
- Differentiate rest vs. action vs. intention tremor
- Blood tests: thyroid function, glucose, electrolytes, liver function
- Brain MRI to rule out structural lesions
- DaTscan (dopamine transporter scan) for Parkinson disease
- Genetic testing for familial essential tremor

240.8 Treatment/Management Overview

- Essential tremor: beta-blockers (propranolol), primidone; deep brain stimulation for refractory
- Parkinson tremor: levodopa, dopamine agonists; deep brain stimulation
- Enhanced physiologic tremor: treat underlying cause
- Cerebellar tremor: physical therapy, weighted limbs; limited medication options
- Botulinum toxin injections for head or voice tremor
- Deep brain stimulation for severe, medication-refractory tremor

Chapter 241

Urinary Frequency

Definition

Increased number voids day without necessarily increasing total urine volume.

Pathophysiology

Lower urinary tract symptoms benign prostatic hyperplasia UTIs diabetes bladder overactivity common causes.

Etiology

- UTI
- Diabetes mellitus
- Benign prostatic hyperplasia
- Overactive bladder
- Fluid intake increase

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluated if new bothersome
- Accompanied by other concerning signs

Related Symptoms

- Urgency nocturia dysuria hematuria pain burning sensation

Note: Always consider serious underlying conditions when evaluating persistent urinary frequency.

Self-Care / Home Management

- Drink adequate water throughout the day unless fluid restriction is medically indicated.
- Avoid bladder irritants including caffeine, alcohol, acidic foods, and artificial sweeteners.
- Practice good hygiene and urinate promptly when the urge arises to prevent urinary stasis.
- Apply a heating pad to the lower abdomen or back for comfort during urinary discomfort.
- Seek medical attention for fever, flank pain, visible blood in urine, or difficulty urinating.

Diagnostic Approach

- History covers urinary symptoms (frequency, urgency, dysuria, hesitancy, stream changes), fever, and flank pain.
- Urinalysis with microscopy and culture is the first-line diagnostic test for urinary symptoms.
- Imaging (renal ultrasound, CT urogram) evaluates for stones, obstruction, structural abnormalities, or masses.
- Urodynamic studies or cystoscopy may be indicated for complex or refractory voiding dysfunction.

Treatment / Management Overview

- Urinary tract infections are treated with targeted antibiotics based on culture sensitivity results.
- Nephrolithiasis management includes hydration, analgesics, medical expulsive therapy, or urologic intervention for large stones.
- Voiding dysfunction may respond to behavioral modification, pelvic floor therapy, or pharmacotherapy.
- Urology referral is indicated for recurrent infections, hematuria, obstructive symptoms, or suspected malignancy.

Chapter 242

Urinary Incontinence

Definition

Involuntary leakage urine affecting quality life stress urge mixed predominant types.

Pathophysiology

Differentiate stress leakage cough sneeze urge sudden strong desire leakage overflow retention prostate neurologic etiologies varying causes require management strategies lifestyle modifications therapeutic interventions surgical options.

Etiology

- Stress weakness women prostate enlargement
- Overactive bladder detrusor overactivity
- Neurogenic bladder spinal cord injury stroke Parkinson MS
- Overflow benign prostatic hyperplasia neurogenic
- Functional mobility limitation cognitive impairment environmental barriers

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluate causing bother embarrassment recurrent UTIs skin breakdown social isolation psychological distress
- Accompanied by other concerning signs

Related Symptoms

- Frequency urgency nocturia dysuria recurrent infections bedwetting dribbling reduced stream hesitancy post-void residual sensation

Note: Always consider serious underlying conditions when evaluating persistent urinary incontinence.

Self-Care / Home Management

- Drink adequate water throughout the day unless fluid restriction is medically indicated.
- Avoid bladder irritants including caffeine, alcohol, acidic foods, and artificial sweeteners.
- Practice good hygiene and urinate promptly when the urge arises to prevent urinary stasis.
- Apply a heating pad to the lower abdomen or back for comfort during urinary discomfort.
- Seek medical attention for fever, flank pain, visible blood in urine, or difficulty urinating.

Diagnostic Approach

- History covers urinary symptoms (frequency, urgency, dysuria, hesitancy, stream changes), fever, and flank pain.
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- Imaging (renal ultrasound, CT urogram) evaluates for stones, obstruction, structural abnormalities, or masses.
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Treatment / Management Overview

- Urinary tract infections are treated with targeted antibiotics based on culture sensitivity results.
- Nephrolithiasis management includes hydration, analgesics, medical expulsive therapy, or urologic intervention for large stones.
- Voiding dysfunction may respond to behavioral modification, pelvic floor therapy, or pharmacotherapy.
- Urology referral is indicated for recurrent infections, hematuria, obstructive symptoms, or suspected malignancy.

Chapter 243

Urinary Retention

Definition

Inability to partially or completely empty the bladder, classified as acute (sudden and painful) or chronic (gradual onset with less discomfort but significant residual urine volume).

Pathophysiology

Acute retention typically results from mechanical obstruction (BPH, urethral stricture, stone, clot) or neurological causes (spinal cord injury, post-operative). Chronic retention involves detrusor underactivity or bladder outlet obstruction with compensatory detrusor hypertrophy, leading to high post-void residuals, overflow incontinence, and upper tract damage risk.

Etiology

- Benign prostatic hyperplasia (BPH) — most common cause in men
- Urethral stricture
- Medication side effects (anticholinergics, antihistamines, decongestants)
- Post-operative (anesthesia, opioids, pelvic surgery)
- Neurological conditions (spinal cord injury, multiple sclerosis, diabetic neuropathy)
- Severe constipation (fecal impaction)
- Bladder stones or tumors
- Pelvic organ prolapse (cystocele) in women
- Prostate cancer or bladder cancer
- Phimosis or paraphimosis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Complete inability to urinate (emergency requiring catheterization), painful distension, or acute retention with fever
- Accompanied by other concerning signs

Related Symptoms

- Urinary frequency
- Hesitancy
- Weak urinary stream
- Overflow incontinence
- Abdominal distension

Note: Always consider serious underlying conditions when evaluating persistent urinary retention.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 244

Urine Odor

Definition

Unusual or strong smell of urine, which may be sweet, foul, fishy, or ammonia-like, reflecting dietary, metabolic, infectious, or medication-related causes.

Pathophysiology

Normal urine has a mild odor from byproducts of metabolism. Strong ammonia odor results from bacterial urea breakdown (UTI or prolonged urine stasis). Sweet or fruity odor suggests ketones (diabetic ketoacidosis, starvation) or maple syrup urine disease. Fishy odor indicates bacterial vaginosis or trichomonas (contaminating urine sample) or trimethylaminuria. Asparagus causes a characteristic sulfurous odor from asparagusic acid metabolism.

Etiology

- Urinary tract infection (ammonia or foul odor)
- Dehydration (concentrated urine with strong ammonia)
- Dietary: asparagus, garlic, onions, curry, coffee
- Diabetic ketoacidosis (sweet or fruity odor)
- Medications and supplements (B vitamins, sulfonamides)
- Liver failure (musty odor)
- Maple syrup urine disease (sweet, maple syrup odor in infants)
- Phenylketonuria (musty or mousy odor)
- Trimethylaminuria (fishy odor, rare metabolic disorder)
- Bacterial vaginosis or trichomonas (contaminating urine sample)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Urine odor with fever, flank pain, dysuria, blood in urine, or unusual odor persisting without clear dietary cause
- Accompanied by other concerning signs

Related Symptoms

- Frequent urination
- Painful urination
- Urinary urgency
- Dehydration
- Vaginal discharge

Note: Always consider serious underlying conditions when evaluating persistent urine odor.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 245

Vaginal Bleeding

245.1 Definition

Vaginal bleeding is bleeding from the vagina that may be normal (menstruation) or abnormal. Abnormal vaginal bleeding includes bleeding between periods, postmenopausal bleeding, heavy menstrual bleeding, or bleeding after intercourse.

245.2 Pathophysiology

Normal menstrual bleeding results from the shedding of the endometrial lining under the influence of declining progesterone. Abnormal bleeding can arise from hormonal imbalances (anovulation), structural abnormalities (fibroids, polyps), pregnancy complications, infection, or malignancy. Postmenopausal bleeding is caused by atrophic endometrium, hormone therapy, endometrial hyperplasia, or endometrial cancer.

245.3 Etiology

- Pregnancy-related: implantation bleeding, ectopic pregnancy, miscarriage, placental abruption
- Hormonal: anovulation, perimenopause, thyroid disorders, hyperprolactinemia
- Structural: uterine fibroids, endometrial polyps, cervical polyps, adenomyosis
- Infectious: cervicitis, endometritis, pelvic inflammatory disease
- Malignancy: cervical cancer, endometrial cancer, vaginal cancer
- Medications: anticoagulants, tamoxifen, hormonal contraceptives
- Trauma: sexual assault, foreign body, vaginal laceration
- Postmenopausal bleeding: atrophic vaginitis, endometrial atrophy, endometrial hyperplasia or carcinoma

245.4 Indications for Evaluation

- Bleeding between menstrual periods
- Heavy menstrual bleeding (soaking through pads/tampons hourly)
- Bleeding after intercourse (postcoital bleeding)
- Bleeding after menopause
- Bleeding during pregnancy
- Bleeding accompanied by pelvic pain or cramping
- Clots larger than 1 inch in diameter
- Signs of anemia: fatigue, lightheadedness, pallor

245.5 Related Symptoms

- Heavy menstrual bleeding (menorrhagia)
- Irregular periods
- Postmenopausal bleeding
- Pelvic pain
- Anemia
- Uterine fibroids
- Endometrial polyps
- Cervical cancer

245.6 Self-Care/Home Management

- Keep a menstrual diary to track bleeding patterns
- Use pads or tampons and change frequently
- Take over-the-counter pain relievers if needed
- Maintain iron-rich diet if bleeding is heavy
- Avoid aspirin (increases bleeding risk)
- Seek medical attention for severe or persistent bleeding

245.7 Diagnostic Approach

- Thorough history: menstrual pattern, pregnancy status, medications
- Pregnancy test
- Pelvic examination with speculum and bimanual exam
- Complete blood count for anemia
- Coagulation profile if bleeding disorder suspected
- Pelvic ultrasound (transvaginal) for structural evaluation
- Endometrial biopsy for postmenopausal bleeding or abnormal uterine bleeding
- STI testing

245.8 Treatment/Management Overview

- Treat underlying cause
- Hormonal therapy: combined oral contraceptives, progestins, GnRH agonists
- NSAIDs for heavy menstrual bleeding
- Tranexamic acid to reduce bleeding
- Surgical: hysteroscopy with polypectomy/myomectomy, endometrial ablation, hysterectomy
- Dilation and curettage (D&C) for acute heavy bleeding
- Iron supplementation for anemia

Chapter 246

Vaginal Discharge

Definition

Fluid exiting vagina varying amount color odor texture consistency normal physiological process abnormal indicating infection inflammation malignancy requiring evaluation.

Pathophysiology

Normal discharge clear white thin-smelly cyclic related estrogen levels ovulation pregnancy postpartum menses; abnormal discharge malodor yellow-green gray frothy thick clumpy bloody accompanied itching burning pain dyspareunia dysuria suggests bacterial vaginosis candidiasis trichomoniasis gonorrhea chlamydia cervicitis endometritis cervical cancer atrophic vaginitis foreign body retained tampon.

Etiology

- Bacterial vaginosis
- Yeast infections Candida
- Trichomoniasis sexually transmitted
- Cervicitis STDs
- Atrophic vaginitis postmenopausal

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent new onset malodor color change bloody associated pain itching burning requires gynecological examination culture testing appropriate management indicated
- Accompanied by other concerning signs

Related Symptoms

- Vaginal itching burning
- Pelvic pain abnormal bleeding
- Dyspareunia dysuria
- Lower abdominal pain
- Urinary frequency

Note: Always consider serious underlying conditions when evaluating persistent vaginal discharge.

Self-Care / Home Management

- Track symptoms including cycle dates, flow characteristics, and associated pain using a diary or app.
- Apply a heating pad to the lower abdomen for menstrual cramp relief.
- Practice good genital hygiene and avoid douching or scented products which can disrupt normal flora.
- Over-the-counter pain relievers (ibuprofen, naproxen) are effective for menstrual pain and can reduce heavy bleeding.
- Seek evaluation for abnormal bleeding, pelvic pain, unusual discharge, or breast changes.

Diagnostic Approach

- History includes menstrual history, sexual activity, contraceptive use, pregnancy status, and associated symptoms.
- Pelvic examination (speculum and bimanual) assesses the cervix, uterus, and adnexa for tenderness, masses, or discharge.
- Pregnancy testing is essential for any reproductive-age woman with abnormal bleeding or pelvic symptoms.
- Ultrasound (transabdominal or transvaginal) evaluates the endometrium, myometrium, ovaries, and adnexal structures.

Treatment / Management Overview

- Hormonal contraceptives (oral, injectable, intrauterine) are first-line treatment for many menstrual disorders.
- NSAIDs reduce prostaglandin-mediated pain and bleeding in dysmenorrhea and menorrhagia.
- Specific diagnoses require targeted therapy: antibiotics for pelvic infection, endometrial ablation for refractory bleeding, surgery for fibroids or endometriosis.
- Referral to gynecology is indicated for complex cases, infertility, or suspected malignancy.

Chapter 247

Vaginal Dryness

Definition

Insufficient vaginal lubrication causing discomfort, irritation, and pain during sexual activity or daily activities, most commonly due to decreased estrogen levels.

Pathophysiology

Estrogen maintains vaginal mucosal thickness, elasticity, and lubrication via cervical and Bartholin gland secretions. Estrogen decline (menopause, breastfeeding, medications) leads to mucosal atrophy, decreased blood flow, reduced secretions, and increased vaginal pH predisposing to infection. Vaginal dryness is symptomatic of the genitourinary syndrome of menopause (GSM).

Etiology

- Menopause (natural or surgical)
- Breastfeeding or postpartum period
- Medication effects (antihistamines, antidepressants, chemotherapy, tamoxifen)
- Sjogren syndrome (autoimmune)
- Radiation therapy to pelvis
- Hormonal contraception (some types)
- Perimenopause
- Oophorectomy (surgical removal of ovaries)
- Anxiety or inadequate arousal
- Chemotherapy or endocrine therapy for breast cancer

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Vaginal dryness with painful intercourse, itching, burning, or recurrent urinary tract infections
- Accompanied by other concerning signs

Related Symptoms

- Painful sex (dyspareunia)
- Vaginal itching
- Urinary frequency
- Recurrent UTIs
- Vaginal irritation

Note: Always consider serious underlying conditions when evaluating persistent vaginal dryness.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 248

Vaginal Irritation and Infection

Definition

Inflammation of the vaginal mucosa characterized by itching, burning, abnormal discharge, odor, and discomfort, most commonly caused by infectious or irritant triggers.

Pathophysiology

Vaginal ecosystem disruption alters normal lactobacillus-dominated flora, increasing pH and allowing pathogen overgrowth. Common pathogens: *Candida albicans* (fungal), *Gardnerella vaginalis* with other anaerobes (bacterial vaginosis), *Trichomonas vaginalis* (parasitic). Non-infectious causes include chemical irritants, allergic reactions, and atrophic changes.

Etiology

- Vulvovaginal candidiasis (yeast infection)
- Bacterial vaginosis
- Trichomoniasis (sexually transmitted)
- Chemical irritants (soaps, douches, detergents, spermicides)
- Atrophic vaginitis (menopause-related)
- Allergic reactions (latex, lubricants, contraceptive devices)
- Poor hygiene or excessive hygiene
- Tight non-breathable clothing
- Diabetes mellitus (increased glucose promotes yeast)
- Antibiotic use disrupting normal flora

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Vaginal irritation with abnormal discharge, fever, pelvic pain, or recurrent episodes
- Accompanied by other concerning signs

Related Symptoms

- Vaginal discharge
- Itching
- Dysuria
- Painful sex
- Abnormal odor

Note: Always consider serious underlying conditions when evaluating persistent vaginal irritation and infection.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 249

Vaginal Prolapse (Pelvic Organ Prolapse)

249.1 Definition

Vaginal prolapse is the descent of the vaginal walls or uterus into the vaginal canal. It occurs when the pelvic floor muscles and connective tissue are weakened or damaged. Types include cystocele (bladder), rectocele (rectum), and uterine prolapse.

249.2 Pathophysiology

The pelvic floor is a hammock-like structure of muscles and fascia that supports the pelvic organs. Weakening from childbirth, age, or chronic straining allows the bladder, rectum, uterus, or vaginal apex to descend. The prolapse is graded I-IV based on descent relative to the hymen. Symptoms result from the bulging mass and associated organ dysfunction.

249.3 Etiology

- Vaginal childbirth (most common risk factor)
- Chronic constipation and straining
- Chronic cough (COPD, smoking)
- Obesity
- Heavy lifting
- Aging and menopause (estrogen loss weakens tissues)
- Connective tissue disorders
- Prior pelvic surgery (hysterectomy)

249.4 Indications for Evaluation

- Sensation of a bulge or pressure in the vagina
- Feeling of something falling out of the vagina
- Lower back pain or pelvic pressure
- Urinary symptoms: incontinence, frequency, urgency, incomplete emptying
- Difficulty with bowel movements (need to splint the vagina)
- Sexual discomfort
- Vaginal discharge or bleeding
- Symptoms worsen with standing, lifting, or prolonged activity and improve when lying down

249.5 Related Symptoms

- Urinary incontinence (stress or urge)
- Overactive bladder
- Frequent urination
- Constipation
- Rectocele (difficulty with bowel movements)
- Cystocele (bladder prolapse)

249.6 Self-Care/Home Management

- Pelvic floor exercises (Kegel exercises) daily
- Maintain a healthy weight
- Avoid heavy lifting and straining
- Treat constipation with high-fiber diet and hydration
- Use vaginal estrogen cream if postmenopausal
- Consider a vaginal pessary for support
- Quit smoking to reduce chronic cough

249.7 Diagnostic Approach

- Pelvic examination with Valsalva to assess descent
- Pelvic organ prolapse quantification (POP-Q) staging
- Urodynamic testing if urinary symptoms are present
- Ultrasound or MRI for complex cases

249.8 Treatment/Management Overview

- Pelvic floor physical therapy
- Pessary fitting (vaginal support device)
- Vaginal estrogen therapy
- Surgery: sacrocolpopexy, uterosacral ligament suspension, anterior/posterior repair, colpocleisis for severe cases

Chapter 250

Vertigo

Definition

Spinning sensation self environment appears moving despite stationary position.

Pathophysiology

Peripheral vestibular BPPV labyrinthitis Meniere versus central stroke MS cerebellar etiatis distinguished by associated neurologic findings.

Etiology

- BPPV (most common)
- Vestibular neuritis
- Meniere disease
- Stroke/TIA
- Multiple sclerosis

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than First episode or with neurologic deficits warrants immediate medical evaluation
- Accompanied by other concerning signs

Related Symptoms

- Nausea/vomiting
- Hearing loss/tinnitus
- Nystagmus
- Ataxia
- Double vision

Note: Always consider serious underlying conditions when evaluating persistent vertigo.

Self-Care / Home Management

- Ensure a safe environment to prevent injury during episodes (remove hazards, avoid driving or operating machinery).
- Keep a symptom diary documenting triggers, duration, frequency, and associated features.
- Practice stress reduction techniques (deep breathing, meditation, adequate sleep) as stress often exacerbates neurological symptoms.
- Avoid alcohol, caffeine, and recreational drugs which can worsen many neurological conditions.
- Seek immediate evaluation for sudden-onset, severe, or progressive neurological symptoms.

Diagnostic Approach

- Detailed neurological history includes onset pattern, progression, triggers, associated symptoms, and functional impact.
- Comprehensive neurological examination assesses mental status, cranial nerves, motor/sensory function, reflexes, coordination, and gait.
- Neuroimaging (CT, MRI) is often indicated to evaluate for structural, vascular, or demyelinating causes.
- Specialized testing (EEG for seizures, nerve conduction studies for neuropathy, lumbar puncture for meningitis) is guided by clinical suspicion.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic pharmacotherapy may include analgesics, anticonvulsants, antidepressants, or disease-modifying agents as appropriate.
- Physical, occupational, and speech therapy play important roles in functional recovery and rehabilitation.
- Referral to neurology is indicated for unexplained, progressive, or treatment-refractory neurological symptoms.

Chapter 251

Vision Blurred

Definition

Loss sharpness objects appear out focus transient persistent.

Pathophysiology

Refractive error ocular media opacity retinal optic nerve neurological causes all possible ranging benign vision correction needs emergent neurological ophthalmological emergencies requiring prompt differentiation.

Etiology

- Refractive error myopia hyperopia astigmatism presbyopia
- Cataracts
- Glaucoma
- Diabetic retinopathy
- Migraine aura

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluate persistent sudden change visual field defects floaters flashes pain associated redness discharge photophobia double vision diplopia
- Accompanied by other concerning signs

Related Symptoms

- Eye pain headaches dizziness balance problems speech difficulty facial weakness arm numbness drooping mouth sudden onset suggest stroke TIA emergency evaluation required

Note: Always consider serious underlying conditions when evaluating persistent vision blurred.

Self-Care / Home Management

- Rest the eyes using the 20-20-20 rule: every 20 minutes, look at something 20 feet away for 20 seconds.
- Apply artificial tears or lubricating eye drops for dryness — avoid drops that claim to 'get the red out' which can cause rebound redness.
- Use a warm compress for blepharitis or styes; use a cold compress for allergic conjunctivitis or swelling.
- Remove contact lenses immediately and switch to glasses if eye irritation develops.
- Seek urgent evaluation for eye pain, vision changes, light sensitivity, or chemical exposure.

Diagnostic Approach

- History covers onset, duration, visual changes, pain character, discharge, contact lens use, and trauma.
- Visual acuity testing and slit-lamp examination are essential components of the eye evaluation.
- Fluorescein staining identifies corneal abrasions or ulcers; tonometry measures intraocular pressure.
- Fundoscopic examination evaluates the retina, optic disc, and vasculature for pathology.

Treatment / Management Overview

- Infectious conjunctivitis is treated with appropriate topical antibiotics (bacterial) or antihistamines (allergic).
- Dry eye disease management includes artificial tears, punctal plugs, topical cyclosporine, and lifestyle modifications.
- Corneal abrasions heal with antibiotic ointment and pressure patching; deep ulcers require ophthalmology referral.
- Sudden vision loss, retinal detachment, or acute angle-closure glaucoma are ophthalmologic emergencies requiring immediate specialist care.

Chapter 252

Vomiting

Definition

Forceful expulsion stomach contents coordinated brainstem reflex mechanism involves retching abdominal contraction.

Pathophysiology

Emesis protective eliminating toxic substances leads dehydration electrolyte aspiration distinguishes regurgitation heartburn.

Etiology

- Acute gastroenteritis viral bacterial food poisoning
- Intestinal obstruction volvulus
- Increased intracranial pressure tumor meningitis
- Metabolic disorders hypercalcemia ketoacidosis
- Postoperative anesthesia recovery pregnancy

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Projectile bilious coffee-ground material obstipation dehydration hemodynamic instability urgent evaluation
- Accompanied by other concerning signs

Related Symptoms

- Dehydration electrolyte imbalance esophagitis tear aspiration pneumonia weight loss chest pain upper GI bleeding

Note: Always consider serious underlying conditions when evaluating persistent vomiting.

Self-Care / Home Management

- Maintain hydration with clear fluids (water, oral rehydration solutions, clear broths) in small, frequent amounts.
- Eat bland, easy-to-digest foods (bananas, rice, applesauce, toast) and avoid fatty, spicy, or dairy-heavy meals during recovery.
- Rest the digestive system by eating smaller meals more frequently rather than large meals.
- Over-the-counter remedies may help: antacids for heartburn, loperamide for diarrhea, fiber supplements for constipation.
- Seek medical attention if symptoms persist beyond 48 hours, or if accompanied by severe pain, fever, or bloody stools.

Diagnostic Approach

- History covers onset, duration, stool frequency/character, associated symptoms (pain, fever, nausea), dietary triggers, and travel history.
- Physical examination includes abdominal auscultation, palpation for tenderness/masses, and assessment of hydration status.
- Stool studies (culture, ova/parasites, occult blood, fecal calprotectin) help identify infectious or inflammatory causes.
- Endoscopy or colonoscopy is indicated for chronic symptoms, unexplained weight loss, or concerning features.

Treatment / Management Overview

- Supportive care with hydration and dietary modification is first-line for most acute gastrointestinal symptoms.
- Probiotics may help restore gut flora balance after infectious diarrhea or antibiotic use.
- Specific pharmacotherapy depends on diagnosis: antiemetics for vomiting, antispasmodics for cramping, antibiotics for bacterial infection.
- Chronic conditions (IBS, IBD, GERD) require specialist management with tailored medical and lifestyle interventions.

Chapter 253

Vomiting and Nausea

253.1 Definition

Nausea is the unpleasant sensation of the urge to vomit, while vomiting (emesis) is the forceful expulsion of gastric contents through the mouth. Both are common symptoms with many causes, ranging from benign to life-threatening.

253.2 Pathophysiology

The vomiting reflex is coordinated by the vomiting center in the medulla oblongata, which receives input from four sources: the chemoreceptor trigger zone (CTZ) in the area postrema (sensitive to toxins and drugs), the vestibular system (motion sickness), the gastrointestinal tract (via vagal afferents), and higher cortical centers (emotions, smells, pain). Activation of the vomiting center triggers a coordinated sequence: salivation, palatal elevation, diaphragmatic contraction, and gastric relaxation.

253.3 Etiology

- Gastrointestinal: gastroenteritis, gastritis, peptic ulcer, pancreatitis, cholecystitis, appendicitis, bowel obstruction
- Pregnancy: morning sickness (nausea gravidarum), hyperemesis gravidarum
- Medications: chemotherapy, antibiotics, opioids, NSAIDs, anticonvulsants
- Motion sickness
- Inner ear disorders: labyrinthitis, Meniere disease, vestibular neuritis
- Migraine headache
- Metabolic: uremia, diabetic ketoacidosis, adrenal insufficiency, hypercalcemia
- Infections: viral, bacterial, parasitic (not just GI)
- Central nervous system: increased intracranial pressure, meningitis, tumor, concussion
- Psychiatric: anxiety disorders, bulimia nervosa
- Postoperative nausea and vomiting

253.4 Indications for Evaluation

- Nausea with or without vomiting
- Duration and frequency of episodes
- Vomiting blood (hematemesis) - emergency
- Bilious (green) or fecal-colored vomitus - suggests obstruction
- Abdominal pain, fever, or diarrhea
- Headache, stiff neck, or photophobia
- History of head trauma
- Inability to keep fluids down (dehydration risk)
- Weight loss

253.5 Related Symptoms

- Abdominal pain
- Diarrhea
- Dehydration
- Dizziness (with vestibular causes)
- Headache (migraine)
- Weight loss

253.6 Self-Care/Home Management

- Eat small, frequent meals of bland foods (BRAT diet: bananas, rice, applesauce, toast)
- Avoid strong odors, spicy or greasy foods
- Sip clear fluids (water, clear broth, electrolyte solutions) frequently
- Ginger (tea, candied, or supplements) can reduce nausea
- Peppermint aromatherapy
- Acupressure wristbands (P6 point)
- Rest and avoid sudden head movements

253.7 Diagnostic Approach

- History and physical examination
- Pregnancy test in women of reproductive age
- Basic metabolic panel, liver enzymes, amylase, lipase
- Abdominal imaging (X-ray, CT, ultrasound) based on suspected cause
- Upper endoscopy if upper GI source suspected

253.8 Treatment/Management Overview

- Antiemetic medications: ondansetron, promethazine, metoclopramide, prochlorperazine
- IV fluids for dehydration
- Treat underlying cause (antibiotics for infection, migraine therapy, chemotherapy antiemetics)
- Electrolyte replacement
- Nutritional support for prolonged symptoms

Chapter 254

Warning Signs During Pregnancy

Definition

A set of signs and symptoms during pregnancy that may indicate serious complications requiring urgent medical evaluation, including preeclampsia, placental abruption, preterm labor, or infection.

Pathophysiology

Pregnancy induces significant physiological changes in the cardiovascular, renal, respiratory, and immune systems. Warning signs reflect pathological decompensation: hypertension and proteinuria indicate preeclampsia (endothelial dysfunction, placental ischemia); vaginal bleeding suggests placental abruption or placenta previa; decreased fetal movement indicates fetal distress; severe abdominal pain may signal placental abruption, uterine rupture, or appendicitis; fever suggests chorioamnionitis or other infection.

Etiology

- Preeclampsia (hypertension, proteinuria, headache, visual changes, RUQ pain)
- Placental abruption (vaginal bleeding, abdominal pain, uterine tenderness)
- Placenta previa (painless vaginal bleeding in third trimester)
- Preterm labor (regular contractions, pelvic pressure, back pain before 37 weeks)
- Chorioamnionitis (fever, uterine tenderness, foul amniotic fluid)
- Gestational diabetes complications (polyuria, polydipsia, blurred vision)
- Deep vein thrombosis (unilateral leg swelling, pain, warmth)
- Pulmonary embolism (sudden shortness of breath, chest pain, hemoptysis)
- Miscarriage (vaginal bleeding, cramping in first trimester)
- Ectopic pregnancy (sharp abdominal/pelvic pain, shoulder pain, fainting)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Any vaginal bleeding in pregnancy, severe headache with visual changes, sudden swelling of face/hands, decreased fetal movement, rupture of membranes, or signs of preterm labor before 37 weeks
- Accompanied by other concerning signs

Related Symptoms

- Abdominal pain
- Vaginal bleeding
- Headache
- Swelling
- Fever

Note: Always consider serious underlying conditions when evaluating persistent warning signs during pregnancy.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 255

Watery Eyes (Epiphora)

Definition

Excessive production of tears or impaired drainage of tears from the eye, causing tears to spill onto the face.

Pathophysiology

Epiphora results from either tear overproduction (reflex tearing from dry eye, irritation, or inflammation) or impaired tear drainage (nasolacrimal duct obstruction, punctal stenosis, eyelid malposition). Reflex tearing from dry eye is the most common mechanism: ocular surface dryness triggers a neural reflex for excessive tear production, which overwhelms normal drainage.

Etiology

- Dry eye syndrome (reflex tearing)
- Nasolacrimal duct obstruction (congenital or acquired)
- Punctal stenosis or occlusion
- Ectropion or entropion (eyelid malposition)
- Allergic conjunctivitis
- Viral conjunctivitis
- Corneal foreign body or abrasion
- Eyelash irritation (trichiasis)
- Sinusitis or nasal congestion (impaired drainage)
- Medication side effects (some glaucoma drops, chemotherapy)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Persistent watery eyes with pain, discharge, vision changes, or recurrent eye infections
- Accompanied by other concerning signs

Related Symptoms

- Red eye
- Eye pain
- Blurred vision
- Eye discharge
- Nasal congestion

Note: Always consider serious underlying conditions when evaluating persistent watery eyes.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 256

Weakness

Definition

Decreased muscle power strength perform voluntary movements distinguish fatigue subjective tiredness objective reduced force generation measured medical examination physical therapy assessment

Pathophysiology

Differentiate proximal distal symmetric asymmetric pattern determines differential diagnosis neuromuscular endplate junction disorders peripheral nerve root spinal cord neuromuscular junction muscular metabolism metabolic derangement electrolyte imbalance central nervous system lesions neuromuscular transmission defects

Etiology

- Neuromuscular myasthenia gravis Lambert-Eaton syndrome
- Myopathies inflammatory polymyositis inclusion body genetic muscular dystrophy
- Peripheral neuropathies diabetic alcohol compressive radiculopathies spinal stenosis
- Central processes stroke tumor neurodegeneration ALS Parkinson multiple sclerosis
- Metabolic electrolyte abnormalities hypokalemia hyperkalemia hypocalcemia thyroid dysfunction anemia

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Progressively unexplainable weakness walking difficulty breathing support posture require neurological evaluation EMG serum CK autoimmune panel imaging indicated urgent assessment
- Accompanied by other concerning signs

Related Symptoms

- Fatigue exertion pattern muscle atrophy fasciculations reflexes diminished gait clumsiness balance problems falls

Note: Always consider serious underlying conditions when evaluating persistent weakness

Self-Care / Home Management

- Prioritize adequate rest and sleep to support the body's healing and recovery processes.
- Maintain a balanced diet with regular meals and stay well-hydrated.
- Monitor symptoms and track any changes in severity, frequency, or associated features.
- Avoid overexertion and gradually increase activity levels as tolerated.
- Seek medical evaluation if symptoms persist, worsen, or interfere significantly with daily function.

Diagnostic Approach

- Comprehensive history and physical examination guide the diagnostic workup based on the specific symptom presentation.
- Laboratory testing (CBC, metabolic panel, inflammatory markers) helps screen for underlying systemic conditions.
- Imaging studies are tailored to the suspected anatomic region and pathology.
- Referral to appropriate specialists is guided by the differential diagnosis established through initial evaluation.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic management may include pharmacotherapy, lifestyle modifications, and supportive care.
- Multidisciplinary care is often beneficial for complex or chronic symptoms.
- Close follow-up ensures treatment effectiveness and allows adjustment of the management plan as needed.

Chapter 257

Weight Gain (Menopausal Weight Change)

Definition

Unintentional increase in body weight, typically with a shift toward abdominal (visceral) fat distribution, occurring during the perimenopausal and postmenopausal transition.

Pathophysiology

Menopausal weight gain results from several mechanisms: declining oestrogen reduces basal metabolic rate by approximately 50-100 kcal per decade; oestrogen deficiency increases visceral adipose tissue deposition by altering lipoprotein lipase activity; insulin sensitivity decreases, promoting fat storage; loss of lean muscle mass (sarcopenia) further reduces resting energy expenditure. Sleep disruption elevates cortisol and ghrelin while reducing leptin, increasing appetite and cravings. Fat distribution shifts from gynaecoid (hip/thigh) to android (abdominal) pattern, increasing cardiometabolic risk.

Etiology

- Oestrogen decline (reduced metabolic rate, increased visceral fat)
- Age-related sarcopenia (loss of muscle mass)
- Insulin resistance and glucose intolerance
- Sleep disruption (altered ghrelin, leptin, and cortisol)
- Thyroid dysfunction (hypothyroidism)
- Medication side effects (SSRIs, antipsychotics, beta-blockers, corticosteroids)
- Reduced physical activity (joint pain, fatigue)
- Emotional eating (mood swings, depression, stress)
- Cushing syndrome (cortisol excess, rare)
- Genetic predisposition to obesity

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Rapid or excessive weight gain (more than 5 kg in 6 months), weight gain with fatigue, cold intolerance, or new onset of metabolic syndrome features
- Accompanied by other concerning signs

Related Symptoms

- Fatigue
- Joint pain
- Bloating
- Shortness of breath
- Mood swings

Note: Always consider serious underlying conditions when evaluating persistent weight gain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 258

Weight Loss

Definition

Unintentional decrease in body weight exceeding caloric intake; clinically significant >5

Pathophysiology

Results from decreased appetite malabsorption increased metabolic demand combinations thereof requires systematic evaluation after excluding physiological weight loss.

Etiology

- Depression mood disorders most common
- Malignancy GI lung lymphoma
- Hyperthyroidism
- Chronic infection TB HIV
- Malabsorption syndromes

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than Evaluate promptly especially elderly unexplained
- Accompanied by other concerning signs

Related Symptoms

- Anorexia early satiety abdominal pain night sweats fatigue

Note: Always consider serious underlying conditions when evaluating persistent weight loss.

Self-Care / Home Management

- Prioritize adequate rest and sleep to support the body's healing and recovery processes.
- Maintain a balanced diet with regular meals and stay well-hydrated.
- Monitor symptoms and track any changes in severity, frequency, or associated features.
- Avoid overexertion and gradually increase activity levels as tolerated.
- Seek medical evaluation if symptoms persist, worsen, or interfere significantly with daily function.

Diagnostic Approach

- Comprehensive history and physical examination guide the diagnostic workup based on the specific symptom presentation.
- Laboratory testing (CBC, metabolic panel, inflammatory markers) helps screen for underlying systemic conditions.
- Imaging studies are tailored to the suspected anatomic region and pathology.
- Referral to appropriate specialists is guided by the differential diagnosis established through initial evaluation.

Treatment / Management Overview

- Treatment is directed at the underlying cause identified through diagnostic evaluation.
- Symptomatic management may include pharmacotherapy, lifestyle modifications, and supportive care.
- Multidisciplinary care is often beneficial for complex or chronic symptoms.
- Close follow-up ensures treatment effectiveness and allows adjustment of the management plan as needed.

Chapter 259

Wheezing

Definition

High-pitched whistling sound during breathing typically on exhalation indicating narrowed airways.

Pathophysiology

Caused by turbulent airflow through constricted bronchi; varies from intermittent to continuous monophonic vs polyphonic helps localize obstruction.

Etiology

- Asthma
- Chronic obstructive pulmonary disease (COPD)
- Bronchitis
- Heart failure (cardiac asthma)
- Foreign body aspiration

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Symptoms lasting more than persistent or accompanied by respiratory distress requires prompt evaluation
- Accompanied by other concerning signs

Related Symptoms

- Chest tightness
- Shortness of breath
- Cough
- Use of accessory muscles
- Prolonged expiration

Note: Always consider serious underlying conditions when evaluating persistent wheezing.

Self-Care / Home Management

- Maintain adequate hydration with water, warm tea, or broth to thin secretions.
- Use a humidifier or steam inhalation to soothe irritated airways and loosen mucus.
- Rest and avoid exposure to smoke, dust, and other respiratory irritants.
- Over-the-counter remedies such as saline nasal spray, cough drops, or honey (for adults) may provide symptomatic relief.
- Monitor for warning signs including high fever, difficulty breathing, chest pain, or worsening symptoms.

Diagnostic Approach

- History focuses on onset, duration, character of symptoms (productive vs dry cough, nasal discharge color), fever pattern, and exposure history.
- Physical examination includes vital signs, lung auscultation, nasal inspection, and oropharyngeal examination.
- Targeted testing may include rapid antigen/ PCR for respiratory viruses, chest X-ray, pulse oximetry, and sputum culture.
- Allergy testing or pulmonary function tests are reserved for chronic or recurrent cases.

Treatment / Management Overview

- Symptomatic management includes antitussives, expectorants, decongestants, antihistamines, and saline irrigation as appropriate.
- Bacterial infections require appropriate antibiotic therapy based on suspected pathogen and local resistance patterns.
- Inhaled bronchodilators or corticosteroids are used for reactive airway disease or asthma exacerbations.
- Referral to pulmonology or ENT is indicated for chronic, recurrent, or treatment-refractory cases.

Chapter 260

White Tongue

Definition

White discoloration or coating on the dorsal surface of the tongue, which may be patchy or diffuse, resulting from accumulation of debris, fungal overgrowth, or epithelial abnormalities.

Pathophysiology

White tongue most commonly results from oral thrush (*Candida albicans* overgrowth), which penetrates the superficial epithelial layer producing creamy white plaques that can be scraped off. Oral hairy leukoplakia (EBV-associated) presents with adherent white corrugated patches on the lateral tongue in immunocompromised individuals. Leukoplakia (premalignant) and lichen planus produce adherent white plaques that cannot be scraped off.

Etiology

- Oral thrush (candidiasis) — most common
- Poor oral hygiene with bacterial/food debris coating
- Oral lichen planus (reticular form)
- Leukoplakia (premalignant white patch)
- Oral hairy leukoplakia (EBV, HIV-associated)
- Dehydration or dry mouth (xerostomia)
- Smoking or tobacco use (smoker's keratosis)
- Geographic tongue (benign migratory glossitis)
- Mouth breathing (drying of tongue surface)
- Autoimmune conditions (lupus, Sjogren syndrome)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- White patch on tongue that does not scrape off, persists more than 2 weeks, or is associated with pain, bleeding, or difficulty swallowing
- Accompanied by other concerning signs

Related Symptoms

- Bad breath
- Dry mouth
- Oral thrush
- Taste changes
- Sore throat

Note: Always consider serious underlying conditions when evaluating persistent white tongue.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

- A thorough history and physical examination are the first steps in evaluation.
- Laboratory tests (blood work, urinalysis) may be ordered based on clinical suspicion.
- Imaging studies (X-ray, ultrasound, CT, MRI) may be indicated when structural pathology is suspected.
- Specialist referral is arranged when the diagnosis remains uncertain or requires specific expertise.

Treatment / Management Overview

- Treatment targets the underlying cause identified through diagnostic evaluation.
- Supportive care and symptom management are provided while awaiting definitive therapy.
- Medications, lifestyle modifications, and physical therapies may be prescribed as appropriate.
- Follow-up ensures treatment effectiveness and allows timely adjustments to the care plan.

Chapter 261

Wound Care (Wound Management)

261.1 Definition

Wound care involves the assessment, cleaning, and management of skin wounds to prevent infection and promote healing. Wounds range from minor cuts and abrasions to complex surgical wounds and chronic ulcers.

261.2 Pathophysiology

Wound healing occurs in four phases: hemostasis (clot formation), inflammation (immune response), proliferation (new tissue formation), and remodeling (scar maturation). Proper wound care optimizes these phases by keeping the wound clean, moist, and protected. Factors that impair healing include infection, ischemia, foreign bodies, malnutrition, and chronic diseases.

261.3 Etiology

- Acute wounds: cuts, lacerations, abrasions, punctures, surgical incisions
- Chronic wounds: pressure sores, diabetic foot ulcers, venous stasis ulcers, arterial ulcers
- Thermal wounds: burns, frostbite
- Traumatic wounds: avulsions, degloving injuries, crush injuries
- Infected wounds: abscesses, cellulitis with drainage

261.4 Indications for Evaluation

- Any break in the skin requiring cleaning and protection
- Wounds with active bleeding that requires control
- Wounds with dirt, debris, or foreign material
- Signs of infection: redness, warmth, swelling, pus, red streaks
- Chronic or non-healing wounds
- Surgical wounds with drains or staples

261.5 Related Symptoms

- Scar formation
- Cellulitis (wound infection)
- Tetanus (risk with contaminated wounds)
- Abscess formation
- Impetigo (superficial infection)

261.6 Self-Care/Home Management

- Clean the wound gently with mild soap and water or saline
- Remove debris with tweezers cleaned with alcohol
- Apply antibiotic ointment and cover with sterile bandage
- Change dressing daily or when wet/soiled
- Keep the wound moist but not wet
- Do not pick at scabs
- Monitor for signs of infection

261.7 Diagnostic Approach

- Clinical assessment of wound size, depth, and characteristics
- Wound culture if infection suspected
- X-ray to rule out foreign body or underlying fracture
- Ankle-brachial index for leg wounds to assess arterial supply

261.8 Treatment/Management Overview

- Cleaning and debridement of devitalized tissue
- Appropriate wound dressing selection (hydrocolloid, foam, alginate)
- Negative pressure wound therapy for complex wounds
- Antibiotics for infected wounds
- Tetanus prophylaxis (tetanus toxoid or immune globulin)
- Surgical closure for appropriate wounds
- Management of underlying cause (offloading for pressure sores, compression for venous ulcers)

Chapter 262

Wrist Injury

262.1 Definition

Wrist injuries involve damage to the bones, ligaments, tendons, or nerves of the wrist joint. The wrist is a complex joint composed of the radius, ulna, and eight carpal bones. Common injuries include fractures, sprains, and tendonitis.

262.2 Pathophysiology

The wrist absorbs and transmits forces from the hand to the forearm. Fall onto an outstretched hand (FOOSH) is the classic mechanism of injury, causing fractures (distal radius, scaphoid) or ligament sprains. The scaphoid bone has a tenuous blood supply, and fractures can lead to avascular necrosis. Repetitive motions cause tendonitis (De Quervain tenosynovitis) or carpal tunnel syndrome from nerve compression.

262.3 Etiology

- Fall onto an outstretched hand (FOOSH): most common mechanism
- Sports injuries: gymnastics, skating, football, basketball
- Direct blow to the wrist
- Repetitive motion: typing, assembly work, weightlifting
- Osteoporosis (increases fracture risk in older adults)
- Motor vehicle accidents

262.4 Indications for Evaluation

- Pain, swelling, and tenderness around the wrist
- Decreased range of motion
- Bruising or deformity

- Numbness or tingling in the hand (possible nerve involvement)
- Weakness in grip

262.5 Related Symptoms

- Wrist fracture (distal radius, scaphoid)
- Carpal tunnel syndrome
- De Quervain tenosynovitis
- Wrist sprain
- Arthritis of the wrist

262.6 Self-Care/Home Management

- Apply ice and elevate the wrist
- Immobilize with a splint or brace
- Take OTC pain relievers
- Rest and avoid activities that worsen pain
- Be cautious of scaphoid fractures - any wrist pain in the anatomical snuffbox warrants evaluation

262.7 Diagnostic Approach

- Physical examination: inspection, palpation, range of motion, neurovascular assessment
- X-ray: standard wrist views and scaphoid view

262.8 Treatment/Management Overview

- Fracture: closed reduction and casting vs. open reduction with internal fixation
- Ligament injury: immobilization vs. surgical repair
- Tendonitis: rest, ice, NSAIDs, physical therapy, corticosteroid injection

Chapter 263

Wrist Pain

Definition

Pain or discomfort localized to the wrist joint and surrounding structures, commonly resulting from overuse, trauma, or inflammatory conditions.

Pathophysiology

The wrist's complex anatomy of eight carpal bones, multiple ligaments, and flexor/extensor tendons makes it susceptible to injury. Pain localizes anatomically: radial side (De Quervain tenosynovitis, scaphoid fracture, thumb arthritis), ulnar side (TFCC tear, extensor carpi ulnaris tendinopathy), dorsal (ganglion cyst, intersection syndrome), volar (carpal tunnel syndrome).

Etiology

- De Quervain tenosynovitis
- Carpal tunnel syndrome
- Wrist sprain or fracture (scaphoid, distal radius)
- Osteoarthritis or post-traumatic arthritis
- Ganglion cyst
- Rheumatoid arthritis
- Gout or pseudogout
- Ligamentous injury (TFCC tear)
- Tendinopathy (extensor carpi radialis, flexor carpi ulnaris)
- Kienböck disease (avascular necrosis of lunate)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Wrist pain after injury, inability to move the wrist, severe swelling, numbness in fingers, or fever
- Accompanied by other concerning signs

Related Symptoms

- Hand numbness
- Finger pain
- Swelling
- Weak grip
- Snapping sensation

Note: Always consider serious underlying conditions when evaluating persistent wrist pain.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

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Chapter 264

Yellow Tongue

Definition

Yellowish discoloration of the dorsal tongue surface, usually resulting from accumulation of bacteria, debris, or staining on elongated filiform papillae.

Pathophysiology

Yellow tongue typically arises from overgrowth of chromogenic bacteria or fungi (often *Candida*) on the tongue dorsum, especially when filiform papillae are elongated (hairy tongue). Elongation occurs when normal desquamation is impaired by poor oral hygiene, dry mouth, soft diet, or medications. The yellow color may also result from staining by food, tobacco, or mouthwashes containing oxidizing agents.

Etiology

- Poor oral hygiene (bacterial overgrowth)
- Dry mouth (xerostomia)
- Dehydration
- Smoking or tobacco use
- Mouthwashes containing peroxide or witch hazel
- Oral thrush (*Candida* overgrowth, can appear yellow)
- Coffee, tea, or food staining
- Antibiotic use (altered oral flora)
- Fever or illness (reduced oral clearance)
- Jaundice (rare — sclera also yellow, bilirubin elevated)

Indications for Evaluation

Seek care if:

- Severe or worsening symptoms
- Yellow tongue persisting more than 2 weeks despite good oral hygiene, or with jaundice, pain, or difficulty eating
- Accompanied by other concerning signs

Related Symptoms

- Bad breath
- White tongue
- Dry mouth
- Taste changes
- Oral thrush

Note: Always consider serious underlying conditions when evaluating persistent yellow tongue.

Self-Care / Home Management

- Rest and avoid activities that may aggravate the condition.
- Maintain adequate hydration and a balanced diet.
- Over-the-counter remedies may provide symptomatic relief (consult a pharmacist or doctor).
- Monitor symptoms and seek professional advice if they persist or worsen.
- Avoid self-medicating with prescription medications without medical guidance.

Diagnostic Approach

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